ARKITE

Software Manual

Version 2026.1

About this document

Purpose of the document

This document aims to enable both new and experienced users of the Arkite software to translate their standard operating procedures into a project and as such configure their workstation. The manual contains an overview of all basic and advanced software features.

Note

- *For instructions on installing and mounting the sensor, please refer to the **Arkite hardware installation guide**.*
- *For instructions on installing the Arkite software, please refer to the **Arkite software installation guide**.*
- *This document describes the features that are available in the 2026.1 version of the Arkite software.*

Table of Contents

1. About the Arkite software 7

1.1 Purpose of the Arkite software 7
1.2 How does the Arkite 3D sensor work? (Validate and Vision license only) 8
1.3 How does the Arkite Vision sensor work? (Vision license only) 8
1.4 Installing the Arkite software 9
1.5 Starting the Arkite software for the first time 9
1.6 Arkite User Interface Overview 15
1.7 Help and Support 16

2. Calibration 19

2.1 About calibration 19
2.2 Calibrating your workstation 20

2.2.1 Calibrating a workstation with Arkite 3D sensor using the calibration wizard (Validate and Vision license only) 20
2.2.2 Calibrating a workstation with Arkite Vision sensor (Vision license only) 25
2.2.3 Calibrating a workstation with Arkite RGB sensor 27
2.2.4 Calibrating a workstation without sensor (Guide license) 30

2.3 Other calibration options 31
2.4 Finetuning Calibration 32

3. Projects 33

3.1 About projects 33
3.2 Creating a project 33
3.3 Managing projects 34
3.4 Versioning 35

3.4.1 About versioning 35
3.4.2 Versioning settings 38
3.4.3 Creating a new project version 39
3.4.4 Approving or rejecting a project 41
3.4.5 Version history 44

3.5 Working on projects with multiple authors 47
3.6 Switching workstations 52

4. Jobs 55

4.1 About jobs and job steps 55
4.2 Configuring jobs 56
4.3 Configuring steps 61

4.3.1 Adding steps 61
4.3.2 Setting up your step structure 61
4.3.3 Step types 70
4.3.4 Testing your job 80
4.3.5 Job selection and job execution 81

4.4 Managing visualizations for your job steps 82

4.4.1 Visualization types 83
4.4.2 Visualization properties 84
4.4.3 Adding visualizations 108
4.4.4 Editing visualizations 110

4.4.5 Managing visualization or DMX light colors 112
4.4.6 Managing text and image instructions 116
4.5 Working with conditions 118
5. Detections 126
5.1 About detections 126
5.2 Project-Wide detections versus Job-Specific detections 126
5.3 Detection types 126
5.4 Detection states 127
5.5 Adding detections 129
5.6 Teaching and finetuning detections (Validate and Vision license only) 134
5.6.1 Optimize your sensor stream view 134
5.6.2 Teaching detections 135
5.6.3 Test a detection 148
5.6.4 Reteach a detection 148
5.6.5 Finetune a detection 149
5.6.6 Managing visualizations for the detection 152
5.7 Managing detections 158
5.8 Managing materials 159
5.9 Working with smart tools 161
5.10 Working with reflective tape (Validate and Vision license only) 166
5.11 Working with stacked picking bins (Validate and Vision license only) 167
5.12 Working with mobile picking racks (Validate and Vision license only) 168
5.13 CAD/CAM detections (Validate and Vision license only) 169
5.13.1 About CAD/CAM detections 169
5.13.2 Adding CAD/CAM detections 169
5.13.3 Example 1: no transformation 170
5.13.4 Example 2: rotate and translate 171
6. Processes 173
6.1 About processes 173
6.2 Generating Job Selection processes 173
6.3 Adding a new process 175
6.4 Overview of process start triggers 176
6.4.1 Alarm Clock 176
6.4.2 Project Loaded 177
6.4.3 Receive Communication 177
6.4.4 Timer 178
6.4.5 Watchdog 179
6.4.6 Variable Changed 179
6.5 Configuring process steps 182
6.5.1 Process step types 182
6.5.2 Adding process steps 183
6.5.3 Organizing your process steps 184
6.5.4 Testing your process and process steps 185
6.6 Managing visualizations for your process steps 185
6.7 Working with conditions 186
6.8 Working with tasks 186
6.9 Managing processes 191
7. Connectivity 194
7.1 About connectivity 194

7.2 Displays ... 194
7.2.1 What? ... 194
7.2.2 Display types ... 194
7.2.3 Adding a display ... 195
7.2.4 Managing displays ... 197
7.3 Communication modules ... 197
7.3.1 What? ... 197
7.3.2 Communication module types ... 198
7.3.3 Adding a communication module ... 214
7.3.4 Configuring lights for a Pick To Light (DMX) communication module ... 216
7.3.5 Configuring a step for the communication module to send data ... 217
7.3.6 Configuring a step to wait for the receival of data from the communication module ... 226
7.3.7 Configuring a trigger for the communication module to start a process when receiving data ... 228
7.3.8 Following up on the communication that has been exchanged ... 229
7.3.9 Managing communication modules ... 230
7.4 Sensors ... 230
8. Experience levels ... 233
8.1 About experience levels ... 233
8.2 Managing experience levels ... 233
8.3 Experience level settings ... 234
8.4 Selecting the experience level ... 236
9. Variables ... 238
9.1 About variables ... 238
9.2 Variable types ... 238
9.3 Managing variables ... 240
9.4 Watching variables ... 241
10. Recordings (Validate and Vision license only) ... 243
10.1 About recording ... 243
10.2 Creating recordings ... 243
10.3 Managing recordings ... 244
10.4 Using recordings ... 245
11. Server management ... 248
11.1 About server management ... 248
11.2 Connecting to a server ... 248
11.3 Workstation overview ... 249
11.4 Managing licenses ... 250
11.5 Managing server backups ... 254
11.6 Cleaning up server resources ... 256
12. Dashboarding ... 257
12.1 About dashboarding ... 257
12.2 Consulting dashboards ... 257
12.3 Available dashboards ... 257
13. Logging ... 262
13.1 About logging ... 262

13.2 Consulting log tables in the Arkite software 262
13.3 Changing the logging database, the logging database and the cleanup of old data 264
13.4 Enabling audit logging 265
13.5 Changing logging settings 266
13.6 Adding logging yourself 267
13.7 Exporting log data 268

14. Images 270

14.1 About images 270
14.2 Managing images 270

15. Color templates 276

15.1 About color templates 276
15.2 The Detection Colors template 276
15.3 Managing color templates 277

16. Languages and Translations 280

16.1 About languages and translations 280
16.2 Managing languages 280
16.3 Managing Translations 281

17. User management 283

17.1 About user management 283
17.2 Managing Authentication Settings 283
17.3 Managing roles 285
17.4 Managing local users 288

18. The Arkite API 292

18.1 About the Arkite API 292
18.2 API examples 292
18.3 More information 292

1. About the Arkite software

1.1 Purpose of the Arkite software

Arkite offers the ultimate operator guidance platform. It transforms workstations into a digital and interactive environment.

The Arkite software allows you to configure the platform and adapt it to your specific working environment. Through the Arkite software, you can translate your standard operating procedures into a project without any programming knowledge.

Three license types are available for using Arkite's Operator Guidance platform:

- **Arkite Guide** license:

  Once your project has been configured, Arkite's operator guidance platform will be able to provide your operators with real-time picking and assembly instructions through Augmented Reality.

  The system cannot verify if a step was done correctly. The operator will always have to confirm he has executed a step and press the **Next** button on the touch screen, a hardware button or a foot pedal to continue to the next step.

  All features from this license are also included in the **Arkite Validate** and **Arkite Vision** licenses.

- **Arkite Validate** license:

  When using this license, your Arkite Operator Guidance Platform is equipped with an Arkite 3D sensor. The system will verify if a step is executed correctly. It will automatically continue to the next step after a step has been done correctly, and will warn your operators in case of any error.

  All features from this license are also included in the **Arkite Vision** license.

  Throughout this document, features that are only available when a workstation is used in combination with a sensor are tagged with a note **'Validate and Vision license only'**.

- **Arkite Vision** license**:**

  When using this license, your Arkite Operator Guidance Platform is equipped with an additional high definition infrared vision sensor. You can then benefit from Arkite's smart vision features.

  - Product inspection during assembly
  - Improved presence detection for small objects (> 0,5 cm)
  - Object orientation

  Features that are only available when a workstation is used in combination with an Arkite Vision sensor are tagged with a note **'Vision license only'**

  Tip

  For an overview of different applications of Arkite's operator guidance platform, have a look at the references on our website:

  https://arkite.com/references/.

1.2 How does the Arkite 3D sensor work? (Validate and Vision license only)

Technology

The Arkite 3D sensor used by Arkite's operator guidance platform is a Time-of-Flight sensor. Combining this ToF-sensor with intelligent algorithms, the Arkite software can verify the presence of objects and validate operating procedures.

The sensor uses two types of streams to capture data:

- Depth stream
- Infrared stream

By default, the Arkite software will automatically select the stream with the best contrast. However, you can overrule this selection and force the software to use a specific stream.

Depth stream

The sensor is sending out infrared light, invisible to the human eye. The IR light reflects on all objects. For each pixel, the time is measured for the light to travel from the sensor to the object and back. Then the distance of each pixel to the sensor is calculated. This results in a 3D image or point cloud. Virtual boxes are created on top of this real-world 3D image. The activity of an operator or the presence of an object in the box will lead to a detection trigger. By combining all these triggers in the correct sequence, the Arkite software can validate operating procedures.

Infrared stream

The reflected infrared light beams will also create an infrared image. The reflection intensity of each pixel is measured. The more IR light is captured by a pixel, the higher the value will be. A matte black surface will return a near 0 value, a highly reflective surface will return values up to 9000.
This results in a 2D image containing distinctions of color and material of objects.

A specific application of the infrared stream is edge detection. When using this application, the algorithm will specifically look for edges in the infrared image to validate the presence of a certain object. This detection is mainly used for tooling.

1.3 How does the Arkite Vision sensor work? (Vision license only)

What?

When you are using Arkite's Operator Guidance Platform with a Vision licence, the platform is equipped with an additional high-definition infrared Vision sensor.

How does it work?

The Arkite Vision sensor captures high-quality image and transmits the image data to the Arkite PC. The Arkite software processes the images and applies different algorithms to them to achieve accurate and efficient image analysis that can be used for the smart vision features.

1.4 Installing the Arkite software

What?

The Arkite software is split in a workstation and a server component. The server component, containing all data and resources for multiple clients and projects, should preferably run on a separate windows server to obtain higher security, maintainability and data integrity.

Installation instructions

For instructions on installing the Arkite software, please refer to the **Arkite software installation guide**.

1.5 Starting the Arkite software for the first time

What?

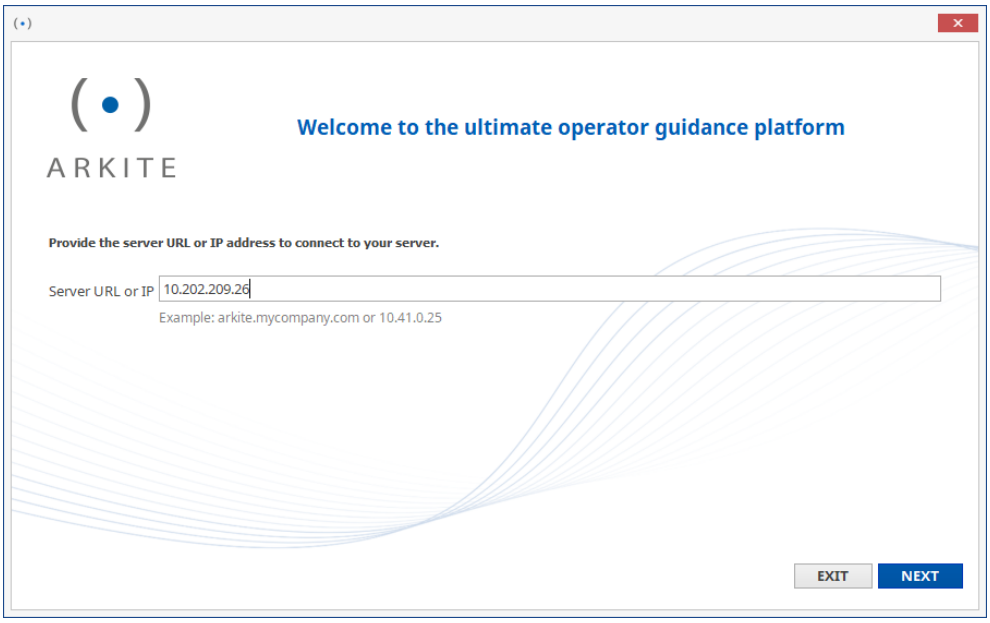
The Arkite software automatically starts after the workstation has been turned on. If it does not, you can double-click the Arkite icon on the desktop.

Connect to the server

When you open the Arkite software for the first time, a welcome wizard appears.

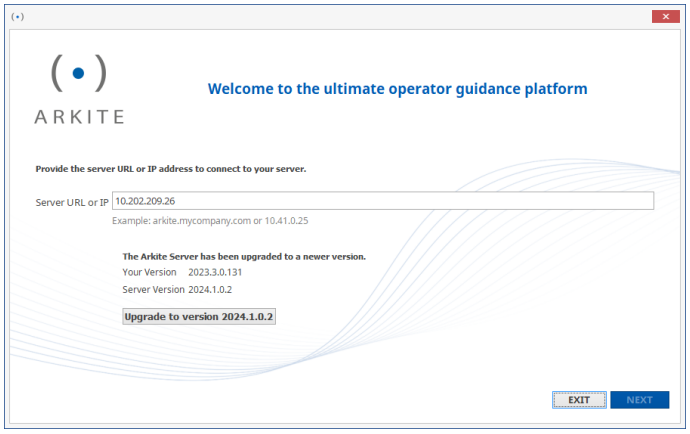
In first screen of this wizard:

1. In the **Server URL or IP** field, enter the URL of IP address of the server where the server component was installed.
2. Click **Next**.



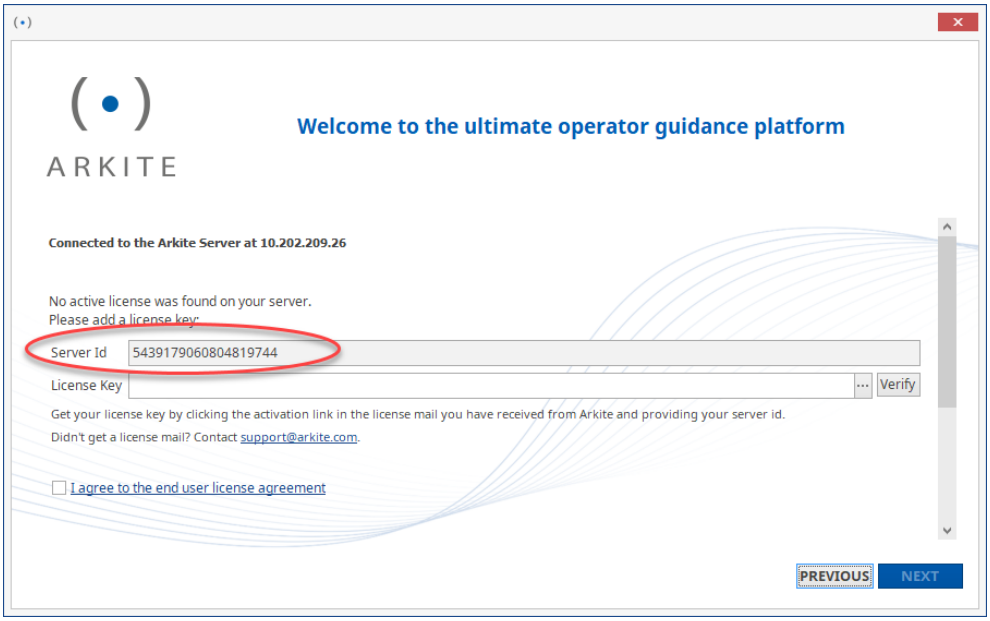
Note

- *If the server component is installed on your workstation, fill in **localhost**.*
- *If the software version on your workstation is not the same as the one on your server, you will first need to upgrade. A button will be shown allowing you to download and install the latest software version.*



Enter your server license key

If your workstation is the first one to connect to the server, you will need to provide a license key in the next screen of the wizard.



Note

If another workstation already connected to the server before and provided the server license key, this step won't appear. You will immediately be taken to the next step in the wizard.

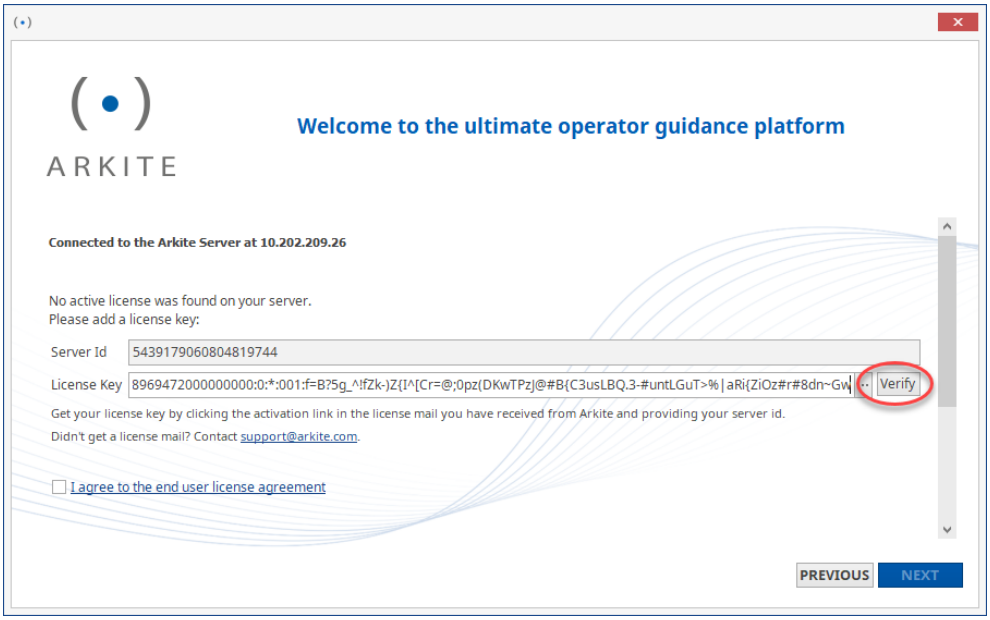
To obtain your server license key, click the activation link you have received in an e-mail from Arkite, fill in the webform and click **Submit**. If you haven't received an activation link, please contact support@arkite.com.

You will need to provide your **Server Id** (you can copy this from the information in the screen in the wizard).

After a few minutes, a license key will be generated and sent to your e-mail. Once you have received this e-mail:

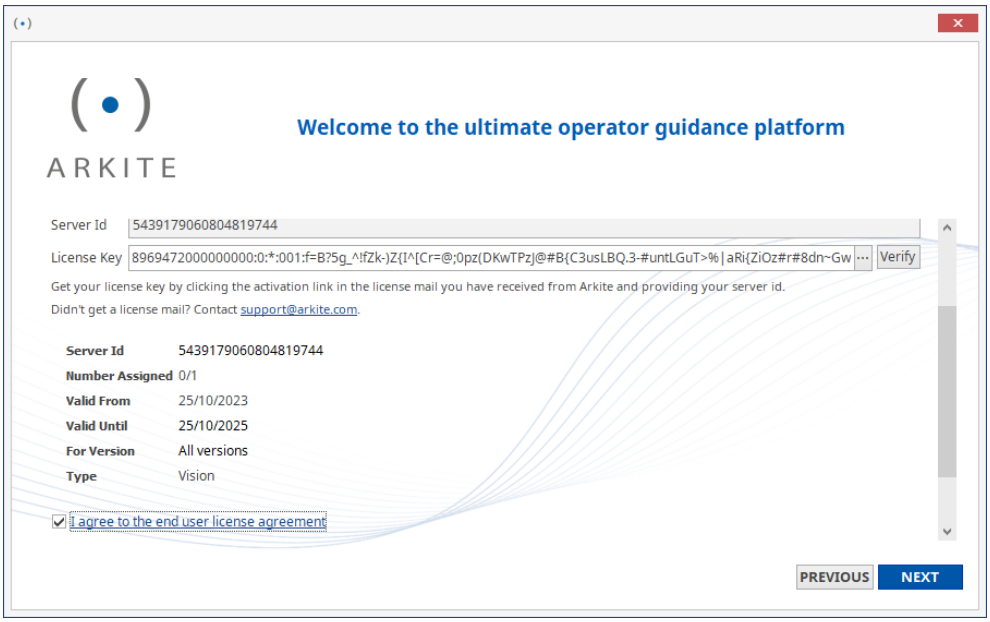
1. Copy the license key from the e-mail or download the **.key** file in attachment of that e-mail and save it on the workstation.
2. In the wizard, paste the license key in the **License Key** field or click the **...** button next to the field and select the **.key** file you have just saved.

3. Click **Verify**.



The information related to your server license appears. You will see:

- The **number of workstations** the license can be used for, and how many of those have already been assigned to the license
- The **validity** of your license
- The **software version** for which the license is applicable
- The **type** of your license: **Guide**, **Validate** or **Vision**.



4. Check the **I agree to the end user license agreement** checkbox.
5. Click **Next**.

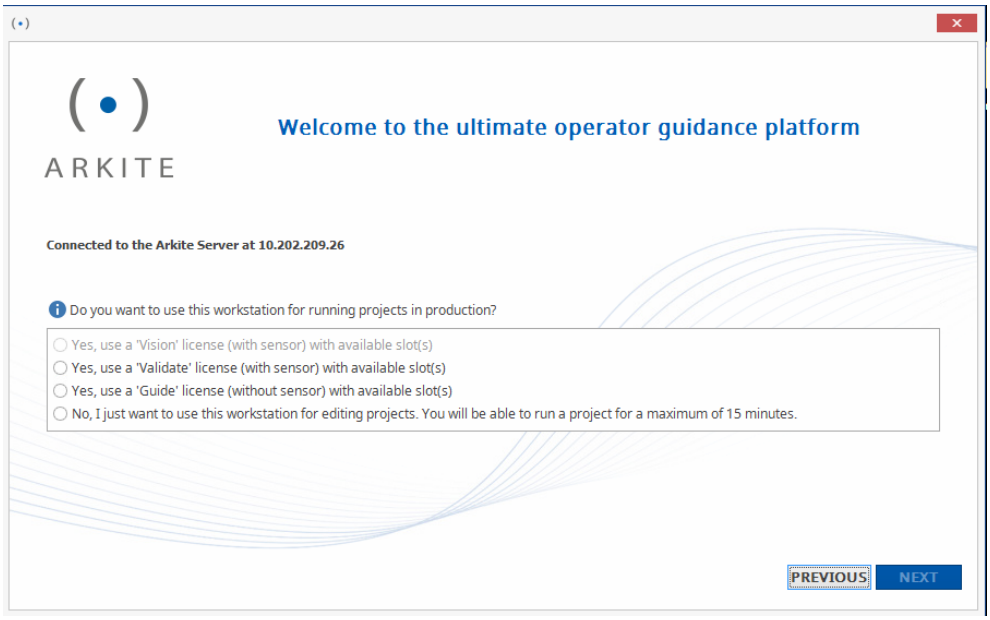
Note

- *In case you have a perpetual license, your license key is version specific. When you install a new version of the Arkite software, you will have to request a new license key. Recurring licenses can be used for any version of the software.*
- *Between three and two months before your license expires, you will receive a quotation from Arkite for the renewal of your contract. Once you have placed the purchase order, you will receive a new license key. You can then add this new license key as described above, without assigning any workstations to it yet. For more information, consult the section* ***How to renew a license key*** *on page 254.*
- *Your Arkite software license is linked to the server on which you install the server component. In order to move your Arkite installation to a different server while retaining your license, you should execute the steps as described in the procedure* ***How to move your Arkite installation to a different server****.*

Assign your workstation to a license

In the next step of the wizard, you can assign your workstation to a license, on the condition that a license key with available slots is present for the server you are connected to. If license keys for different license types are available, you will be able to choose which license type to use.

Indicate if you want to assign your workstation to a license and click **Next**.



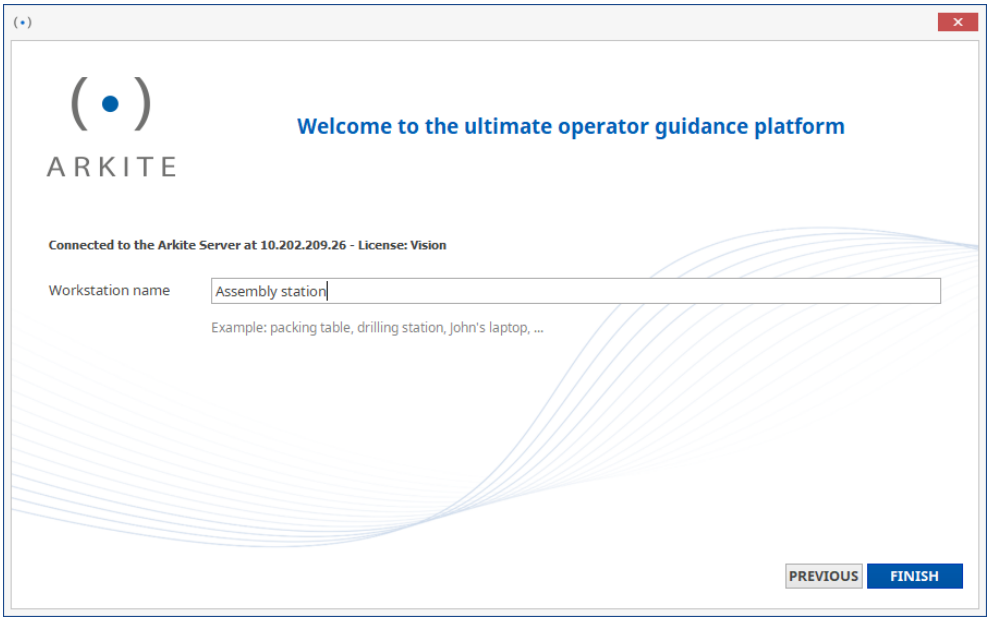
Note

- *If you have a Vision license, you will only be able to add workstations to it that are* ***Vision Ready****. If your workstation is not* ***Vision Ready****, the option will be disabled. In that case, finish the installation without assigning your workstation to a license. Then follow the steps described in the procedure* ***How to make your workstation Vision Ready*** *(available on the customer portal). The number of workstations you can make* ***Vision Ready*** *is limited to the number of workstations for which you have bought a Vision license. If you already have reached this number, your request will be denied.*
- *If your workstation is* ***Vision Ready****, you will not be able to assign it to a* ***Validate*** *or* ***Guide*** *license.*

- *If no licenses with open slot(s) are available or if you choose to not assign your workstation to a license, you can use your workstation as an **editor**. You will be able to prepare and configure projects on them, but you cannot use them to run a project. For testing purposes, you will be able to run your project for a maximum of 15 minutes. This way, you can prepare your project on your laptop at your desk.*
- *When your workstation is an editor, you can also edit a project for an another workstation connected to the same server. To do so, you can use the **Switch Workstation** feature. This way, you can limit the production downtime on your workstation as a minimum. For more information, refer to section **3.6 Switching workstations** on page 52.*
- *At any moment, you can add or remove license keys and change the workstations that have been assigned to them. For more information, refer to section **11.4 Managing licenses** on page 250.*

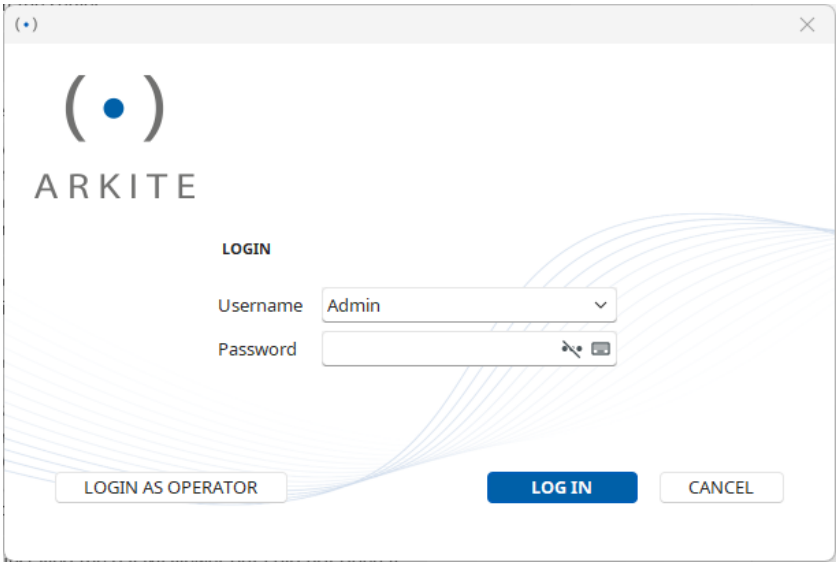
Provide a name for your workstation

In the next step of the wizard, enter a name for your workstation in the **Workstation name** field and click **Next**. Choose a name that will make it easy for you to identify your workstation.



Log in

A login dialog appears.

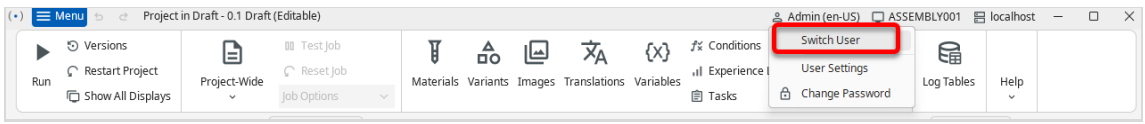


Provide the password for the **Admin** user. By default, this is 'Arkite3600'. Next, click **Login**.

Note

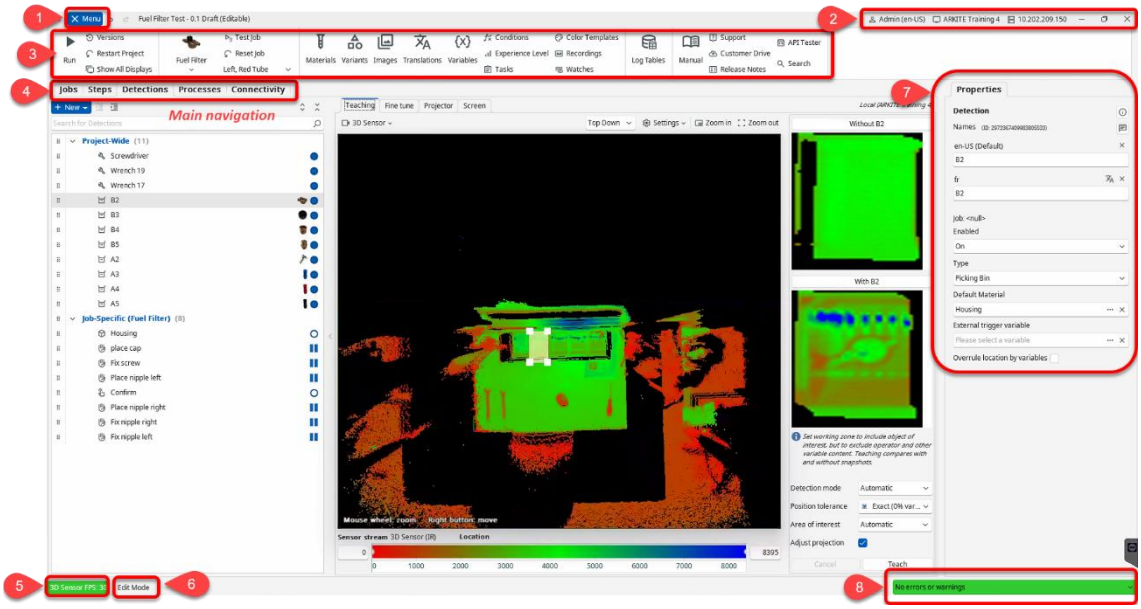
- *You can click the button to make the password you've entered visible and verify if you haven't made any typing mistakes.*
- *You can click the button to display an online keyboard, in case you are working with a touch screen for example.*
- *If you do not provide a password for the Admin user within 8 seconds, you will log in as an operator, with read-only access to the Arkite software. You can also click the **Login as Operator** button to do so immediately.*
- *Once you have logged in with the Admin user, you can set up other users and roles at your convenience. For more information, see chapter **17 User management** on page 283.*

To login with another user, click your username in the top bar and select **Switch User**.



1.6 Arkite User Interface Overview

The image below gives an overview of the Arkite software user interface:



1. The **Menu** gives access to features that apply to the Arkite software as a whole:
   - The overview of projects
   - Server Settings
   - Dashboarding
   - Workstation Overview
   - User management features
   - Workstation calibration
2. The **top bar** provides you with information about:
   - The user that is currently logged in
   - Your workstation's name
   - The server you are connected to

   It also allows you to switch user or workstation (in case you are logged in as an editor) or update your server connection settings.
3. The **ribbon** gives access to features that apply to the project as a whole or are used across the project
4. The **Main** navigation gives access to the features that are key for building your project:
   - Jobs
   - Job Steps
   - Detections
   - Processes
   - Connectivity: Displays, Sensors and Communication Modules

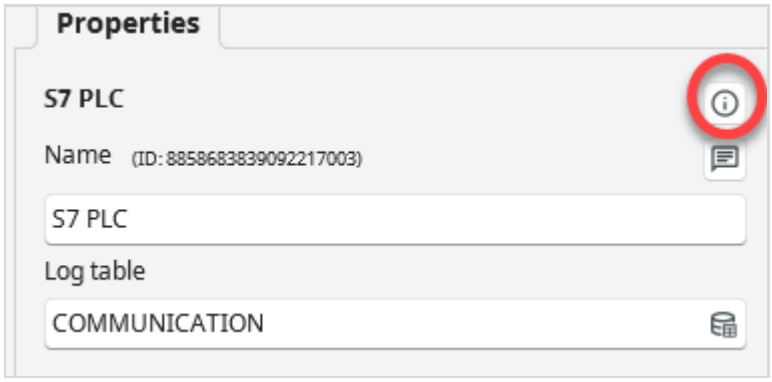
5. The **Sensor status** indicates the status of your sensor(s).
6. The **Project mode** indicates whether your project is in **Run** or **Edit** mode.
7. The **Properties** pane shows the properties of the component that is currently selected.
8. The **Project warnings** indicate whether there are any errors or warnings into account.

1.7 Help and Support

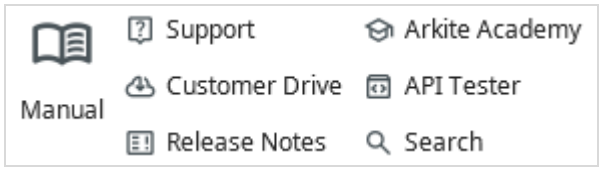
Help in the Arkite software

You can find a lot of help within our software:

- For many features, you will see a ⓘ button. Clicking this button will give you specific information for that feature.



- In the ribbon, you can find buttons to:



  - Open the manual – you can also use the F1 key.
  - Navigate to the customer portal, containing more extended documentation (internet connection required)
  - Open the release notes
  - Navigate to Arkite Academy, an online training platform to support you in getting acquainted with Arkite's Operator Guidance Platform.
    Through short, hands-on training modules, you can learn about the platform's setup and installation, the essentials of our software and connectivity with different external devices and communication protocols. The available content on Arkite Academy will be gradually expanded.
    Arkite Academy is available in English and French.
  - Open the API Tester page

    Note
    *For more information about the Arkite API, refer to chapter **18 The Arkite API** on page 292.*

Contact our support team

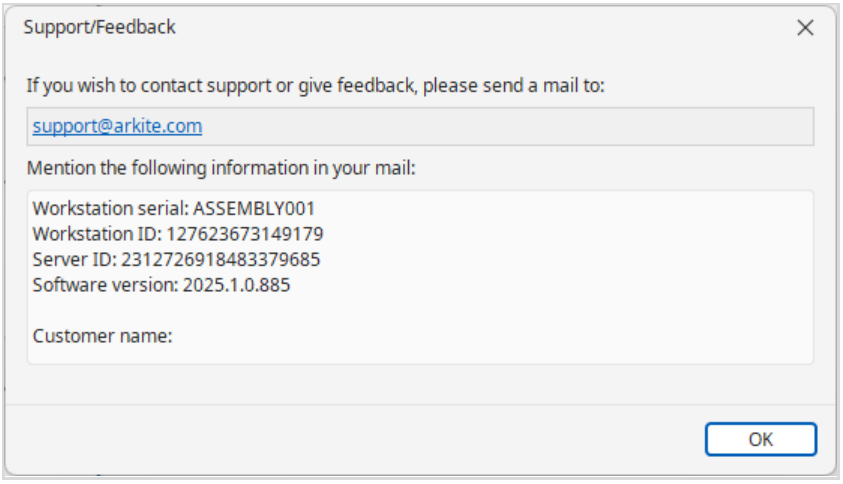
If you still have questions after reading this manual, our team is happy to support you.

To obtain support, you can:

- Create a ticket on our portal
- Send an e-mail to support@arkite.com in which you mention:
  - your Workstation Serial Number
  - your Workstation ID
  - your Server ID
  - the software version you are using
  - your customer name

Tip

*If you have trouble finding this information, click the **Support** button in the ribbon. A window with the required information will appear.*



Our team will get back to you as soon as possible.

TeamViewer

On every workstation delivered by Arkite, TeamViewer has been installed. This software allows our support people to access the workstation and provide you with direct support when needed.

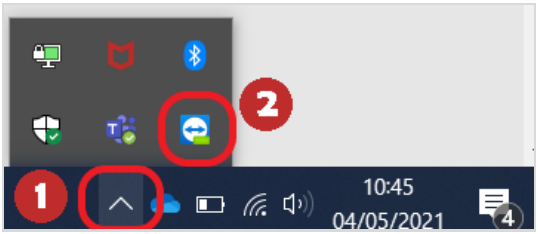
TeamViewer automatically starts up when the workstation is turned on.

If you are using the Arkite software your own workstation, you also need to install Team Viewer using the link below:

https://www.teamviewer.com/download/windows/

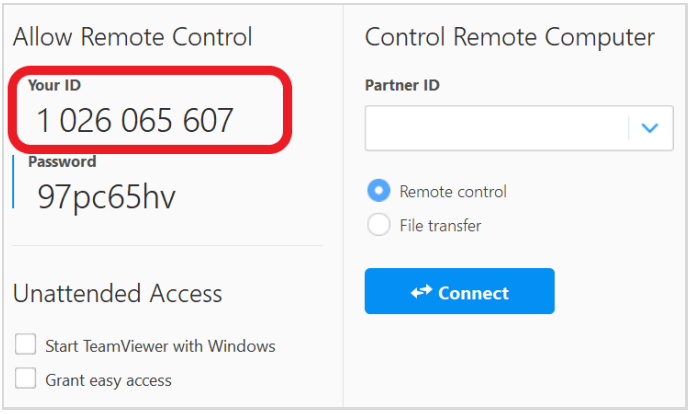
To open TeamViewer:

1. Click the upward arrow in the lower right corner of your screen.
2. Click the TeamViewer icon.



TeamViewer opens.

3. Provide our support team with your ID. Make sure the workstation is connected to the internet.



Note

You can also use TeamViewer yourself in order to take control of the workstation remotely. You can install TeamViewer using the link below:

https://www.teamviewer.com/download/windows/

After the installation, enter the TeamViewer ID and password of the workstation.

2. Calibration

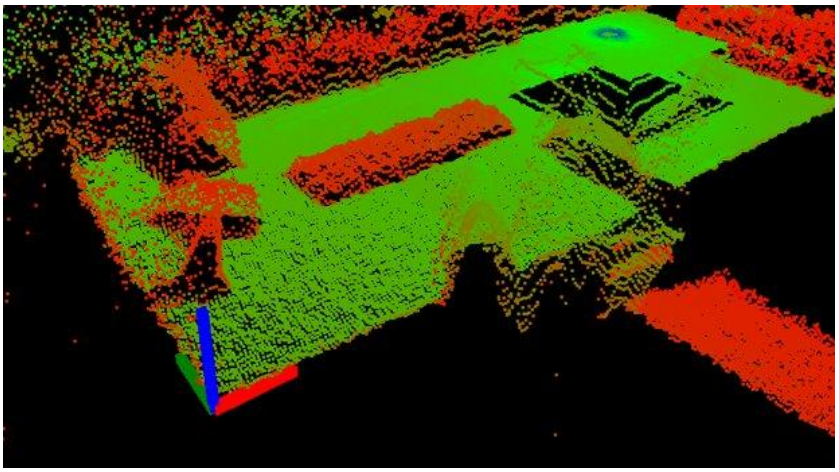
2.1 About calibration

Before you can start configuring your project, you will have to calibrate your workstation. Calibration defines the coordinate system (mm) and origin for all detections.

If you are using the system with Arkite 3D sensor (**Validate** or **Vision** license), both the sensor and the projector need to be calibrated.

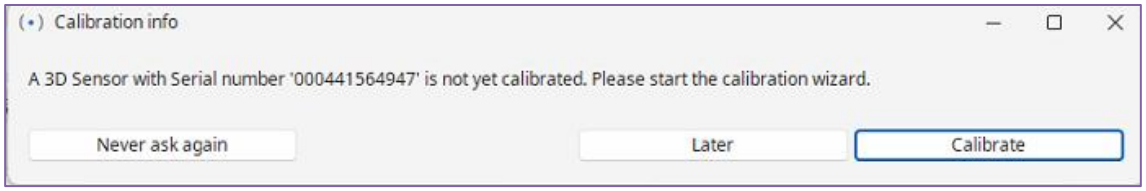
When using a Vision license, you will additionally need to calibrate the **Arkite Vision** sensor.

During sensor calibration, each XYZ coordinate is translated to a pixel in the sensor image with a corresponding RGB value.



Projector calibration defines the coordinate system that is used to define where projections need to be shown on the workbench.

If calibration has not taken place yet on a particular workstation, a popup will appear.



Calibration is one-time operation, but after the calibration the projector and the sensor need to remain fixed relative to the workbench. When the sensor and/or projector have moved, you have to recalibrate.

2.2 Calibrating your workstation

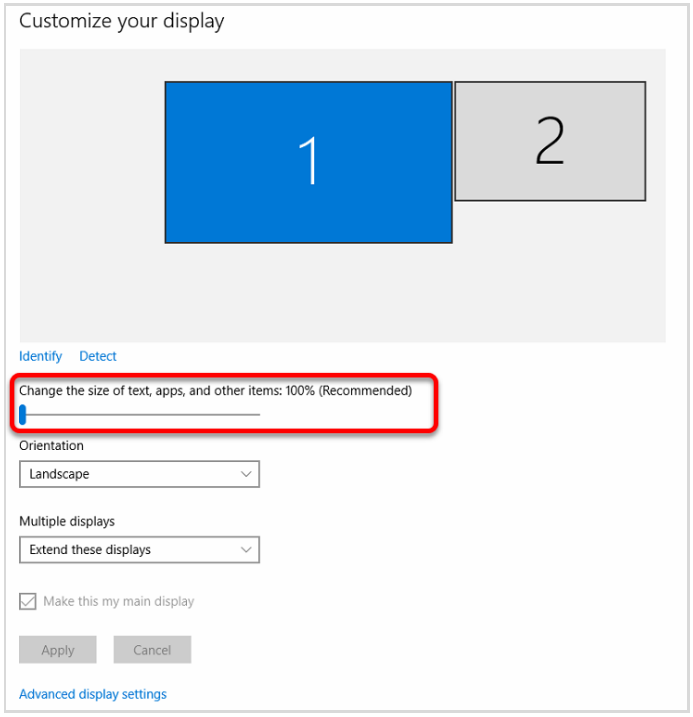
2.2.1 Calibrating a workstation with Arkite 3D sensor using the calibration wizard (Validate and Vision license only)

What?

The Arkite software contains a calibration wizard that will take you through all the required steps to calibrate your sensor and projector.

Note

For an optimal result, the scaling settings for your projector in Windows should be configured at 100%. Please verify before you calibrate. In order to do so, right-click in your desktop and select ***Display Settings****.*

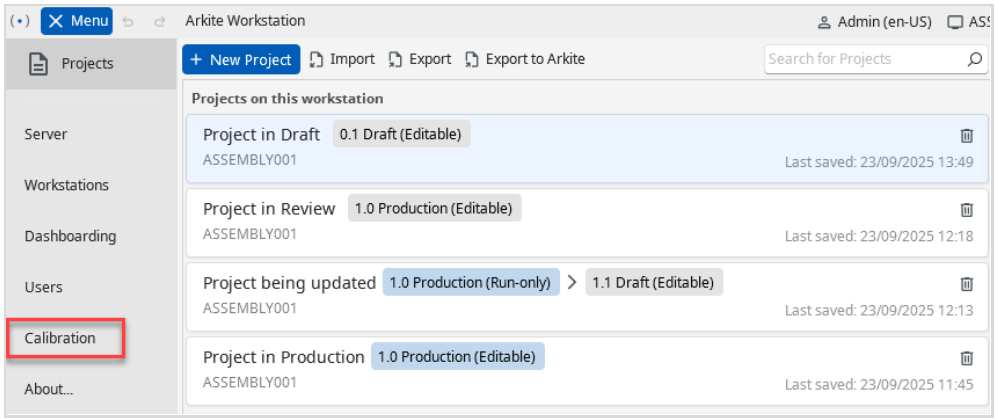


Steps

To use the calibration wizard, execute the following steps:

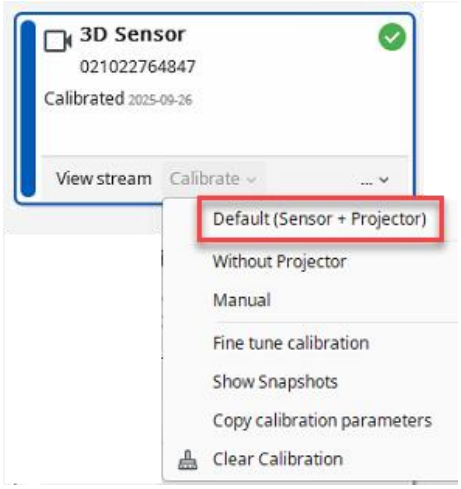
1. If calibration has not taken place yet on your workstation, click **Calibrate** in the **Calibration info** popup.

   When recalibrating, click **Menu** and then **Calibration**.



Note

*When you are in a project, you can also navigate to the **Connectivity** tab, select the **3D sensor** and select **Default Calibration (Sensor+Projector)** from the **Calibrate** menu.*



*When you have created the project before calibrating, a **Projector** display is added with the screen resolution of the device you are working on (e.g. your laptop). This may not match with the actual projector resolution, used during calibration. This will cause projections to be faulty. You can fix this by updating the navigating to the **Connectivity** tab, selecting the **Projector** card and updating the **Resolution** property to the correct value in the **Properties** pane.*

2. For the initial calibration, select **Fully Restart Calibration** and click **Next**.

   To calibrate additional sensors or projectors, use the other options.

3. Select the **3D Sensor** and click **Next**.

4. Check the projector settings:

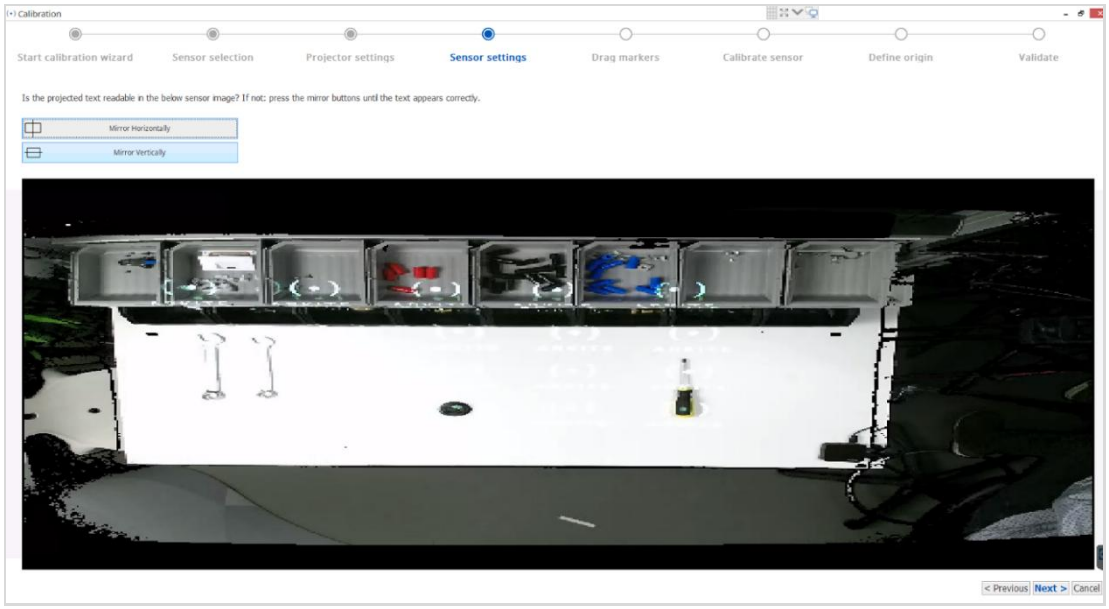
   - Verify if the Arkite logo is projected. If not, change the output to which the projector is connected.

- Verify if the Arkite logo is projected in the correct orientation on the workbench. Rotate and/or mirror if required.
- Optionally, check the **Enable Height** checkbox.
  That is useful when you have many detections above the calibration surface, or if your detections are dynamically updated by an external system.

  Note

  - *For other use cases, don't enable this setting, since it lowers the calibration accuracy.*
  - *Currently, the **Enable Height** option is only available for Optoma ZH420, Optoma ZW400 and Optoma ZH403 projectors.*
- When finished, click **Next**.

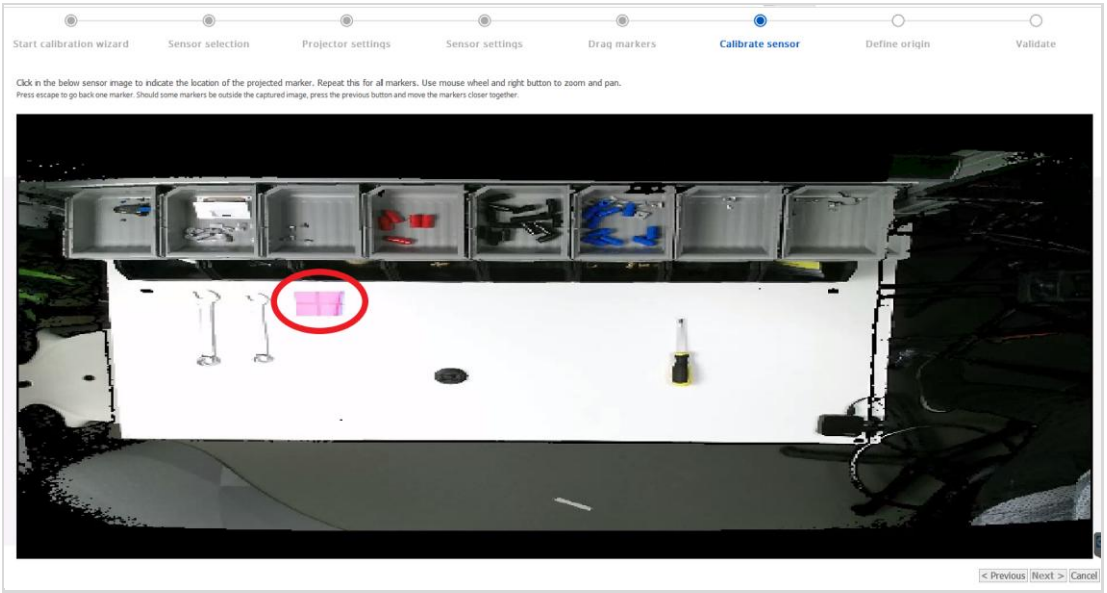
5. Make sure the sensor image matches the orientation on the workbench. Apply mirroring if required. When finished, click **Next**.



6. Verify whether the four calibration markers are projected correctly on the workbench. Drag them into a position where they are on a flat surface (i.e. the table, not a tool or picking bin), as far from each other as possible.



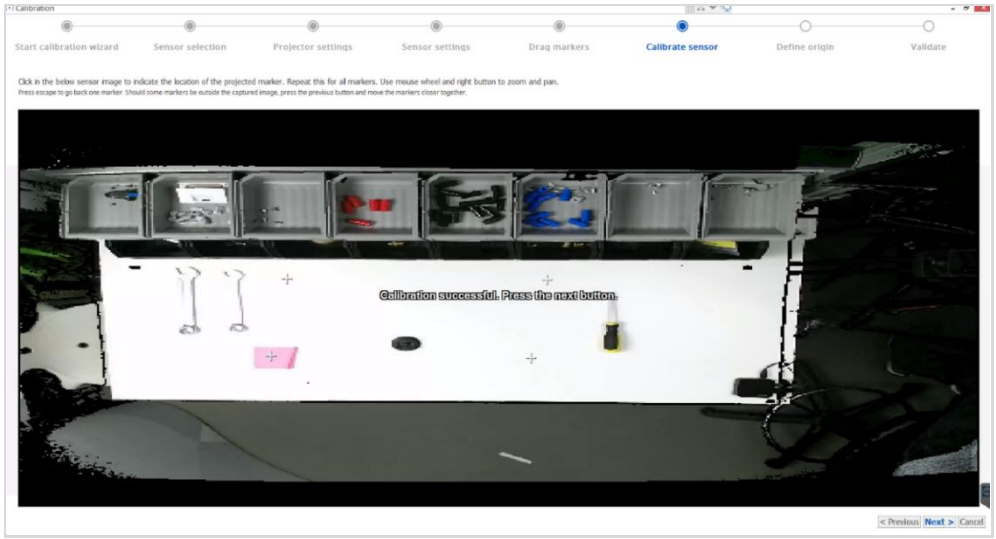
7. Click the center of each marker in the projected image.
   Be as precise as possible. Use the mouse wheel and the right mouse button to zoom and pan.



Tip

If a marker is difficult to see, you may add a piece of paper on that location.

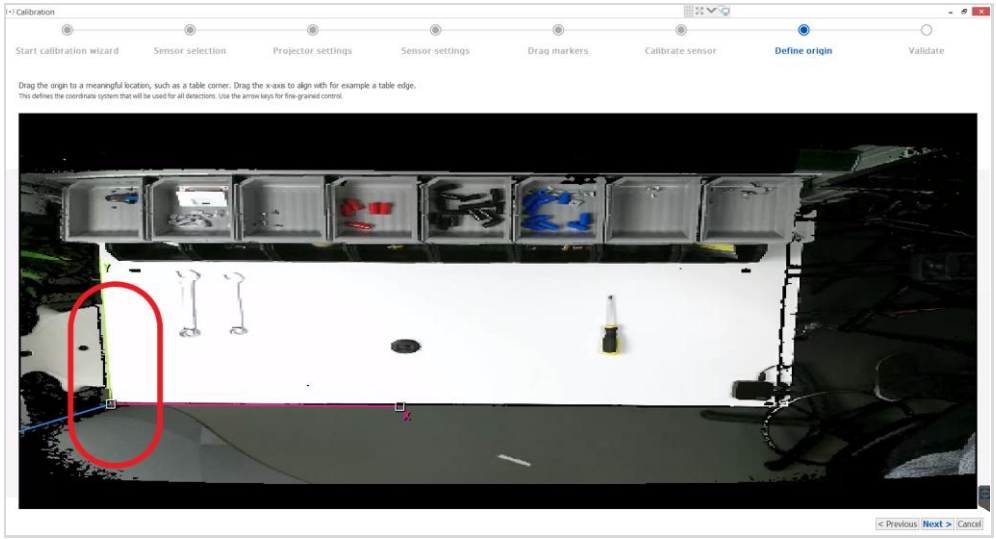
8. If calibration was successful, this will be indicated on the screen. Click **Next**.



Note

In case calibration was not successful, a message will appear indicating the calibration is invalid. If the calibration is invalid because of one or more calibration marker(s) for which no depth value could be measured, the respective marker(s) will be indicated by a red circle. Click the ***Escape*** *button once or more until you reach the invalid marker. Click the marker(s) again. If that doesn't work, click* ***Previous*** *to return to the previous step and drag the markers to another location.*

9. Define the origin of your image by dragging it to a meaningful location, e.g. the corner of the workbench. Click **Next**.



Note

With the workstation, you will also receive a small aluminum plate with the Arkite logo. You should screw the plate onto the workbench at the location where you want define your origin. This will help you to calibrate, or to re-calibrate in case of issues.



10. Verify if the projection on your workbench matches the position on the sensor image. If it does, click **Next**. If not, click **Previous** to adjust your calibration.

Result

Your sensor and projector have been calibrated. If you have multiple sensors or projectors, run the calibration wizard again to calibrate them as well.

2.2.2 Calibrating a workstation with Arkite Vision sensor (Vision license only)

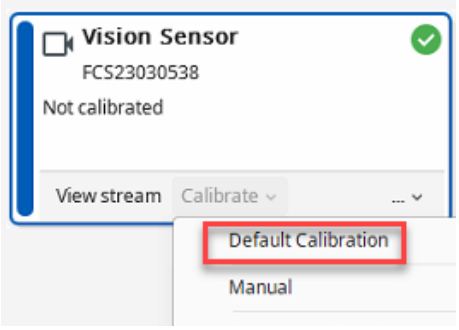
What?

When you are using the Arkite Operator Guidance Platform with a Vision license, also your Arkite Vision sensor will need to be calibrated.

Steps

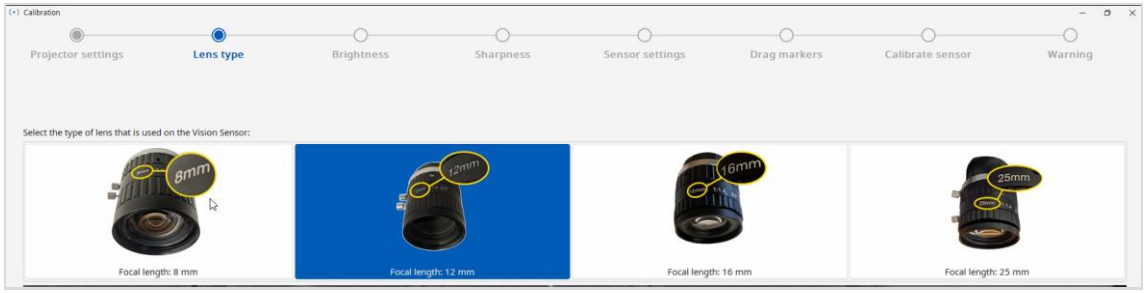
Follow the steps in the Arkite Vision sensor calibration wizard. When using your workstation for the first time, the calibration wizard will pop up automatically after you have finished the Arkite 3D sensor and projector calibration wizard.

If not, you can start the wizard by navigating to the **Connectivity** tab, select the **Arkite Vision Sensor** and select **Default Calibration** from the **Calibrate** menu.



To use the calibration wizard, execute the following steps:

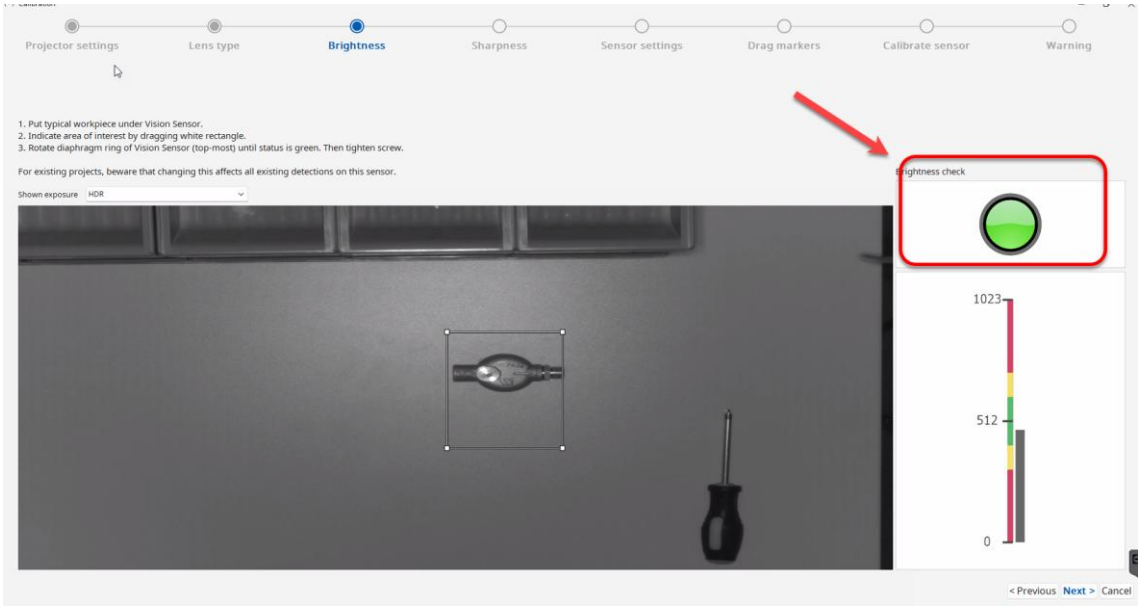
1. Check the projector settings:
   - Verify if the Arkite logo is projected. If not, change the output to which the projector is connected.
   - Verify if the Arkite logo is projected in the correct orientation on the workbench. Rotate and/or mirror if required.
   - Click **Next**.
2. Select the type of lens that is used on your Arkite Vision Sensor and click **Next**. The lens types below are supported:
   - 8 mm
   - 12 mm
   - 16 mm
   - 25 mm



Note

You can find this information on the lens.

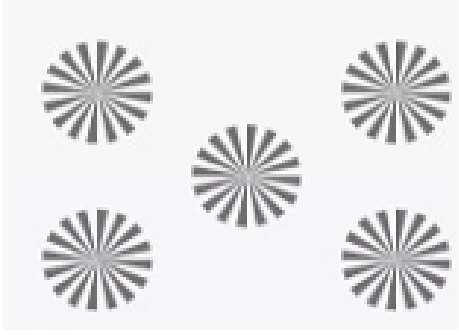
3. Adjust the brightness of the Arkite Vision sensor:
   - Put a typical workpiece under the sensor.
   - Rotate the diaphragm ring (=the top ring) of the Vision sensor until the status is green.



   - Tighten the screw.
   - Click **Next**.

4. Take the 5 Arkite Calibration documents you have received with your hardware. Arrange them on your workbench so the Siemens stars have a configuration like the one below..

   Rotate the focus ring (=the bottom ring) of the Vision sensor until the image is sharp.



Note

If you don't have the Arkite Calibration documents anymore or if you didn't receive them, you can download a pdf version from the corresponding step in the Vision calibration wizard in our software..

5. Verify whether the four calibration markers are projected correctly on the workbench. Drag them into a position where they are on a flat surface (i.e. the table, not a tool or picking bin), as far from each other as possible.
6. Verify if the text on the Arkite Calibration document is projected in the correct, readable orientation on the workbench. Rotate and/or mirror if required.
7. Click the center of each marker in the projected image.
8. Because the Arkite Vision sensor uses infrared, the markers are by default not visible in the captured image. Therefore, use the crosshair markers on the Arkite Calibration document and align them with the projected markers to make them visible in the captured image.
   Be as precise as possible. Use the mouse wheel and the right mouse button to zoom and pan.

Result

Your Arkite Vision sensor has been calibrated.

2.2.3 Calibrating a workstation with Arkite RGB sensor

What?

If you have a Validate or Guide license and you want to be able to detect barcodes or texts through OCR recognition, you need an Arkite RGB sensor. The Arkite RGB sensor needs to be calibrated as well.

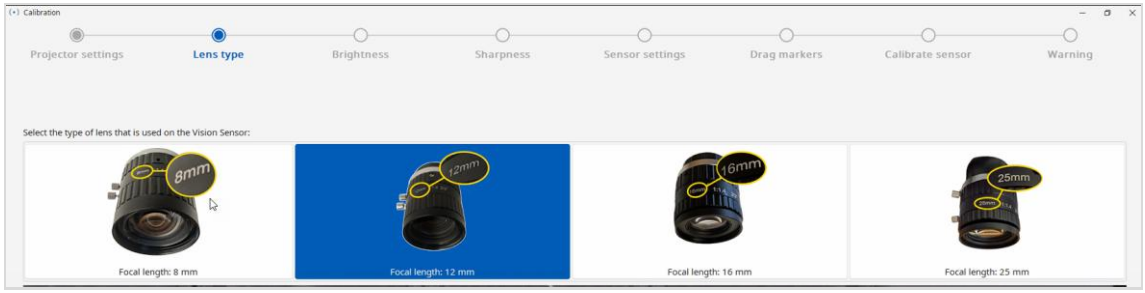
Steps

Follow the steps in the Arkite RGB sensor calibration wizard. When using your workstation for the first time, the calibration wizard will pop up automatically.

If not, you can start the wizard by navigating to the **Connectivity** tab, select the **Arkite RGB Sensor** and select **Default Calibration** from the **Calibrate** menu.

To use the calibration wizard, execute the following steps:

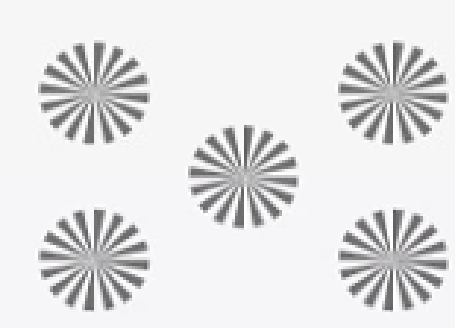
1. Check the projector settings:
   - Verify if the Arkite logo is projected. If not, change the output to which the projector is connected.
   - Verify if the Arkite logo is projected in the correct orientation on the workbench. Rotate and/or mirror if required.
   - Click **Next**.
2. Select the type of lens that is used on your Arkite RGB Sensor and click **Next**. The lens types below are supported:
   - 8 mm
   - 12 mm
   - 16 mm
   - 25 mm



Note

You can find this information on the lens.

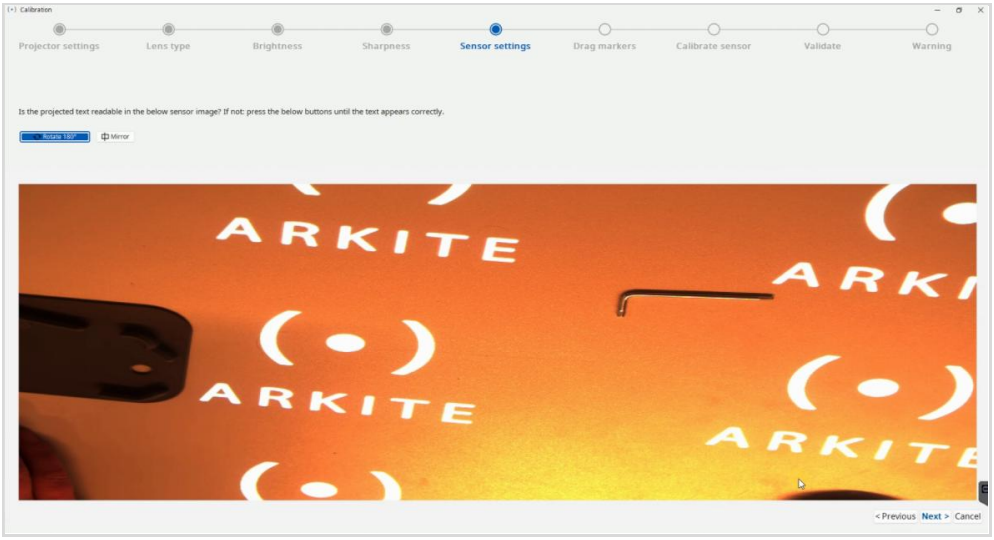
3. Adjust the brightness of the Arkite RGB sensor:
   - Rotate the diaphragm ring (=the top ring) of the RGB sensor to the lowest number (= the largest aperture).
   - Tighten the screw.
   - Click **Next**.
4. Take the 5 Arkite Calibration documents you have received with your hardware. Arrange them on your workbench so the Siemens stars have a configuration like the one below.
   Rotate the focus ring (=the bottom ring) of the Arkite RGB sensor until the image is sharp.



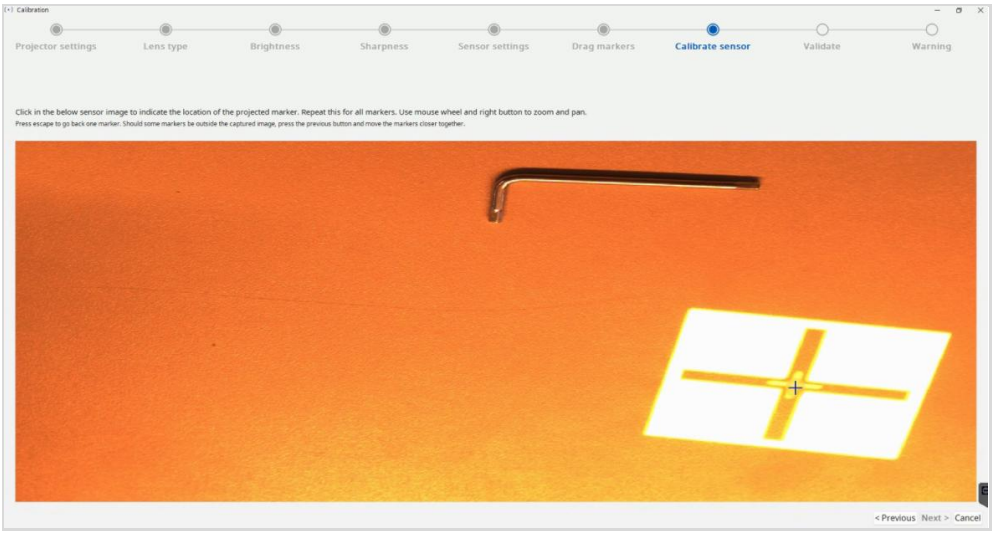
Note

If you don't have the Arkite Calibration documents anymore or if you didn't receive them, you can download a pdf version from the corresponding step in the calibration wizard in our software.

5. Verify if the text that is projected on the workbench is readable in the sensor image. Rotate and/or mirror if required.



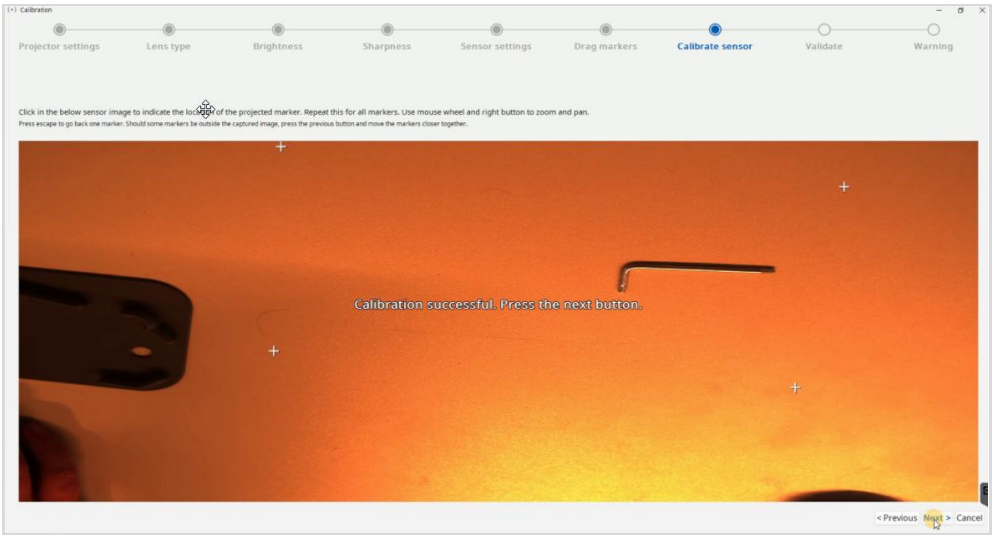
6. Verify whether the four calibration markers are projected correctly on the workbench. Drag them into a position where they are on a flat surface (i.e. the table, not a tool or picking bin), as far from each other as possible.
7. Click the center of each marker in the projected image.
   Be as precise as possible. Use the mouse wheel and the right mouse button to zoom and pan.



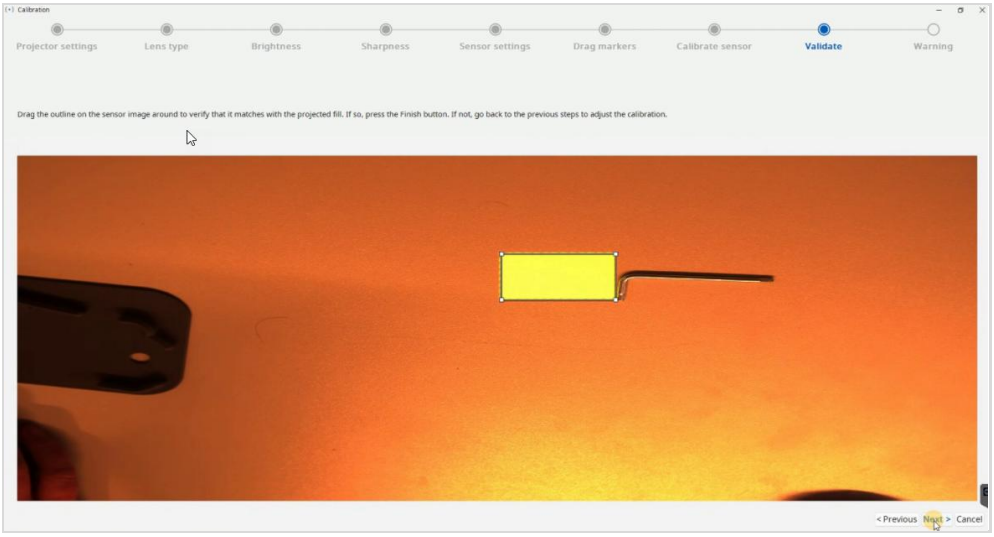
Note

*In case one of the markers is not visible in the captured sensor image, click the **Previous** button and move the markers closer together.*

8. If calibration was successful, this will be indicated on the screen. Click **Next**.



9. Verify if the projection on your workbench matches the position on the sensor image. If it does, click **Next**. If not, click **Previous** to adjust your calibration.



Result

Your Arkite RGB sensor has been calibrated.

2.2.4 Calibrating a workstation without sensor (Guide license)

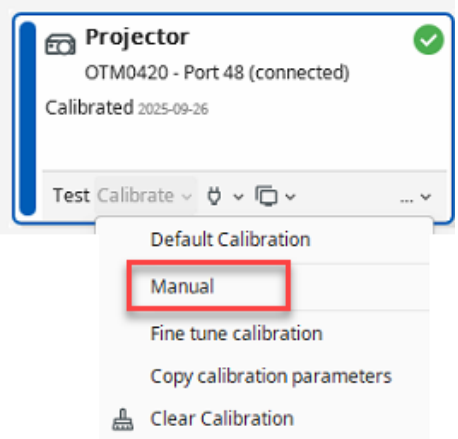
What

If you are using the system without sensor (**Guide** license), only the projector needs to be calibrated by entering coordinates in mm.

Steps

1. Open a project.
2. Navigate to the **Connectivity** tab.
3. Select the **Projector** card.

4. In the **Calibrate** menu, select **Manual**.



5. Follow the wizard and enter the coordinates of the markers in mm.

2.3 Other calibration options

Apart from the calibration wizard, a few other options are available in the **Calibration** menu of the **3D Sensor, Vision Sensor** sensor and **Projector** cards on the **Connectivity** tab:

- **3D Sensor:**
  - **Without projector:**
    Use this option to calibrate the sensor only.

    Tip

    *When adding detections at your desk based on a recording, first select a recording and use this calibration option. For more information about using recordings, refer to chapter **10 Recordings (Validate and Vision license only)** on page 243.*
  - **Manual:**
    Calibrate a sensor by entering coordinates in mm.
- **Vision Sensor (Vision license only):**
  - **Default Calibration:**
    Calibrate the Vision Sensor based on an already calibrated projector that covers the same area.
  - **Manual:**
    Calibrate a vision sensor by entering coordinates in mm.
- **Projector:**
  - **Default Calibration:**
    Calibrate the projector based on an already calibrated 3D sensor that covers the same area.
  - **Manual:**
    Calibrate a projector by entering coordinates in mm.

Note

- *To delete the calibration of a sensor or projector, click the **Calibration** menu on the corresponding card and select **Clear calibration**.*

- *To consult the parameters that have been used during calibration of a sensor or projector, click the **Calibration** menu on the corresponding card and select **Copy calibration parameters**. You can then paste them in a text editor.*

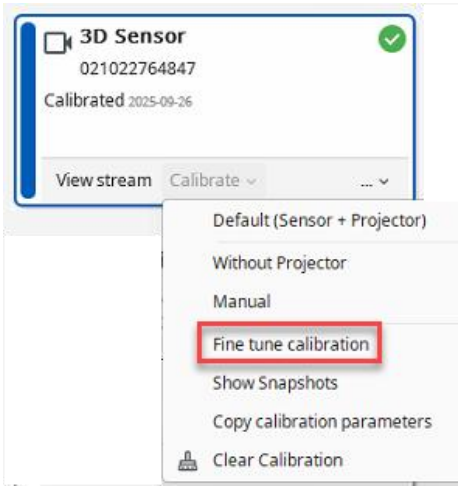
2.4 Finetuning Calibration

What?

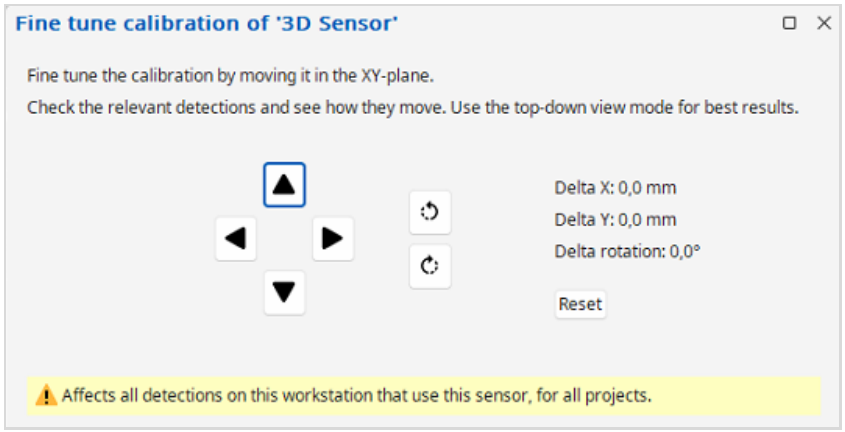
In case you had to recalibrate your sensor and/or projector, for example when a broken sensor or projector was replaced or when the mounting structure has changed, there might be a small offset in the projections or small deviations causing not all detections to work properly.

Steps

1. Navigate to the **Connectivity** tab.
2. Select the card for the sensor or projector for which you want to finetune the calibration.
3. In the **Calibrate** menu, select **Fine tune calibration**.



4. Navigate to the **Detections** tab, select the relevant detections and use the arrows in the **Fine tune calibration** popup window to move them.



Warning

Be aware that the modifications you make will affect all detections using the sensor in all projects on your workstation.

3. Projects

3.1 About projects

Projects contain all the information that is required to support your jobs. A **job** is a work task that an operator performs on a workstation, for example an assembly, a kitting application, a quality inspection etc., or any other work task for which you want to guide your operators through augmented reality work instructions.

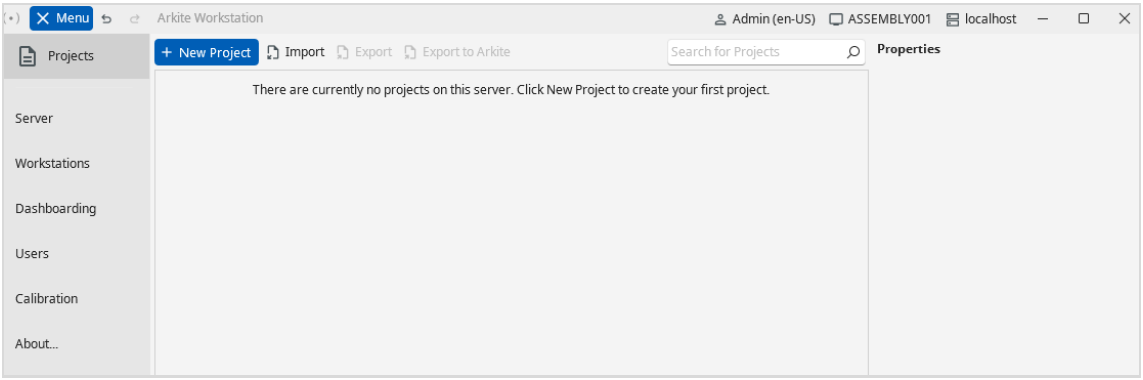
Only one project can be loaded at the same time on a workstation. Therefore, make sure to manage all jobs that the workstation is used for within the same project.

If you have different production lines in which an entirely different set of jobs should be supported, you can create a different project for each production line.

3.2 Creating a project

What?

When you open the Arkite software for the first time, no projects will be available. The first thing you need to do is to create a project.



Steps

To create a new project, execute the following steps:

1. On the **Projects** tab, click **New Project**.
2. In the **Properties** pane, update the project properties:
   - In the **Name** field, provide an appropriate name for your project.

     Note

     *Next to the **Name** label, the object ID for the project is shown. You can copy it to use it when working with the Arkite API. For more information about the Arkite API, refer to chapter **18 The Arkite API** on page 292.*

   - By clicking the **Comment** icon next to the **Name** property, you can provide a short explanation of the purpose of the project.

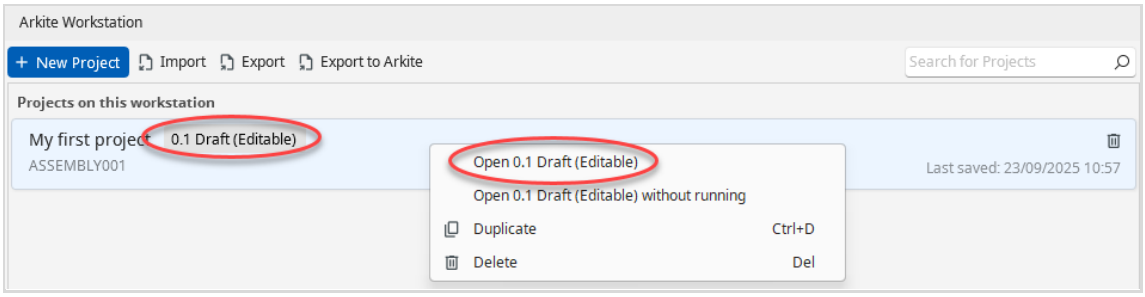
   Note

   *If you want to load the project automatically when starting the Arkite software as a run-only user, check the **Auto open on this workstation on startup** checkbox.*

*This checkbox is only available for projects that have a production version, since projects that only have a **Draft** version are only available to Admin users.*

Result

Your project is added to the project list and you are ready to go. To start configuring your project, click the **0.1 Draft** button next to the project name or right-click the project and select **Open**.



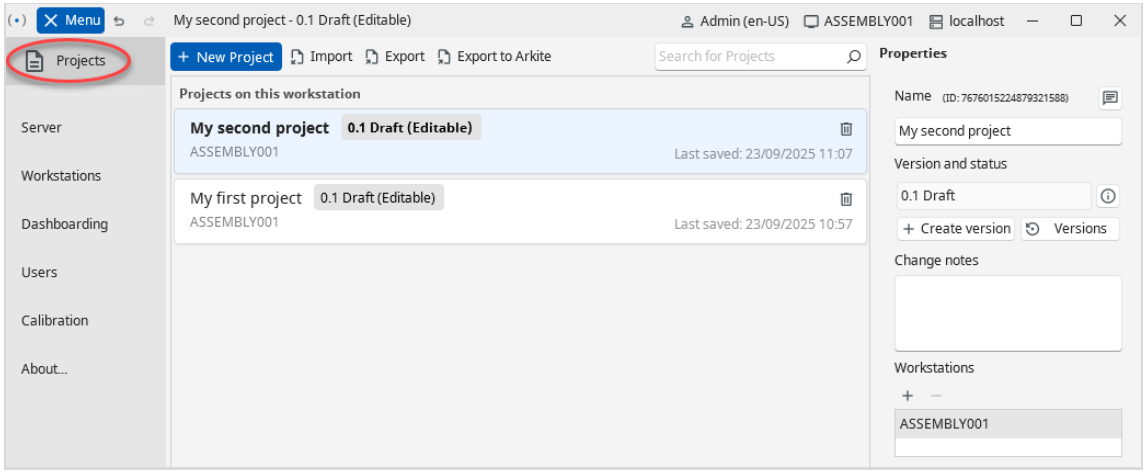
Note

*In case of issues with your project, for example a communication loop, you can open your project without running it to troubleshoot. To do so, right-click the project and select **Open without running.***

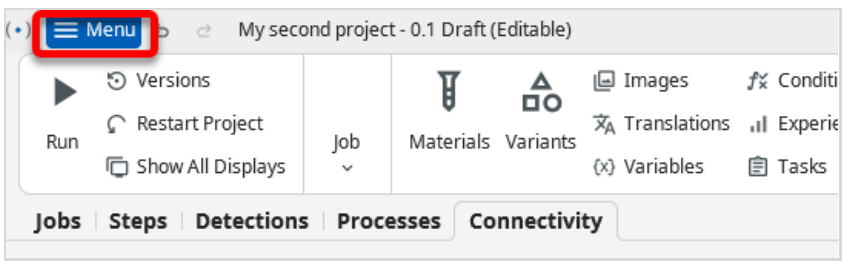
3.3 Managing projects

What?

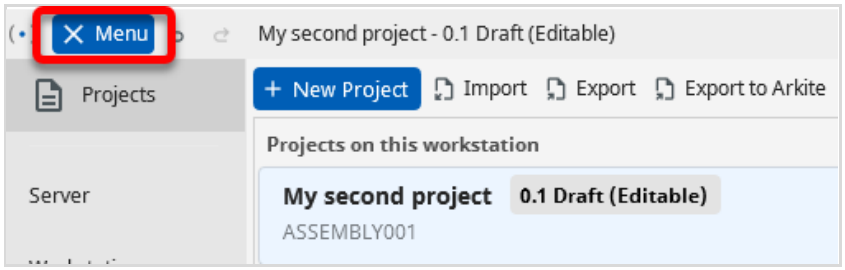
You can find an overview of all projects that are available on your workstation in the **Projects** menu. Only one project can be loaded on the workstation at the same time. That project is rendered in **bold**.



When you are working in a project, you can always return to your project overview by clicking the **Menu** button.



To return to your project, close the Menu.



How to manage projects

On the **Projects** tab, you can:

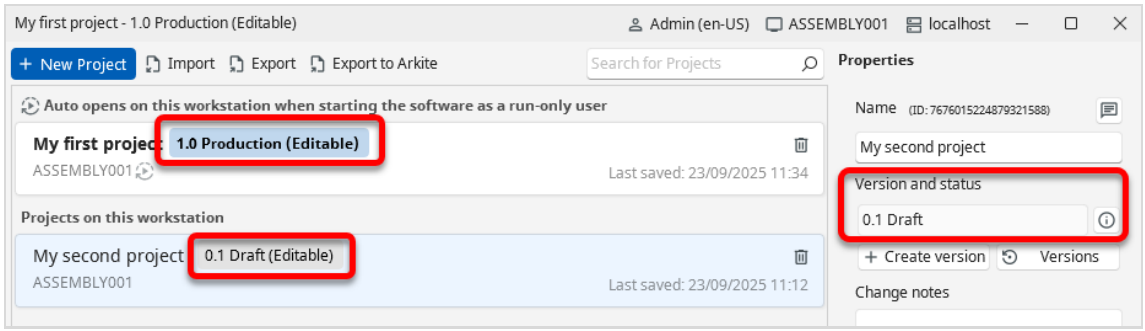
- Click **Import** to import a project prepared on another server
- Select a project and click **Export** to make it available for use on another server.
- Select a project and click **Export to Arkite** to provide a copy to Arkite for support.
- Right-click a project and select **Duplicate** to duplicate it.
- Right-click a project and select **Delete** or click the button in the project row to delete it.

3.4 Versioning

3.4.1 About versioning

What?

Every project has a specific version and status at all times. The current version and status of a project is shown next to each project in the project overview as well as in the **Properties** pane.



Project versions

A project's version consists of a major and a minor version number.

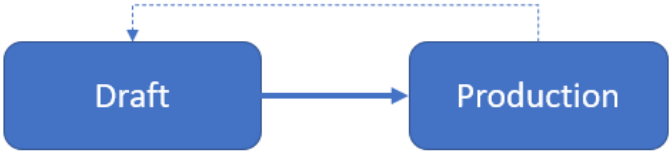
Every time a new version is created, the minor version number increases with one digit, e.g. from 0.1 to 0.2.

When the project status is changed to **Production**, the major version number increases with one digit and the minor version number is reset to 0, e.g. from 0.4 to 1.0.

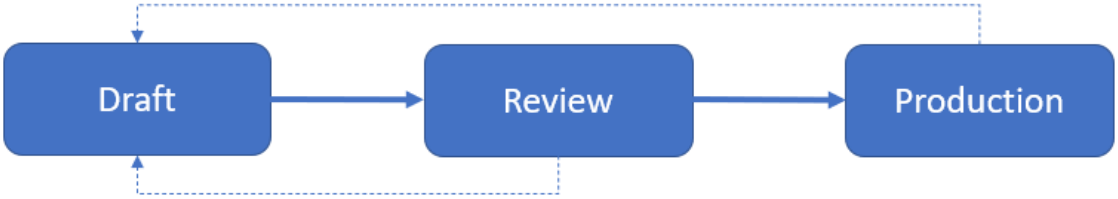
Project workflows

The Arkite software supports different project workflows:

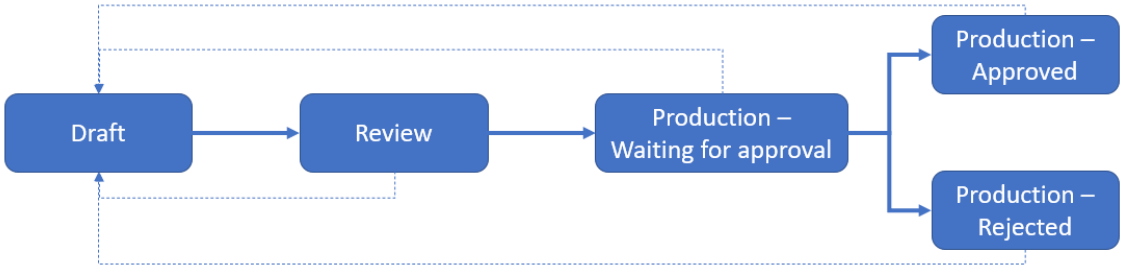
- The most simple workflow only discriminates between projects that are still being configured and projects that are being used in production. Once the project has been finalized, the project status is changed from **Draft** to **Production**.



- In a more advanced setup, a **Review** status is added to the workflow. In this status, the project is validated on all workstations before it is being put into **Production**.



- In the most advanced workflow, a project needs to be approved by one or more approvers before it can be used in production.



Note

For more information on how to define which workflow is implemented, refer to section ***3.4.2 Versioning settings*** *on page 38.*

Project statuses

A project can have the statuses in the table below. Depending on the project workflow that is implemented, one or more statuses can be skipped.

Status	Description
Draft	The project is being prepared offsite.
Review	The project is being validated on all workstations.
Production	The project is ready to be used in production.
Production – Awaiting approval	The project is ready to be used in production, but still needs to be approved by one or more approvers.
Production – Approved	The project is ready to be used in production and has been approved by all required approvers.

Status	Description
Production – Rejected	The project was ready to be used in production but was rejected by at least one approver.

Note

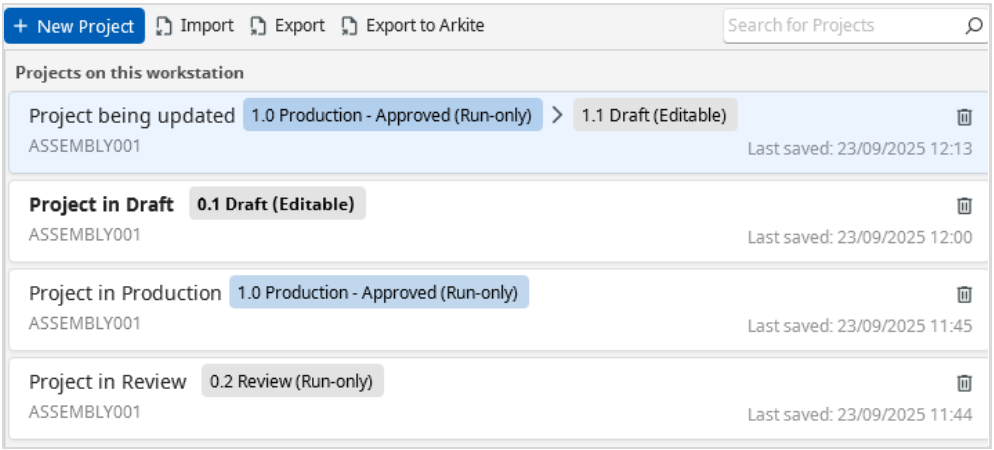
- *In the workflows in which the **Review** status is used, only projects with status **Draft** can be edited or removed.*
- *In the **Draft-Production** workflow, the **Admin** user can still edit a project in status **Production** as long as no new **Draft** version has been created.*

A project needs to be in status **Production – Approved** in order to be used in production. Operators (i.e. users without permissions to create or manage projects in the Arkite software) will only be able to open and run projects in status **Production – Approved**.

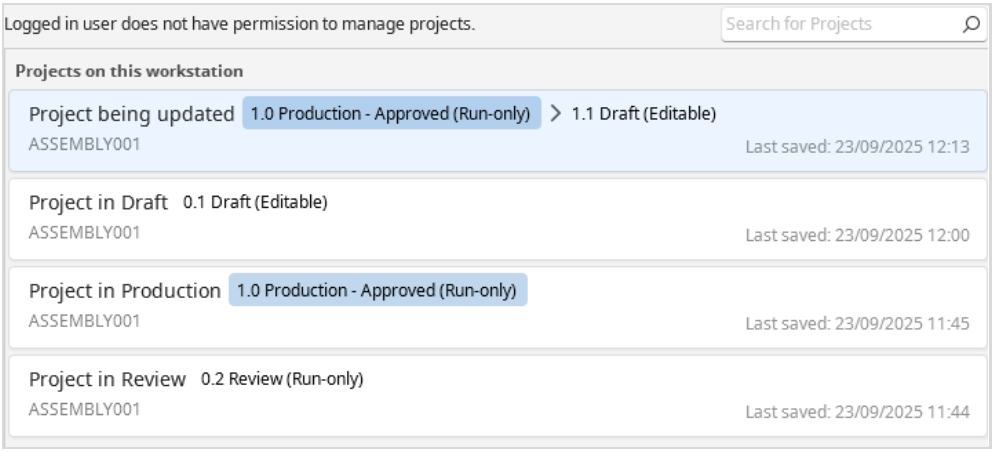
As an example, consider a case in which four projects are available:

- a project in status **Draft** (version 0.1)
- a project in status **Review** (version 0.2)
- a project in status **Production – Approved** (version 1.0)
- a project for which a production version exists, but is currently being updated (status **Draft**, version 1.1)

Users with admin permissions will be able to see and open all projects. For the project with version 1.1, he can choose to open the production version or the current draft version.



Operators can only open the production version of a project. Draft and review versions of a project are not clickable.



3.4.2 Versioning settings

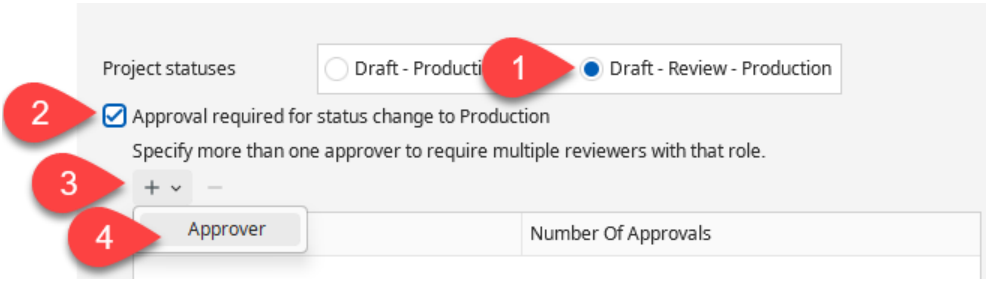
What?

In the versioning settings, an admin user can define which project workflow is used.

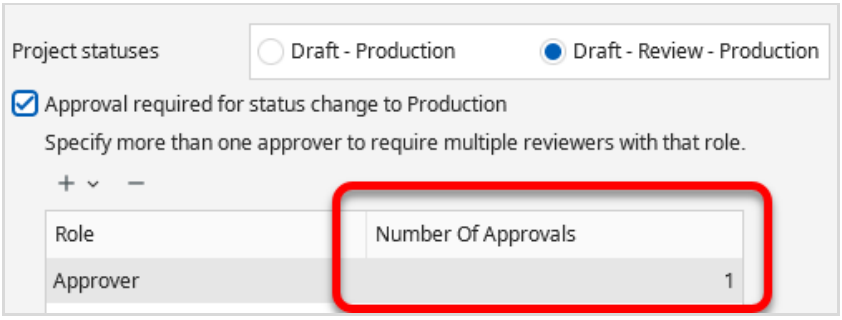
How to manage versioning settings

To manage the versioning settings, execute the steps below:

1. Click **Menu**.
2. Click **Server** and then **Settings**.
3. Under **Versioning**, select the workflow you want to be used:
   - If you only want to use the statuses **Draft** and Production, select the **Draft – Production** radio button.
   - If you want to use the status **Review** too, select the **Draft – Review – Production** radio button.
4. In case you want one or more users to approve a project before it is used in production:
   - Check the **Approval required for status change to Production** checkbox.
   - Click the **Add** button and select the role of the user that needs to approve the project.



   - To define multiple approvers with different roles, add all the required roles.
   - To define multiple approvers with the same role, change the number of approvers in the **Number of Approvals** column.



Note

*The role you add here should have the **Approve Projects** permission. By default, the **Approver** role has this permission.*

*For more information on managing roles and users, refer to chapter **17 User management** on page 283.*

3.4.3 Creating a new project version

What?

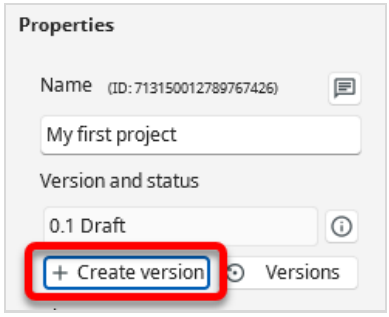
You can create a new project version and update its status:

- From the project overview
- From within a project

How to create a new project version from the project overview

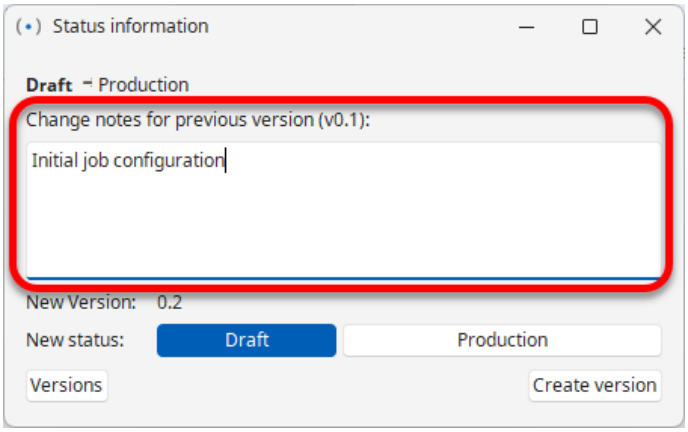
To create a new project version from the project overview, execute the steps below:

1. Click **Menu**.
2. Select the project for which you want to create a new version and/or update the status.
3. In the **Properties** pane under **Version and status**, click **Create version**.

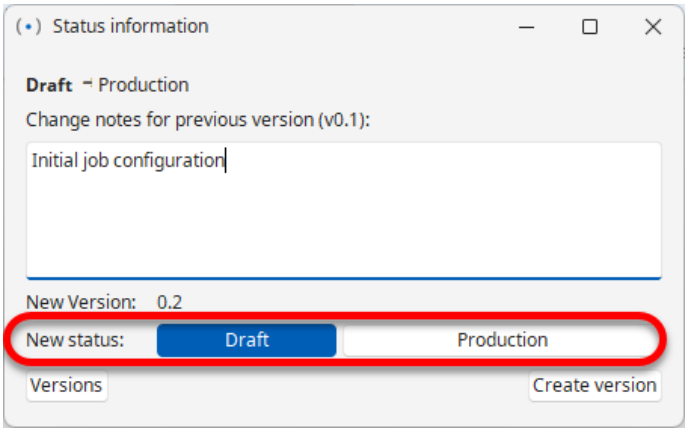


The **Status Information** window appears.

4. In the **Change notes for previous version** text field, describe which changes have been made to the project in the current version.



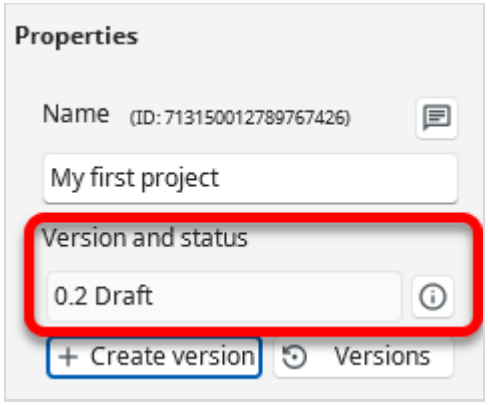
5. Under **New Status**, select the project's status.



Note

The statuses you can select here depend on the project's current status and on the project workflow that you have defined.

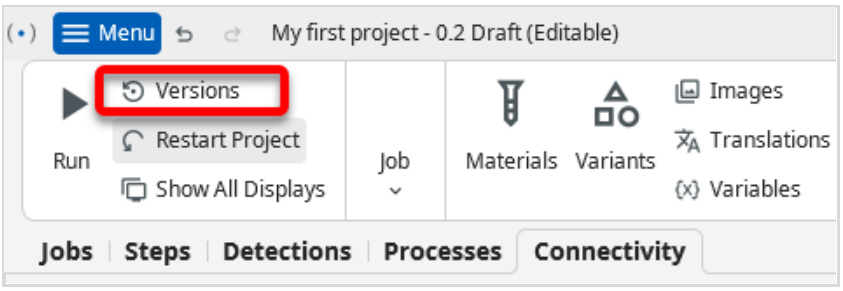
6. Click **Create Version**.
   A new version with a new version number is created.



How to create a new project version from within the project

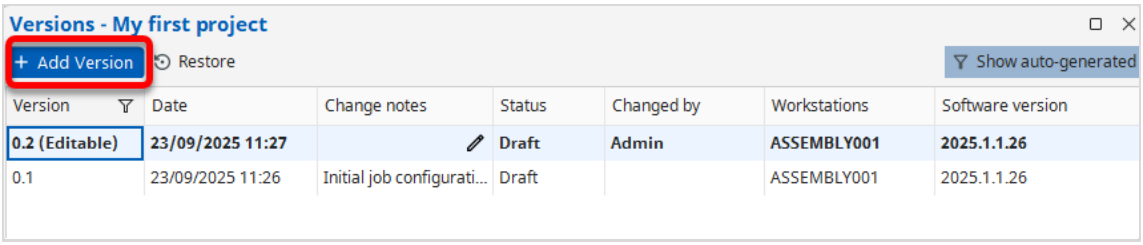
To create a new project version from within the project, execute the steps below:

1. In the ribbon, click **Versions**.



   The **Versions** window appears.

2. Click **Add Version**.



   The **Status Information** window appears.

3. In the **Change notes for previous version** text field, describe which changes have been made to the project in the current version.

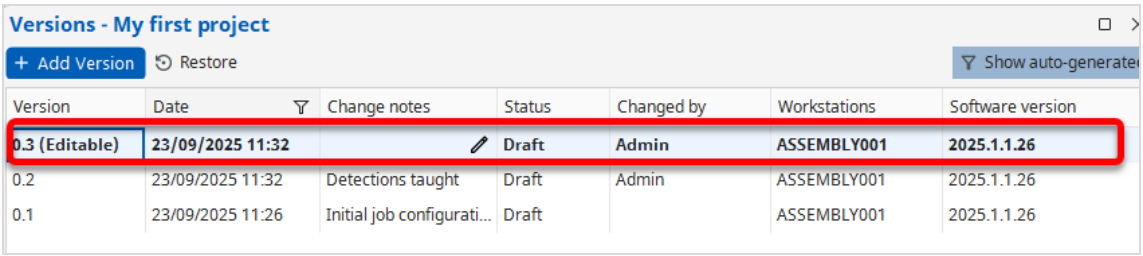
4. Under **New Status**, select the project's status.

   Note

   The statuses you can select here depend on the project's current status and on the project workflow that you have defined.

5. Click **Create Version**.

   A new version with a new version number is created.



3.4.4 Approving or rejecting a project

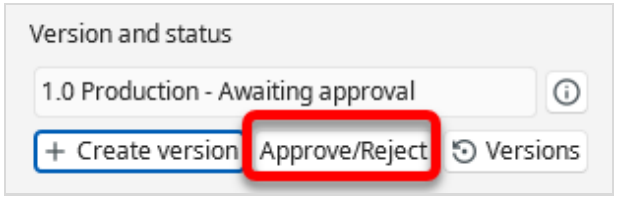
What?

In case the project workflow with approval is configured, one or more users with a role which has the **Approve Projects** permission need to log in to the Arkite software and approve the project, before it can be used in production.

How to approve a project

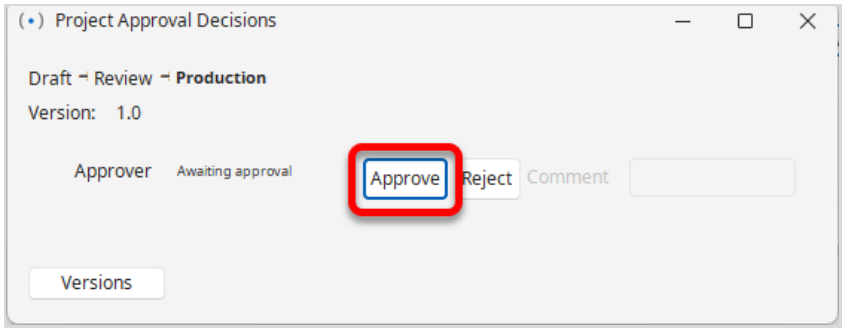
To approve a project, execute the steps below:

1. Log in to the Arkite software with a user that has a role with the **Approve Projects** permission. By default, the **Approver** role has this permission.
2. Select the project with status **Production – Waiting for Approval**.
3. In the **Properties** pane under **Version and status**, click **Approve / Reject**.



The **Project Approval Decisions** window appears.

4. To approve the project, click **Approve**.



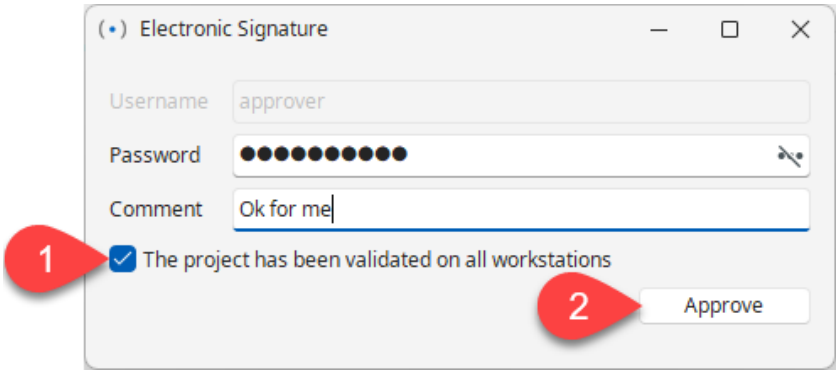
The **Electronic Signature** window appears.

5. Enter your password and, optionally, a comment.

Note

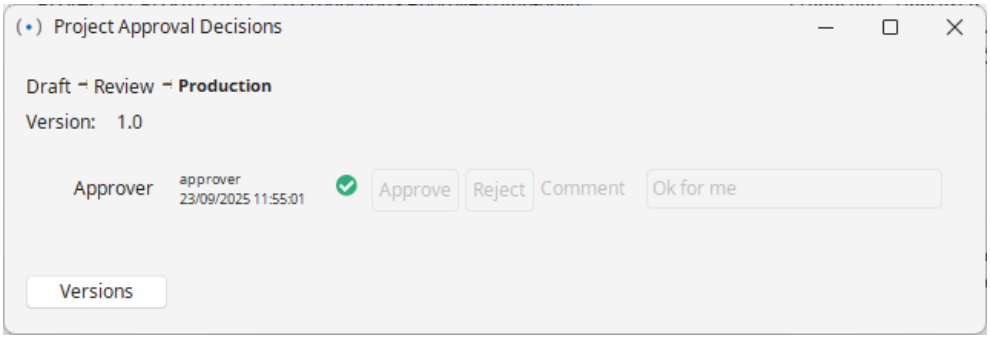
You can click the button to make the password you've entered visible and verify if you haven't made any typing mistakes.

6. Check the **The project has been validated on all workstations** checkbox.
7. Click the **Approve** button.



The approval is registered.

If you were the only or the last approver, the status is changed to **Production – Approved**.

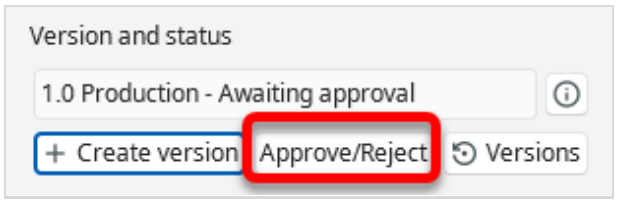


If more approvals are required, the status remains in **Production – Waiting for Approval**.

How to reject a project

To reject a project, execute the steps below:

1. Log in to the Arkite software with a user that has a role with the **Approve Projects** permission. By default, the **Approver** role has this permission.
2. Select the project with status **Production – Waiting for Approval**.
3. In the **Properties** pane under **Version and status**, click **Approve / Reject**.



The **Project Approval Decisions** window appears.

4. To reject the project, click **Reject**.



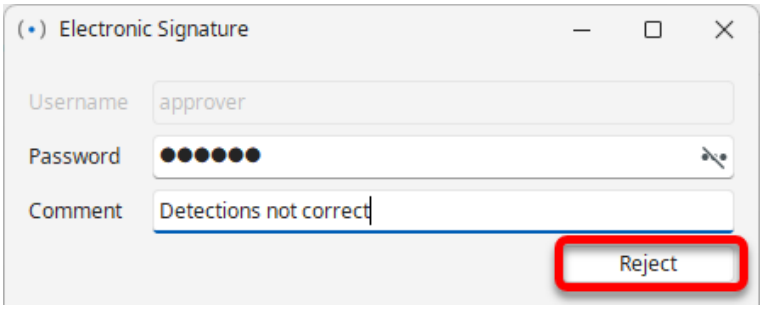
The **Electronic Signature** window appears.

5. Enter your password and a reason for the rejection in the **Comment** field.

   Note

   You can click the button to make the password you've entered visible and verify if you haven't made any typing mistakes.

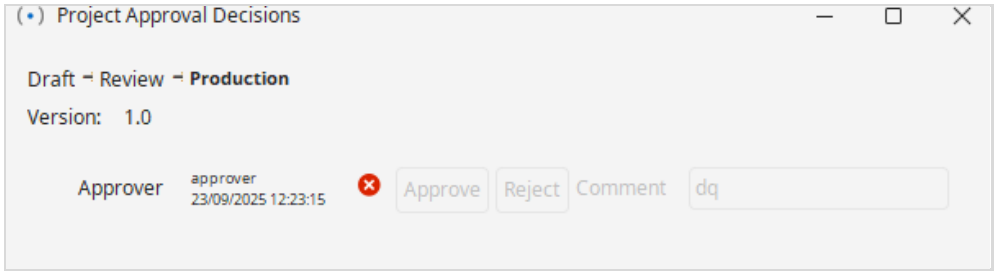
6. Click the **Reject** button.



The rejection is registered.

The status is changed to **Production – Rejected**.

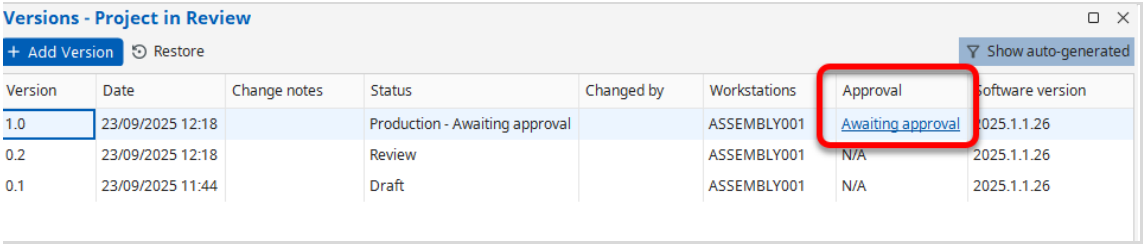
Any additional approvers will no longer be able to approve the project at this stage.



Note

It is also possible to approve or reject a project from the project's version history:

- *Select the project you want to approve or reject.*
- *In the **Properties** pane under **Status**, click **Versions**.*
- *In the **Approval** column, click the text **Awaiting approval**.*
- *Approve or reject the project as explained in the section above.*



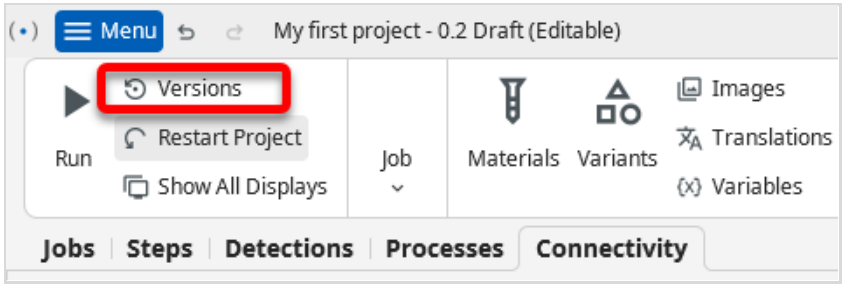
3.4.5 Version history

What?

The Arkite software keeps a history of all previous versions of the project. You can consult it and restore a previous version if required.

How to consult the version history

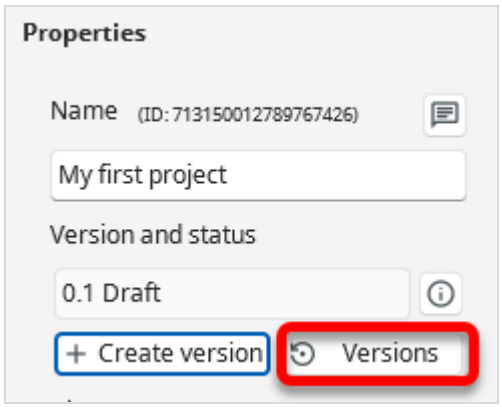
To consult the version history, click **Versions** in the ribbon.



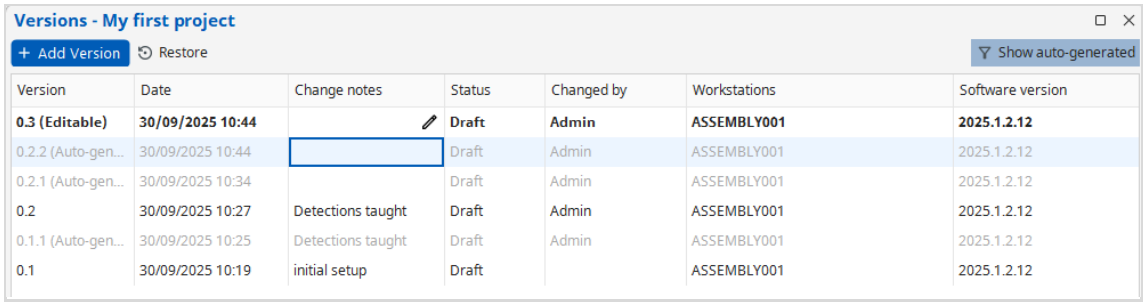
Note

You can also open the version history from the project overview:

- *Select the project for which you want to see the version history.*
- *In the **Properties** pane under **Version and status**, click **Versions**.*



In the **Versions** screen, you get an overview of all previous versions of a project, with their timestamp, change notes and status.

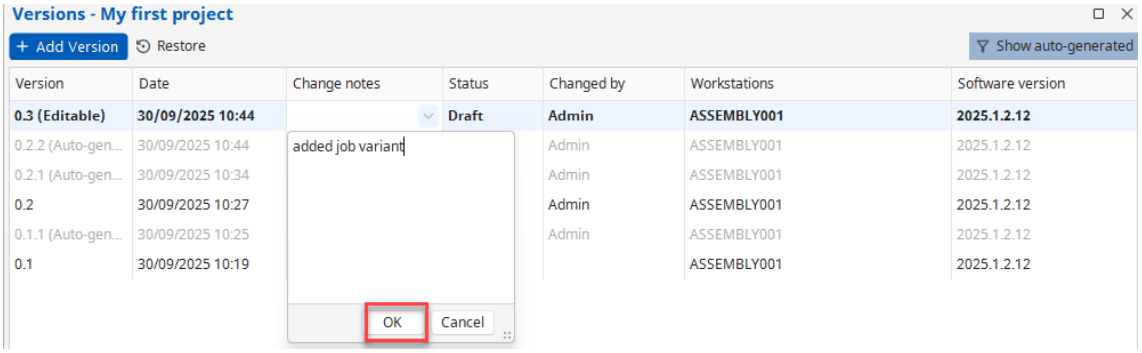


Version	Date	Change notes	Status	Changed by	Workstations	Software version
0.3 (Editable)	**30/09/2025 10:44**		**Draft**	**Admin**	**ASSEMBLY001**	**2025.1.2.12**
0.2.2 (Auto-gen...	30/09/2025 10:44		Draft	Admin	ASSEMBLY001	2025.1.2.12
0.2.1 (Auto-gen...	30/09/2025 10:34		Draft	Admin	ASSEMBLY001	2025.1.2.12
0.2	30/09/2025 10:27	Detections taught	Draft	Admin	ASSEMBLY001	2025.1.2.12
0.1.1 (Auto-gen...	30/09/2025 10:25	Detections taught	Draft	Admin	ASSEMBLY001	2025.1.2.12
0.1	30/09/2025 10:19	initial setup	Draft		ASSEMBLY001	2025.1.2.12

To read the complete change notes for a version, click the corresponding cell in the **Change Notes** column.

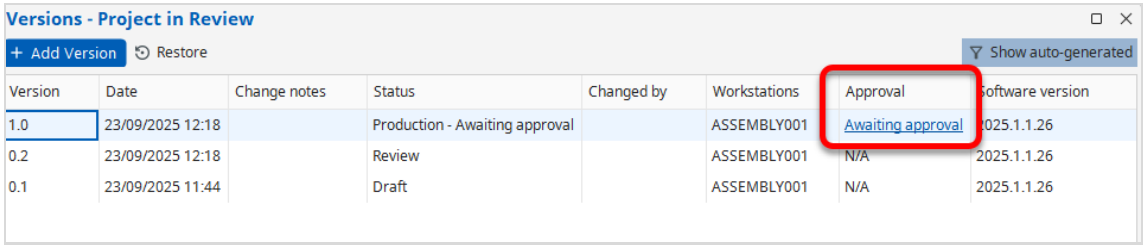
For the current version, you can edit the change notes:

- Click the cell in the **Change Notes** column
- Enter your text
- Click **OK**.



Projects in status **Production – Awaiting Approval** can also be approved or rejected from here, provided the user that is logged in has the required role:

- In the **Approval** column, click the text **Awaiting approval**.
- Approve or reject the project as explained in section **3.4.4 Approving or rejecting a project** on page 41.

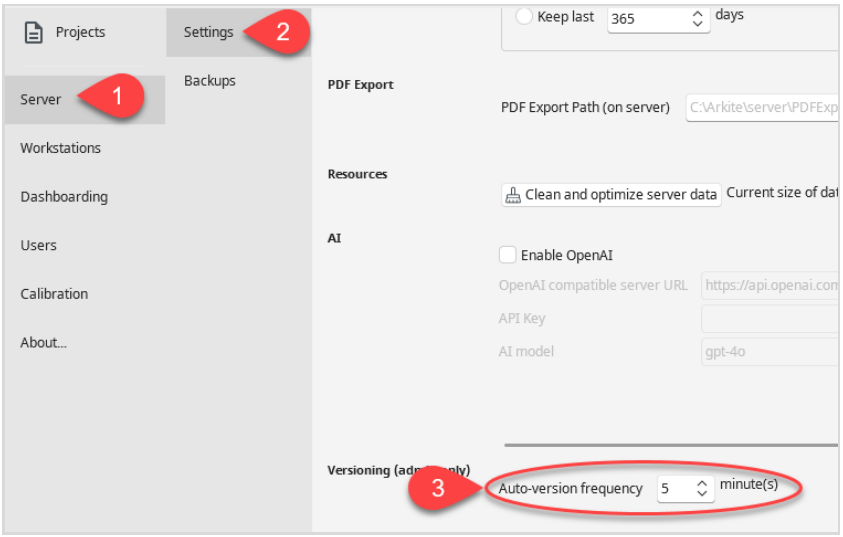


While you are working on your project, the Arkite software automatically generates a version every five minutes (provided there have been any changes). This way, you can always return to a very recent version of your project when something goes wrong.

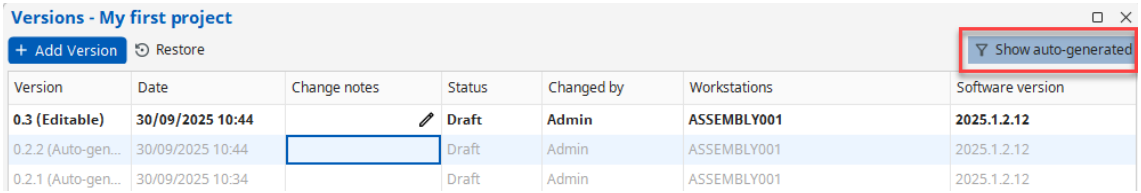
Note

You can configure the frequency with which the Arkite software auto-generates a version. To do so:

- *Click **Menu**.*
- *Click **Server** and then **Settings**.*
- *Under **Versioning (admin only)**, update the value in the **Auto-version frequency** field.*

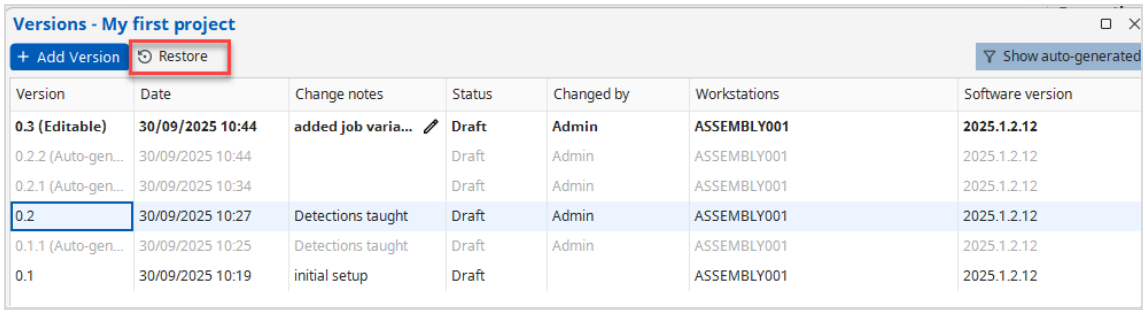


To include the versions that have been automatically generated, click the **Show Auto-generated** button.

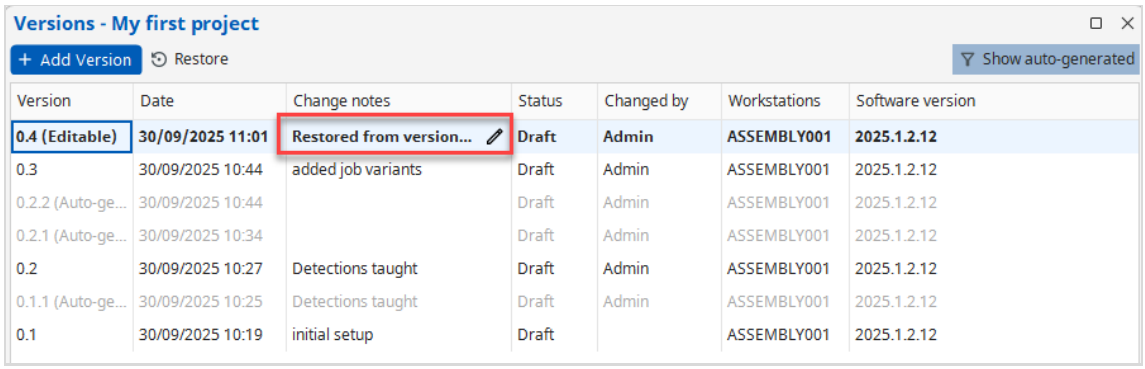


How to restore a previous version

To go back to a previous version, select the version and click **Restore**.



A new draft version of the project is created. In the change notes, it's indicated from which version the restore has been made.



How to delete a previous version

To delete a previous version, right-click the version and select **Delete**.

> ***Note***
>
> *If the workflow with required approval is configured, it is not possible to delete any of the previous versions of a project.*

3.5 Working on projects with multiple authors

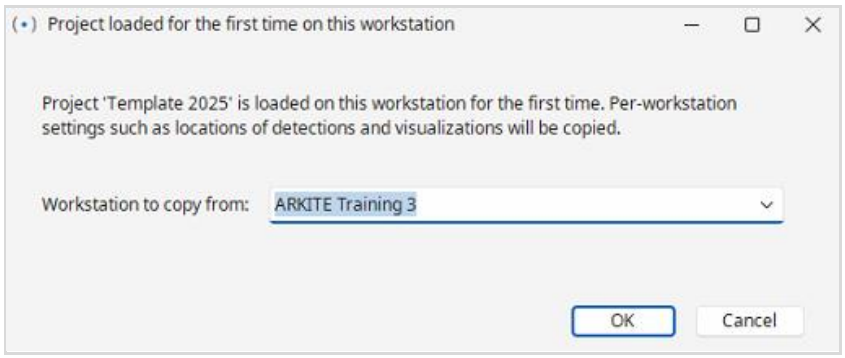
What?

In your project overview, you have access to all projects on any workstation connected to the same server. This also allows you to work on the same project with multiple authors, and to use the same project on multiple workstations. If for example the same or a very similar assembly process is taking place on several workstations, the project will be stored on the server and can be edited and maintained from different workstation.

Opening a project created on another workstation

In your **Projects** overview under the **Menu**, you will see all projects that are on the server, also the ones that were created on another workstation.

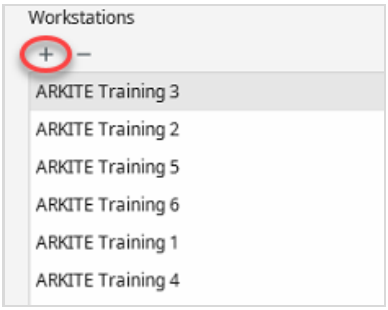
When opening a project for the first time on your workstation, the complete project is copied from another workstation, including all workstation specific settings and properties such as visualizations and the location of detections. The popup below will appear when opening the workstation. Select the workstation you want to copy from and click **OK**.



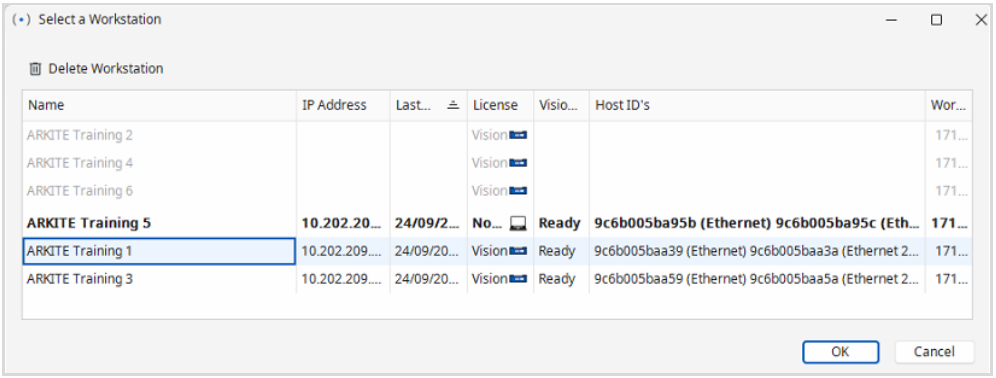
Note

If you want to open the project on an editor, i.e. a workstation without a license, the system will by default switch to another workstation. If you want to open the project on the editor, you first need to add the editor to the workstations in the project properties:

- *Select the project*
- *In the **Properties** pane under **Workstations**, click the **Add** button.*



- *Select a workstation to copy from and click **OK**.*

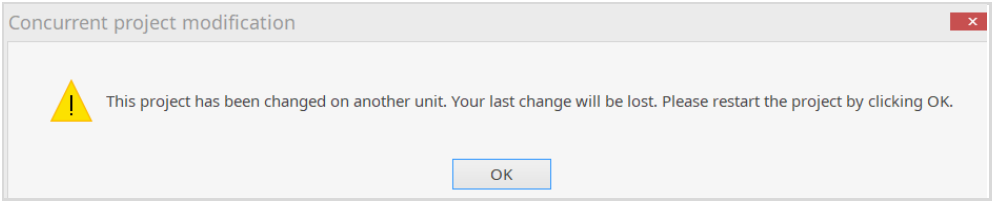


Editing a project created on another workstation

From that moment, you can start editing the project yourself. Anything you add or change, will also be immediately available on the other workstation.

Note

If the project is being edited on several workstations simultaneously, it's possible the project is changed on another workstation before your change can be applied. In that case, the pop up below will appear. Your change will be reverted, the project will be restarted.



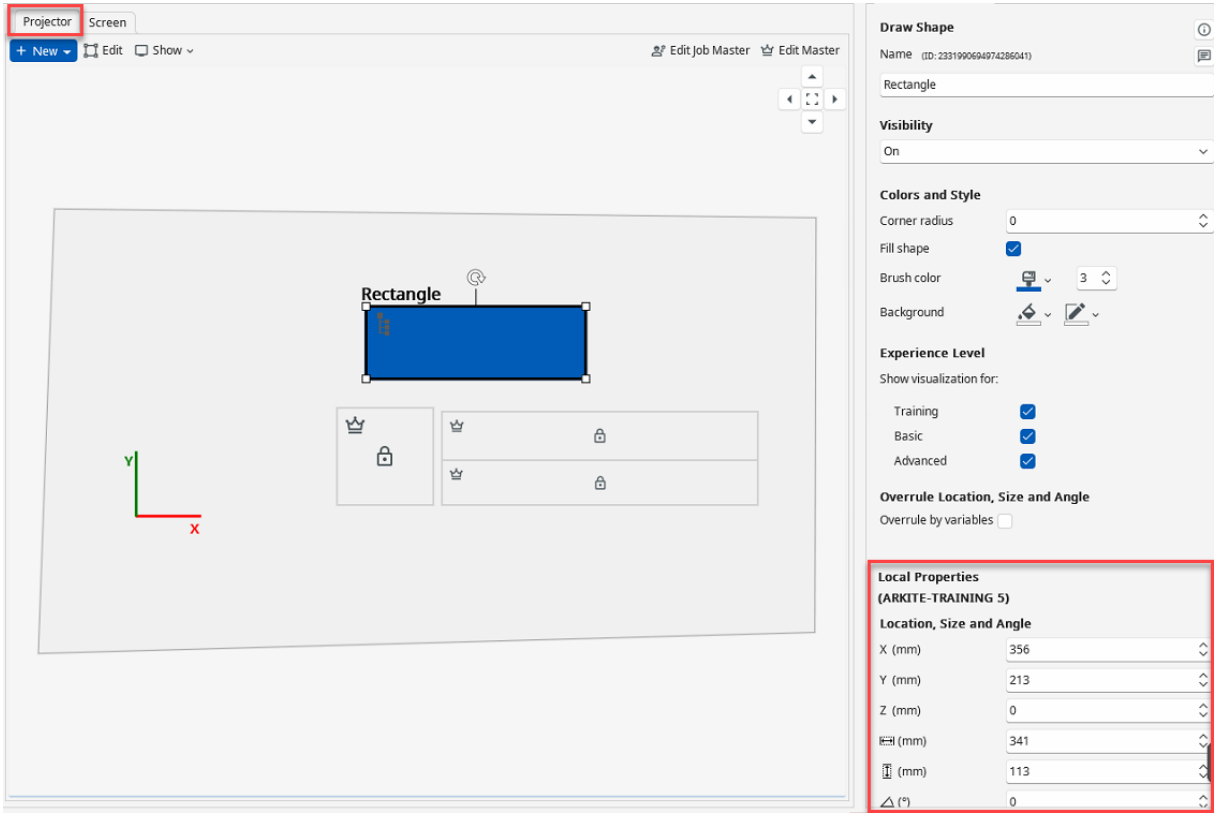
Global properties versus workstation-specific properties

Some properties and settings are **global**. This means they are valid for the project as such and are always exactly the same on each workstation. This applies for example for detections. If you add a detection, change its name or delete it, this will have an immediate effect on all other workstations.

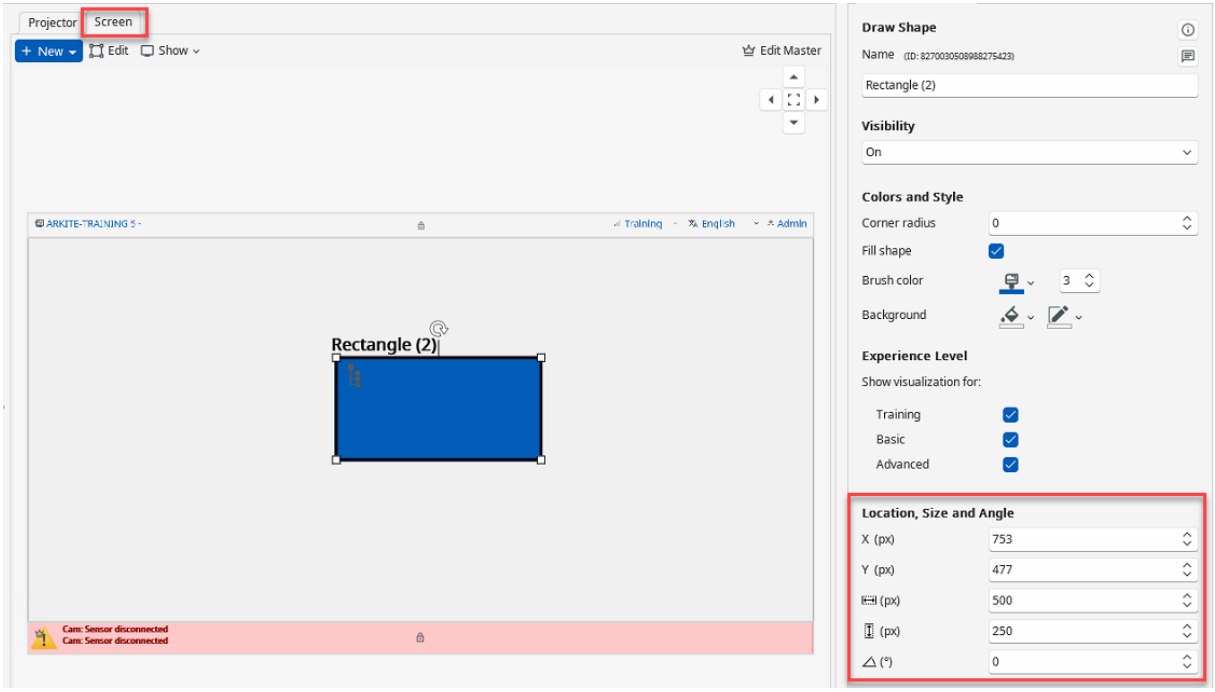
Some properties and settings are **workstation specific**. This means they can be different on every workstation. This applies for example for the **location** of a detection. If you add a detection and teach it, the detection and its location is copied on all other workstations. However, if you change the location afterwards, e.g. by reteaching it, the location will only change on that particular workstation.

In the user interface, it's always indicated if something is global or local/workstation specific.

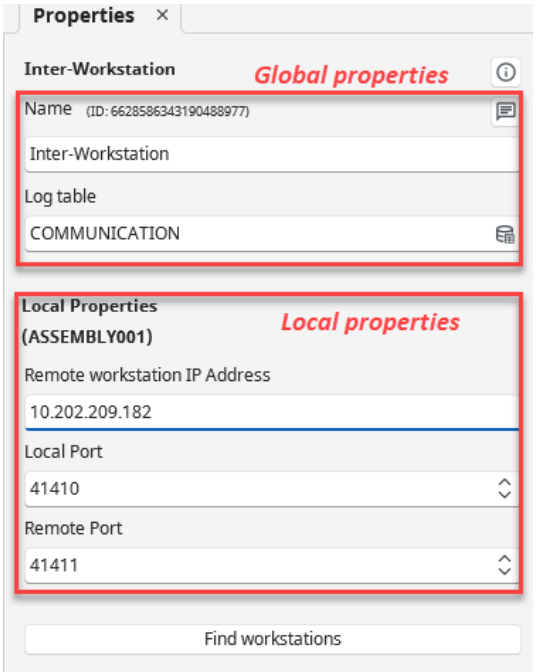
For example, the location of visualizations on a **Projector** display is local or workstation specific, as you can see on the screenshot below.



The location of visualizations on a **Screen** display is global however, as you can see on the screenshot below.



Some elements have properties that are global and others that are local. In the screenshot below, you can see this for an **Inter-Workstation** communication module:



The table below gives an overview of the most important global and workstation specific aspects of a project:

Global	Local / Workstation specific
Project name, description and status	Location of a detection (**Validate** license only)

Detections	Location of visualizations on a **Projector** display
Processes	Display Port
Process steps	Value of a non-shared variable
Text and image instructions	
Displays	
Location of visualizations on a **Screen** display	
Variables	
Value of a shared variable	

Sharing variables

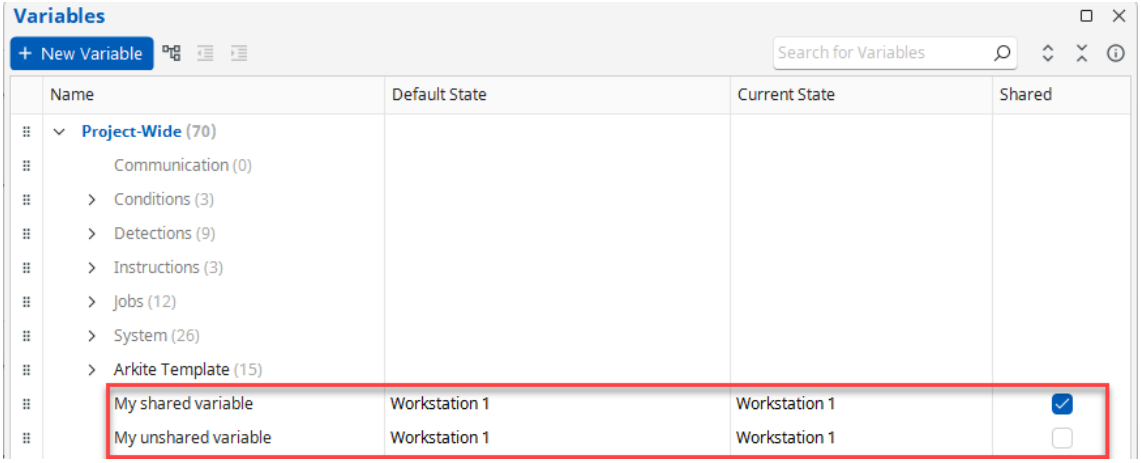
For manual variables and communication variables, you have the choice to synchronize the value across different workstations or to manage them independently. In order to synchronize values for a variable, check the **Shared** checkbox in the **Variables** window.

You can share variables while running the same project on different workstations, but you can also share variables across projects.

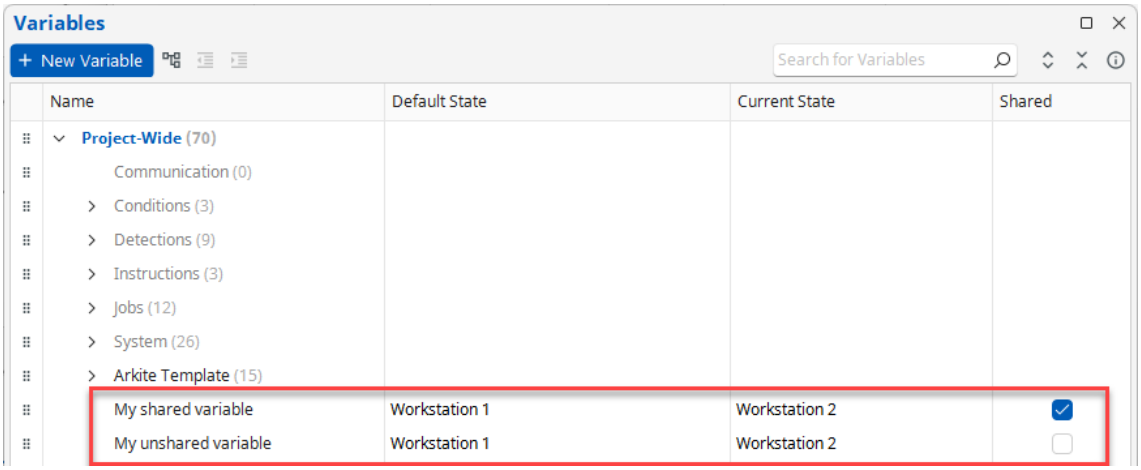
In the example below, two variables have been created, one shared variable and one unshared variable. They both have the value **Workstation 1**.

Be aware that:

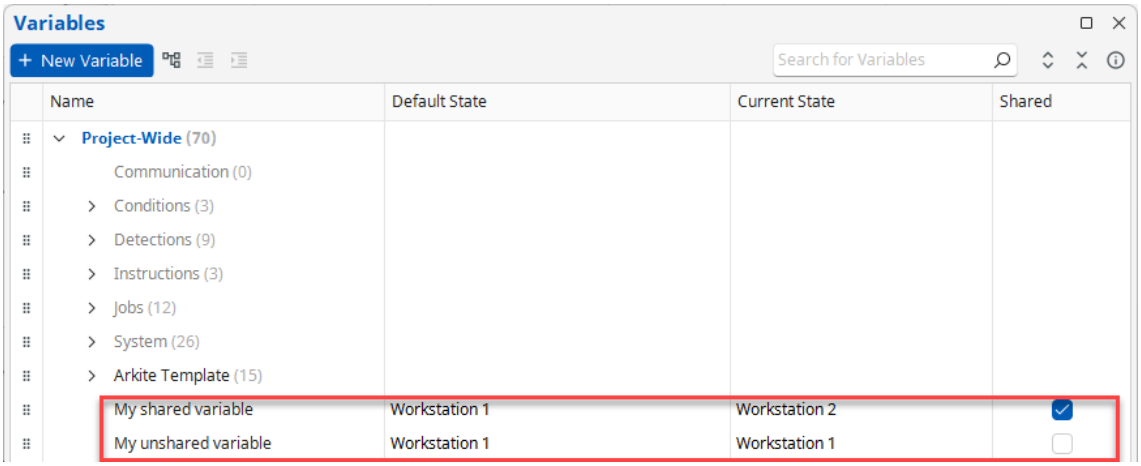
- If you're working in the same project, creating the variables on one workstation will also create them on the other workstations.
- If you're working in different projects, you need to create the variables in both projects on both stations, and mark them as shared accordingly.



Then, the value of both variables is updated to **Workstation 2** on one of the workstations.



On the other workstations, the value of the shared variable is updated to **Workstation 2** as well. The value of the unshared variable hasn't changed however: it is still **Workstation 1**.



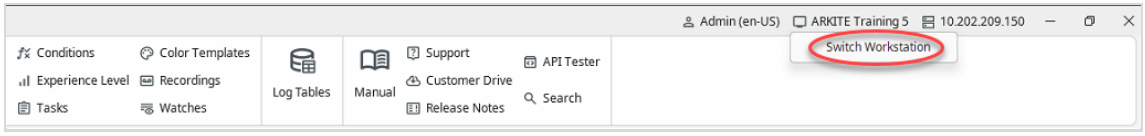
3.6 Switching workstations

What?

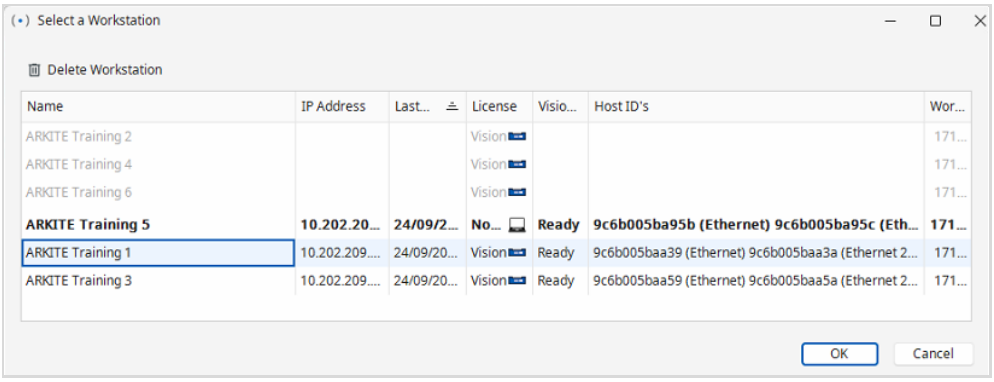
When you are working on an editor, i.e. a workstation that has not been assigned to a license, you can switch to another workstation on the same server. From then, your editor will 'impersonate' that workstation: all workstation specific properties from that workstation will be loaded, and all local properties you edit will be changed on that workstation and that workstation only. Also the license of that workstation will be used.

How to switch workstations

In the top bar, click your workstation name and select **Switch Workstation**.



In the **Select a Workstation** window, select the workstation of your choice and click **OK**.



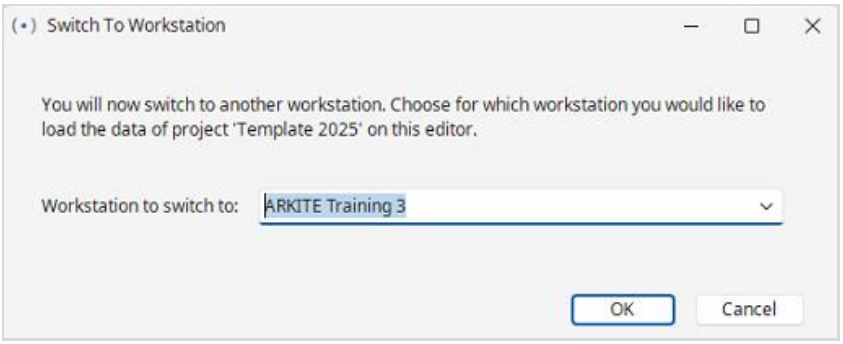
Note

If a project is open when you are switching workstations, the project will be closed after switching. You need to re-open it.

Automatic switch of workstation when opening a project

When you are working on an editor and open a project that has never been opened on that editor before, the system will automatically switch to another workstation.

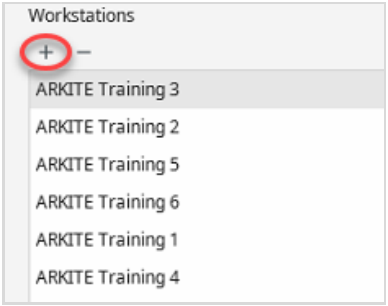
The **Switch to Workstation** pop-up will appear. Select the workstation you want to switch to and click **OK**.



Note

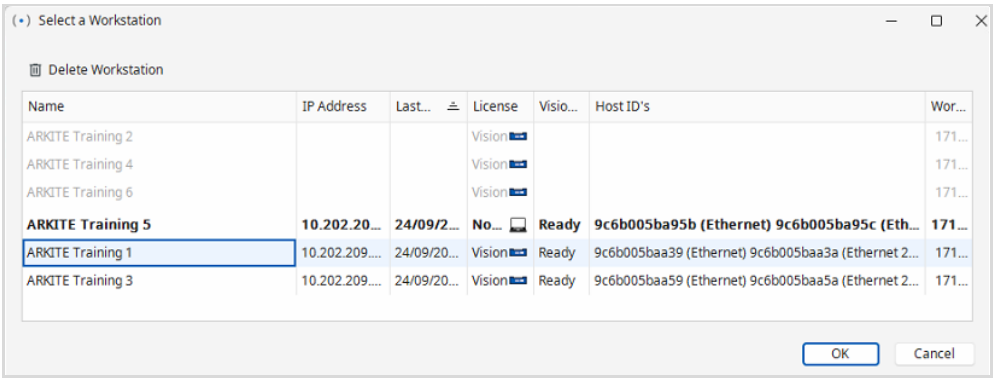
If you want to open the project on your editor instead of switching to another workstation, you first need to add the editor to the workstations in the project properties:

- *Select the project*
- *In the **Properties** pane under **Workstations**, click the **Add** button.*

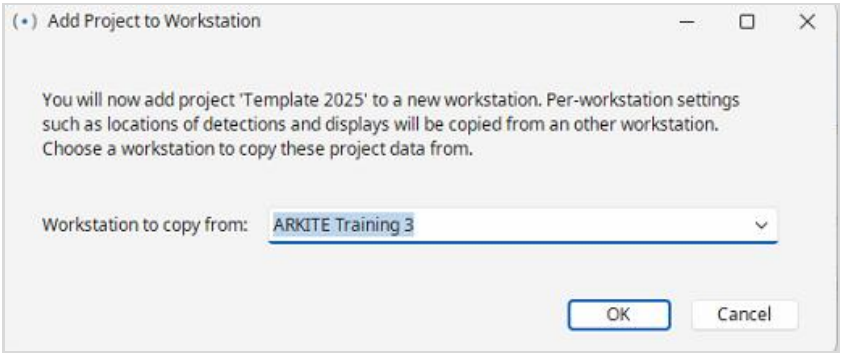


- *In the **Select a Workstation** window, select the workstation you want to add to the project and click **OK**.*

By default, your own workstation is selected.



- *Select a workstation to copy from and click **OK**.*



4. Jobs

4.1 About jobs and job steps

Jobs

A **job** is a work task that an operator performs on a workstation, for example an assembly, a kitting application, a quality inspection etc., or any other work task for which you want to guide your operators through augmented reality work instructions.

Steps

A job will consist of different steps, some of which can be optional, based on one or more properties of the product to assemble.

Job Options

Job options allow you to define different properties for which the assembly steps in your job assembly can be slightly different.

Example

Suppose you're assembling a particular car model. Some options for the job assembly could be:

- *The **fuel type**: petrol, diesel, electric, ...*
- *The **transmission**: manual, automatic, ...*
- *The **color**: blue, black, red, ...*

Job Variants

The combination of different job options results in different **job variants**.

Example

Suppose you're assembling a particular car model with options color, fuel type and transmission.

Some variants for the job assembly could be:

- *a blue car with manual transmission that rides on diesel.*
- *a black car with automatic transmission that rides on petrol.*

4.2 Configuring jobs

Adding a new job

To add a new job, execute the steps below:

1. Navigate to the **Jobs** tab.
2. Click the **New Job** button.
3. Configure the job's properties in the **Properties** pane:

Property	Description
Name	The name for the job. By clicking the **Comment** icon next to the **Name** property, you can add a few words of explanation regarding the use of the job. ***Note*** *Next to the **Name** label, the object ID for the detection is shown. You can copy it to use it when working with the Arkite API.*
Image	Provide an image for the job. It will be used in the project ribbon and on the job selection screen. Click the icon to select or change the image. Click the icon to delete the image.
Product Identifier	Optionally, add an identifier for your product, e.g. a serial number.
Expected Cycle Time	Provide the estimated time it takes to assemble one product. The expected cycle time is stored in the variables **Expected Cycle Time** (format HH:mm:ss) and **Expected Cycle Time Sec** (format ss). This allow you to easily display **the Expected Cycle Time** on a **Projector** or a **Screen** using a Text visualization. The Expected Cycle time is also used in the **Cycle Time** dashboard. For more information about dashboarding, refer to chapter **12 Dashboarding** on page 257. ***Note*** *This cycle time can be overruled by a specific expected cycle time for each **job variant.***
Options	Options allow you to define different properties for which the assembly steps in your job assembly can be slightly different. Click the button to add an option. A popup window will open where you can provide a name and different values for the option. Select an option and – click the icon to edit the option's name or values. – click the icon to delete the option.

Configuring a job option

In the **Options** popup window, you can configure your job option.

In the **Name** field, enter or edit the name for your option, for example fuel type, color or transmission.

Under **Values**, enter all values the option can have.

For example, for the option fuel type, the values could be:

- Petrol
- Diesel
- Hybrid
- Plug-in Hybrid

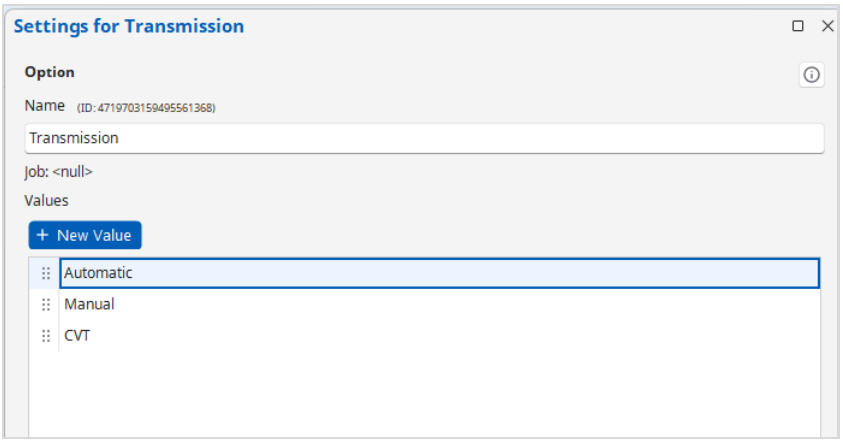
- Electric

Click **New Value** to add a value.

Double-click on a value to edit it.

Right-click a value and select **Delete** to delete it.

Example



Configuring job variants

To add a variant:

1. Click the **Variants** button in the ribbon or the **n variants** link in the job overview.
2. In the **Job Variants** popup window, click the **New Variant** button.
   A job variant is added to the table with variants, automatically generating a **Variant Code** and selecting the first unused combination of option values. All of these data can be edited.

Property	Description
Variant Code	A unique code for the job variant. The code is generated automatically, but you can edit it. ***Warning*** *Variant codes must be unique across all jobs. For example, you can not have a variant code A-1 both in Job X and in Job Y.*
Variant Name	Provide a longer description for the job variant. The name is generated automatically, but you can edit it.
Option	For each of the available job options, a column is shown. Select the applicable value for each option.
image	Optionally, select an image for the job variant. It will be used in the project ribbon and on the job selection screen. ***Note*** *If you don't select an image here, the general job image will be used.*

Property	Description
Expected Cycle Time	Optionally, provide the estimated time it takes to assemble one product for the job variant. The expected cycle time is stored in the variables **Expected Cycle Time** (format HH:mm:ss) and **Expected Cycle Time Sec** (format ss). This allow you to easily display **the Expected Cycle Time** on a **Projector** or a **Screen** using a Text visualization. The Expected Cycle time is also used in the Cycle Time dashboard. For more information about dashboarding, refer to chapter **12 Dashboarding** on page 257. ***Note*** *If you don't enter a cycle time here, the default expected cycle time that is configured in the properties of the job will be used.*

To generate a job variant for all possible combinations of option values, click **Generate all**.

Example

In a case where we have two options with each 3 possible values and one option with 2 possible values, a total of 18 job variants will be generated.

To duplicate a job variant, right-click the job variant and select **Duplicate.**

To delete one or more job variant, right-click the job variant and select **Delete.** To select multiple rows to duplicate or delete at the same time, use Shift-Click or Ctrl-Click.

Importing and exporting job variants

To export your list of job variants to a CSV file, click **Export to CSV**. The export will include the **Variant Code, Variant Name, Variant Expected Cycle Time**, and all **Job Option** columns. The generated CSV file uses a **comma (,)** as the separator.

You can also add job variants by importing a CSV file.

To import the CSV file:

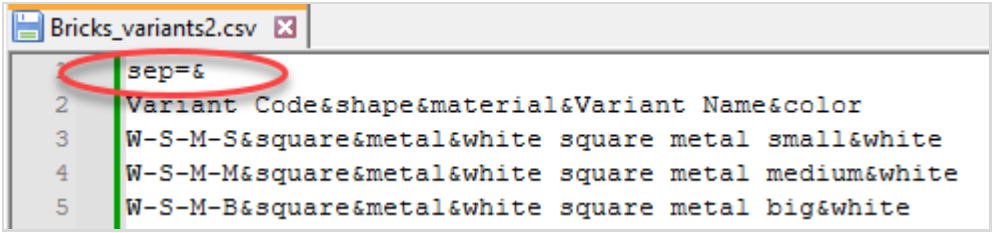
1. Click the **Import from CSV** button.
2. Select the CSV file of your choice.

Note

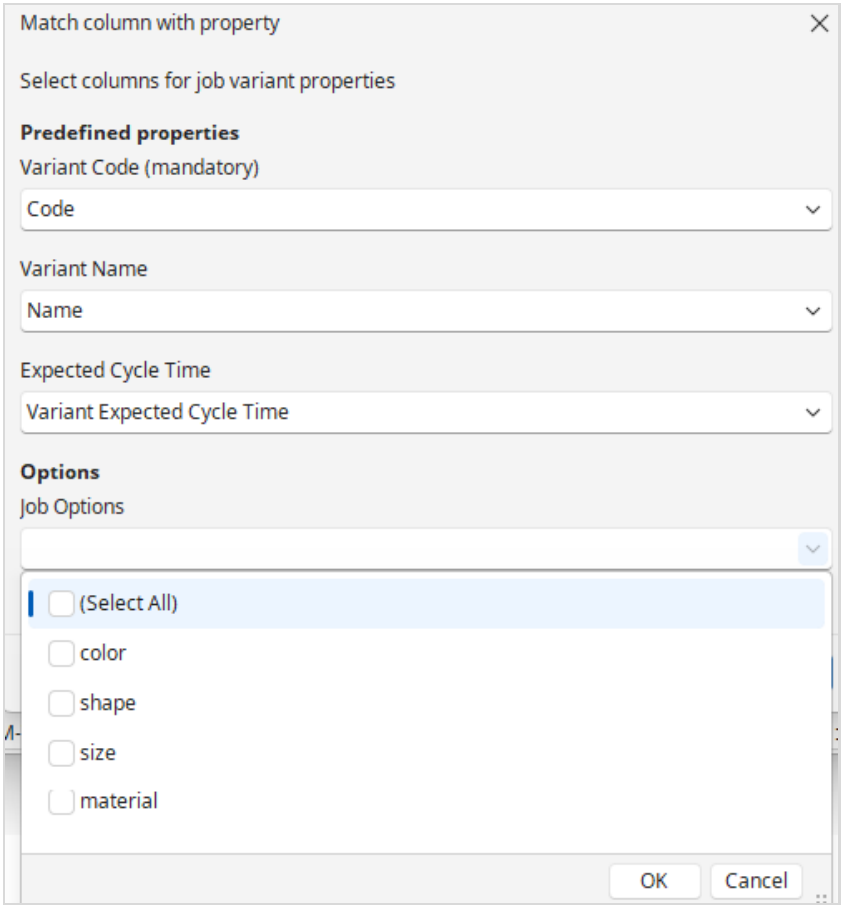
The CSV import automatically detects the following separators:

- *Comma (,)*
- *Semicolon (;)*
- *Colon (:)*
- *Pipe (|)*
- *Tab*

If your CSV file uses a different separator, open the file in a text editor first and add ***sep=[separator]*** *as the first line of the file. Arkite will then import the file correctly. In the example below, & is defined a separator.*



3. In the **Match column with property** window, you can indicate which column from the CSV file you want to use for the **Variant Code**, **Variant Name** and **Expected Cycle Time**. You can also indicate one or more columns to use as **Job Options**.



Note

- *The **Variant Code** property is mandatory. By default, the Arkite software will suggest to match it with the first column in the CSV file.*
- *If the CSV file contains variants with a variant code that already exists in your project, these variants will be overwritten with the data from the CSV file. The variants will not be duplicated.*
- *If you don't match the **Variant Name** property with a column or if the CSV contains an empty variant name for a variant, a variant name will be generated automatically based on the job option values.*
- *If you match columns with the **Job Options** property for which no options exist yet in your project, the job options will automatically generated with all values that occur in the CSV file.*
- *For existing job options, values that occur in the CSV file, but not in the project will be automatically added.*

4. Click **OK**.

The job variants from the CSV file will be imported into your project.

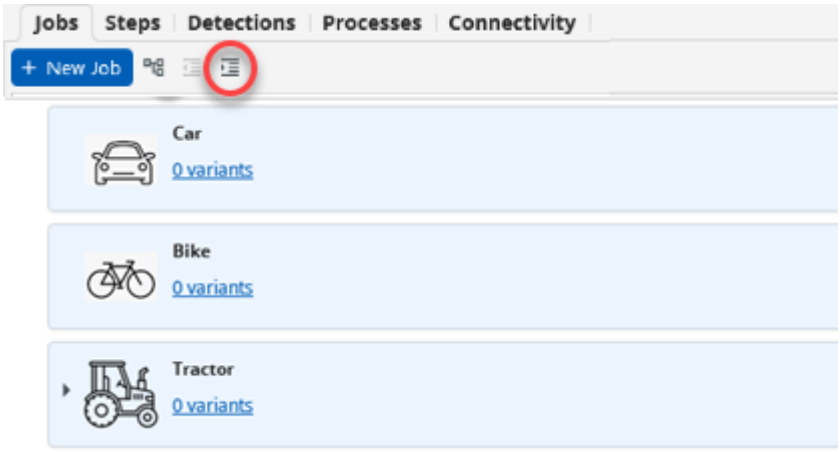
Grouping jobs

You can organize your jobs in different job groups, to create some structure and have a better overview.

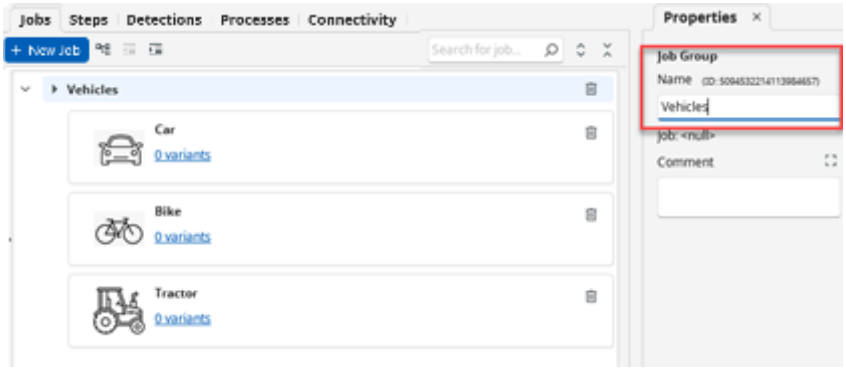
To create a job group, click the **Group** icon.



You can also select the different jobs you want to add to the job group (using Shift-click or Ctrl-click) and then click the **Indent** button.



A job group is created. In the **Properties** pane, you can define a name for the job group.



You can drag additional jobs into the job group using your mouse.

To delete a job group, click the icon on the right.

You can add different levels of job groups by using the **Indent** and **Unindent** buttons.

Deleting jobs

To delete a job, select the job and click the icon on the right.

4.3 Configuring steps

About configuring steps

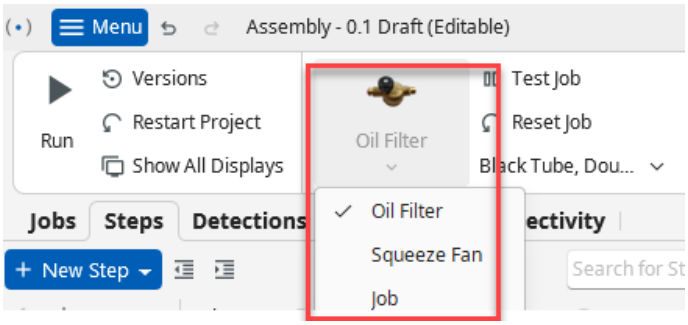
A job will consist of different steps, some of which can be optional, based on one or more properties of the product to assemble.

On the **Steps** tab, you can define all the steps in your job assembly.

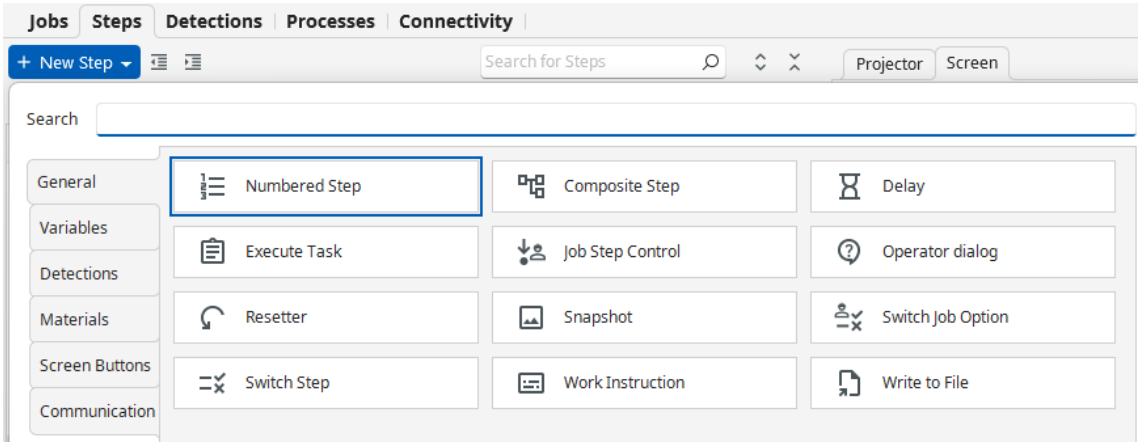
4.3.1 Adding steps

To add a new step, execute the steps below:

1. Navigate to the **Steps** tab.
2. In the ribbon, select the job for you which you want to add steps.



3. Under **Steps**, click the **New Step** button and select the step type you want to add.



Note

*You can also enter the step type in the **Search** field to find it.*

4.3.2 Setting up your step structure

About setting up your step structure

Several possibilities are available to apply a structure your assembly steps, with the purpose of creating a better overview for your operators, manage step order or repetition or mark steps as optional.

The sections below provide more information about each of the possibilities below:

- Numbered steps

- Composite steps
- The **Switch Job Option** step
- The **Switch** step
- The **Check Quality** step with rework flows

Numbered steps

A **numbered step** is a logical step in your job assembly from the viewpoint of the operator. It can consist of multiple (detection) steps in the Arkite software.

Example

*The operator step **Fix a screw** can consist of the detection steps below:*

- *Take screwdriver*
- *Pick screw*
- *Tighten screw*
- *Place screwdriver*

Using numbered steps, you can add structure to your job assembly and divide it into different subassemblies. You can create different levels of numbered steps.

Example

In our demo fuel filter assembly job, the subassembly where the cap is assembled onto the fuel filter housing could be assembled like this:

1. *Assemble cap*
   - 1.1. *Place the cap*
     - *Pick cap*
     - *Place cap*
   - 1.2. *Fix the cap*
     - *Take screwdriver*
     - *Pick screw*
     - *Tighten screw*
     - *Place screwdriver*

The numbered steps in your job assembly can be displayed on the operator screen using a **Numbered Steps List** visualization. It serves as a table of contents, intended to help the operator in gaining insight into the assembly process. The Arkite project template contains a preconfigured **Numbered Steps List**, but you can also add and configure one yourself. For more information, refer to section **Numbered Steps List** on page 107.

Composite steps

A composite step is a step that consists of two or more child steps. These child steps can be individual steps or other composite steps, that in their turn also consist of child steps.

In the **Properties** pane for a composite step, you can define in which order and how often the child steps need to be executed.

Property	Description
Name	The name for the process step.

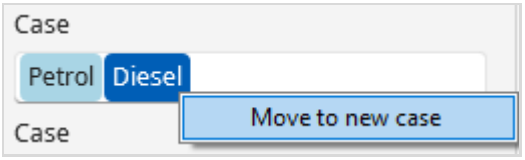
Property	Description
	Note *Next to the **Name** label, the object ID for the process step is shown. You can copy it to use it when working with the Arkite API.*
Child step order	Define in which order the child steps are executed: – **Sequential order (icon):** The child steps are executed in the order in which they appear. This is the default option. – **Any order (icon - Validate and Vision licenses only):** It does not matter in which order the child steps are executed. However, all child steps must be executed for the process to continue to the next step. – **One arbitrary step (icon):** Only one of the child steps should be executed for the process to continue to the next step. – **All steps simultaneously (icon - Validate and Vision licenses only):** All steps should be executed at exactly the same time for the process to continue to the next step. For example, you can use this type of step to check if all tools are present when you start the assembly, or if you want an operator to press two virtual buttons at the same time to confirm an action, as a safety measure. ***Note*** *Do not use the **Any order and All steps simultaneously** options when using a **Guide** license, because in that case there is no way to verify which child steps are executed by the operator. When running a project that contains such steps on a system without sensor, these steps will be treated by the system as sequential steps.*
Child step instruction separator	Only available if the **Child step order** is **Any order**, **One arbitrary step** or **All steps simultaneously**. Define which separator word should be used to concatenate the instructions for the different child steps: – **Auto:** Use the default separator. For **Any order** and **All steps simultaneously**, the default is ***and***. For **One arbitrary step**, the default is ***or***. – **And:** Use **and** as a separator. – **Or:** Use **or** as a separator. ***Example*** *If you have a composite step, configured as Any Order with two child steps **Take Screwdriver** and **Pick Screw**, by default the system will show the instruction **Take Screwdriver and Pick Screw**.* ***Note*** *Instead of concatenating the instructions for the different child steps, you can also define a custom instruction on the level of the composite step.*
Repetition	Indicate if the composite step should be repeated: – **None:** The composite step is executed only once. After execution of the last child step, the process continues with the step after the composite step. – **Repeat:** The step is repeated in an endless loop. After the execution of the last child step, the first child step is executed again. ***Note*** *This feature should be used in combination with a condition: the composite step is repeated as long as the condition is valid. For more information about working with conditions, refer to section **4.5 Working with conditions** on page 118.*

Property	Description
	– **Repeat number of times:** The step is repeated a specified number of times (see below). – **Repeat for each element:** The step is repeated for all elements in a specified list (see below).
From index / To index	If you have indicated the step should be repeated a number of times, enter here the index to start counting from and the index after which the loop should be stopped. The value should be numeric. You can either enter a fixed value or use the value of a variable. To do so, click the {x} icon and select the variable of your choice. Alternatively, you can also type the variable name between double braces, i.e. *{{Variable name}}*. By default, the **From index** is 1. ***Example*** *If the **From index** is 1 and the **To index** is 5, the step will be repeated 5 times.* *If the **From index** is 2 and the **To index** is 5, the step will be repeated 4 times.* *You can use these properties to save the index when a loop is stopped before it is completed, and pick up the loop again at a later stage to complete it.*
List of elements to loop over	If you have indicated the step should be repeated for all elements in a specified list, enter the list here. You can also specify a variable that contains the list. To do so, click the {x} icon and select the variable containing the list. Alternatively, you can also type the variable name between double braces, i.e. {{Variable name}}.
List type	If you have indicated the step should be repeated for all elements in a specified list, enter the list type here: – Json Array – List with separator
Separator	If you have indicated the step should be repeated for all elements in a specified list of type **List with separator**, enter the separator here.
Variable for current element	If you have indicated the step should be repeated for all elements in a specified list, this variable will be updated with the current element.
Variable for count	If you have indicated the step should be repeated for all elements in a specified list, this variable will be updated with the total number of elements in the list.
Variable for index	If you have indicated the step should be repeated for a specified number of times or for all elements in a specified list, this variable will be updated with the current iteration.

The Switch Job Option step

The **Switch Job Option** step allows you to set up different assembly flows for the different values of a job option.

Property	Description
Name	The name for the **Switch Job Option** step. By clicking the **Comment** icon next to the **Name** property, you can add a few words of explanation regarding the use of the step. ***Note*** *Next to the **Name** label, the object ID for the process step is shown. You can copy it to use it when working with the Arkite API.*

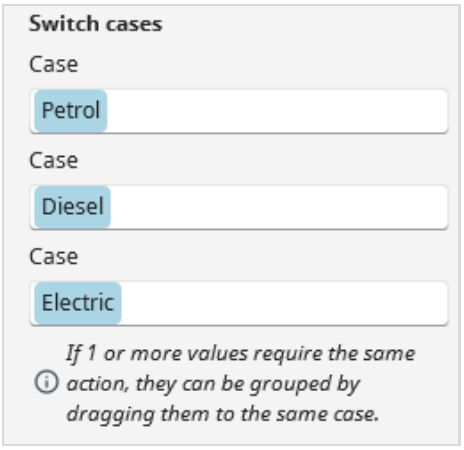
Property	Description
Job Option	Select the job option for which you want to create different assemblies.
Switch Cases	By default, a case will be available for every possible value of the job option. In your steps tree, there will be a branch for each value. Under each branch you can add the required steps for that value. In case more values require exactly the same actions, you can group these values by dragging them into the same case. Doing so will also merge the branches in the steps tree. If you want to undo the merging of a case, you can right-click one of the values and click **Move to New Case.** 

Example

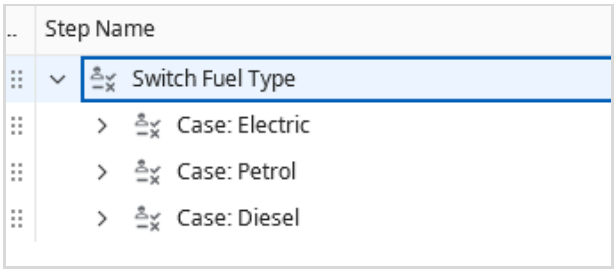
Suppose you're assembling a particular car model. One job option could be the car's fuel type (petrol, diesel, electric).

*The **Switch Job Option** step allows you to easily set up specific steps for each fuel type.*

The screenshot below show the default configuration, with a separate branch for each fuel type.

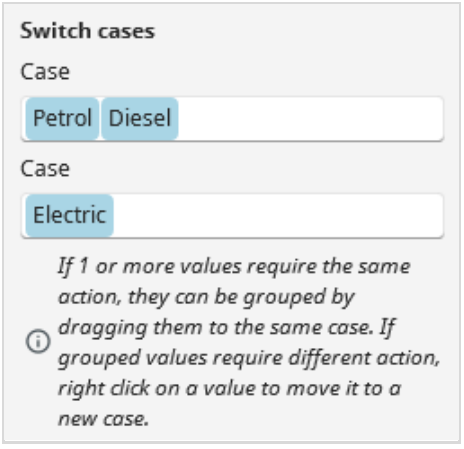


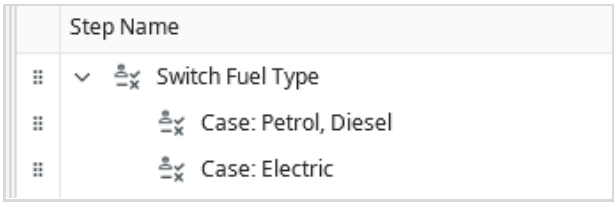
Step Properties



Steps tree

*In case **Petrol** and **Diesel** require the same steps, you can drag them into the same case in the step properties, resulting in the configuration below.*



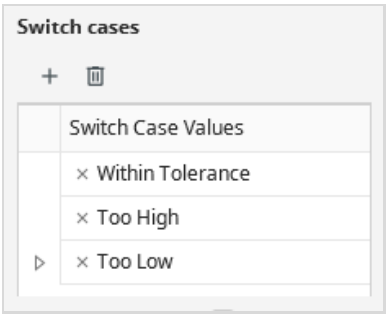


Step Properties Steps tree

The Switch step

The **Switch** step allows you to set up different cases with substeps. Only one of the cases will be executed, depending on the value of the variable you assigned to the switch.

To set up a switch, configure the properties below:

Property	Description
Name	The name for the **Switch** step. By clicking the **Comment** icon next to the **Name** property, you can add a few words of explanation regarding the use of the step. ***Note*** *Next to the **Name** label, the object ID for the process step is shown. You can copy it to use it when working with the Arkite API.*
Variable for Switch	The state of this variable will determine which case will be executed.
Switch Cases	Enter the value(s) for which you want to execute different substeps. Type a value and press the **Enter** button to submit it. You can add multiple values for one case. By default, two cases are available, but you can add more cases by clicking the + button or remove cases by selecting them and clicking the button. ***Example*** *In the example below, three cases are defined based on the value of the **Torque Result** variable. Different steps will be executed if the value is either **Within Tolerance**, **Too High** or **Too Low**.* 
Add Default Case	Check this property to add a case to execute for any value of the **Switch Variable** that is not assigned to another case.

For each of the cases, a subpath is created within the **Switch** step. In each of these subpaths, you can add steps defining what needs to be done in the situation at hand.



The Check step with rework flows (Vision license only)

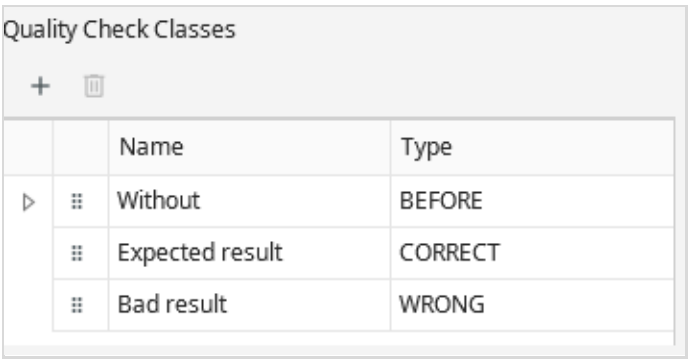
For the detection types below, you can add a check step with rework flows:

- Quality Check
- Object with Quality Check
- Barcode
- Text

When adding a **Check** step, **Rework Flow** substeps are automatically created for each possible class linked to the detection.

By default, the three classes below are linked to an ***object with quality check*** *detection*:

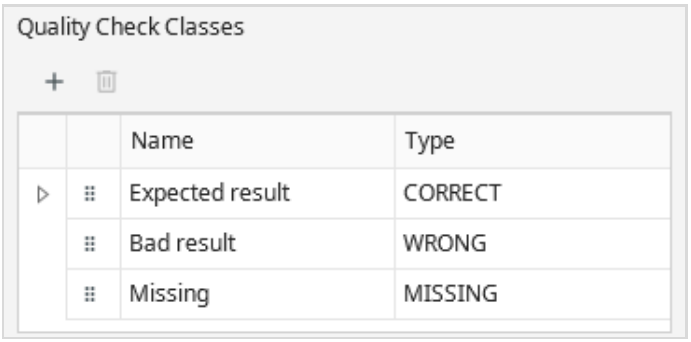
- The **Without** class
- The **Expected Result** class
- The **Bad Result** class



By default, the three classes below are linked to a ***quality*** *detection*:

- The **Expected Result** class

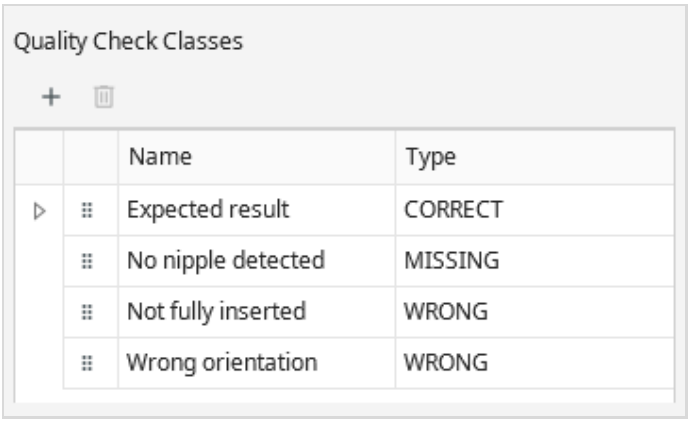
- The **Bad Result** class
- The **Missing** class



Note

*In the **Properties** of the quality detection or the object with quality check detection, you can add more classes or rename existing classes. Under **Quality Check Classes**, click the + button and provide a name and a type for your class. A new class is typically created if you want to show a different error text or give different rework instructions.*

In the example below, two classes have been created for specific WRONG situations, and the class with MISSING type has been renamed to match the specific situation.



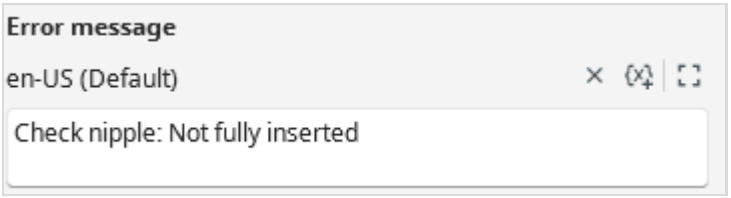
The three classes below are linked to a ***text*** *or* ***barcode*** *detection*:

- The **Expected Result** class
- The **Bad Result** class
- The **Missing** class

Classes for texts or barcodes cannot be renamed or edited. For texts or barcodes without validation, all detected texts/codes are considered to be an expected result.

For each **Rework Flow** substep, you can:

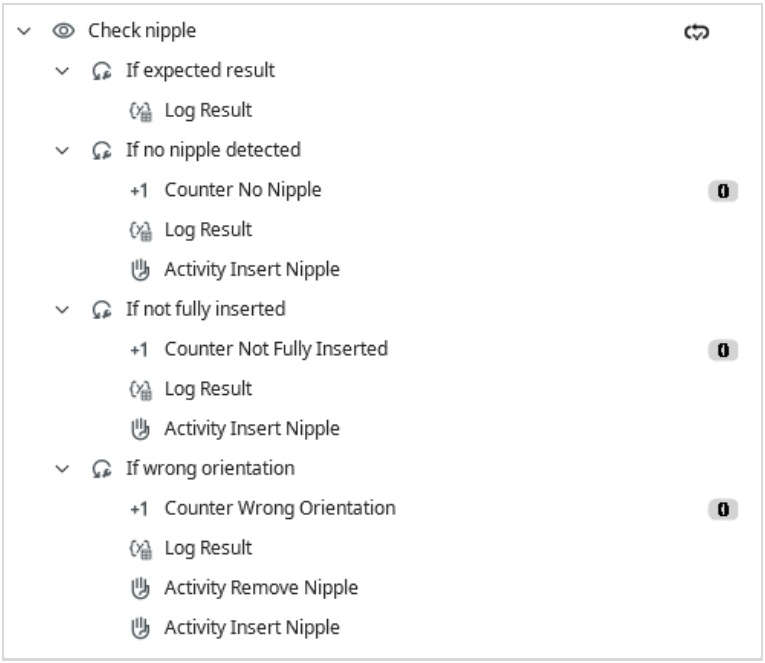
- add process steps defining what needs to be done in the situation at hand
- for Quality Check Classes of type WRONG or MISSING: define an error message in the **Properties** of the **Rework Flow** step.



When the **Check Quality** step is triggered, the corresponding **Quality** or **Object with Quality Check** detection will be evaluated. Depending on the result of this evaluation, the corresponding rework flow will be selected. The process steps in this path will be activated and in case of a result of type WRONG or MISSING, the error message you have defined will be displayed in red on both the **Projector** and the **Screen**.

In the example below:

- For each of the classes, the status of a variable will be logged.
- In case of a result of type **Missing** or **Wrong**, the number of the errors will be counted and specific detections with customized instructions will be triggered



What happens after the execution of the rework flow, depends on the initial evaluation of the quality check:

- If the evaluation resulted in a class of type CORRECT, the system will execute the steps in the rework flow and then move to the next step in the process.
- If the evaluation resulted in a class of type WRONG or MISSING, the system will execute the steps in the corresponding rework flow and then re-evaluate the Quality detection. That process will continue until the **Expected Result** is reached.

How to organize your steps

You can organize your steps as described below:

- Right-click a step and select **Duplicate** to duplicate it.
- Select a process step and drag it with your mouse to change its order.

- To delete a process step, you can:
  - Select it and press the **Delete** key on your keyboard
  - Right-click the step and select **Delete**.
- Right-click a step, select **Move** and then **To job** to move the step to another job.
  Select the job you want to move the step to and click **OK**.
- Right-click a step, select **Move** and then **To process** to move the step to a process.
  Select the process you want to move the step to and click **OK**.

- Create composite steps:
  - Click **New Step** and select **Composite Step**.
    A composite step is created. Any new step you add while this group is selected will be added as a child step of this step. You can also drag existing steps into the composite step.
  - Select one or more steps and click the  button. This way, a composite step can also be created.
- Click the  button to expand the steps tree and see all child steps.
- Click the  button to collapse the steps tree and only see the upper level.

4.3.3 Step types

This section contains an overview of all types of steps that are available to you to build your job assembly. They are divided in the categories below:

- **General steps**: some basic, often used steps.
- **Variables**: steps that use the values of a variable as an input or an output.
- **Detections**: steps related to the detections you have added
- **Materials:** steps related to the materials you have configured.
- **Screen buttons:** steps related to the buttons you have added to the touch screen.
- **Communication:** steps related to the communication modules you have configured.
- **Extra**: a number of advanced, less often used steps.

General steps

The table below gives an overview of all process step types that are available in the **General steps** subtab:

Step type	Description
Numbered step	A logical step in your job assembly from the viewpoint of the operator. It can consist of multiple (detection) steps. The numbered steps in your job assembly can be displayed on the operator screen using a **Numbered Steps List** visualization. It serves as a table of contents, intended to help the operator in gaining insight into the assembly process. ***Note*** *For more information, refer to section **Numbered steps** on page 62.*

Step type	Description
Composite step	A step that can contain a number of child steps. It allows you to control the order in which child steps need to be executed or to repeat the child steps a number o times. ***Note*** *For more information, refer to section **Composite steps** on page 62.*
Delay	Set a certain time in milliseconds for the job to wait.
Execute task	A number of steps coming from a task. Tasks allow you to re-use the same steps across jobs. ***Note*** *For more information, refer to section **6.8 Working with tasks** on page 186.*
Job Control	Go to a specific step in your job. You can either select the step or type the name of the step in a field. ***Note*** *If the field contains a value that does not match with a step name, the **Job Control** step will not be executed and an error will be added in the error tray.* *If the field contains the name of the **Job Control** step itself, the **Job Control** step will not be executed.* *If the field contains a value that matches with multiple step names, the first step with a matching step name will be activated.* When using the step in a process, you can also choose to: – Abort the active job – Go to the next step in the active job – Go to the previous step in the active job, on the condition this previous step is either a detection step or a work instruction.
Operator Dialog	Show a popup on the operator screen, asking the operator to confirm an action or to make a choice. The operator's choice is saved in a response variable. Combine this step with a **Switch** step using the response variable.

Step type	Description
Process Control	Control a particular process. You can either select the process or type the name of the process in a field. ***Note*** *If the field contains a value that does not match with a process name, the **Process Control** step will not be executed and an error will be added in the error tray.* You can choose to: – Reset a process – Pause a process – Pause and reset a process – Start a (paused) process – the process will continue from the step where it has been paused – Restart a process, i.e. reset and start it – Go to the next step in a process – Go to the previous step in a process ***Note*** *Going to a **Previous** step is only possible if that previous is step is a detection step or work instruction.* – Go to a specific step in a process. You can either select the step or type the name of the step in a field. ***Note*** *If the field contains a value that does not match with a step name, the **Process Control** step will not be executed and an error will be added in the error tray.* *If the field contains the name of the **Process Control** step itself, the **Process Control** step will not be executed.* *If the field contains a value that matches with multiple step names, the first step with a matching step name will be activated.*
Resetter	Reset a variable or entity to its default state.
Snapshot	Takes a snapshot from the selected feed and stores it as a *.jpeg or a Data URI. ***Note*** *For making RGB product snapshots, we recommend using the Arkite RGB sensor. When connecting an Arkite RGB sensor, the exposure time is set by default to 30 milliseconds.* *To set another exposure time, navigate to the **Connectivity** tab and select the sensor card. In the **Properties** pane under **Exposure**, uncheck the **Automatic** checkbox and in the **Exposure time (ms)** property, set the exposure time of your choice.*
Switch Job Option	Step that allows you to set up different assembly flows, for the different values of a job option. ***Note*** *For more information, refer to section **The Switch Job Option step** on page 64.*
Switch Step	Step that allows you to set up different assembly flows, based on the value of a variable. Only one flow will be executed, depending on the value of the variable you assigned to the switch. ***Note*** *For more information, refer to section **The Switch step** on page 66.*

Step type	Description
Work Instruction	Provide additional text and/or image instructions. ***Note*** *The process will never automatically continue after a **Work Instruction** step, Therefore you always need to provide a **Next** or **Confirm** button on the screen or combine it with a **Job Control** step.*

Variables

The table below gives an overview of all process step types that are available in the **Variables** subtab:

Step type	Description
Counter	Increment or decrement the value of a variable by 1
Find and Replace String	Allows you to look for a text string within a fixed or dynamic text and replace it with another text string. ***Example*** *Suppose you want to flatten a multi-line Windows-style string into a single line, separating the original lines with a |.* *Add a **Find and Replace String** step with the configuration below:* ***Input:** (the variable containing) the original multi-line string* ***Find string:** \r \n* ***Replace string:** |* ***Variable for result:** typically the same variable that contained the original multi-line string, but you could also select another variable.*
Function	Allows you to calculate or evaluate a function with fixed numbers and/or dynamic content based on the value of one or more variables. For a full overview of the supported functions, please consult the framework documentation. ***Note*** *Always use . as the decimal separator to separate the integer part from the fractional part of a number.* *To define the order of several operations, brackets () can be used.*
Set Variable	Update the value of a variable or a list of variables. In the **State to Set** field, enter the value you want to set for the selected variable(s). To insert dynamic content, click the {x} icon at the location where you want to include the content and select the variable containing the dynamic content. Alternatively, you can also type the variable name between double braces, i.e. *{{Variable name}}*. You can also use a variable within a variable. The variable you want to update can also be dynamic: by selecting the **By Name** option you can provide a variable name with fixed and dynamic parts. This allows you to update variables in a loop. To do so, embed the step in a composite step that is repeated for a number of elements. ***Example*** *If you provide the name **Orientation_x{{index}}**, the variable that is set will depend on the value of the **index** variable. If the value of the **index** variable is 01 at the time of execution of the step, the variable **Orientation_x01** will be set.*
Split String by Index	Extracts a fragment of a string in a variable based on: – **index:** the position of the first character of the string – **length:** the number of characters to extract, starting from the index

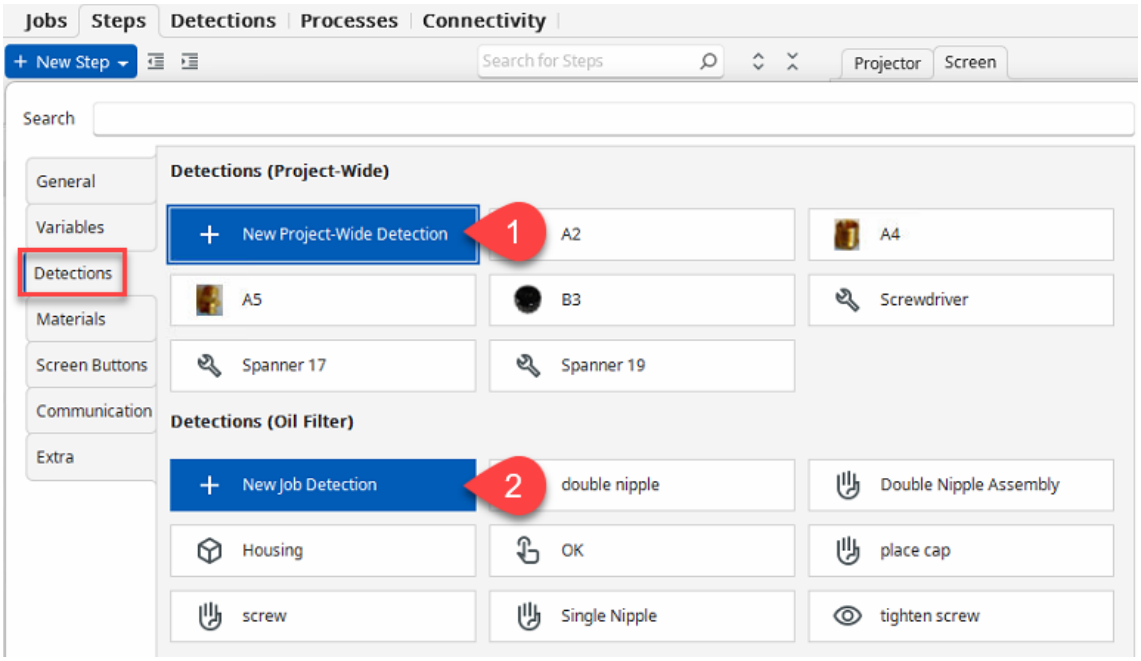
Step type	Description
Split String by Argument	Extracts a fragment of a string in a variable based on: – **split symbol:** a character used to split the string – **index:** the number of occurrences of the split symbol after which you want to extract the fragment Use the **Variable** property for the string that needs to be split, the **Variable for Result** property for the extracted fragment and the **Variable for remainder** property for what is left of the string after extracting the fragment you needed. ***Example*** *If the string you want to split is* ***"Arkite;is;a;great;system;for;guiding;operators"*** *and you want to retain only the word* ***great****, then you can use ; as a split symbol and 4 as index.* *In that case, the variable indicated in the* ***Variable for remainder*** *property will contain what is left from the string, i.e.* ***Arkite;is;a;system;for;guiding;operators"***
String Formatter	Compiles a text string containing fixed text and/or dynamic text based on the value of one or more variables. To insert dynamic text, click the {x} icon at the location where you want to include the text and select the variable containing the dynamic text. Alternatively, you can also type the variable name between double braces, i.e. *{{Variable name}}*. You can also use a variable within a variable. For example, if the value of a variable **Index** is 01, then *{{Variable{{Index}}}}* will resolve into the value of the variable **Variable01**. ***Tip*** *For long text strings, click the*  *icon to enlarge the* ***String to format*** *text box.*
String Parser	Searches a JSON or XML string for the names of specified variables and stores the corresponding values of the first match as the current state of those variables.
Variable changed	Check the value of a particular variable. Depending on the specific settings, the job will continue if: – the variable has a particular value – the variable does not have a particular value – the variable's value matches another variable's value – the variable's value does not match another variable's value – the variable's value is changed
Variable Logger	Log the current state of one or more variables.

Detections

At a certain point in your assembly, you can integrate a step in which an action is done with a detection.

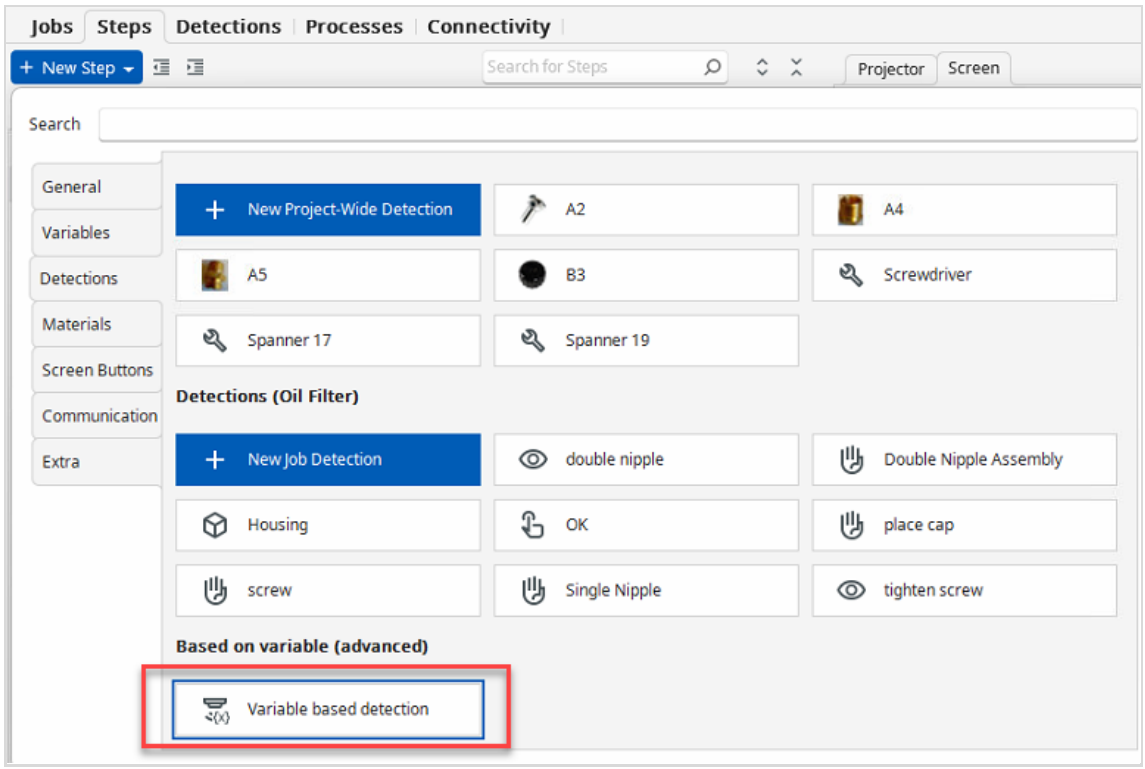
However, before you can do so, you first need to add the detection:

- Click the **New Job Detection** button in the **Detections** subtab to add a detection you only want to use in the job you are configuring.
- Click the **New Project-Wide Detection** button in the **Detections** subtab to add a detection you want to use in all the jobs in your project.

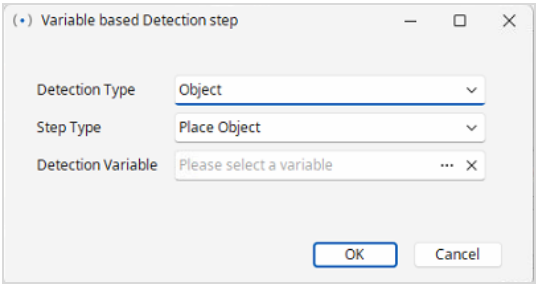


For more information regarding the next steps, refer to section **5.5 Adding detections** on page 129.

You can also add a step in which the detection that is used depends on the value of a variable.



In the popup that appears, select a detection type and action, and specify a variable in the **Variable for detection name** property. If this variable contains a value that does not match with a detection in the project, or with a detection that is not taught, the project will be paused. An error will be shown in the error tray and logged in the variable configured in the **Variable for error message** property.



The actions that are available to be used in a process step depend on the detection type. The table below provides you with an overview of available actions per detection type.

Detection type	Available actions
Object	– **Place:** put the object at the location where it should be. – **Take:** remove the object from the location where it should be.
Tool	– **Place:** put the tool at the location where it should be. – **Take:** remove the tool from the location where it should be.
Picking bin	– **Place bin:** put the picking bin at the location where it should be. – **Take bin:** remove the picking bin from the location where it should be. – **Pick:** take material from the picking bin ***Note*** *If you are adding a pick step, it is often better to use the material from the **Materials** subtab, rather than the picking bin. If at some point, the material is linked to a different picking bin, your pick step will be automatically updated.*
Activity	**Activity**: perform the activity.
Virtual button (Validate and Vision license only)	**Press**: press the virtual button.
Quality Check (Vision license only)	**Check:** perform the quality check. When adding a **Check Quality** step, substeps are automatically created for each possible outcome of the quality check. In these substeps, you can create rework flows for each of the outcomes. For more information, refer to section **The Check step with rework flows (Vision license only)** on page 67.

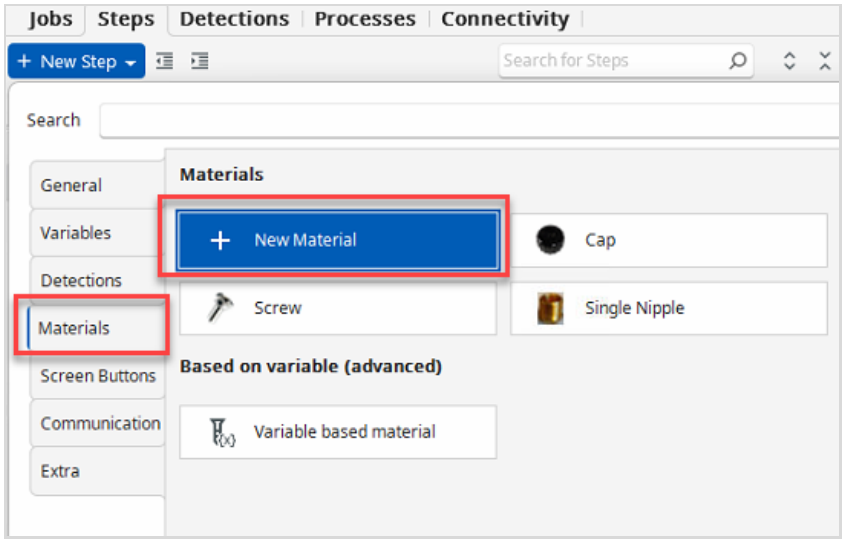
Note

*When a detection is not taught, the step cannot be completed: the action that is required to trigger the detection is not available. If teaching the detection is not possible (when using the system with a **Guide** license), you always need to provide a **Next** or **Confirm** button on the screen or combine it with a **Job Control** step.*

Materials

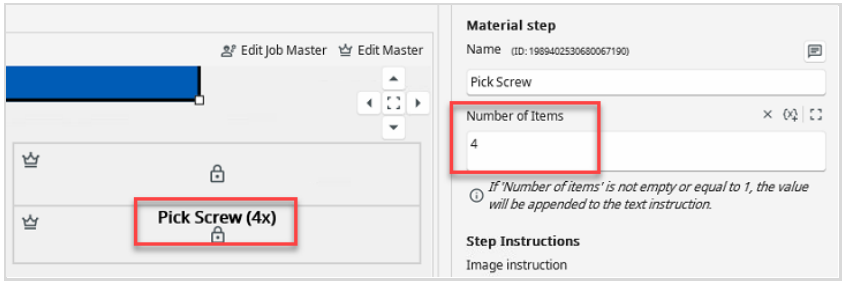
At a certain point in your assembly, you can integrate a step in which a material is picked from a picking bin.

However, before you can do so, you first need to add the material. Click the **Add Material** button in the **Materials** subtab.



For more information regarding the next steps, refer to section **5.8 Managing materials** on page 159.

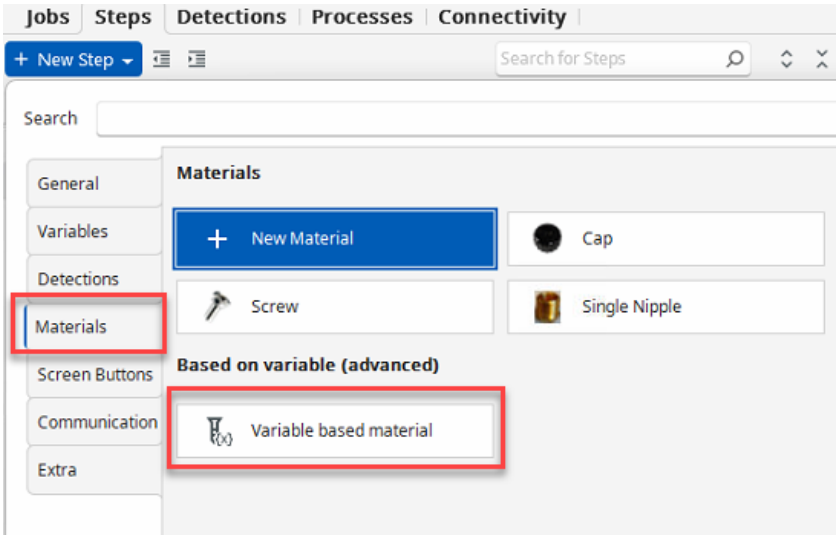
In the **Number of Items** property, you can indicate how many pieces need to be picked from the picking bin. If this property contains a value that is more than one, the number of items will automatically be appended to the text instruction that is displayed to the operator.



Note

Be aware the sensor cannot see or verify if the correct number of items is actually picked from the picking bin. It will just validate the picking action itself.

You can also add a step in which the material that is picked depends on the value of a variable, defined in the **Variable for material name** property. If this variable contains a value that does not match with a material in the project, contains a material that is not linked to a picking bin or is linked to a picking bin that is not taught, the project will be paused. An error will be shown in the error tray and logged in the variable configured in the **Variable for error message** property.



Note

*When there is no picking bin defined for the material, or if the defined picking bin is not taught, the step cannot be completed: the action that is required to trigger the detection is not available. If teaching the picking bin is not possible (when using the system with a **Guide** license), you always need to provide a **Next** or **Confirm** button on the screen or combine it with a **Job Control** step.*

Screen buttons

At a certain point in your assembly, you can integrate a step in which a screen button on the touch screen is pressed.

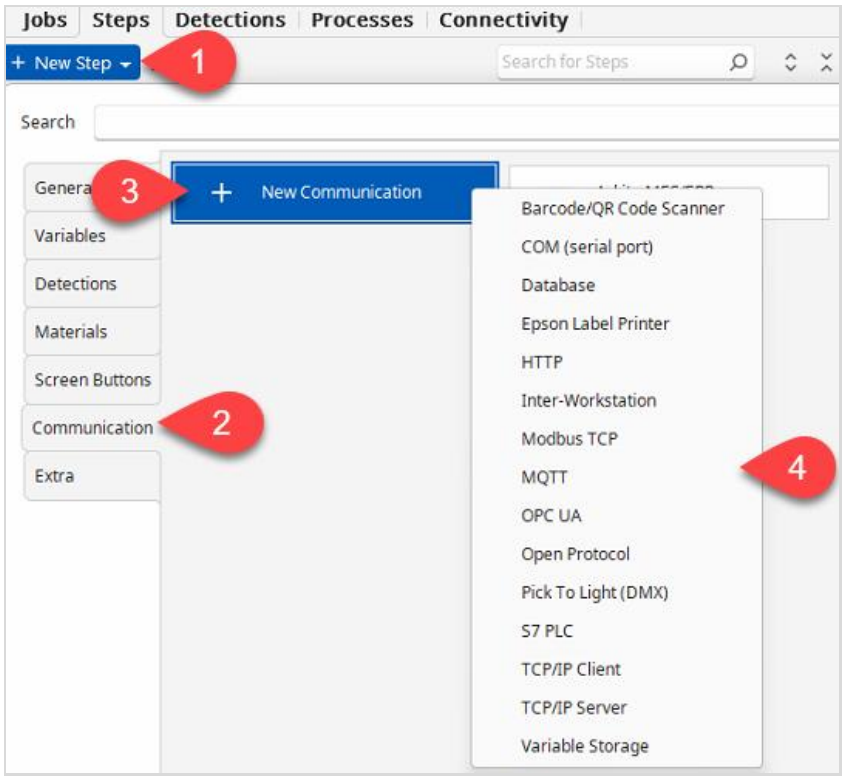
However, before you can do so, you first need to add the button on a **Screen** display. Until you have done this, the **Screen buttons** subtab in the overview of process steps will be empty.

For more information, refer to section **4.4.3 Adding visualizations** on page 108.

Communication

At a certain point in your assembly you can integrate a step in which a communication is sent.

However, before you can do so, you first need to configure a communication module. Click the **New Communication** button in the **Communication** subtab and select a module of your choice.



For more information regarding the next steps, refer to section **7.3.3 Adding a communication module** on page 214.

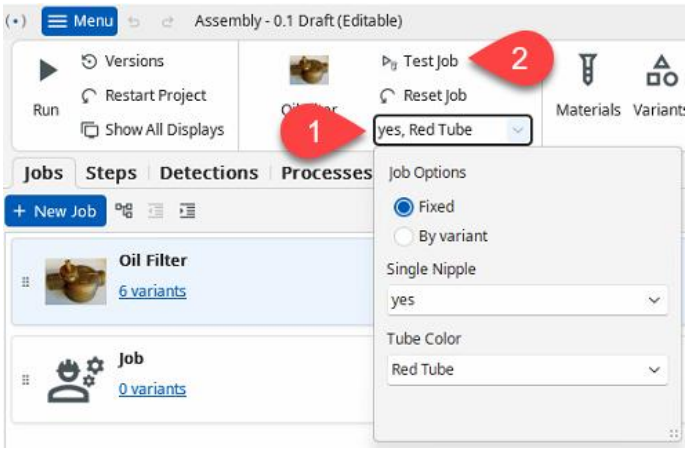
Extra

The table below gives an overview of all step types that are available in the **Extra** subtab:

Process step type	Description
File Mover	Move an existing file to a new location or changes the name of that file.
Read from File	Read data from a file on a workstation and store them in a variable in Arkite.
Sound	Plays a sound file in .wav format.
Write to File	Write data and store them in a file on the workstation.

4.3.4 Testing your job

You can test your job by selecting the job variant or the job option values of your choice in the ribbon and then clicking the **Test Job** button.



The icons for every step indicate the progress of your job:

Icon	Description
○	The step has not been started yet.
● ◑	The step is currently active (Run mode). For composite steps, the part of the circle that is filled indicates the percentage of the child steps that is executed.
◍ ◍	The step is currently active, but is suspended because the project is not running (Edit mode). For composite steps, the part of the circle that is filled indicates the percentage of the child steps that is executed.
●	The step has been completed.
◌	The step will be ignored due to variant and condition logic.

Depending on the step type, the icon can be circular, square or triangular:

Icon	Description
○	The step is linked to a trigger. The job will continue to the next step after the trigger has been completed. Examples of such steps are **Detection, Material** or **Wait for Communication** steps.
▽	The step will complete instantly. The job will continue to the next step immediately. You can't associate any visualizations with this step, because they would never be displayed anyway. Examples of such steps are **Set Variable** or **Send Communication** steps.

Icon	Description
☐	The step is blocking the process – in its current state, the job will never continue to the next step. This can have several causes: – A detection is not taught (**Detection** step) – A detection is disabled (**Detection** step) – There is no picking bin defined for a material (**Material** step) – There is no picking bin taught for a material (**Material** step) – There is no trigger defined for the step (**Work Instruction** step)

Note

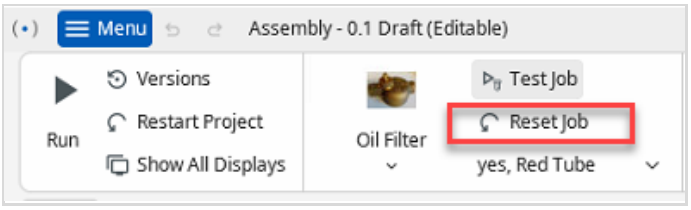
*On an editor, i.e. a workstation without a valid **Guide**, **Validate** or **Vision** license, you will be able to run a project for a maximum of 15 minutes. After that, it will automatically revert to **Edit** mode.*

To force the job to go to the next step, click the **Next** button.

To return to the previous step, click the **Previous** button. Going to a **Previous** step is only possible if that previous is step is a detection step or work instruction.

To go to a specific step, select the step and click the **Go To** button.

To reset the job and start again from the beginning, click the **Reset Job** button in the ribbon.



4.3.5 Job selection and job execution

The Operator Job Selection process

When you create a project, on the **Processes** tab, by default one process called **Operator Job Selection** will be available that is started when the project is loaded.

The **Operator job selection** process consists of three major steps in which an operator:

- Selects a job from a gallery on the touch screen
- Selects a variant from a gallery on the touch screen
- Enters a quantity on the touch screen

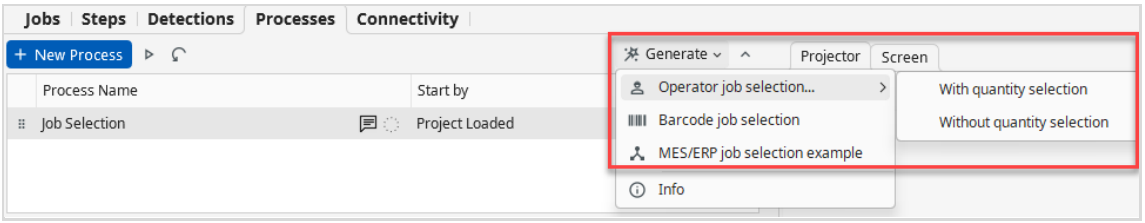
After the selection, the selected job variant is executed the specified number of times. After the last iteration, the job selection is automatically started again.

When you add new jobs or new job variants, these are automatically added to the galleries on the selection screens.

Warning

If you haven't configured any job variants, the job will be executed skipping all optional steps (if there are any).

With the **Generate** menu, you can add a job selection process yourself.



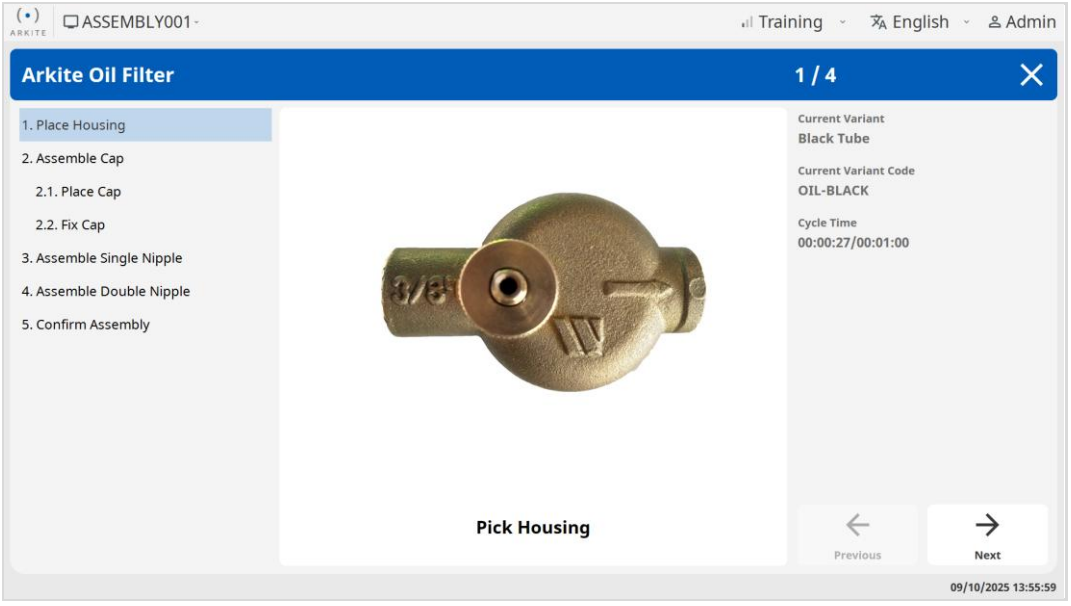
For more information, refer to section **6.2 Generating Job Selection processes** on page 173.

The Job Screen Template

While a job is being executed, a screen with the information below is displayed:

- A menu bar allowing you to switch languages, experience level or users, to turn your projectors on and off, to shut down the system or to restart the project.
- A **Numbered Steps List**, providing the operator insight into the assembly process
- Information about the current job, such as selected variant, quantity, cycle time and expected cycle time.

Of course, you can add additional information if needed. Click **Edit Job Master** to do so.



4.4 Managing visualizations for your job steps

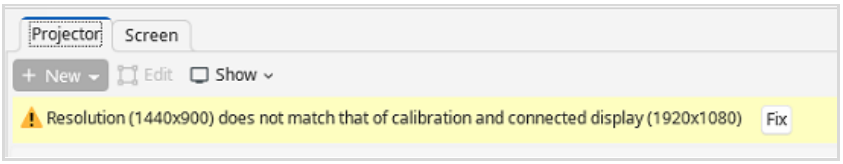
What?

After creating your job steps, you can manage the visualizations that will be displayed when those steps are active, on a connected projector or screen.

You will see a subtab for every display that is available. By default, a **Projector** subtab will always be available.

Note

- *When you have created the project before calibrating, a **Projector** display is added with the screen resolution of the device you are working on (e.g. your laptop). This may not match with the actual projector resolution, used during calibration. This will cause projections to be faulty. A message on the **Projector** subtab will warn you for this. Click **Fix** and update the **Resolution** property to the correct value in the projector's **Properties** pane.*



- *To add more displays, refer to section **7.2.3 Adding a display** on page 195.*

When creating a process, the Arkite software will automatically generate visualizations to display text and image instructions.

Additionally, you may also create more visualizations to display.

When the **Experience Level** functionality is enabled, you can add experience level buttons to a **Screen** display to allow an operator to select his experience level.

4.4.1 Visualization types

Following types of visualizations are available in the Arkite software:

- **Draw shape:**
  A geometrical figure. It can be a (rounded) rectangle, an ellipse, a line, a polygon or an arrow.
- **Text:**
  Visualization that allows you display a text to the user. The text can contain both fixed and dynamic parts, based on the value of a variable.
- **Image:**
  Visualization that allows you to display a fixed or dynamic image or video to the user.
- **Screen button:**
  Adds a clickable/pressable button to a **Screen** display. Allows the user to set a variable and/or control a process or job in your project by doing so.
- **Progress bar:**
  Visualization that shows the progress of the currently active job or of a process. Alternatively, it can also be based on two custom user defined values.
- **Operator Input:**
  Visualization that allows you to add an input field to a **Screen** display, in which the operator can type information. The information is stored in a variable.
- **Keyboard:**
  Visualization that displays an on-screen keyboard onto a **Screen** display. The operator can use it to enter information in an **Operator Input** visualization if no physical keyboard is available.
- **Sensor stream (Validate and Vision license only):**
  Visualization that displays a specific set of detections with their current state.
- **Gallery:**

Visualization that displays a number of tiles with text and images in a grid view on a **Screen** display. Operators can tap a tile to select it, and depending on the selection you can store data in a variable.

- **Combo Box:**
  Visualization that displays a dropdown list with a number of text items on a **Screen** display. Operators can select an item from the list, and depending on the selection you can store data in a variable.
- **Table:**
  Visualization that displays a table with customizable rows and columns on a **Screen** display. Optionally, operators can select a row in the table, and depending on the selection you can store data in a variable.
- **Status Bar:**
  Shows the operator an error message when there is an issue with a communication module, with a sensor, screen or projector, or when the server is offline.

  Note
  *A status bar is added by default to the bottom of every **Screen** display. Its location and look and feel can be edited, and it can also be deleted if not required.*
- **Menu Bar:**
  Visualization with preconfigured features for the operator:
  - Turn projector(s) on or off (if an IP address for the projector has been configured)
  - Restart the project
  - Shut down the system
  - Change the experience level (if experience level is enabled)
  - Change the instruction language (if multiple languages are added to the project)
  - Switch the user

  By default, an Arkite project contains a **Menu Bar** on the **Screen** display already.
- **Numbered Steps List:**
  Visualization that displays the numbered steps from a job that is being executed on a **Screen** display. The step that is currently active is highlighted. It serves as a table of contents, intended to help the operator in gaining insight into the assembly process. The Arkite project template contains a preconfigured **Numbered Steps List**, but you can also add and configure one yourself.

4.4.2 Visualization properties

The sections below provide you with an overview of the properties that can be defined for each visualization type.

First, the properties that are available for each visualization type are described:

- Name property
- Visibility properties
- Color properties
- Experience Level properties
- Location, Size and Angle properties

Next, the additional specific properties for each visualization type are described.

Name property

The **Name** property allows you to define a specific name for the visualization. By clicking the **Comment** icon next to the **Name** property, you can add a few words of explanation regarding the use of the visualization.

Note

*Next to the **Name** label, the object ID for the visualization is shown. You can copy it to use it when working with the Arkite API. For more information about the Arkite API, refer to chapter **18 The Arkite API** on page 292.*

Visibility properties

For each visualization, the properties in the table below can be configured in the **Visibility** section:

Property	Description
Visibility	Select one of the options below from the **Visibility** dropdown list: – **On**: the visualization will always be visible – **Blink**: the visualization will blink. Use the **Blink Interval** property to define how fast the visualization will blink. – **Based on Variable**: the visibility of the visualization will be based on the current state of the variable provided in the **Variable for visibility** property.
Variable for visibility	Only available if the **Based on Variable** option is selected in the **Visibility** dropdown list. The state of this variable will determine whether the visualization will be visible: – If the state of the variable is ON, TRUE or 1, the visualization will be visible. – If the state of the variable is OFF, FALSE or 0, the visualization will be hidden. – If the state of the variable is BLINK, the visualization will blink. Use the **Blink Interval** property to define how fast the visualization will blink. – If the variable has another state than the ones mentioned above, the visualization will be visible.
Blink Interval (ms)	Only available if the **Blink** option is selected in the **Visibility** dropdown list. Interval in ms with which the visualization will blink.

Note

- *When the visibility is based on a variable, the visualization will always be visible in the UI and when your project is in **Edit** mode. In **Run** mode, the visibility as specified by the variable will be used.*
- *Visualizations that are linked to a detection rather than a process step have some additional **Visibility** properties that define for which **Detection State(s)** they will be visible. For more information, refer to section **5.6.6 Managing visualizations for the detection** on 152.*

Color properties

For each visualization, the properties in the table below can be configured in the **Colors** section:

Property	Icon	Description
Background Color		Set the visualization's background color.
Background Border Color		Set the visualization's background border color.

Note

- *For some visualizations, additional color properties can be set. These are explained in the sections below.*
- *For each color property, you can either set a fixed color, select two alternating colors, define the color based on the value of a variable or define the color based on a template. For more information, refer to section* ***Managing visualization or DMX light colors*** *on page 112.*

Experience Level properties

In case the **Experience Level** functionality is enabled, a check box is available for each experience level. This way, you can indicate for which experience level(s) the visualization should be shown. By default, every visualization is shown for all experience levels.

Note

For more information on the Experience Level functionality, refer to chapter ***8 Experience levels*** *on page 233.*

Overrule location, size and angle properties

The location, size and angle properties of some visualizations can set to be dynamic by checking the **Overrule by variables** checkbox. This checkbox is available for:

- On a **Projector** display, and
- Linked to a process step (so not linked to a Detection)

Note

Visualizations that are linked to a dynamic detection will move and rotate with that detection.

In the **Location Variables** properties, you can define the variables that contain the X, Y and Z coordinates of the visualization (in mm from the origin). These values define the coordinates of the lower left point of the visualization's minimum bounding box.

In the **Size Variables** properties, you can define the variables that contain the length and width of the visualization, i.e. the size in the X, and Y direction respectively.

Note

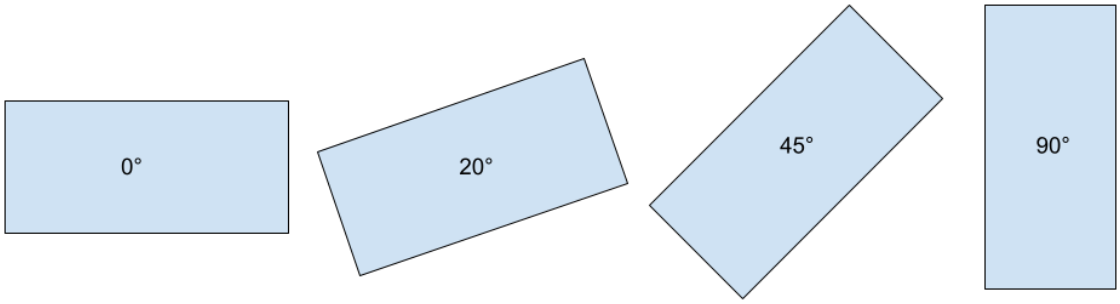
When clicking the ***Set Location Variables to Default*** *button, the values of the variables will be set to the visualization's coordinates as defined in the Arkite software.*

When clicking the ***Set Size Variables to Default*** *button, the values of the variables will be set to the visualization's size as defined in the Arkite software.*

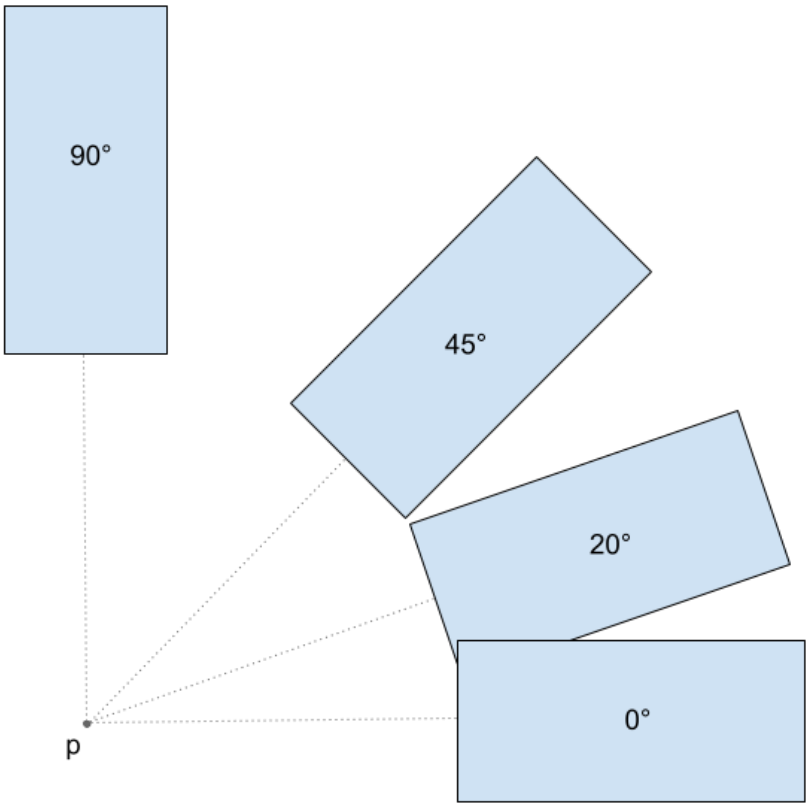
In the **Angle Variable** property, you can define a variable that contains an offset to the visualization's default angle. When the value of this variable changes, the visualization will rotate according to the variable's value.

For example, when the visualization as it is drawn in the software has an angle of 25 degrees in relation to the X-axis and the Angle Variable has a value of 15, the visualization will be rendered in an angle of 40 degrees in relation to the X-axis.

By default, the visualization will rotate around its centre, as shown in the image below:



When checking the **Off-center pivot point** property, you can define the X and Y coordinate of a pivot point around which the visualization should rotate. For example, in the image below, the detection rotates around a point p.



Location, Size and Angle properties

For each visualization, the properties in the table below can be configured in the **Location, Size and Angle** section:

Property	Description
X	Position on the X-axis of the lower left point of the visualization's minimum bounding box. In mm for visualizations on a **Projector** display, in pixels for visualizations on a **Screen** display.

Property	Description
Y	Position on the Y-axis of the lower left point of the visualization's minimum bounding box. In mm for visualizations on a **Projector** display, in pixels for visualizations on a **Screen** display.
Z	Position on the Z-axis of the lower left point of the visualization's minimum bounding box. In mm for visualizations on a **Projector** display, in pixels for visualizations on a **Screen** display.
Width	Width of the visualization's minimum bounding box. In mm for visualizations on a **Projector** display, in pixels for visualizations on a **Screen** display.
Height	Height of the visualization's minimum bounding box. In mm for visualizations on a **Projector** display, in pixels for visualizations on a **Screen** display.
Angle	Angle of the visualization in relation to the X-axis 0° = parallel to the X-axis.

Note

You can also change a visualization's location, size and angle using your mouse or by selecting the visualization and move it with your keyboard arrows. To move fast, use Ctrl-arrow, to move very slow, use Shift-arrow.

Draw shape

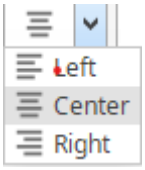
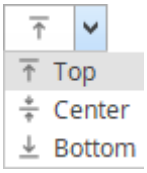
A **draw shape** visualization has the properties in the table below:

Property	Icon	Description
Corner Radius		For rectangles only. Allows you to add rounded corners to your rectangle.
Fill shape		If checked, the shape will be filled with the brush color. If unchecked, the shape will be transparent and the background color will be visible.
Brush Color		If the **Fill Shape** property is unchecked, a brush on the outside of the shape will have this color. If the **Fill Shape** property is checked, the entire shape will have this color. ***Note*** *You can either set a fixed color, select two alternating colors, define the color based on the value of a variable or define the color based on a template. For more information, refer to section **Managing visualization or DMX light colors** on page 112.*
Brush Size		Thickness of the brush

Text

A **text** visualization has the properties in the table below:

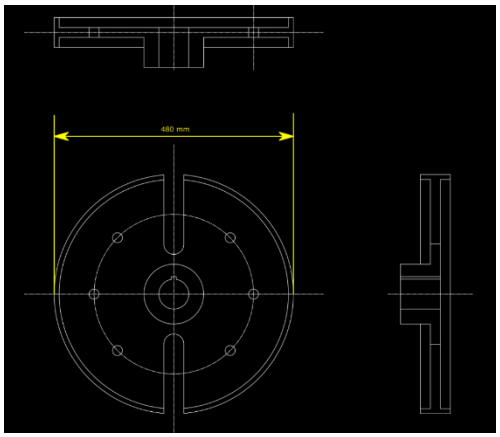
Property	Icon	Description
Text Font Size		The font size of the displayed text.

Property	Icon	Description
Text Color		The font color of the text. ***Note*** *You can either set a fixed color, select two alternating colors, define the color based on the value of a variable or define the color based on a template. For more information, refer to section* ***Managing visualization or DMX light colors*** *on page 112.*
Horizontal Text Align		The horizontal alignment of the text. This can be either left, center or right.
Vertical Text Align		The vertical alignment of the text. This can be either top, center or bottom.
Wrap text/shrink to fit		If the **Wrap text** radio button is selected (default option), text will automatically move to the next line when it reaches the edge of the text box. If there is more text than can be shown in the text box, an ellipsis at the end of the text will indicate so. If the **Shrink to fit** radio button is selected, the text font size will be adapted so the full contents of the text box can be displayed. Text will not automatically move to the next line.
Text		Provide the text you want to display, in any available language. To insert dynamic text, click the {x} icon at the location where you want to include the text and select the variable containing the dynamic text. Alternatively, you can also type the variable name between double braces, i.e. *{{Variable name}}*. You can also use a variable within a variable. For example, if the value of a variable **Index** is 01, then *{{Variable{{Index}}}}* will resolve into the value of the variable **Variable01**. ***Tip*** *For long texts, click the*  *icon to enlarge the text box.* ***Note*** *To add or remove languages, refer to section* ***16.2 Managing languages*** *on page 280.*
Show Text in PDF		If checked, the text visualization will be shown in the PDF export.

Image

An **image** visualization has the properties in the table below:

Property	Description
Image Source	Specify where to find the image that should be displayed. – **Project Images:** display an image available in the project images, reachable through the **Images** button on the ribbon, by selecting it. – **Image name:** display an image available in the project images, reachable through the **Images** button on the ribbon, based on the image name. – **File Name:** display an image that is stored on the workstation, based on the file name. – **SVG:** display an image based on SVG content. – **Data URI:** display an image based on a data URI.
Image	Only available if the **Project Images** option is selected in the **Image Source** property. Click the icon to select the image that should be displayed.
Image Name	Only available if the **Image name** option is selected in the **Image Source** property. The name of the image that should be displayed. The image should be available in the project images, reachable through the **Images** button on the ribbon. To use the value of a variable, click the icon and select the variable of your choice. Alternatively, you can also type the variable name between double braces, i.e. *{{Variable name}}.*
File Name	Only available if the **File name** option is selected in the **Image Source** property. The file name, with or without file extension, of the image that should be displayed. The image should be available in the folder specified in the **Search local folder** variable. To use the value of a variable, click the icon and select the variable of your choice. Alternatively, you can also type the variable name between double braces, i.e. *{{Variable name}}.*
SVG Content	Only available if the **SVG** option is selected in the **Image Source** property. The SVG (Scalable Vector Graphics) content for the image to be displayed. To use the value of a variable, click the icon and select the variable of your choice. Alternatively, you can also type the variable name between double braces, i.e. *{{Variable name}}.*
Data URI	Only available if the **Data URI** option is selected in the **Image Source** property. The Data URI of the image to be displayed. The expected format is *[data].* To use the value of a variable, click the icon and select the variable of your choice. Alternatively, you can also type the variable name between double braces, i.e. *{{Variable name}}.*

Property	Description
Search local folder	Only available if the **File Name** option is selected in the **Image Source** property. The folder in which the system will look to find the image to display. It does not look into subdirectories. The folder is local in that it looks for files in the specified folder on the workstation on which the project happens to be running. So different workstations may have a folder with the same name, but it may contain different files. Click the icon to select the relevant folder on your workstation. You can also use different folders on different workstations by storing the local folder name in a variable. Click the {x} icon and select the variable of your choice. Alternatively, you can also type the variable name between double braces, i.e. *{{Variable name}}*. ***Example*** *Typically, this feature can be used to display a snapshot that was taken in a **Snapshot** process step. To make things work, use the same folder you have specified in the **Snapshot folder** property in the **Snapshot** process step. In the **Variable** property, use the same variable you have specified in the **Name** or the **Output last snap shot name to variable** property in the **Snapshot** process step.*
	The horizontal alignment of the image or video. This can be either left, center or right.
Maintain ratio	If checked, the image or video will always have the same width/height ratio, so it will not be stretched to fill up all available space.
True scale (1:1)	Allows to project images on a surface using scale 1:1, which is especially useful to project CAD drawings. Both vector (SVG and STL) and image formats (PNG, TIFF) are supported. Rendering on scale ignores the **Maintain ratio** and **Align** properties, as the coordinates in the source file are respected 1:1. The accuracy that can be achieved mostly depends on the amount of distortion introduced by the projector lens. The distortion depends on the type of projector, but typical values are +/- 0.5%. If the projector is not perpendicular to the surface, the margin increases with the angle. To achieve best results, accurate calibration of the projector is a must, and preferably manual projector calibration (from the ribbon) is used. ***Note*** *For more information on manual projector calibration, refer to **2.3 Other calibration options** on page 31.* As an example, a simple CAD drawing can be found in various formats in: *C:\Program Files\Arkite\Arkite Workstation\Help\CAD* 
Offset X (mm)	Specifies the offset of the image on the X axis. Only available if **True Scale (1:1)** is enabled.
Offset Y (mm)	Specifies the offset of the image on the Y axis. Only available if **True Scale (1:1)** is enabled.

Property	Description
Show Image in PDF	If checked, the image or a thumbnail of the video will be shown in the PDF export.

You can let the software render images based on SVG content. An example use case is where feedback from a vision system provides areas to highlight, by writing such SVG content to a variable via the API.

To render images based on SVG content:

- Check the **SVG** option in the **Image Source** property
- Check the **True Scale (1:1)** property.
  This allows you to use the world coordinate system (mm) that was established during calibration, i.e. the coordinates in mm on your table or surface.

  Note

  *If you don't check the **True Scale (1:1)** property, whatever content you provide will be displayed in the center of your **Image** visualization element.*
- In the **SVG Content** property, specify the SVG (Scalable Vector Graphics) content for the image to be displayed, or the variable containing the SVG content.
  You can use standard SVG elements, some typical examples are <circle>, <ellipse>, <line>, <polyline>, <polygon>, <text> or <rect> (for a rectangle).
  For a full overview of SVG elements and their properties, see https://www.w3schools.com/graphics/svg_intro.asp.
  Encoding for SVG elements is UTF-8.

  Note 1

  While writing your SVG content, take into account the guidelines below:

  - *Make sure the coordinates you provide in the SVG contents are within the coordinates of the **Image** visualization element, otherwise you contents will not be visible.*
  - *Use single quotes rather than double quotes. This simplifies their use in the API.*
  - *Use the style attribute rather than the individual presentation attributes. See https://www.w3.org/TR/SVG/styling.html#StyleAttribute for more information.*
  - *For colors within the style attribute, you can either use rgb, rgba or hex color codes. Named colors are also supported.*
  - *Be aware that the Y-axis in the Arkite coordinate system is oriented in the opposite direction (upwards from the origin) than the one in the SVG coordinate system (downwards from the origin). When writing SVG content, the Arkite coordinate system is used, unless you use a full SVG document.*

Note 2

*If desired, you can also use a full SVG document, but in that case you need to take into account that the coordinate system as specified in the SVG document is used as a source and converted to the Arkite coordinate system (also taking care of the orientation of the Y-axis). By default, SVG units are in pixels, and if not otherwise scaled, 1 unit = 1 pixel. To ensure that your SVG document is rendered correctly, include an explicit **width** and **height** attribute ensuring the scaling is set to 1 unit = 1 mm.*
*For example, an SVG document with **viewBox="2200 0 3350 325"** defines a coordinate system starting at (2200, 0) with a width of 3350 units and a height of 325 units. To ensure the correct scaling, include **width="3350mm" height="325mm"**.*

Below, you can find some examples with the corresponding result:

Example – SVG element, rectangle with blue fill, pink border and 50% opacity

Code

```
<rect x='50' y='20' width='150' height='150'
style='fill:blue;stroke:pink;stroke-width:5;opacity:0.5' />
```

Result



Code explanation

- *The **width** and **height** attributes of the **<rect>** element define the height and the width of the rectangle*
- *The **style** attribute is used to define CSS properties for the rectangle*
- *The CSS **fill** property defines the fill color of the rectangle*
- *The CSS **stroke** property defines the color of the border of the rectangle*
- *The CSS **stroke-width** property defines the width of the border of the rectangle*
- *The CSS **opacity** property defines the opacity value for the whole element (legal range: 0 to 1)*

Example – SVG element, yellow circle

Code

```
<circle style='fill: yellow' cx='100' cy='50' r='30'/>
```

Result

Code explanation

- *The **cx** and **cy** attributes define the x and y coordinates of the center of the circle. If cx and cy are omitted, the circle's center is set to (0,0)*
- *The **r** attribute defines the radius of the circle*

Example – SVG element, red polyline

Code

```
<polyline points='0,40 40,40 40,80 80,80 80,120 120,120 120,160'
style='fill:white;stroke:red;stroke-width:4' />
```

Result



Code explanation

The **points** attribute defines the list of points (pairs of x and y coordinates) required to draw the polyline.

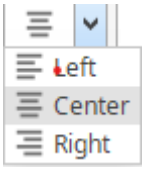
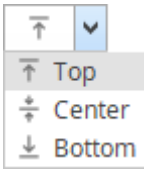
Screen button

A **screen button** visualization has the properties in the table below:

Property	Icon	Description
Show confirmation dialog		If checked, a confirmation dialog is displayed in a pop up window before the action(s) defined in the **Action** properties are executed. If the operator presses **OK**, the action(s) are executed. If the operator presses **Cancel,** the action(s) are ignored and the confirmation dialog closes.
Dialog caption		Provide the text you want to show in the confirmation dialog header/caption, in any available language. To insert dynamic text, click the {x} icon at the location where you want to include the text and select the variable containing the dynamic text. Alternatively, you can also type the variable name between double braces, i.e. *{{Variable name}}*.
Dialog message		Provide the text you want show in the confirmation dialog, in any available language. To insert dynamic text, click the {x} icon at the location where you want to include the text and select the variable containing the dynamic text. Alternatively, you can also type the variable name between double braces, i.e. *{{Variable name}}*.
Set Variable		Check this checkbox if you want the state/value of a variable to change if the button is clicked.
Variable to Set		Variable that should be changed when clicking the button.

Property	Icon	Description
State to Set		Variable state that should be set when clicking the button. Enter the value you want to set for the selected variable(s). To insert dynamic content, click the {x} icon at the location where you want to include the content and select the variable containing the dynamic content. Alternatively, you can also type the variable name between double braces, i.e. *{{Variable name}}*. You can also use a variable within a variable. For example, if the value of a variable **Index** is 01, then *{{Variable{{Index}}}}* will resolve into the value of the variable **Variable01**.
Process Control		Check this checkbox if you want to use the button to perform an action on a particular process within your project.
Process / Based on process name		Select one of the options below: – **Process:** select a specific process in the **Process** property – **Based on process name:** enter the name of the process that needs to be manipulated
Process		Select the process you want to control with the button. Only available if the **Process** option is selected.
Process Name		Enter the name of the process that needs to be manipulated. Only available if the **Based on process name** option is selected. To insert dynamic content, click the {x} icon at the location where you want to include the content and select the variable containing the dynamic content. Alternatively, you can also type the variable name between double braces, i.e. *{{Variable name}}*. ***Tip*** *For long texts, click the*  *icon to enlarge the text box.* ***Note*** *If the field contains a value that does not match with a process name, the **Process Control** step will not be executed and an error will be added in the error tray.* *If the field contains a value that matches with multiple process names, the process that was added first will be selected. It is a good practice to make sure your process names are unique.*
Action Type		Select the action you want to be performed on the selected process. You can choose to: – Reset a process – Pause a process – Pause and reset a process – Start a (paused) process – the process will continue from the step where it has been paused – Restart a process, i.e. reset and start it – Go to the next step in a process – Go to a specific step in a process. – Go to the previous step in a process ***Note*** *Going to a **Previous** step is only possible if that previous is step is a detection step or work instruction. In other cases, a screen button configured to take you to the previous step will be disabled.*
Step/Based on step name		Select one of the options below: – **Step:** select a specific step in the **Step** property – **Based on step name:** enter the name of the step that needs to be activated Only available if the 'Go to specific step' option is selected in the **Action Type** property.

Property	Icon	Description
Step		Select the step you want to activate. Only available if the 'Go to specific step' option is selected in the **Action Type** property and if the **Step** option is selected.
Step Name		Enter the name of the step that needs to be activated. Only available if the 'Go to specific step' option is selected in the **Action Type** property and if the **Based on step name** option is selected. To insert dynamic content, click the {x} icon at the location where you want to include the content and select the variable containing the dynamic content. Alternatively, you can also type the variable name between double braces, i.e. *{{Variable name}}*. ***Tip*** *For long texts, click the*  *icon to enlarge the text box.* ***Note*** *If the field contains a value that does not match with a step name, no action will be executed and an error will be added in the error tray.* *If the field contains a value that matches with multiple step names, the first step with a matching step name will be selected.*
Job Control		Check this checkbox if you want to use the button to navigate to another step in the currently active job.
Action Type		Select the action you want to be performed. You can choose to: – Abort a job – Go to the next step in the job – Go to a specific step in the job – Go to the previous step in the job ***Note*** *Going to a **Previous** step is only possible if that previous is step is a detection step or work instruction. In other cases, a screen button configured to take you to the previous step will be disabled.*
Step/Based on step name		Select one of the options below: – **Step:** select a specific step in the **Step** property – **Based on step name:** enter the name of the step that needs to be activated Only available if the **Go To** option is selected in the **Action Type** property.
Step		Select the step you want to activate. Only available if the **Go To** option is selected in the **Action Type** property and if the **Step** option is selected.
Step Name		Enter the name of the step that needs to be activated. Only available if the **Go To** option is selected in the **Action Type** property and if the **Based on step name** option is selected. To insert dynamic content, click the {x} icon at the location where you want to include the content and select the variable containing the dynamic content. Alternatively, you can also type the variable name between double braces, i.e. *{{Variable name}}*. ***Tip*** *For long texts, click the*  *icon to enlarge the text box.* ***Note*** *If the field contains a value that does not match with a step name, no action will be executed and an error will be added in the error tray.* *If the field contains a value that matches with multiple step names, the first step with a matching step name will be selected.*

Property	Icon	Description
Text Font Size		The font size of the displayed button label.
Text Color		The font color of the button label. ***Note*** *You can either set a fixed color, select two alternating colors, define the color based on the value of a variable or define the color based on a template. For more information, refer to section* ***Managing visualization or DMX light colors*** *on page 112.*
Horizontal Text Align		The horizontal alignment of the button text. This can be either left, center or right.
Vertical Text Align		The vertical alignment of the button text. This can be either top, center or bottom.
Button text		Provide the text label you want show on the button, in any available language. To insert dynamic text, click the {x} icon at the location where you want to include the text and select the variable containing the dynamic text. Alternatively, you can also type the variable name between double braces, i.e. *{{Variable name}}*. You can also use a variable within a variable. For example, if the value of a variable **Index** is 01, then *{{Variable{{Index}}}}* will resolve into the value of the variable **Variable01**. ***Tip*** *For long texts, click the*  *icon to enlarge the text box.* ***Note*** *To add or remove languages, refer to section* ***16.2 Managing languages*** *on page 280.*
Image Source		Specify where to find the image that should be displayed on the button. – **Project Images:** display an image available in the project images, reachable through the **Images** button on the ribbon, by selecting it. – **Image name:** display an image available in the project images, reachable through the **Images** button on the ribbon, based on the image name. – **File Name:** display an image that is stored on the workstation, based on the file name. – **SVG:** display an image based on SVG content. – **Data URI:** display an image based on a data URI.
Image		Only available if the **Project Images** option is selected in the **Image Source** property. Click the  icon to select the image that should be displayed.

Property	Icon	Description
Image Name		Only available if the **Image name** option is selected in the **Image Source** property. The name of the image that should be displayed on the button. The image should be available in the project images, reachable through the **Images** button on the ribbon. To use the value of a variable, click the {x} icon and select the variable of your choice. Alternatively, you can also type the variable name between double braces, i.e. *{{Variable name}}*.
File Name		Only available if the **File name** option is selected in the **Image Source** property. The file name, with or without file extension, of the image that should be displayed on the button. The image should be available in the folder specified in the **Search local folder** variable. To use the value of a variable, click the {x} icon and select the variable of your choice. Alternatively, you can also type the variable name between double braces, i.e. *{{Variable name}}*.
SVG Content		Only available if the **SVG** option is selected in the **Image Source** property. The SVG (Scalable Vector Graphics) content for the image to be displayed on the button. To use the value of a variable, click the {x} icon and select the variable of your choice. Alternatively, you can also type the variable name between double braces, i.e. *{{Variable name}}*.
Data URI		Only available if the **Data URI** option is selected in the **Image Source** property. The Data URI of the image to be displayed on the button. The expected format is *[data]*. To use the value of a variable, click the {x} icon and select the variable of your choice. Alternatively, you can also type the variable name between double braces, i.e. *{{Variable name}}*.
Search local folder		Only available if the **File Name** option is selected in the **Image Source** property. The folder in which the system will look to find the image to display. It does not look into subdirectories. The folder is local in that it looks for files in the specified folder on the workstation on which the project happens to be running. So different workstations may have a folder with the same name, but it may contain different files. Click the 🗀 icon to select the relevant folder on your workstation. You can also use different folders on different workstations by storing the local folder name in a variable. Click the {x} icon and select the variable of your choice. Alternatively, you can also type the variable name between double braces, i.e. *{{Variable name}}*.
Alignment		Define the alignment of the image on the button. By default, your image will be shown in the **Center** of the button. The other options can be interesting if you want to show both an image and a text on the button. For example, if you choose **Top**, your button text will be automatically shown below the button image.
Maintain ratio		If checked, the image will always have the same width/height ratio, so it will not be stretched to fill up all available space.

Property	Icon	Description
Enabled		Select one of the options below from the **Enabled** dropdown list: – **On**: the screen button will always be enabled. The operator will be able to click or press it, and the action(s) defined in the **Action** properties will be executed. – **Off**: the screen button will always be disabled. The button will have a reduced opacity. No actions will be executed when the operator tries to click/press it. – **Based on Variable**: the enabled/disabled state of the screen button will be based on the current state of the variable provided in the **Variable for enabled** property.
Variable for enabled		Only available if the **Based on Variable** option is selected in the **Enabled** dropdown list. The state of this variable will determine whether the screen button will be enabled: – If the state of the variable is ON, ENABLED or 1, the screen button will be enabled. – If the state of the variable is OFF, DISABLED or 0, the screen button will be disabled.
Button Fill Color		The fill color of the button. ***Note*** *You can either set a fixed color, select two alternating colors, define the color based on the value of a variable or define the color based on a template. For more information, refer to section **Managing visualization or DMX light colors** on page 112.*
Button Border Color		The border color of the button. ***Note*** *You can either set a fixed color, select two alternating colors, define the color based on the value of a variable or define the color based on a template. For more information, refer to section **Managing visualization or DMX light colors** on page 112.*

Progress bar

A **progress bar** visualization has the properties in the table below:

Property	Icon	Description
Based on		– If the **Variable** radio button is selected, the progress bar visualization will be based on the values defined by the **Variable for current value** and **Maximum value** properties. – If the **Process** radio button is selected, the progress bar visualization will reflect the progress of a process in your project. The current value is the active step, the maximum value is the total number of steps in the process. – If the **Job** radio button is selected, the progress bar visualization will reflect the progress of the currently active job. The current value is the number of finished numbered steps, the maximum value is the total number of numbered steps in the job.
Variable for current value		Only available if the **Variable** option is selected in the **Based on** property. the Specifies the variable for which the state will be used as current value. The state of the variable should be numeric.

Property	Icon	Description
Maximum value		Only available if the **Variable** option is selected in the **Based on** property. Specifies the maximum value (100%) of the progress bar. The value should be numeric. You can either enter a fixed value or use the value of a variable. To do so, click the {x} icon and select the variable of your choice. Alternatively, you can also type the variable name between double braces, i.e. *{{Variable name}}*.
Process		Only available if the **Process** option is selected in the **Based on** property. Process for which the progress is reflected.
Progress Color		The color of the progress indicator. ***Note*** *You can either set a fixed color, select two alternating colors, define the color based on the value of a variable or define the color based on a template. For more information, refer to section* ***Managing visualization or DMX light colors*** *on page 112.*
Orientation		Determines the orientation of the progress bar: left to right (Horizontal) or bottom to top (Vertical).
Mirror		When checked the progress bar will be mirrored: right to left in case of **Horizontal** orientation, top to bottom in case of **Vertical** orientation.
Show Percentage		If checked, the current progress is shown on the progress bar, as a percentage of the maximum value.
Text Font Size		Only available if the **Show Percentage** option is selected. Set the font size for the percentage shown on the progress bar.
Text Color		Only available if the **Show Percentage** option is selected. Set the text color for the percentage shown on the progress bar.

Operator Input

A **Operator Input** visualization has the properties in the table below:

Property	Icon	Description
Variable for text input		Variable in which the operator's answer is stored. As soon as the operator starts typing in the field, the variable is updated.
Show Current Value		If checked, the current value of the variable defined in the **Variable for text input** property is shown in the input field.
Password Appearance		If checked, the characters entered in the **Operator Input** field are hidden and displayed as masked symbols (●) instead of the actual text, preventing sensitive information from being visible while typing.
Text Font Size		The font size of the displayed text.
Text Color		The font color of the text. ***Note*** *You can either set a fixed color, select two alternating colors, define the color based on the value of a variable or define the color based on a template. For more information, refer to section* ***Managing visualization or DMX light colors*** *on page 112.*

Note

*If no physical keyboard is available, also add a **Keyboard** visualization to allow the operator to enter text in the **Operator Input** visualization.*

Keyboard

The **Keyboard** visualization has the properties in the table below:

Property	Description
Keyboard Type	Select type of keyboard that will be displayed: – **Alphanumeric**: letters, numbers and some symbols – **Alphabetic:** letters and some symbols only – **Numeric:** numbers, some symbols and arithmetic operators only – **Only Numbers and Backspace**
Dynamic Key	Not available if the **Keyboard Type** is **Only Numbers and Backspace.** For one key on the keyboard, you can configure the character that is linked to it yourself. By default, that is the # character. To use the value of a variable, click the {x} icon and select the variable of your choice. Alternatively, you can also type the variable name between double braces, i.e. *{{Variable name}}*. ***Note*** *If you specify multiple characters in this field or in the value of the linked variable, only the first character will be used by the key.*

Sensor stream (Validate and Vision license only)

A **sensor stream** visualization has the properties in the table below:

Property	Icon	Description
Mirror horizontally		If checked, the sensor stream will be mirrored horizontally to compensate the mirror caused by the depth sensor.
Mirror vertically		If checked, the sensor stream will be mirrored vertically to compensate the mirror caused by the depth sensor.
Detections		Detections that should be displayed in the sensor stream visualization.
Color for 'OFF'		The color for an inactive detection.
Color for 'ON'		The color for an active detection.

Gallery

A **Gallery** visualization has the properties in the table below:

Property	Icon	Description
Type		Select the format for your data input: – Json – XML

Property	Icon	Description
Data list		Enter valid xml or a valid Json array that contains an item for each tile you want to display in the gallery. Each item should contain properties/elements matching the ones you have set for the **Name property**, **Image property** and **Selection property**. To use the value of a variable as the value for a property, click the {x} icon at the location where you want to include the value and select the variable of your choice. Alternatively, you can also type the variable name between double braces, i.e. *{{Variable name}}*.
Name property		Define the property in the Json array or the element in the XML provided in the **Data list** property that contains the text label that should be displayed on the tile. You can also use Json path constructs such as: `product.code` or `$.product.code`: gives the value of the code property of the product `product.productMaterials[1].name`: gives the value of the name property of the second material for the product `$['product.materials']`: gives the value of the product.materials property You can also an expression in the format `tokenize(jsonPath, 'delim')[index]` or `tokenize(xPath, 'delim')[index]` with a zero based index. ***Example*** *Using `tokenize(nested.Value, '|')[1]` on the data `[{ "nested": { "Value": "X|Y|Z" }}]` will return Y.* *Using `tokenize(/root/item/nested/Value, '|')[1]` on the data `<root><item><nested><Value>X|Y|Z</Value></nested></item></root>` will return Y.*
Image property		Define the property in the Json array or the element in the XML provided in the **Data list** property that contains the image that should be displayed on the tile. You can also use Json path constructs such as: `product.code` or `$.product.code`: gives the value of the code property of the product `product.productMaterials[1].name`: gives the value of the name property of the second material for the product `$['product.materials']`: gives the value of the product.materials property You can also an expression in the format `tokenize(jsonPath, 'delim')[index]` or `tokenize(xPath, 'delim')[index]` with a zero based index. ***Example*** *Using `tokenize(nested.Value, '|')[1]` on the data `[{ "nested": { "Value": "X|Y|Z" }}]` will return Y.* *Using `tokenize(/root/item/nested/Value, '|')[1]` on the data `<root><item><nested><Value>X|Y|Z</Value></nested></item></root>` will return Y.*
Image source		Specify where to find the images that should be displayed on the tiles in the **Gallery** visualization. – **Image name:** display a variable image available in the project images, reachable through the **Images** button on the ribbon, based on the image name. – **File Name:** display an image that is stored on the workstation, based on the file name. – **Data URI:** display an image based on a data URI. The expected format is *[data]* .

Property	Icon	Description
Search local folder		Only available if the **File Name** option is selected in the **Image source** property. The folder in which the system will look to find the image to display. It does not look into subdirectories. The folder is local in that it looks for files in the specified folder on the workstation on which the project happens to be running. So different workstations may have a folder with the same name, but it may contain different files.
Fallback image		Optionally, select an image that should be displayed when an image that is specified in the **Image property** in the **Data list** cannot be found.
Variable for selection		Set the variable where you want the data from the selected tile to be stored.
Selection property		Define the property in the Json array or the element in the XML provided in the **Data list** property that contains the data that should be written to the **Variable for selection**. If you don't specify a property or element here, the complete selected item with all of its properties/elements will be stored.
Orientation		Select the gallery's orientation: – **Horizontal:** the different tiles as defined in the **Data List** will be presented from top to bottom, divided over the specified number of rows. – **Vertical:** the different tiles as defined in the **Data List** will be presented from left to right, divided over the specified number of columns.
Number of Rows		Only available if **Horizontal** is selected in the **Orientation** property. Set the number of rows you want the gallery to contain. ***Example*** *For a gallery with 7 tiles and 3 rows, tiles 1-3 will be shown in the first column, tiles 4-6 will be shown in the second column and tile 7 will be shown in the third column on the first row.*
Number of Columns		Only available if **Vertical** is selected in the **Orientation** property. Set the number of columns you want the gallery to contain. ***Example*** *For a gallery with 7 tiles and 3 rows, tiles 1-3 will be shown in the first column, tiles 4-6 will be shown in the second column and tile 7 will be shown in the third column on the first row.*
Background Color		Set the gallery's background color.
Border Color		Set the gallery's border color.
Text font size		Set the font size of the label displayed on the tiles.
Tile height		Set the height (in pixels) of the tiles in the gallery. Only available if **Vertical** is selected in the **Orientation** property. If **Horizontal** is selected, the tile height will depend on the height of the **Gallery** visualization and the **Number of Rows**. They will scale automatically..
Tile width		Set the width (in pixels) of the tiles in the gallery. Only available if **Horizontal** is selected in the **Orientation** property. If **Vertical** is selected, the tile width will depend on the width of the **Gallery** visualization and the **Number of Columns**. They will scale automatically.
(Selected) Tile Label Text Color		Set the color of the label that is displayed on the tile or the selected tile.

Property	Icon	Description
(Selected) Tile Label Background Color		Set the background color of the label that is displayed on the tile or the selected tile.
(Selected) Tile Image Background Color		Set the background color of the tile or the selected tile.

Combo Box

A **Combo Box** visualization has the properties in the table below:

Property	Icon	Description
Type		Select the format for your data input: – Json – XML
Data list		Enter valid xml or a valid Json array that contains an item for each item you want to display in the combo box. Each item in the array should contain properties matching the ones you have set for the **Name property** and **Selection property**. To use the value of a variable as the value for a property, click the {x} icon at the location where you want to include the value and select the variable of your choice. Alternatively, you can also type the variable name between double braces, i.e. *{{Variable name}}*.
Name property		Define the property in the Json array or the element in the XML provided in the **Data list** property that contains the combo box item. You can also use Json path constructs such as: `product.code` or `$.product.code`: gives the value of the code property of the product `product.productMaterials[1].name`: gives the value of the name property of the second material for the product `$['product.materials']`: gives the value of the product.materials property You can also an expression in the format `tokenize(jsonPath, 'delim')[index]` or `tokenize(xPath, 'delim')[index]` with a zero based index. ***Example*** *Using `tokenize(nested.Value, '|')[1]` on the data `[{ "nested": { "Value": "X|Y|Z" }}]` will return Y.* *Using `tokenize(/root/item/nested/Value, '|')[1]` on the data `<root><item><nested><Value>X|Y|Z</Value></nested></item></root>` will return Y.*
Variable for selection		Set the variable where you want the data from the selected item to be stored.
Selection property		Define the property in the Json array or the element in the XML provided in the **Data list** property that contains the data that should be written to the **Variable for selection**. If you don't specify a property or element here, the complete selected item with all of its properties/elements will be stored.
Text Color		Set the text color for the items in the combo box.
Background Color		Set the combo box's background color.

Property	Icon	Description
Border Color		Set the combo box's border color.
Text Font Size		Set the font size for the items in the combo box.

Table

A **Table** visualization has the properties in the table below:

Property	Icon	Description
Type		Select the format for your data input: – Json – XML
Data list		Enter valid xml or a valid Json array that contains an item for each row you want to display in the table. Each item should contain properties/elements matching the ones you have set for the different columns in your table, and in case data selection is allowed, for the **Selection** property. To use the value of a variable as the value for a property, click the {x} icon at the location where you want to include the value and select the variable of your choice. Alternatively, you can also type the variable name between double braces, i.e. *{{Variable name}}.*
Allow Selection		If checked, the operator can select a row from the table.
Selected Row Text Color		Set the text color for the selected row.
Selected Row Fill Color		Set the background color for the selected row.
Variable for selection		Set the variable where you want the data from the selected row to be stored.
Selection property		Define the property in the Json array or the element in the XML provided in the **Data list** property that contains the data that should be written to the **Variable for selection**. If you don't specify a property or element here, the complete selected item with all of its properties/elements will be stored.
Show Headers		If checked, a header row with the column names will be shown.
Text Size		Set the text size for data in the table.
Text Color		Set the text color for data in the table.
Background Color		Set the background color for the rows in the table
Border Color		Set the table's border color.

Property	Icon	Description
Columns		Add a column for every property in the Json array or for every XML element or tag provided in the **Data List** property that you want to show in the table. Click the + button to add a column. A popup window will open where you can provide: – a **Name**: the column name as displayed in the header row if the Show Headers property is checked. – a Json **Property Name**: the property from the Json array you provided in the **Data List** property that you want to show in the column. You can also use Json path constructs such as: `product.code` or `$.product.code`: gives the value of the code property of the product `product.productMaterials[1].name`: gives the value of the name property of the second material for the product `$['product.materials']`: gives the value of the product.materials property You can also an expression in the format `tokenize(jsonPath, 'delim')[index]` with a zero based index. ***Example*** *Using `tokenize(nested.Value, '|')[1]`* *on the data `[{ "nested": { "Value": "X|Y|Z" }}]` will return Y.* – an **XML Element or Tag Name:** the element or tag name in the XML you provided in the **Data List** property that you want to show in the column. You can also an expression in the format `tokenize(xPath, 'delim')[index]` with a zero based index. ***Example*** *Using `tokenize(/root/item/nested/Value, '|')[1]`* *on the data `<root><item><nested><Value>X|Y|Z</Value></nested></item></root>` will return Y.* – the **Column Width**: this is a relative value. The width of all columns will be added up and then recalculated according to the entire table width. ***Example*** *Suppose the width for columns 1 and 2 is 100, the width for column 2 is 50 and the table width is 1000 pixels. In that case the column width would be 400 pixels for columns 1 and 2 and 200 pixels for column 3.* Select column and – click the pencil icon to edit the column's properties. – click the trash icon to delete the column.
Row Height		Set the row height in pixels.

Status Bar

A **status bar** visualization has no additional properties other than the general properties described above.

Menu Bar

A **menu bar** visualization has the properties in the table below:

Property	Icon	Description
Workstation Menu		Indicate if the workstation menu should be shown in the menu bar: – Turn projector(s) on or off (only available if an IP is configured in the **Projector** properties) – Restart the project – Shut down the system. If you select this option, a confirmation dialog will be shown.
Experience Level		Indicate if the **Experience Level** combo box should be shown in the menu bar. Only available if Experience Level is enabled.
Languages		Indicate if the **Languages** combo box should be shown in the menu bar. Only available if the project contains multiple languages.
Switch User		Indicate if the button to switch users should be shown in the menu bar.
Text Color		Set the text color for the items in the menu bar.
Background Color		Set the background color for the menu bar.
Border Color		Set the border color for the menu bar.
Text Font Size		Set the text size for the items in the menu bar.

Numbered Steps List

A **Numbered Steps List** visualization has the properties in the table below:

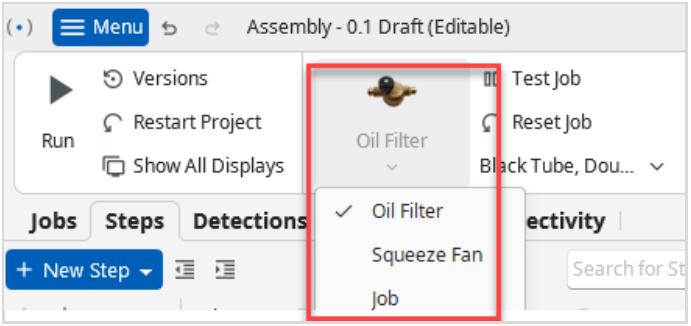
Property	Icon	Description
Row Height		Set the row height in pixels.
Text Size		Set the text size for the numbered steps in the list.
Text Color		Set the text color for the numbered steps in the list.
Background Color		Set the background color for the numbered steps in the list.
Border Color		Set the border color for the Numbered Steps List.
Current Step Text Color		Set the text color for the numbered step that is currently active.

Property	Icon	Description
Current Step Background Color		Set the background color for the numbered steps that is currently active.

4.4.3 Adding visualizations

To add a new visualization, execute the steps below:

1. In the ribbon, select the job for you which you want to add visualizations.

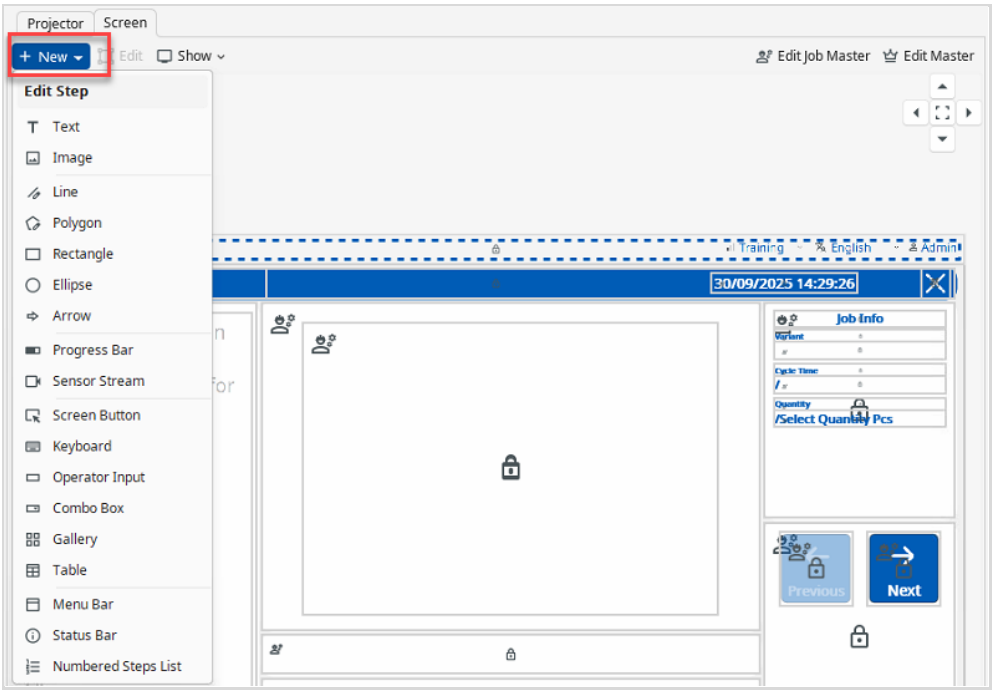


2. In the **Steps** tab, select the step for which you want to add a visualization.

   Note

   If you want to add a visualization that is visible regardless of the process or step, it does not matter which process and step you select.

3. Click the display subtab on which you want to add a visualization, **Projector** or **Screen**.
4. Click the **New** button and select the visualization of your choice.



Tip

*You can also select an existing visualization and click **Duplicate**.*

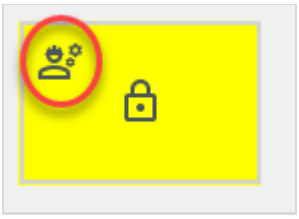
Note

*You can also add a visualizations that are always visible (on the **Master**) or whenever a job is active (on the **Job Master**).*



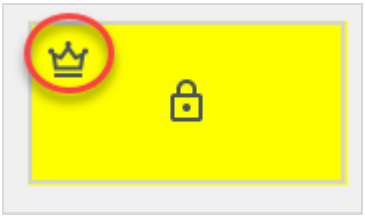
*To add a visualization that is visible whenever a job is active, click **Edit Job Master** first.*

Visualizations on the job master are marked with a job icon:



*To add a visualization that is **always** visible, click **Edit Master** first.*

Visualizations on the master are marked with a crown icon:



5. Define the size and position of the visualization using your mouse.

Tip

While dragging the visualization with your mouse, visual guidelines will appear that will allow you to easily align the visualization. The guidelines below will appear:

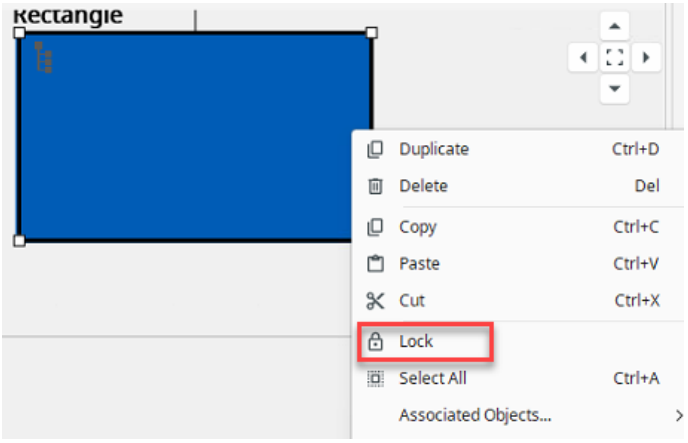
- *Guidelines towards the **middle of the screen** will appear when you reach the middle of the screen with an edge or with the middle of the visualization, both horizontally and vertically.*
- ***Object to object alignment lines** appear to help you to align two visualizations with each other, both horizontally and vertically. This way, you can align the top, the bottom or the middle of the visualizations with each other.*
- *Arrows appear to help you to position two visualizations at the **same distance from the edge or the center** of the screen.*
- *When you have more than two visualizations, arrows appear when the distance between visualizations 1 and 2 is the same as between visualizations 2 and 3. This helps you to ensure there is the same distance between visualizations.*
- *When you are changing the size of an object, lines appear inside the objects when they have the same length or width. This helps you to make the objects have the **same size**.*

6. In the **Properties** pane, update the properties of the visualization if needed.

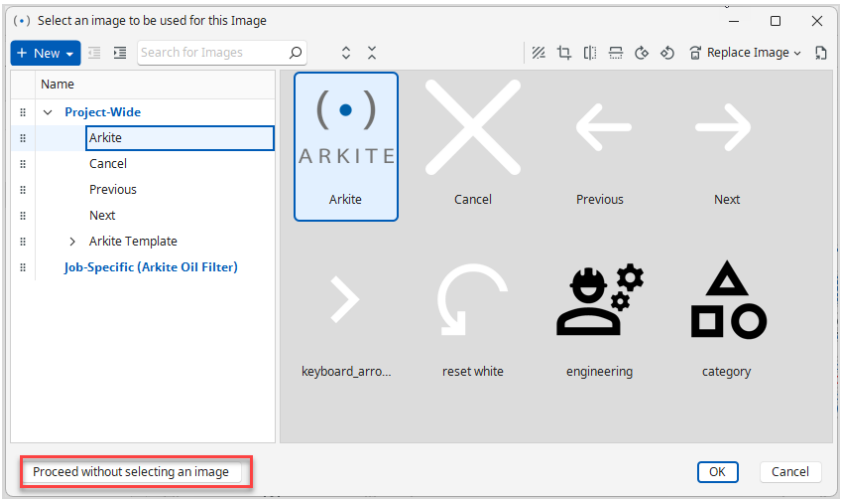
Note

- *The properties that are available depend on the type of visualization you have added. For an overview of visualization types and their properties, refer to sections **4.4.1 Visualization types** on page 83 and **4.4.2 Visualization properties** on page 84.*

- *To protect a visualization from being moved or edited by accident, right-click it and select **Lock**.*



- *When you want to add an **Image** visualization for which you want to use a variable, click **New – Image** and then click **Proceed without selecting an image**. In the properties of the **Image** visualization, you can define the variable to use. For more information, refer to section **4.4.2 Visualization properties** on page 84.*



4.4.4 Editing visualizations

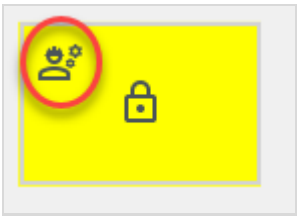
To edit an existing visualization, execute the steps below:

1. In the **Steps** tab, select the step for which you want to edit a visualization.

   Note

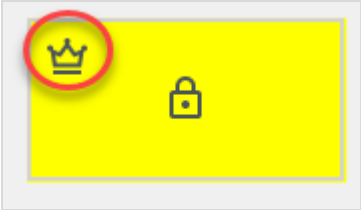
   *To edit a visualization that is visible whenever a job is active, click **Edit Job Master** first.*

   Visualizations on the job master are marked with a job icon:



   *To edit a visualization that is **always** visible, click **Edit Master** first.*

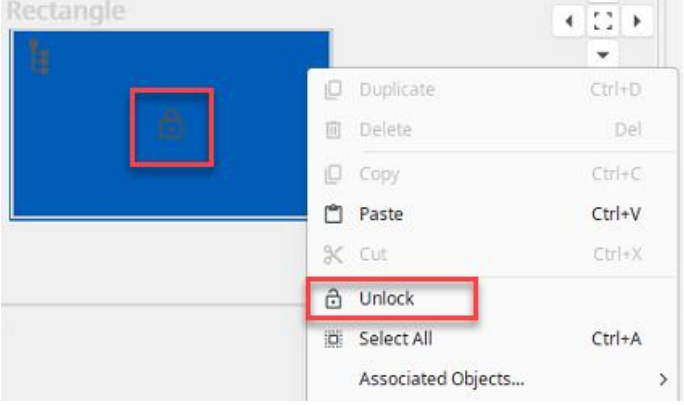
Visualizations on the master are marked with a crown icon:



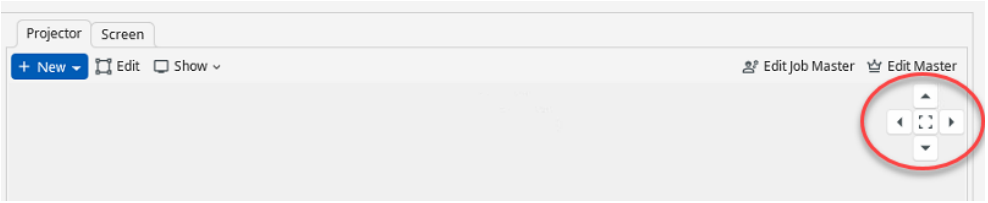
2. Click the display subtab on which you want to edit a visualization, **Projector** or **Screen**.
3. Select the visualization you want to edit.

Note

- *You cannot edit a visualization that is locked. First right-click the visualization and select **Unlock**.*



- *You can select multiple visualizations by holding the **Ctrl** button and clicking the visualizations of your choice with your mouse, or by holding the left mouse button and dragging a rectangle over the visualizations of your choice. Press **Ctrl-A** to select all visualizations.*
- *You can zoom in and out using your mouse wheel. When zoomed in, arrows appear to navigate around in your display.*



4. For draw shapes:
   - Change the size and position of the visualization by dragging the shape or the angular points with your mouse.
   - To change the shape of the visualization, click the **Edit** button above the image and move the angular points.
5. For all shapes, update the properties of the visualization in the **Properties** pane if needed.

Note

*The properties that are available depend on the type of visualization you have added. For an overview of visualization types and their properties, refer to sections **4.4.1 Visualization types** on page 83 and **4.4.2 Visualization properties** on page 84.*

To delete an existing visualization, select it and click **Delete**.

4.4.5 Managing visualization or DMX light colors

Several color properties can be set for a visualization or for a DMX light, in case a DMX device is connected through a Pick To Light (DMX) communication module.

Every color property can be defined in four separate ways:

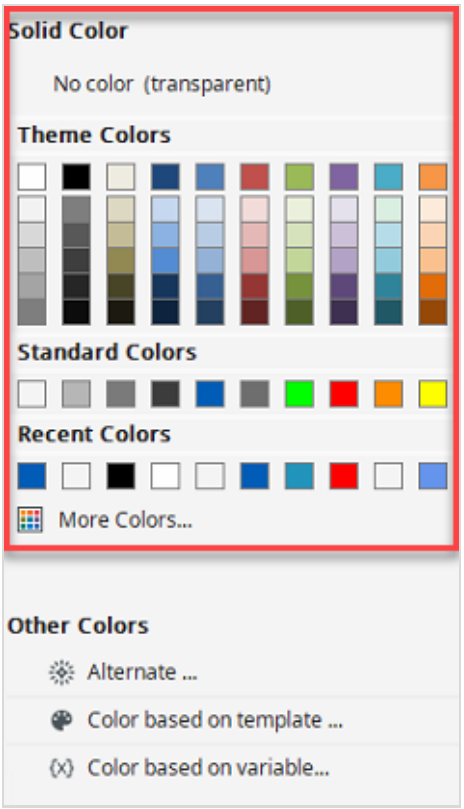
- Setting a fixed solid color
- Selecting two alternating colors
- Defining the color based on the value of a variable
- Defining the color based on a template

Note

- *If you define a color based on the value of a variable, you can manage the color separately from the colors for all other visualizations.*
- *If you define a color based on a template, all visualizations linked to that template will automatically change colors when the template is updated.*

Setting a fixed solid color

If you set a fixed, solid color, the visualization will always have the same color. To do so, click the color icon and select a color from the menu.



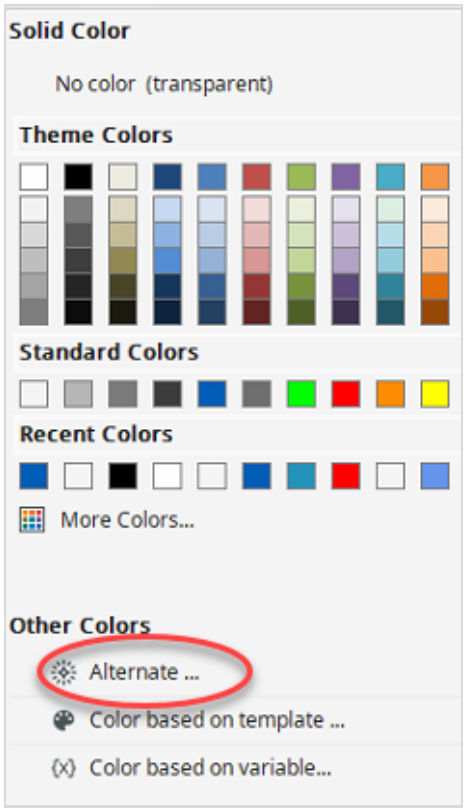
Note

*Click **More Colors...** to define a color based on the RGB or HSB model.*

Selecting two alternating colors

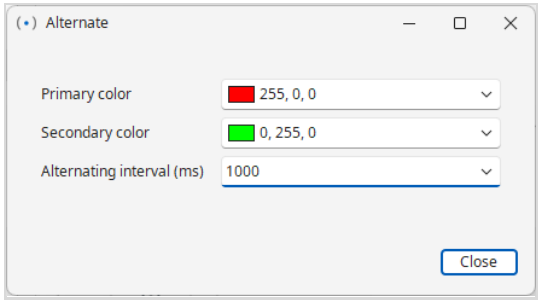
To make the visualization alternate between two colors, execute the following steps:

1. Click the color icon.
2. Under **Other colors**, select **Alternate...**



3. In the **Alternate** window:
   - Select your first color in the **Primary color** dropdown
   - Select your second color in the **Secondary color** dropdown
   - Select an **Alternating interval** in milliseconds (250, 500 or 1000)

   In the example below, the visualization's color will alternate between red and green every second:

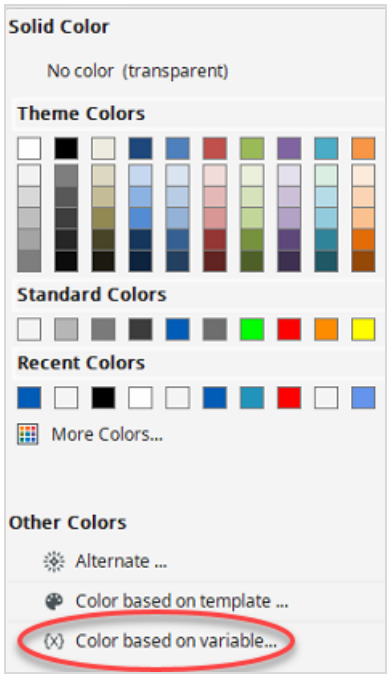


4. Click **Close**.

Defining the color based on the value of a variable

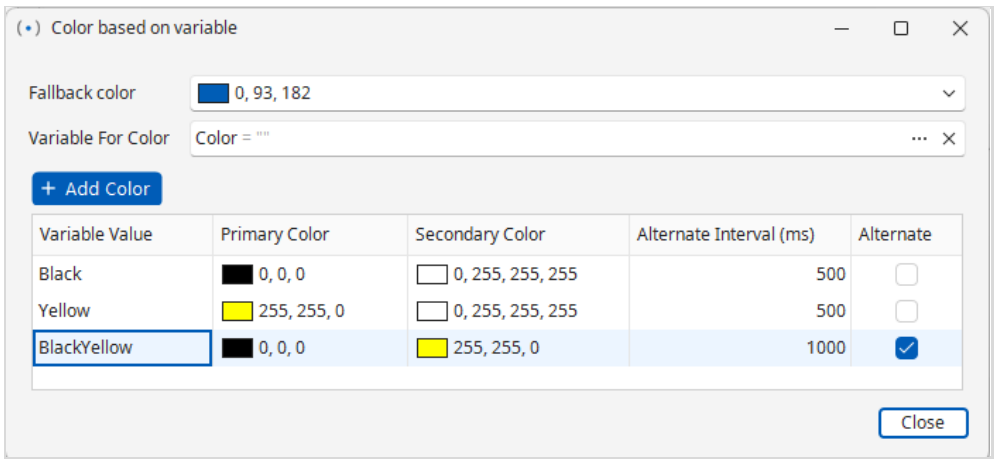
To define the color based on the value of a variable, execute the following steps:

1. Click the color icon.
2. Under **Other colors**, select **Color based on variable...**



3. In the **Color based on variable** window:
   - In the **Fallback color** property, indicate which color the visualization should have if the variable has a value that is not defined.
   - In the **Variable for color** property, specify the variable that contains the values to define the color of the visualization.
   - Define one or more values and the corresponding color(s).

   In the example below, the visualization's color will be yellow when the value of the **Color** variable is *Yellow* and black when the value of the **Color** variable is *Black*. When the value of the **Color** variable is *BlackYellow*, the color will alternate between black and yellow every second. When the **Color** variable has another value, the visualization's color will be blue (i.e. the fallback color).



Note

- *In case the **Alternate** checkbox is checked, the color of the projection will alternate between the **Primary Color** and the **Secondary Color**.*
- *In case the **Alternate** checkbox is not checked, the **Primary Color** will always be used.*

4. Click **Close**.

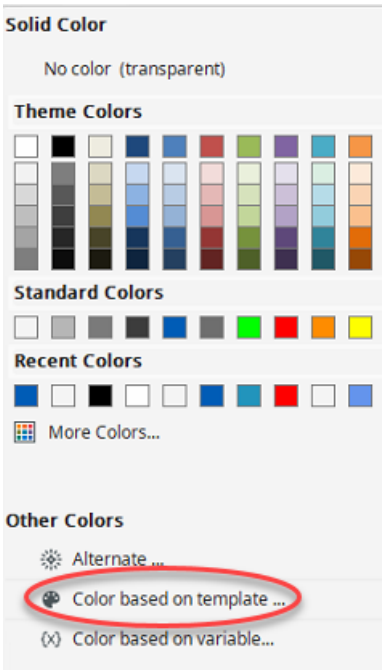
Note

*When there is a color defined for the variable value ON, this color will be used in the UI and when your project is in **Edit** mode. In **Run** mode, the color as specified by the variable will be used.*

Defining the color based on a template

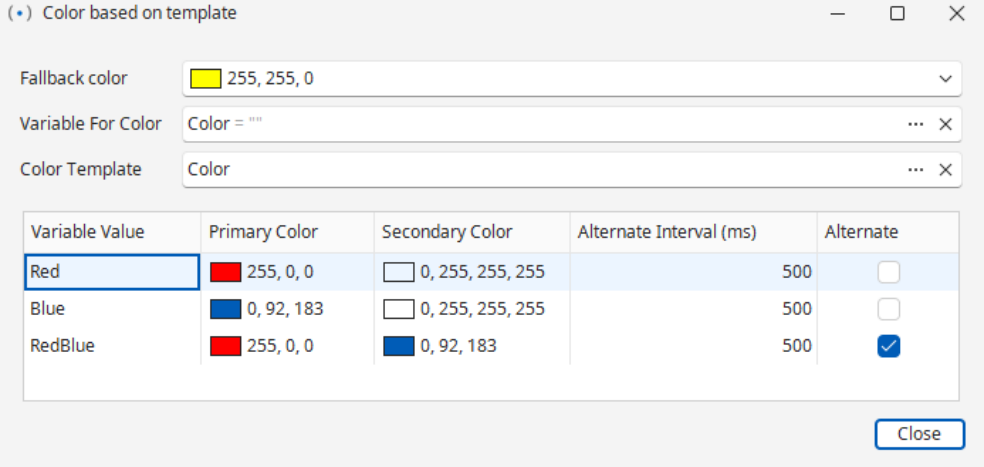
To define the color based on the value of a variable, execute the following steps:

1. Click the color icon.
2. Under **Other colors**, select **Color based on template...**



3. In the **Color based on template** window:
   - In the **Fallback color** property, indicate which color the visualization should have if the template does not define a color.
   - In the **Variable for color** property, specify the variable that contains the values to define the color of the visualization.
   - In the **Color Template** property, select the template to use.

   In the example below, the visualization's color will be red when the value of the **Color** variable is *Red* and blue when the value of the **Color** variable is *Blue*. When the value of the **Color** variable is RedBlue, the color will alternate between red and blue every 500 milliseconds. When the **Color** variable has another value, the visualization's color will be yellow (i.e. the fallback color).



Note

- *In case the **Alternate** checkbox is checked, the color of the projection will alternate between the **Primary Color** and the **Secondary Color**.*
- *In case the **Alternate** checkbox is not checked, the **Primary Color** will always be used.*

4. Click **Close**.

Note

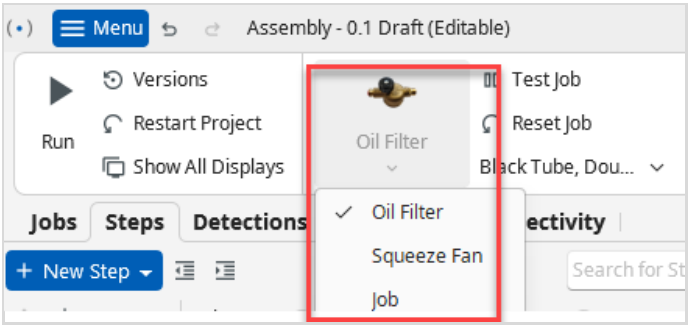
- *When there is a color defined for the variable value ON, this color will be used in the UI and when your project is in **Edit** mode. In **Run** mode, the color as specified by the template will be used.*
- *When you define a color based on a template, you cannot change the values and their corresponding colors in that template. For instructions on how to manage color templates, refer to section **15.3 Managing color templates** on page 277. Be aware that changing a color template changes the color of all visualizations linked to that color template.*

4.4.6 Managing text and image instructions

When creating a job, the Arkite software will generate visualizations to display text and image instructions. At the same time, two variables are created for text and image instructions. These variables will always contain the instructions for the active step and make sure the right instructions are displayed at the right time.

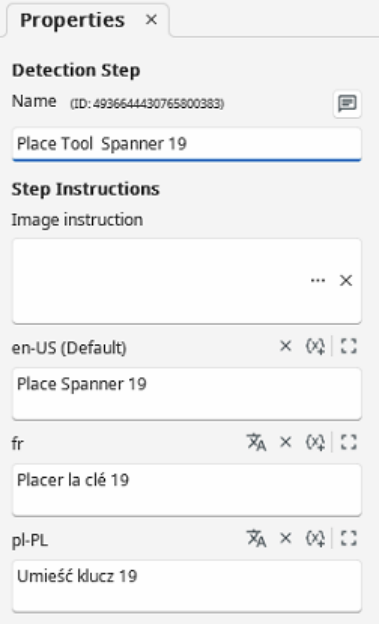
To define the instructions that will be shown for a particular step, execute the steps below:

1. Navigate to the **Steps** tab.
2. In the ribbon, select the job for you which you want to add instructions.



3. Under **Steps**, select the step for which you want to add instructions.

4. In the **Properties** pane, under **Step instructions**, select the image or video you want to be displayed.
5. Provide a text instruction you want to be displayed in all available user languages.
   For detection steps, default instructions are proposed.
   To insert dynamic text, click the {x} icon at the location where you want to include the text and select the variable containing the dynamic text. Alternatively, you can also type the variable name between double braces, i.e. {{Variable name}}. You can also use a variable within a variable. For example, if the value of a variable **Index** is 01, then *{{Variable{{Index}}}}* will resolve into the value of the variable **Variable01**. For long texts, click the ⛶ icon to enlarge the text box.



Note

*To add or remove languages, refer to section **16.2 Managing languages** on page 280.*

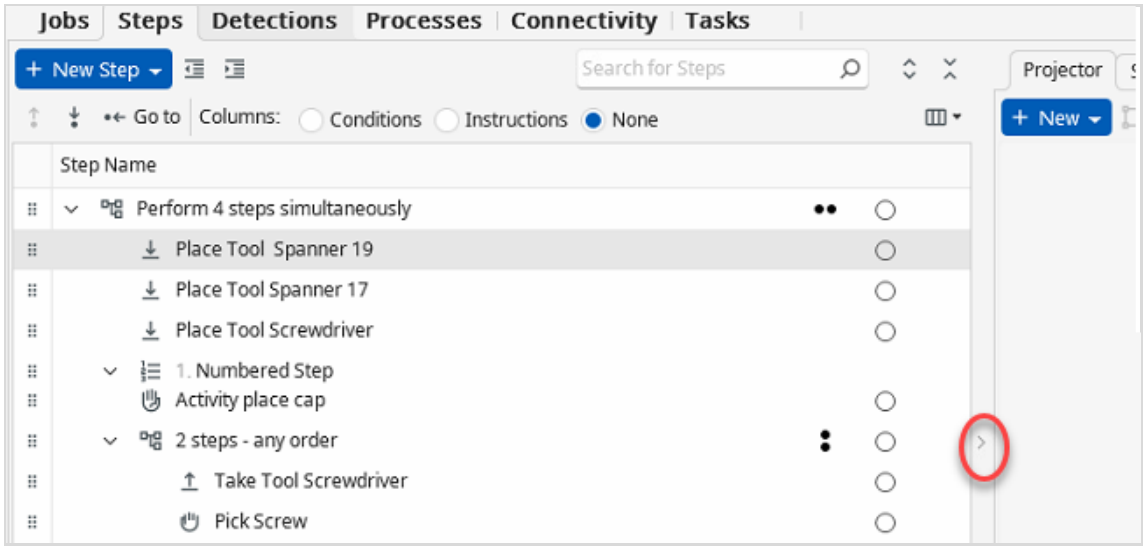
Tip

Click the 文A button to automatically translate text from the default language

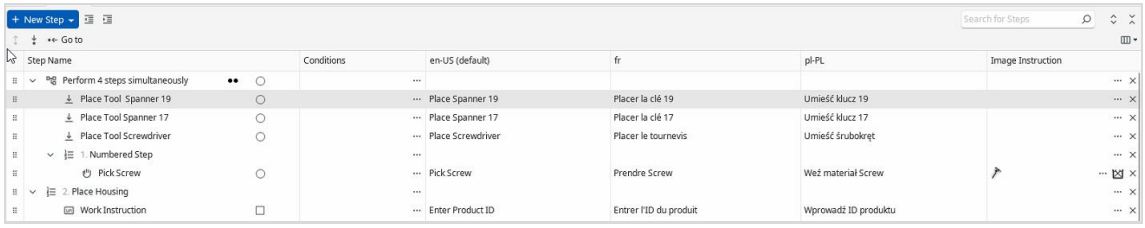
*The translation feature works through an integration with OpenAI. This integration should be enabled. To do so, navigate to **Menu – Server – Settings – AI**. An internet connection is also required.*

Tip

To have an overview of all instructions for all steps in a process, you can also click the right arrow next to the ***Steps*** *overview.*



Additional columns will appear in which you can see and edit text and image instructions for every step.



4.5 Working with conditions

What?

Conditions can be used to specify whether a step should be executed or not. Any individual or composite process step can be linked to one or more conditions. Only if at least one of the conditions holds, the step will be executed.

A condition can consist of one or more subconditions.

Project-Wide versus Job-Specific conditions

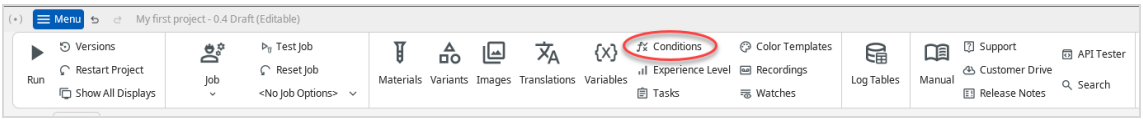
Conditions can be project-wide or job-specific.

Project-wide conditions can be used in every job in your project, while job-specific conditions can be used in one job only, and will only be visible if that job is selected.

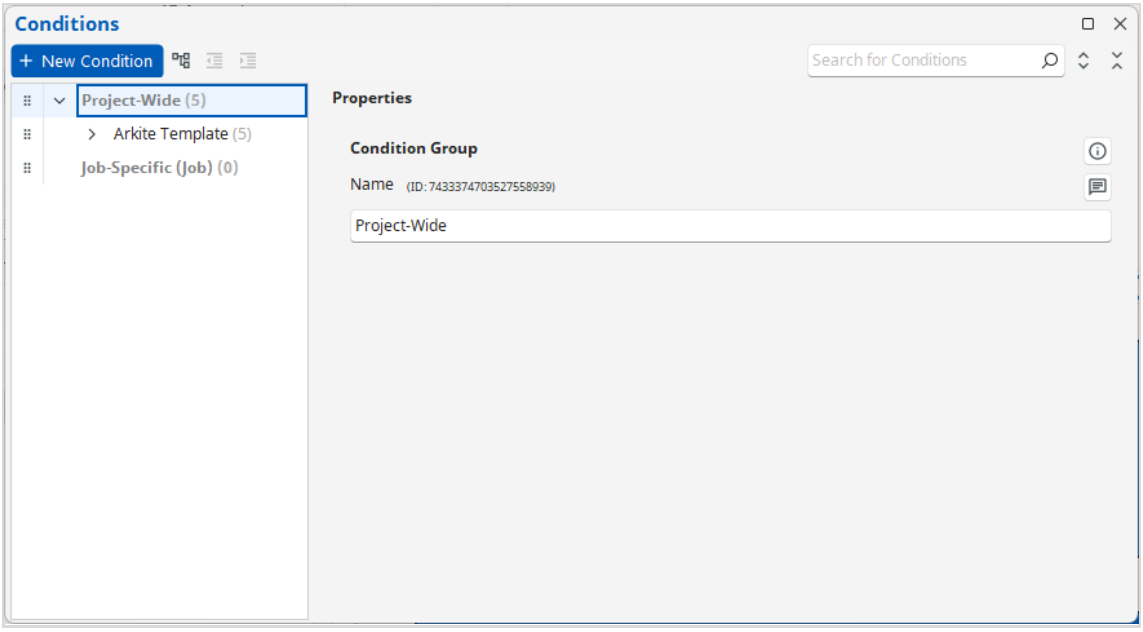
Adding conditions

To add a condition, execute the steps below:

1. In the ribbon, click the **Conditions** button.



   The **Conditions** window appears.

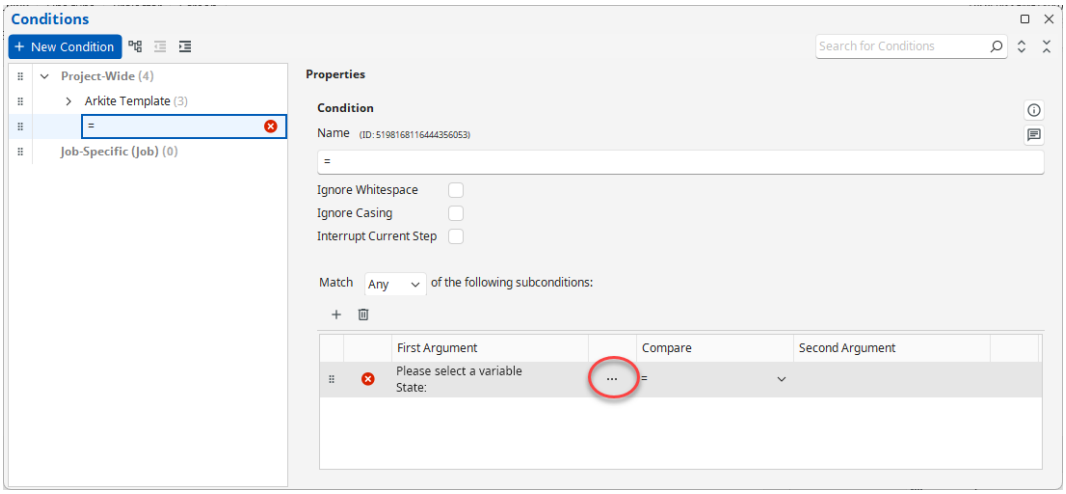


2. To add a project-wide condition, select the **Project-Wide** group.
   To add a job-specific condition, first select the relevant job in the ribbon and then select the **Job-Specific** group.
3. Click the **New Condition** button.
   A condition is added to the list with one subcondition.
4. Set the condition parameters:
   - **Ignore Whitespace:**
     If checked, all spaces will be ignored when evaluating conditions.
     For example, a condition **My Variable = Arkite** will be **True** if the value of **My Variable** is 'Ar kite'.
   - **Ignore Casing:**
     If checked, all casing will be ignored when evaluating conditions.
     For example, a condition **My Variable = Arkite** will be **True** if the value of **My Variable** is 'arkite'.
   - **Interrupt Current Step:**
     If checked, a job or process step that is linked to this condition, will be interrupted when the condition becomes invalid while the step is being executed. For example, a receive communication step will stop waiting for incoming data when its condition no longer holds

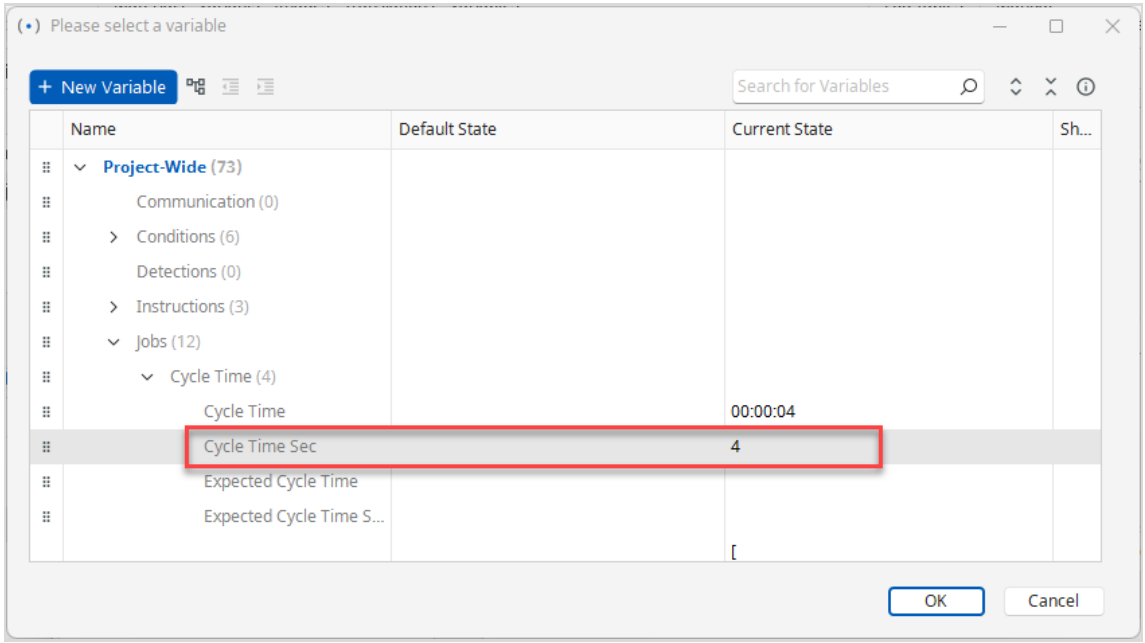
Note

*For Text (OCR) detections, it is recommended to check the **Ignore Whitespace** and **Ignore Casing** properties for the conditions you link. The text reading capability sometimes reads spaces where it should not, and vice versa.*

5. Configure the subcondtion:
   - **First argument**: this should always be a variable.
   - **Compare method**
   - **Second argument**: this can be a variable or a string.
6. To define the first variable, click the **...** button.



7. Select a variable and click **OK**.
   In the example below, we use the variable **Cycle Time Sec**.

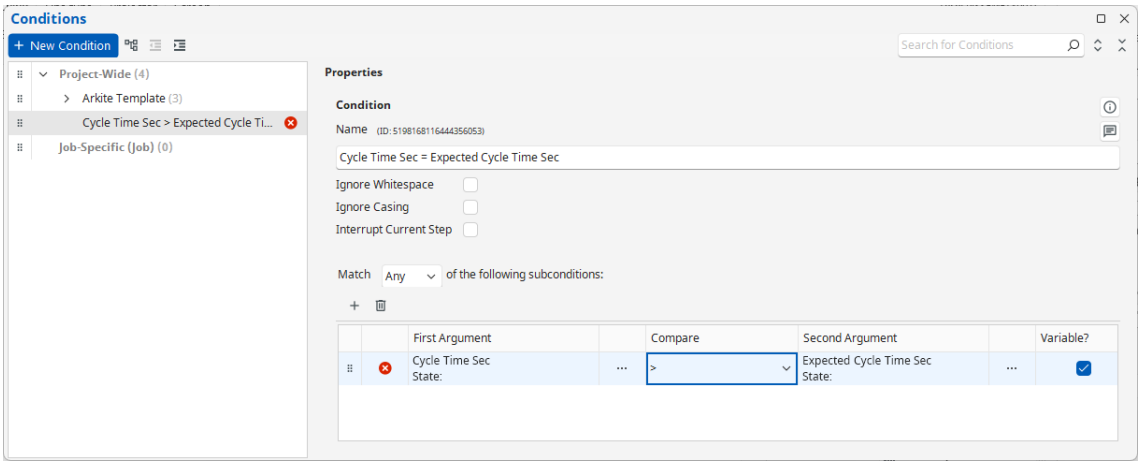


Note

*For more information about working with variables, refer to chapter **9 Variables** on page 238.*

8. Select a compare method.
9. Define the second argument:
   - To compare the state of the variable in the **First Argument** with a fixed string, type it in the **Second Argument** column.
   - To compare the state of the variable in the **First Argument** with the state of another variable, first check the checkbox in the **Variable** column and then in the **Second Argument** column, click the **...** button and select a variable.

     In the example below, we have a created a subcondition where the state of the variable **Cycle Time Sec** needs to be greater than the state of the variable **Expected Cycle Time Sec** for the condition to be true.

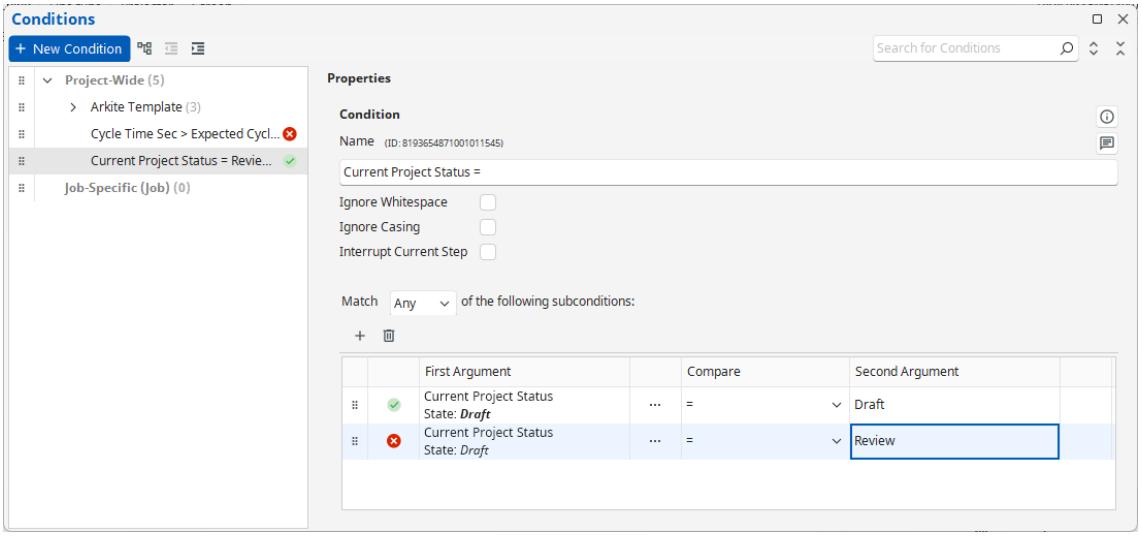


10. Optionally, you can create additional subconditions. You can define whether all, none or any (=at least one) of the subconditions need to be true for the condition to be true.

    Tip

    *If you need to create several subconditions based on the same variable, it can be interesting to duplicate the first subcondition. Right-click the subcondition and select **Duplicate** to do so.*

    In the example below, we have a created a condition where the state of the variable **Current Product Status** needs to be **Draft** (first subcondition) or **Review** (second subcondition) for the condition to be true.



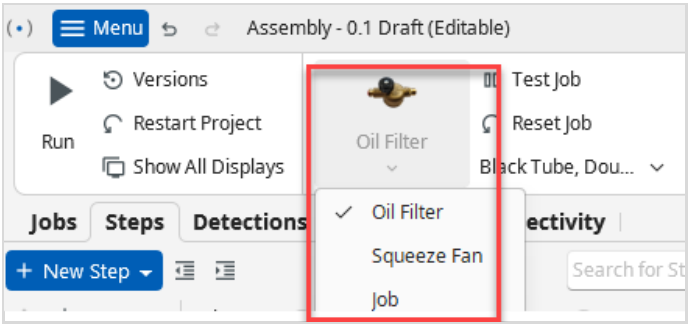
Note

*The **Name** of a condition is automatically defined by the subconditions you create. However, you can edit it.*

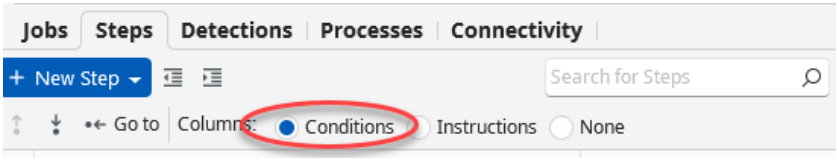
Linking a condition to a step

After you have created a condition, you can link it to a step in your process. To do so, execute the steps below:

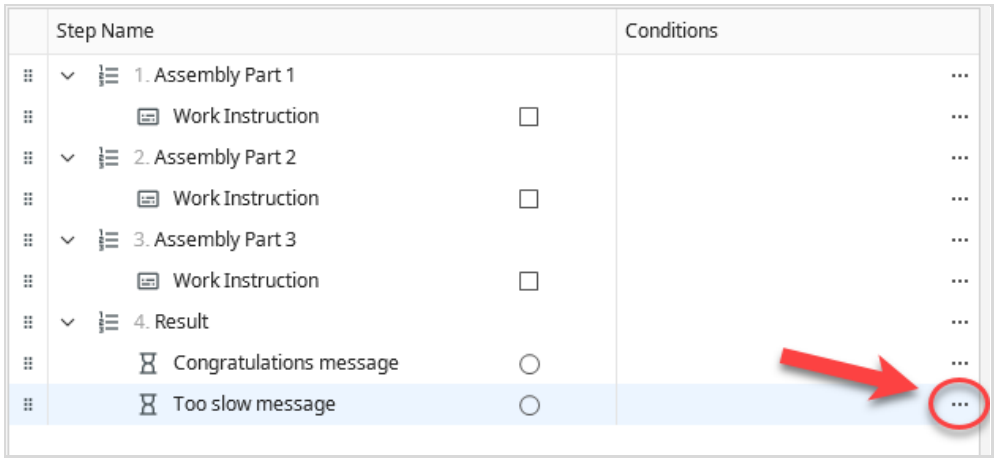
1. Navigate to the **Steps** tab.
2. In the ribbon, select the job that contains steps that you want to link to a condition.



3. In the **Steps** section, select the **Conditions** radio button.



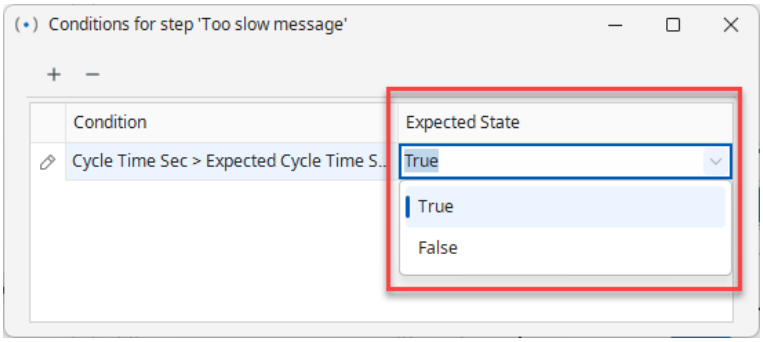
4. Go to the step you want to link to a condition and click the **...** button at the end of the **Conditions** column.



5. In the popup window, click the + button.



6. Select the condition you want to add to the step and click **OK**.
7. In the **Expected State** column, select if you want the step to be executed when the condition is **True** or when it is **False**.



8. Optionally, you can add more conditions to your process step. A step will be executed if at least one of the linked conditions hold.
9. When you have finished, close the popup window.

Note

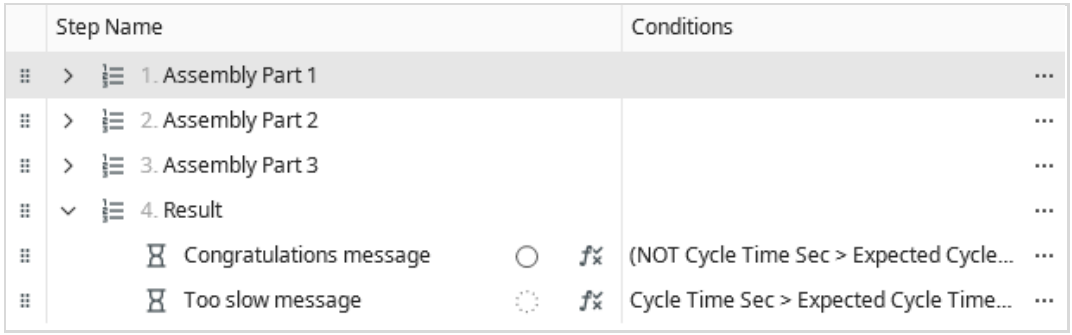
When you link a condition to a composite step, it is automatically applies to all of its child steps as well. This is not always obvious because it is not shown for every specific child step in the user interface. If it ever occurs that a step is not executed and you can't find out why, double-check if none of its parent steps are linked to a condition that prevents it from being executed.

In the example below, we have added two **Delay** steps at the end of the Job steps:

- One which shows a **Congratulations** message to the operator
- One which shows a message to the operator he was too slow

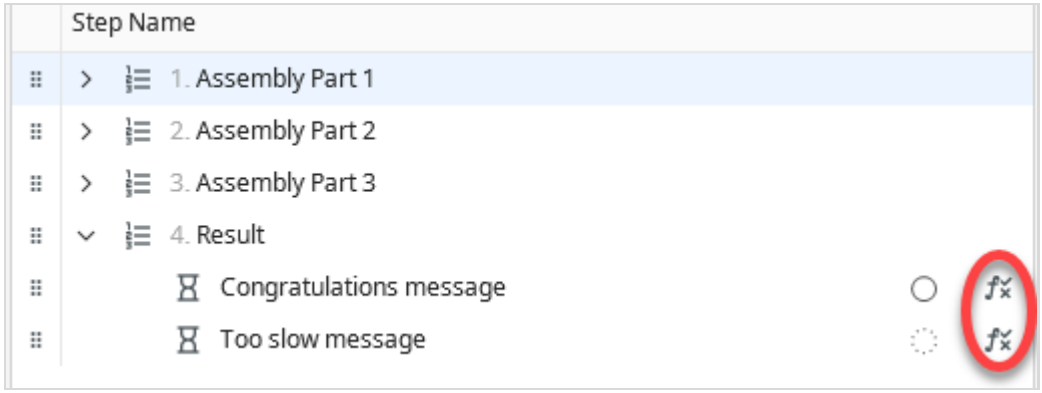
We have linked:

- the step **Congratulations Message** to the condition **Cycle Time > Expected Cycle Time** with **Expected State = False**.
  It will only be shown if cycle time does not exceed the expected cycle time..
- the step **Too Slow Message** to the condition **Cycle Time > Expected Cycle Time** with **Expected State = True**.
  It will only be shown if cycle time exceeds the expected cycle time.



When the **Conditions** column is not shown, you can still see which steps are linked to one or more condition(s) and therefore will not always be executed. They are marked by a f_x icon.

You can also see which steps that are linked to a condition will be executed in the current situation (○ icon), and which will be skipped (◌ icon).

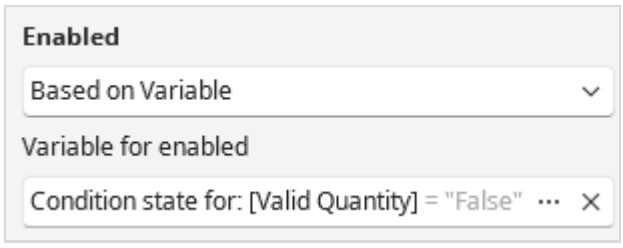


Using condition variables

For every condition you add, a variable will be created automatically as well, that has the condition's current state as value.

You can use these condition variables, for example to define the visibility of a visualization or to enable/disable a button..

To do so, in the **Visibility** or **Enabled** property select **Based on Variable**, and in the **Variable for visibility/enabled** select the condition variable of your choice. In the example below a button will only be enabled when the condition **Valid Quantity** is in state True.



Managing conditions

To manage conditions, click the **Conditions** button in the ribbon.

The **Conditions** window appears.

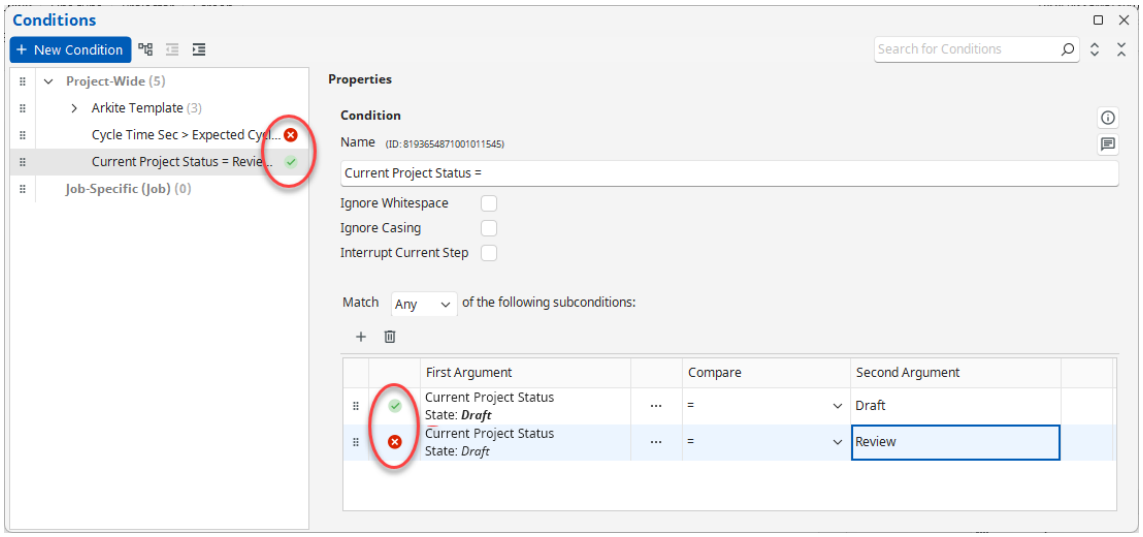
Right-click a condition and select **Duplicate** to duplicate it.

Right-click a condition and select **Delete** to delete it.

To know where a condition is used, right-click the condition and select **Associated Objects.**

Conditions and subconditions that are currently true are indicated with a icon.

Conditions and subconditions that are currently true are indicated with a icon.



5. Detections

5.1 About detections

A detection is a building block in the Arkite software used to make a representation of your workbench. By creating detections, you define which tools and components are used during and what activities take place during your assembly process. Once the detections have been configured in the Arkite software, you can use them to build your assembly process. This allows the system to guide the operator and give him the required instructions.

When using Arkite's operator guidance platform **with Arkite 3D sensor** (**Validate** and **Vision** license), it can also verify certain events or object states. The system will be able to tell your operator whether all required tools and objects are available and what to do with them. It can validate if an activity has been executed correctly and if a tool has been returned to its location.

> ***Note***
>
> *For more information about building your assembly process, refer to chapter **6 Processes** on page 173.*

5.2 Project-Wide detections versus Job-Specific detections

Project-Wide detections

Project-wide detections are detections that can be used in all the jobs in your project.

For example the picking bins on your workbench will typically be defined as project-wide detections.

Job-Specific detections

Job-specific detections are detections that can only be used in one specific job. If another job is selected, you will not be able to see or select such a detection.

For example, activities performed during the assembly of a product will typically be defined as job-specific detections.

5.3 Detection types

In the table below, you can find a list of available detection types:

Detection type	Description
Object	Any component or product on the workbench. When using a **Validate** or **Vision** license (= with Arkite 3D sensor), the system will detect if the object is present or not and can give instructions to place it or take it away.

Detection type	Description
Tool	A tool that is used during the assembly process. When not used, the tool should be at a fixed location. A tool can also have a shadow board: this is an image of the tool that will be shown at the default location of the tool when it's not present. In case of a smart tool, the system can ensure that the tool can only be used during the correct step in the assembly process. When using a **Validate** or **Vision** license (= with Arkite 3D sensor), the system will detect if a tool is present or not and can give instructions to place it or take it away. For smart tools, it will not only ensure the tool is enabled at the right moment during the assembly process, but also at the right location. When the operator wants to use the tool at a wrong place, the tool will not work. When using a **Vision** license (= with Arkite Vision sensor), you can indicate a zone on your workbench in which a tool may be placed. The software will detect where exactly in this zone the tool is located and track it when you put it at a different location within the zone after usage.
Picking bin	A storage unit or picking location, containing materials used in the assembly process. When using a **Validate** or **Vision** license (= with Arkite 3D sensor), the system will detect if the object is present or not and can give instructions to place it or take it away. Additionally, it can also ask to pick a material from the picking bin and verify if the action is executed.
Activity	An activity that is performed during the assembly process. When using a **Validate** or **Vision** license (= with Arkite 3D sensor), the system can verify if an activity has been performed by detecting movement at the defined location. An activity can be linked with a tool.
Virtual button (system with Arkite 3D sensor only)	Only available when using a **Validate** or **Vision** license (= with Arkite 3D sensor)**.** A button that is projected on the workbench. The system can request an operator to press the button in order to make a choice or confirm an action.
Quality Check (system with Arkite Vision sensor only)	Only available when using a **Vision** license (= with Arkite Vision sensor). The expected state of an assembly process at a certain point in time. The system verifies if a (sub)assembly has been executed correctly by comparing the state of the workbench before the assembly step with the expected result after the assembly and, optionally, with images of results that should be considered wrong.
Object with Quality Check (system with Arkite Vision sensor only)	Only available when using a **Vision** license (= with Arkite Vision sensor). Combines the object and the quality check detection. Verifies the presence of an object, and if present, evaluates the correct placing.
Text (OCR)	Reads and optionally validates a text using OCR. When using a **Vision** license, the Arkite Vision sensor can often be used for this detection type. When using a **Validate** or **Guide** license, you should have an Arkite RGB sensor to use this detection type.
Barcode (QR/DM)	Reads and optionally validates a barcode or QR code. When using a **Vision** license, the Arkite Vision sensor can often be used for this detection type. When using a **Validate** or **Guide** license, you should have an Arkite RGB sensor to use this detection type.

5.4 Detection states

Note

*Some detections states are only available when using a **Validate** or **Vision** license (= with Arkite 3D sensor). These are marked with a note **(Validate and Vision license only)** in the table below.*

A detection can have the states below:

State icon	Applicable for	Description
	all detections (Validate and Vision license only)	The detection is currently present on the workbench.
	all detections (Validate and Vision license only)	The detection is currently absent on the workbench.
	all detections (Validate and Vision license only)	The system cannot confirm whether the detection is currently present on the workbench, because something is blocking the sensor's view. Based on the latest state, the system assumes the detection is present.
	all detections (Validate and Vision license only)	The system cannot confirm whether the detection is currently present on the workbench, because something is blocking the sensor's view. Based on the latest state, the system assumes the detection is absent.
	Picking bin (Validate and Vision license only)	Material is being picked from the picking bin.
	all detections except for virtual buttons (Validate and Vision license only)	The detection state is currently not calculated. This is the case for: – Objects with Quality Check, Quality Checks, Text (OCR) and Barcode (QR/DM) detections that are not active (**Run** mode) or not selected (**Edit** mode). – Activities that are not active (**Run** mode) or not selected (**Edit** mode), unless there is an error or shadowboard visualization associated with the detection. – All other detections taught on the Arkite Vision Sensor that are not active (**Run** mode) or not selected (**Edit** mode), unless there is an error visualization associated with the detection.
	All detections	The detection state is currently not calculated because the detection is not enabled in the project. A detection can be disabled if the **Enabled** property of either the detection itself or the detection group it is part of is set to **Based on Variable**, and if the state of that variable is FALSE, OFF or 0.
	Picking bin	A material has been associated with the picking bin. If an image is available for the material, the icon will be replaced with this image.
	Picking bin	No material has been associated with the picking bin.
	Activity	The activity is linked with a tool.
	Tool	The tool has been configured as a smart tool.

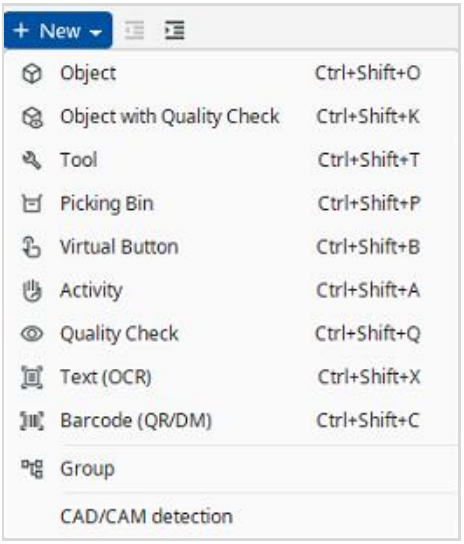
Note

*When using a **Validate** or **Vision** license (= with Arkite 3D sensor), detections that have been added but have not been taught to the system yet will not have any state.*

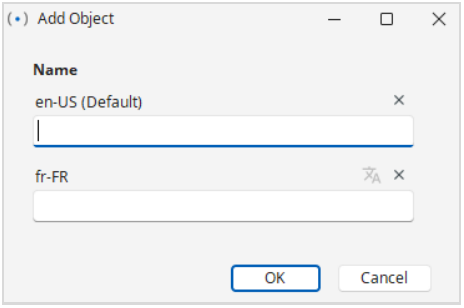
5.5 Adding detections

To add a new detection, execute the steps below:

1. Navigate to the **Detections** tab.
2. To add a project-wide detection, select the **Project-Wide** group.
   To add a job-specific detection, first select the relevant job in the ribbon and then select the **Job-Specific** group.
3. Click the **New** button and select the type of detection you want to add.

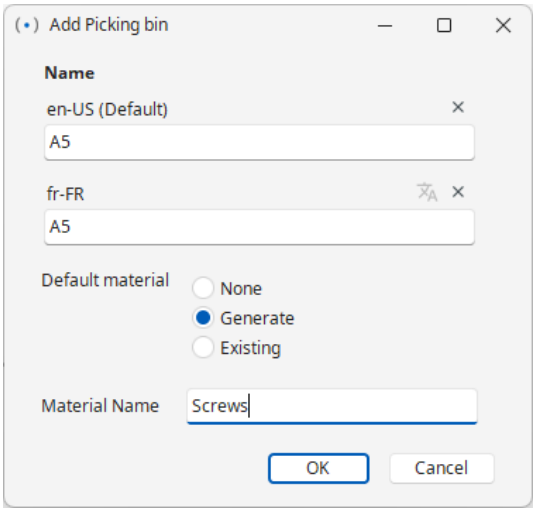


4. For each language in your project, provide a name for the detection and click **OK**.
   Make sure to use a logical name, because this name will be used in the instructions that are generated for the steps using this detection.
   You can use the **Translate** button to generate automatic translations from the default language.



Note 1

*When adding a picking bin, you can choose leave it empty or to associate a new or existing material to it. You can still change this afterwards in the **Materials** window. For more information, refer to section **5.8 Managing materials** on page 159.*



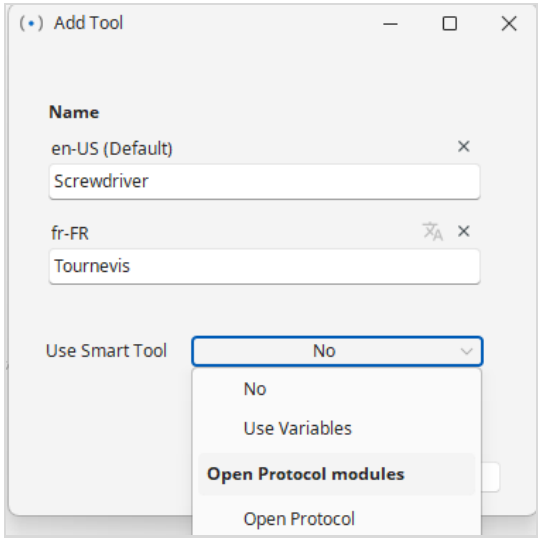
Note 2

*When adding a tool, you should specify whether it is a smart tool or not. In case of a smart tool, you can choose to either use variables of an **Open Protocol** communication module. If you want to use **Open Protocol**, you should add the module first. For more information, refer to section **7.3.3 Adding a communication module** on page 214.*

*Only **one** tool can be linked to a single **Open Protocol** module. To use multiple tools, create multiple **Open Protocol** modules.*

*For more information on configuring your smart tool, refer to section **5.9 Working with smart tools** on page 161.*

*It is always possible to change the option you want to use afterwards. To do so, select the tool in the **Detections** tab and update its **Tool Connection Settings** in the **Properties** tab.*



When using a **Guide** license (=without sensor) a **draw shape** visualization is automatically generated when adding the detection, which will be projected on the workbench when something needs to be done with the object

5. (**Validate** and **Vision** license only) Update the properties in the **Properties** tab if required:
   - **External Trigger Variable:**

It is possible to use an external system, for example a PLC or another sensor than the one provided by Arkite, to validate the presence of a detection. If you want to do so, select the variable containing the information about the detection's presence and the define:

External Trigger ON State:

the value that is expected in the External Trigger Variable to activate the detection. To insert dynamic content, click the {x} icon at the location where you want to include the content and select the variable containing the dynamic content. Alternatively, you can also type the variable name between double braces, i.e. *{{Variable name}}*.

External Trigger ON Time:

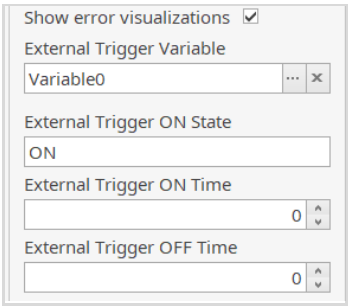
the time in milliseconds that the ON state needs to be reached before the detection is activated.

External Trigger OFF time:

the time in milliseconds the External Trigger variable needs to have another value than the ON state before the detection is deactivated.

Note

In this case, it is not needed to teach the detection. If you do, the teaching will be ignored by the system as long as an external trigger has been configured.



- **Overrule location by variables:**

  It is possible to make the location, size and orientation of a detection dynamic, based on the value of variables. For example, you can teach and finetune the detection in the Arkite software but let the location be defined by the input of an external system through the Arkite API.

  Note

  *For more information on working with the Arkite API, refer to chapter **18 The Arkite API** on 292.*

  In the **Location Variables** properties, you can define the variables that contain the X, Y and Z coordinates of the detection (in mm from the origin). These values define the coordinates of the lower left point of the detection's minimum bounding box.

  In the **Size Variables** properties, you can define the variables that contain the length, width and height of the detection, i.e. the size in the X, Y and Z direction respectively.

  When changing location and size variables properties, all projections linked to the detection as well as the picking boxes for a picking bin will move along.

  The **Projection Z** variable allows you to define a different Z value for your projection than the bottom of your box. If multiple projections are linked to the detection, the **Projection Z** variable defines the Z value of the lowest projection. If left blank, the projection's Z value maintains its existing distance to the bottom of the box.

  The table below contains some examples of how the Z an projection Z location variables behave:

Before setting the variables	Variable values	After setting the variables
Z value for the box = 50 Z value for the projection = 100	Z = 70 Projection Z = 200	Z value for the box = 70 Z value for the projection = 200
Z value for the box = 50 Z value for projection 1 = 100 Z value for projection 2 = 170	Z = 70 Projection Z = 200	Z value for the box = 70 Z value for projection 1 = 200 Z value for projection 2 = 270
Z value for the box = 50 Z value for projection 1 = 100 Z value for projection 2 = 170	Z = 70 Projection Z = (blank)	Z value for the box = 70 Z value for projection 1 = 120 Z value for projection 2 = 190

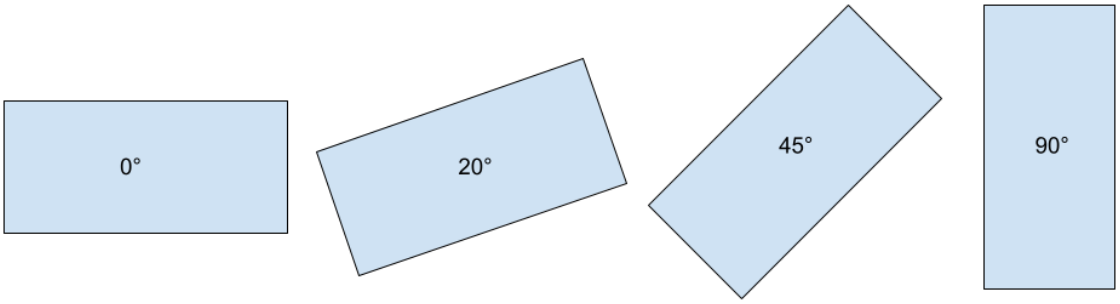
Note

*When clicking the **Set Location Variables to Default** button, the values of the variables will be set to the box's coordinates as defined in the Arkite software.*

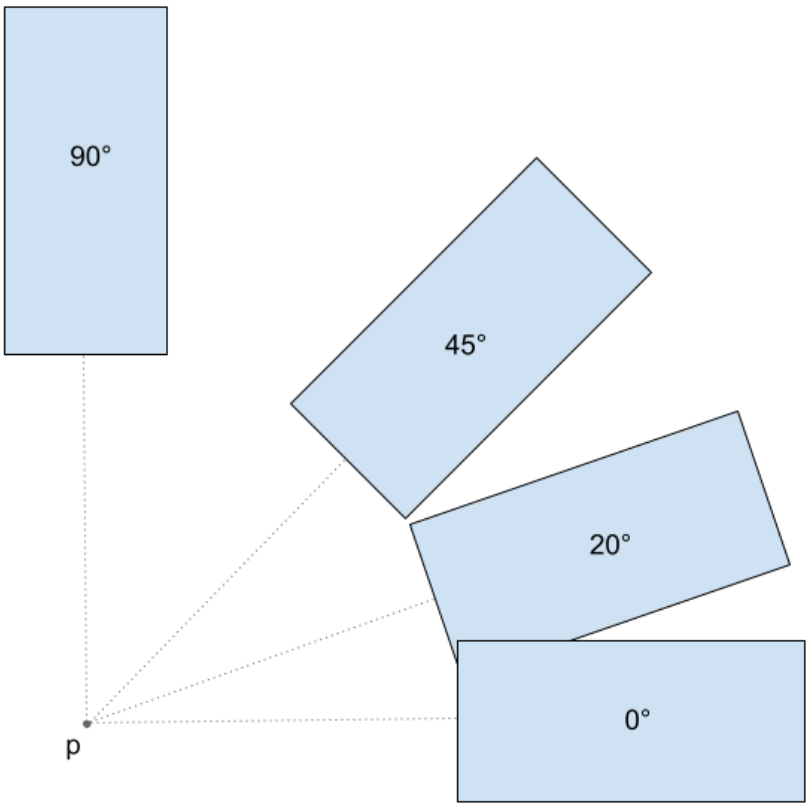
*When clicking the **Set Size Variables to Default** button, the values of the variables will be set to the box's size as defined in the Arkite software.*

In the **Orientation Variable** property, you can define a variable that contains the orientation of a detection. When the value of this variable changes, the detection and all linked visualizations will rotate according to the variable's value.

By default, the detection will rotate around its centre, as shown in the image below:



When checking the **Off-center pivot point** property, you can define the X and Y coordinate of a pivot point around which the detection should rotate. For example, in the image below, the detection rotates around a point p.



For example, in a use case where an operator needs to assemble clamps on different positions on a disk, this feature can be very useful.

- **Pick To Light (DMX)**

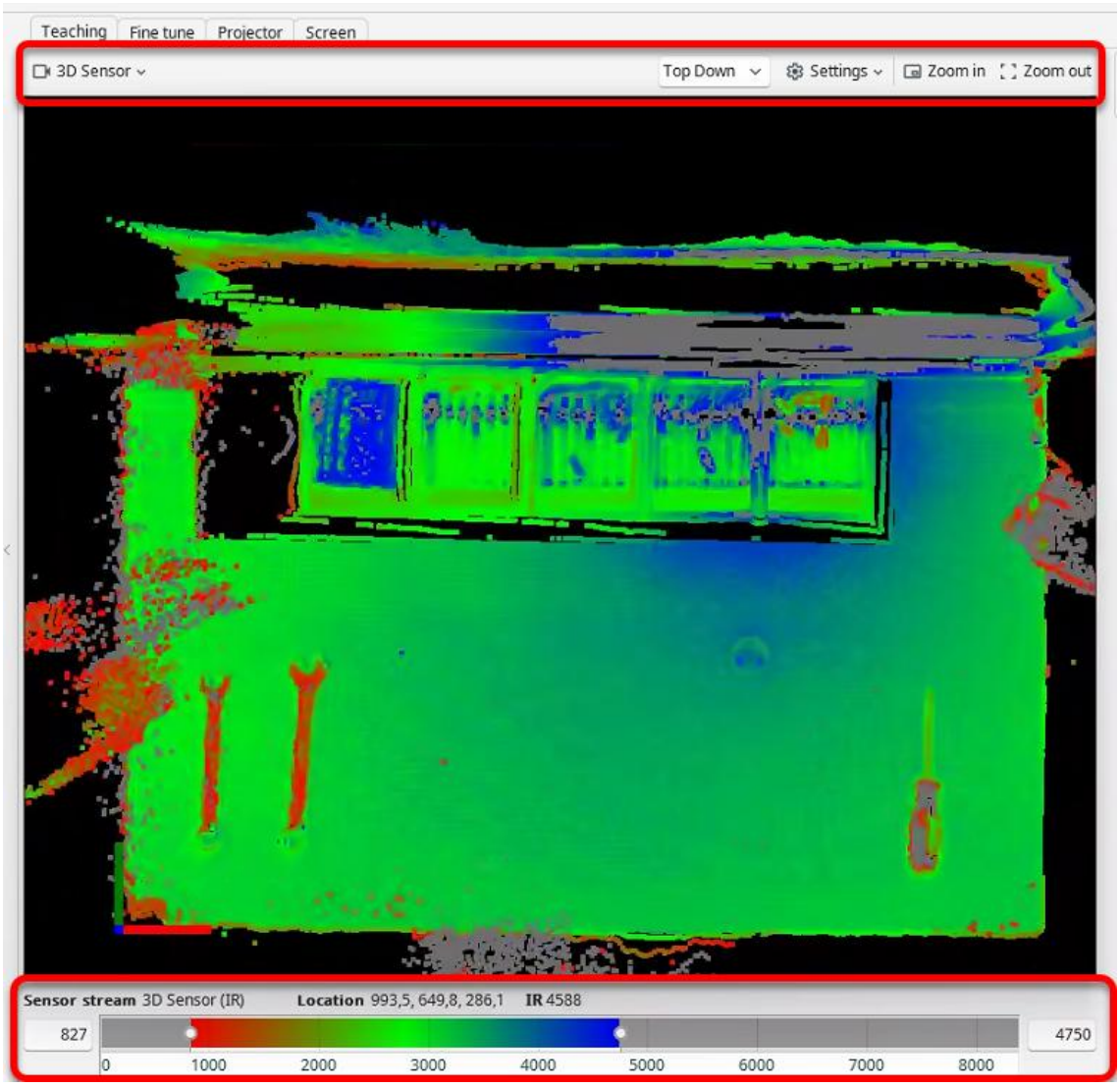
  You can link your detection to a DMX light. To do so, you first need to add a Pick To Light (DMX) communication module and configure the lights on the connected DMX device. For more information, refer to sections **7.3.3 Adding a communication module** on page 214 and **7.3.4 Configuring lights for a Pick To Light (DMX) communication module** on page 216.

  In the **DMX Light** dropdown list, you can select the DMX light you want to link the detection to. In the **DMX Color** menu, you can set the color for the linked DMX light. By default, the color of a DMX light that is linked to a detection changes depending on the status of the detection, based on the **Detection colors** template. To manage the behavior and the colors that are used, refer to section **15.2 The Detection Colors template** on page 276.

5.6 Teaching and finetuning detections (Validate and Vision license only)

5.6.1 Optimize your sensor stream view

Before you start teaching detections, you should make sure you have the best view possible on your workbench. To do so, navigate to the **Detections** tab.



In the **Teaching** subtab, optimize your sensor stream view:

- In the dropdown, select the view that gives the best result for your specific situation:
  - **Top down**: provides an orthographic projection along the Z-axis, so you have a top-down view on your calibration surface without a perspective effect. This view is recommended when you prefer a 2D view, but are working with detections that are high above the calibration surface.
    Use your mouse wheel to zoom in or out
    Hold your right mouse button and move around to shift your view to the left or the right
  - **3D:** provides a 3D view of your workbench.
    Hold your left mouse button and move around to rotate your view

Use your mouse wheel to zoom in or out

Hold your right mouse button and move around to shift your view to the left or the right

- **2D**: provides a 2D view of your workbench's surface

  Use your mouse wheel to zoom in or out

  Hold your right mouse button and move around to shift your view to the left or the right

- Define the zoom level using the **Zoom in** and **Zoom out** buttons.
- In the **Settings** menu, select the stream of your choice (depth or IR) and mirror the screen if required.
- In the **Sensor** menu, select the sensor that should be used for teaching the detection. The selected sensor is shown on the button:
  - For **Quality Checks** and **Objects with Quality Check,** the Arkite Vision sensor will be selected by default (Vision license only)
  - For **Barcode (QR/DM)** and **Text (OCR)** detections

    When using a **Vision** license, the Arkite software will automatically select the Arkite Vision sensor.

    When using a **Validate** or **Guide** license, the Arkite software will automatically select the Arkite RGB sensor.
  - For all other detections, the Arkite 3D sensor will be selected by default. If you are using a Vision license, you can switch to the Arkite Vision sensor. This can be beneficial when teaching small objects.

    Note

    *By default, the **Infrared - Pseudo color stream** will be shown when the Arkite 3D sensor is selected and the **Infrared – Grayscale** stream will be shown when the Arkite Vision sensor is selected.*
- Use the sliders below the image to optimize the visibility of certain objects. Both the red and blue pegs correspond to a certain infrared value and a green gradient connects both.

  Note

  *Apart from the selection you make in the **Sensor** menu, the settings and options described above are only meant to increase the visibility of the objects for the user. Changing these settings will not have any impact on the detection logic.*

5.6.2 Teaching detections

In the section below, instructions are provided on how to teach different types of detections. You will learn how to:

- Teach an object, picking bin, tool or virtual button
- Teach an activity
- Teach a quality check
- Teach a text or barcode

Teach an object, picking bin, tool or virtual button

To teach a detection, execute the steps below:

1. Navigate to the **Detections** tab.

2. Select the detection you want to teach to the system.

   Note

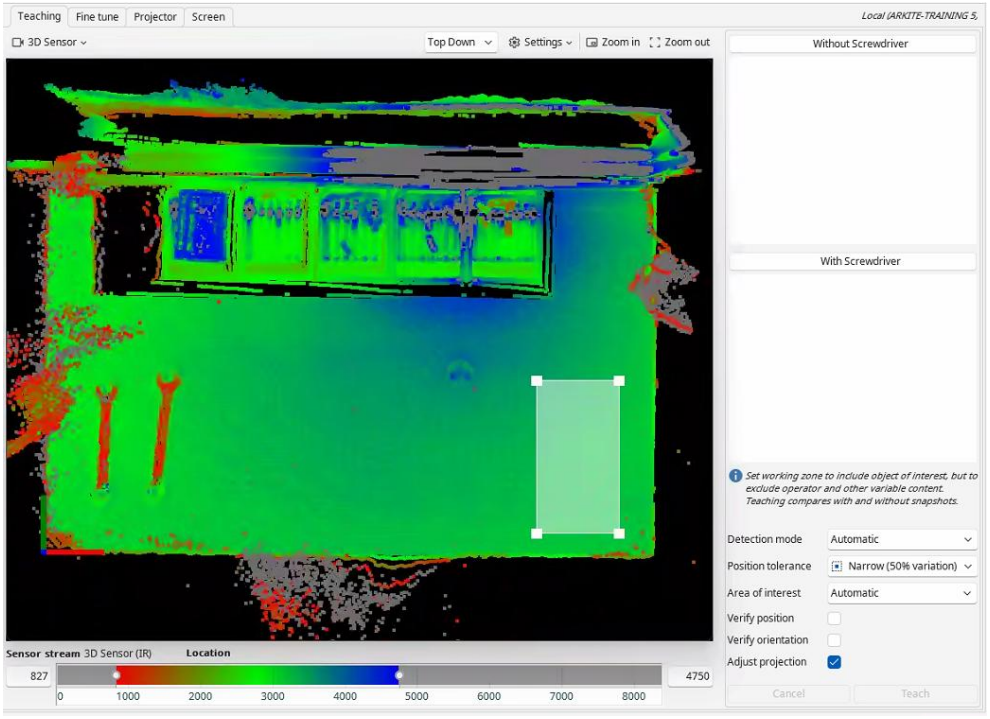
   *By default, the Arkite software will select the Arkite 3D sensor for teaching these detections. When you are using a **Vision** license, you can also select the Arkite Vision sensor from the **Sensors** menu This can be beneficial when teaching small objects.*

   *By default, the **Infrared - Pseudo color stream** will be shown when the Arkite 3D sensor is selected and the **Infrared – Grayscale** stream will be shown when the Arkite Vision sensor is selected.*

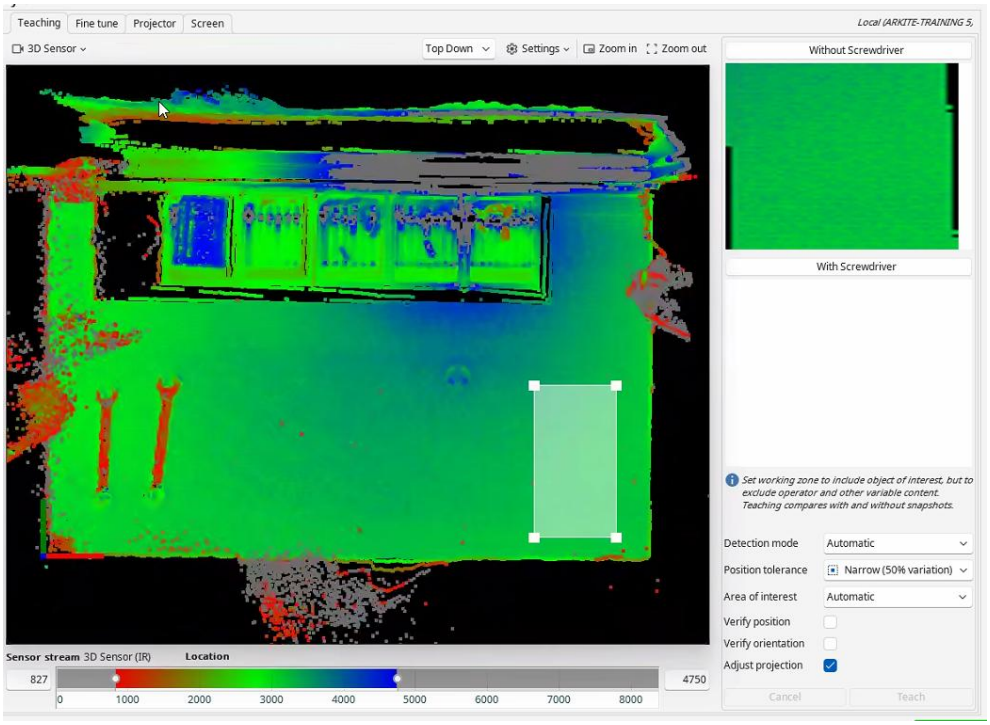
3. Define the working zone by adjusting the white rectangle. Make sure it does not contain any other variable information than the detection you want to teach.

   Note

   *When teaching a tool and using a **Vision** license, you have the option to define a working zone for the location of a tool. The system will then 'follow' the position of that tool within the zone. In such a case, set the white rectangle to the working zone in which the tool can be located.*

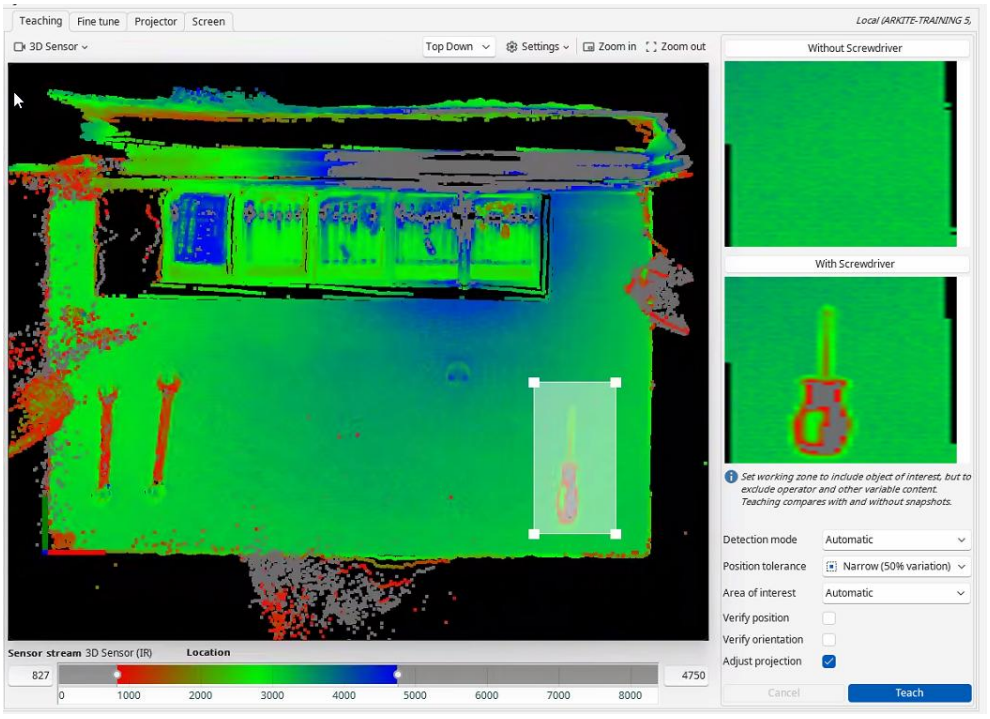


4. Take the detection away and click the **Without...** button.



A snapshot is generated of the area interest, without the detection.

5. Put the detection at its location and click the **With...** button



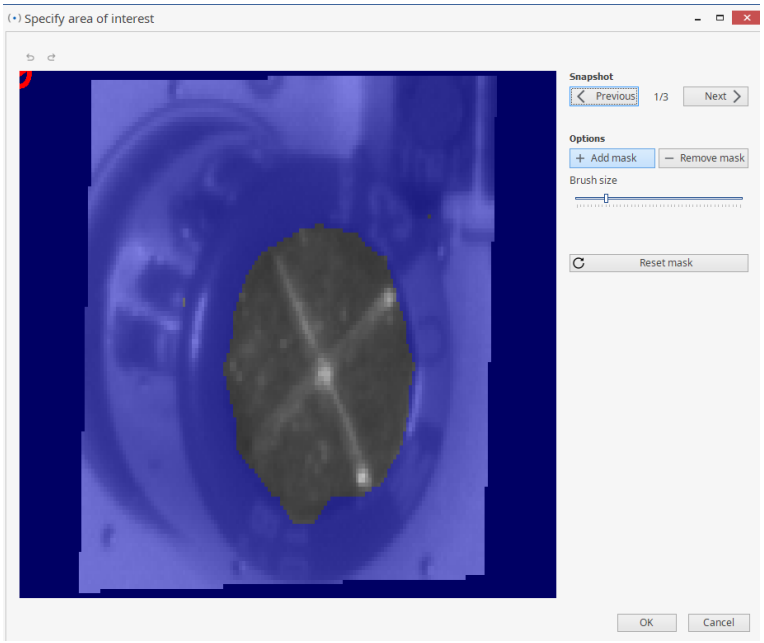
A snapshot is generated of the working zone, containing the detection.

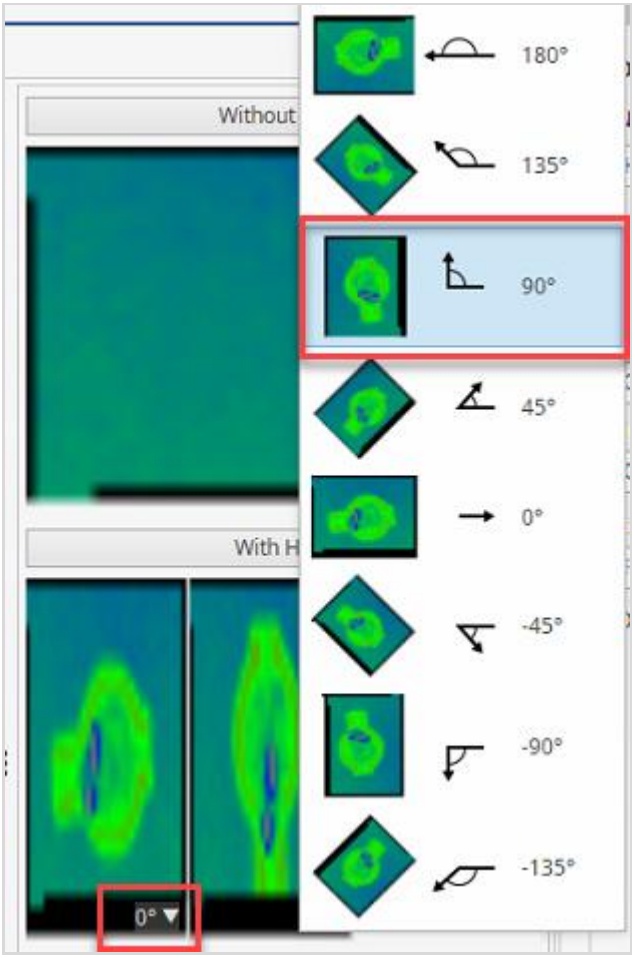
Note

When teaching a virtual a button, use your fist, and narrow the working zone to include your fist only.

6. Optionally, update the settings:

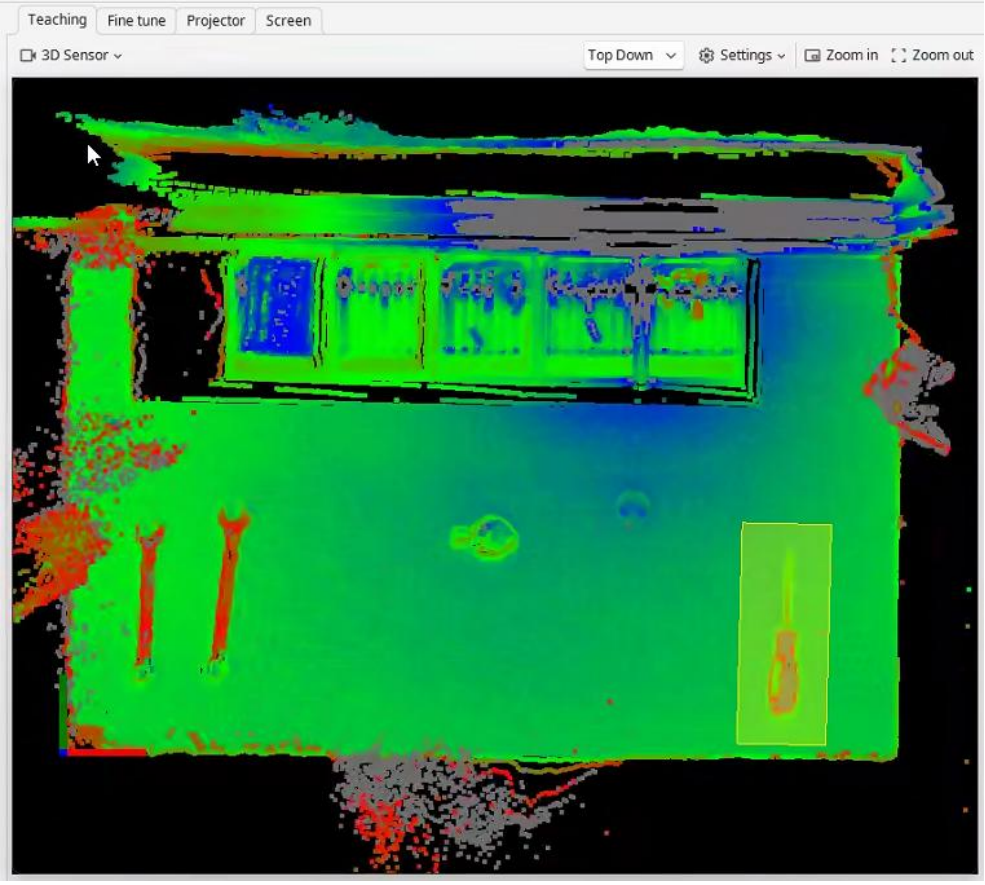
Setting	Description
Detection mode	Select the sensor stream or algorithm that is used for teaching the detection: – **Automatic**: the system will select the stream with the best contrast. This is the default option. – **Depth**: use the depth stream. – **IR:** use the Infrared intensity stream. – **Reflective tape (IR):** in some cases, objects are difficult to detect due to a low contrast or changing light conditions in the working area. This can be solved by replacing the object with a piece of reflective tape with a high and constant IR value. For more information, refer to section **5.10 Working with reflective tape (Validate and Vision license only)** on page 166. – **Infrared edges:** when selecting this option, the IR stream will be used but the sensor will look for edges, in other words for the boundaries of an object where there is a high contrast with the adjacent pixels. It can be used to detect objects regardless their color or variations in infrared lighting conditions. – **Shape based (Vision license only):** use the shape-based matching algorithm. This algorithm will detect the edges of an object, in order to recognize its shape. Color and material is mostly ignored. Use this option to create a quality check that works for the same object in different colors. – **Texture based:** use the correlation-based matching algorithm, based on a Normalized Cross-Correlation (NCC) model. This algorithm looks for patterns in grey values. As opposed to the shape based algorithm, it uses all grey values and not only clear edges. Different color and material will have an impact on the quality check. – **Histogram based (Vision license only):** use the histogram-intersection based matching algorithm. Use this option for checking: – Non-rigid shapes, for example a cable that can be bent in various ways or a dot of glue of which the shape and size may vary. – Objects with different shades/colors. ***Note*** *Not all colors can be distinguished – the infrared intensity value should differ sufficiently. For example, yellow and orange are typically hard to tell apart.*
Position tolerance	Define the accuracy of the size of the box that is created after detection. The options are: – **Wide**: the shape that is 100% larger, i.e. twice the size, than the detected object. – **Narrow:** a shape is created that is 50% larger than the detected object. This is the default option when detecting a tool. This way, the operator does not have to put back a tool at the exact same location. – **Accurate**: a shape is created that is 20% larger than the detected object – **Exact:** a shape is created with the same size as the detected object. This is the default option for other detections.

Setting	Description
Area of interest	Define how the area of interest is defined: – **Automatic:** the system will look for differences within the working zone between the different snapshots. – **Manual:** specify the area of interest yourself. After selecting the **Manual** option, click the **Specify...** button. A popup window will open with an image of the first snapshot. Use the **Previous** and **Next** buttons to browse through snapshots. In the snapshots, you can see the differences as detected by the system.  – Zoom in using your mouse wheel. – Click **Add mask** to ignore particular zones. – Click **Remove mask** to take additional zones into account. – Use the **Brush size** slider to work with more precision if required. – Click **Reset mask** to return to the differences as detected by the system. – Click **OK** to apply your changes.
Verify position (Vision license only)	For tools and objects only. Check this box if you want the software to check if the object or tool is in the exact same position as during teaching. If it's not, arrows will be projected instructing the operator in which sense he should move the tool or object to reach the correct position.

Setting	Description
Verify orientation (Vision license only)	For tools and objects only. Check this box if you want the software to check if the object or tool is in the same orientation as during teaching. If it's not, a projection will be displayed instructing the operator how to change the orientation. ***Note*** *For objects and tools with an irregular form for which the orientation can vary a lot, it is recommended to add multiple **With Object** / **With Tool** snapshots from different angles. For each snapshot, indicate the angle by clicking in the lower left corner of the snapshot and selecting the correct angle.* ***Watch out:*** *Rotate the object **counterclockwise** to indicate an increasing positive angle. For example, a quarter-turn counterclockwise results in an angle of 90 degrees, a quarter-turn clockwise results in an angle of -90*.* 
Adjust projection	Define whether projections should be adjusted when teaching the same quality check a second time. If you have edited the automatically generated projections on the **Projector** tab, it is recommended to uncheck this checkbox and adjust the projections yourself after teaching.

7. Click **Teach**.

For every detection, a box is created with the same size and position as the detected object. This allows the system to verify whether the object is present.



Important

The system can only detect whether an object is present on the location of the detection. It is not able to 'follow' an object and check whether it is present on another location on the workbench. Therefore, it makes sense to integrate a fixture onto the workbench, where the component can be fixated during the process which makes sure that the entire configuration of components and detections are standardized and stay in the same location over time.

*Only for tools and when using a **Vision** license, you have the option to define a working zone for the location of a tool. The system will then 'follow' the position of that tool within the zone.*

Additionally, a **draw shape** visualization is generated on the **Projector** display, which will be projected on the workbench when something needs to be done with the detection.

Note

*By default, the brush color of this visualization changes depending on the state of the detection, based on the **Detection colors** template. To manage the behavior and the colors that are used, refer to section **15.2 The Detection Colors template** on page 276.*

For picking bins, a second box is generated right above the picking bin. This allows the system to detect when something is picked from the picking bin. That works fine, unless you are working with several (rows of) picking bins stacked upon each other, hiding the boxes from the picking bins below them to the sensor. For more information on how to deal with stacked picking bins, refer to section **5.11 Working with stacked picking bins** on page 167.

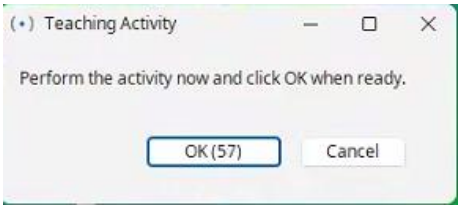
Teach an activity

To teach an activity, execute the steps below:

1. Navigate to the **Detections** tab.
2. Select the activity detection you want to teach to the system.
3. Define the exact area where the activity takes place by adjusting the white rectangle. Make sure it includes your hands since this is where the system will focus on.
4. Optionally, update the settings:

Setting	Description
Detection mode	Select the sensor that is used for teaching the detection: – **Depth**: use the depth stream. – **Reflective tape (IR):** in some cases, objects are difficult to detect due to a low contrast or changing light conditions in the working area. This can be solved by replacing the object with a piece of reflective tape with a high and constant IR value. For more information, refer to section **5.10 Working with reflective tape (Validate and Vision license only)** on page 166.
Adjust projection	Define whether projections should be adjusted when teaching the same quality check a second time. If you have edited the automatically generated projections on the **Projector** tab, it is recommended to uncheck this checkbox and adjust the projections yourself after teaching.

5. Click **Teach**.
   The **Teaching Activity** pop up window appears.
6. Perform the activity and click **OK** when done.



A box is created with the same size and position as the area where the activity takes place. This allows the system to verify whether the activity is being executed.

Important

*The system detects movement in the area where the activity takes place. It is not able to verify whether the activity has been executed correctly unless it is performed with a smart tool. In that case, it can use information such as the torque level as an input. For more information, refer to section **5.9 Working with smart tools** on page 161.*

Teach a quality check or an object with quality check (Vision license only)

To teach a quality check, execute the steps below:

1. Navigate to the **Detections** tab.
2. Select the quality or object with quality check detection you want to teach to the system.

Note

*By default, the Arkite software will select the Arkite Vision sensor for teaching quality checks. Optionally, you can also select the Arkite 3D sensor from the **Sensors** menu. However, most of the time the Arkite Vision sensor will give the best results.*

*By default, the **Infrared - Pseudo color stream** will be shown when the Arkite 3D sensor is selected and the **Infrared – Grayscale** stream will be shown when the Arkite Vision sensor is selected.*

3. Set the working zone around the item to check, such as a screw or a clip, by adjusting the white rectangle.

   Note

   When checking multiple objects, create and teach different quality checks for each of them.

4. Provide snapshots for each of the relevant classes of the quality check.

   By default, the three classes below are linked to a *quality detection*:

   - The **Expected Result** class
   - The **Bad Result** class
   - The **Missing** class

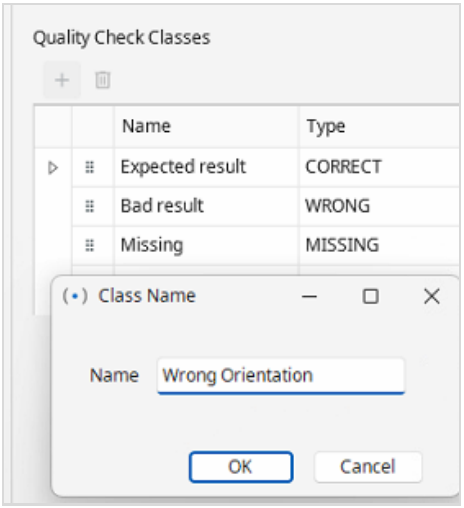
   By default, the three classes below are linked to an *object with quality check detection*:

   - The **Without** class
   - The **Expected Result** class
   - The **Bad Result** class

   For each class, you can provide one or more snapshots.



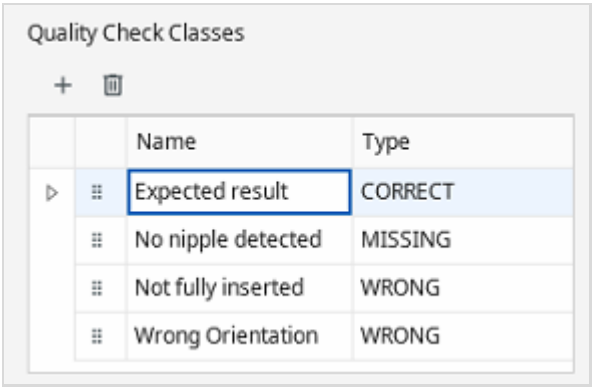
5. (optional) If needed, create additional quality check classes or rename the existing classes. In the **Properties** of the detection, under **Quality Check Classes**, click the + button and provide a name and a type for your class.



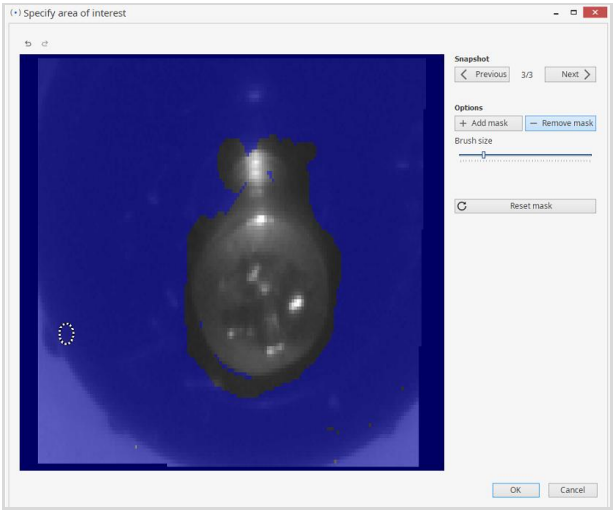
For classes of the type CORRECT, the quality check will be considered successful. For classes of the type WRONG or MISSING, the quality check will be considered unsuccessful. A new class is typically created if you want to show a different error text or give different rework instructions. For more information, see section **The Check step with rework flows (Vision license only)** on page 67.

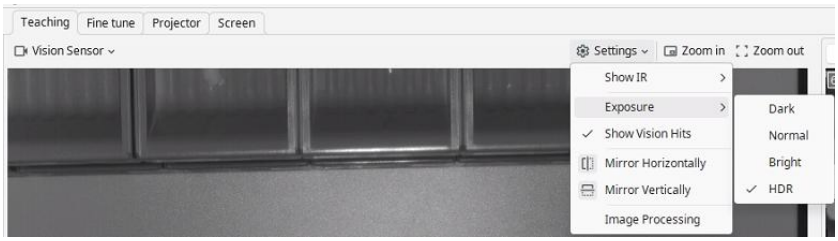
Example

In the example below, two classes of the type WRONG have been created: one for the case in which the nipple is not completely inserted and one for the case in which the nipple was inserted in the wrong direction.



6. Optionally, update the settings:

Setting	Description
Detection mode	Select the algorithm that is used for teaching the detection: – **Automatic**: the system will select the algorithm with the best result. This is the default option. – **Shape based:** use the shape-based matching algorithm. This algorithm will detect the edges of an object, in order to recognize its shape. Color and material is mostly ignored. Use this option to create a quality check that works for the same object in different colors. – **Texture based:** use the correlation-based matching algorithm, based on a Normalized Cross-Correlation (NCC) model. This algorithm looks for patterns in grey values. As opposed to the shape based algorithm, it uses all grey values and not only clear edges. Different color and material will have an impact on the quality check. – **Histogram based:** use the histogram-intersection based matching algorithm. Use this option for checking: – Non-rigid shapes, for example a cable that can be bent in various ways or a dot of glue of which the shape and size may vary. – Objects with different shades/colors. ***Note*** *Not all colors can be distinguished – the infrared intensity value should differ sufficiently. For example, yellow and orange are typically hard to tell apart.*
Position tolerance	Define the accuracy of the size of the box that is created after detection. The options are: – **Wide**: the shape that is 100% larger, i.e. twice the size, than the detected object. – **Narrow:** a shape is created that is 50% larger than the detected object. – **Accurate**: a shape is created that is 20% larger than the detected object – **Exact:** a shape is created with the same size as the detected object. This is the default option for quality checks.
Area of interest	Define how the area of interest is defined: – **Automatic:** the system will look for differences within the working zone between the different snapshots. – **Manual:** specify the area of interest yourself. After selecting the **Manual** option, click the **Specify...** button. A popup window will open with an image of the first snapshot. Use the **Previous** and **Next** buttons to browse through snapshots. In the snapshots, you can see the differences as detected by the system.  – Zoom in using your mouse wheel. – Click **Add mask** to ignore particular zones. – Click **Remove mask** to take additional zones into account. – Use the **Brush size** slider to work with more precision if required. – Click **Reset mask** to return to the differences as detected by the system. – Click **OK** to apply your changes.

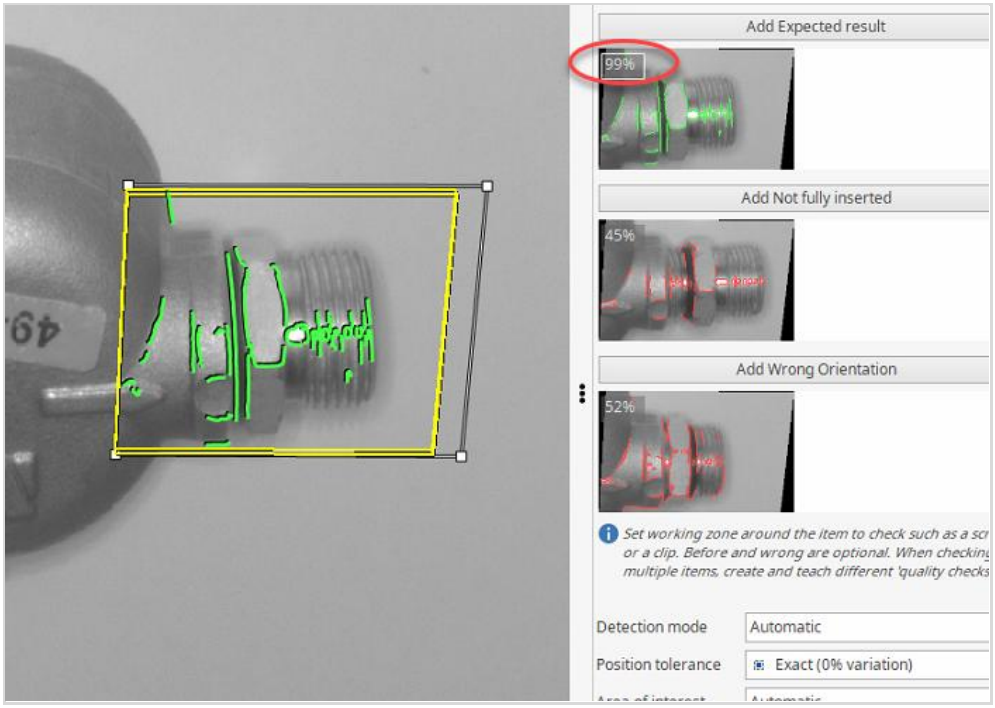
Setting	Description
Exposure mode	Select the camera exposure that will be used for taking the snapshots: – **Automatic**: uses the HDR option – **Dark:** uses a short exposure, resulting in a darker image – **Normal:** uses a medium exposure – **Bright:** uses a long exposure, resulting in a brighter image – **HDR:** High Dynamic Range. The camera captures multiple frames with different exposure times, and combines them into one image, selecting the most appropriate intensity for each part of the image. This allows to perform reliable quality checks on scenes that contain both bright and darker elements. ***Note*** *In the user interface, by default the HDR image is shown. You can select a different exposure in the **Settings** menu.* 
Adjust projection	Define whether projections should be adjusted when teaching the same quality check a second time. If you have edited the automatically generated projections on the **Projector** tab, it is recommended to uncheck this checkbox and adjust the projections yourself after teaching.
Adjust occlusion box	Define whether the occlusion box should be adjusted when teaching the same quality check a second time. If you have edited the automatically generated occlusion box on the **Fine Tune** tab, it is recommended to uncheck this checkbox and adjust the projections yourself after teaching.

7. Click **Teach**.

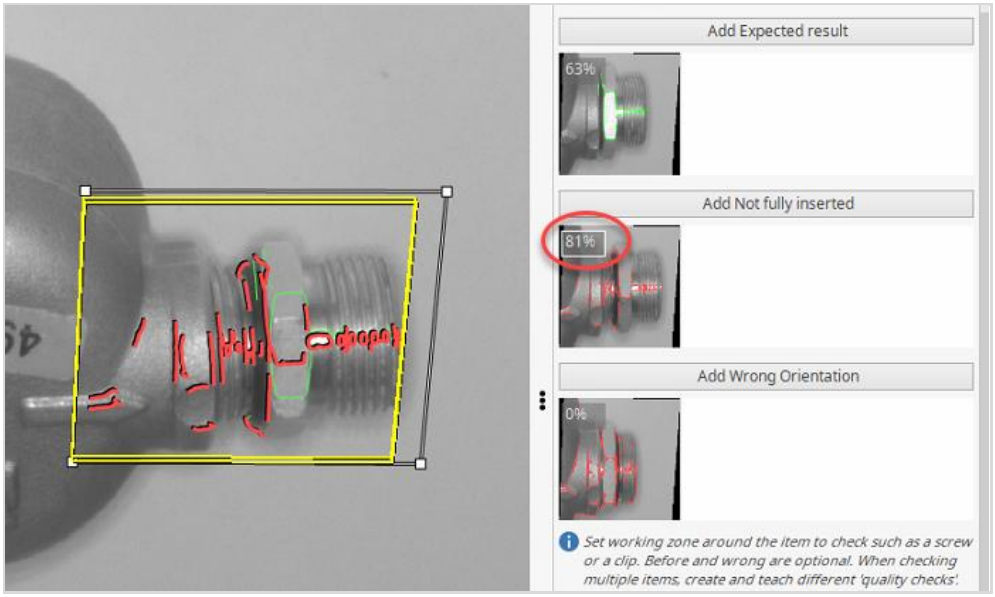
A box is created with the same size and position as the area where quality check needs to be performed. When performing the quality check, the system will compare the live image with all frames for all quality check classes. The quality check will be considered successful when the live image's similarity is the highest for (one of the) frame(s) for a class with type CORRECT and more than the minimum percentage as defined in the **Detection Rate** property on the **Finetune** tab.

Examples

- *In the example below, the quality check will be successful because there is a 99% similarity with one of the **Expected Result** frames (type CORRECT), and there is no higher similarity with one of the other frames.*



- *In the example below, the quality check will be unsuccessful because there is a 81% similarity with one of the frames in class **Not fully inserted** with type WRONG, and there is no higher similarity with one of the other frames.*



Additional to the detection box, an occluder box will also be generated using the Arkite 3D sensor. If the occlusion rate exceeds the minimum rate, the system will wait to evaluate the quality check until the occlusion is gone. This prevents a quality check from being classified as wrong while an operator is still working on it. In case the result is not satisfactory, the occluder box can be edited on the **Finetune** tab.

Teach a text (OCR) or a barcode (QR/DM)

To teach a text or barcode, execute the steps below:

1. Navigate to the **Detections** tab.

2. Select the text or barcode detection you want to teach to the system.

   Note

   *When using a **Vision** license, the Arkite software will automatically select the Arkite Vision sensor for teaching texts and barcodes.*

   *When using a **Validate** or **Guide** license, the Arkite software will automatically select the Arkite RGB sensor for teaching texts and barcodes. It is not possible to use the Arkite 3D sensor for these types of detections.*

3. Set the working zone around text or barcode by adjusting the white rectangle.
4. Click **Add Expected Result.**
5. In the **Properties** pane, specify a **Variable for Result** in which the text or barcode will be stored.
6. Optionally, check the **Validate Code/Text** checkbox, and link the conditions the text or barcode should comply with.

   Note

   *For Text (OCR) detections, it is recommended to check the **Ignore Whitespace** and **Ignore Casing** properties for the conditions you link. The text reading capability sometimes reads spaces where it should not, and vice versa.*

7. Click **Teach.**

Teach detections based on a recording

In order to save time or to work away from the workbench, you can make a recording of all detections and add them later based on the recording. For more information about recordings, refer to chapter **10 Recordings** on page 243.

5.6.3 Test a detection

After a detection has been taught, always test whether the teaching has been successful:

- Place the object or tool and take it away and check if the state of the detection gets updated.
- Pick material from a picking bin and check if the system detects it.
- Press a virtual button and check whether the system detects this.

If you are not satisfied with the result, you can either reteach the detection or finetune the detection.

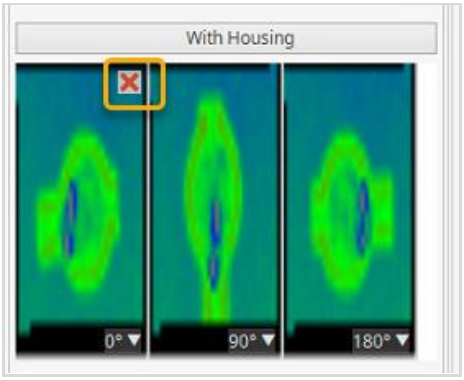
5.6.4 Reteach a detection

If you are not satisfied with a detection, you can decide to reteach it:

1. Under **Detections**, select the detection you want to finetune.
2. In the **Teaching** subtab, define the settings you want to use for reteaching, e.g.
   - Make new **Without** and **With** snapshots

     Note

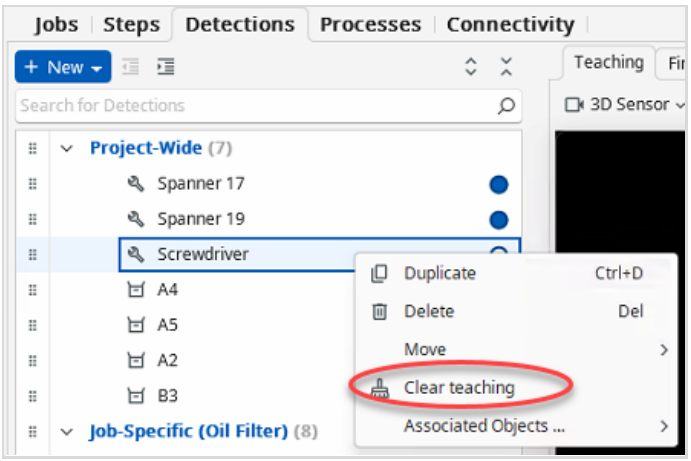
     You can also remove snapshots by clicking the cross in the right-upper corner of the snapshot.



*Change the teaching properties, such as the **Detection mode** or the **Position tolerance**.*

3. Check the **Adjust projection** checkbox if you want to visualizations on the **Projector** tab to be updated along with the new detection.
   - If you haven't changed anything to the visualization generated by the system, you can safely use this option
   - If you have edited the visualization generated by the system or added extra visualizations, it is recommended to uncheck this checkbox, since the system will otherwise undo your changes.
4. Check the **Adjust occlusion box** checkbox if you want to occlusion box to be updated along with the new detection.
   - If you haven't changed anything to the occlusion box generated by the system, you can safely use this option
   - If you have edited the automatically generated occlusion box on the **Fine Tune** tab, it is recommended to uncheck this checkbox, since the system will otherwise undo your changes.
5. Click **Teach** to apply your changes and reteach the detection.
   Click **Cancel** to revert your changes.

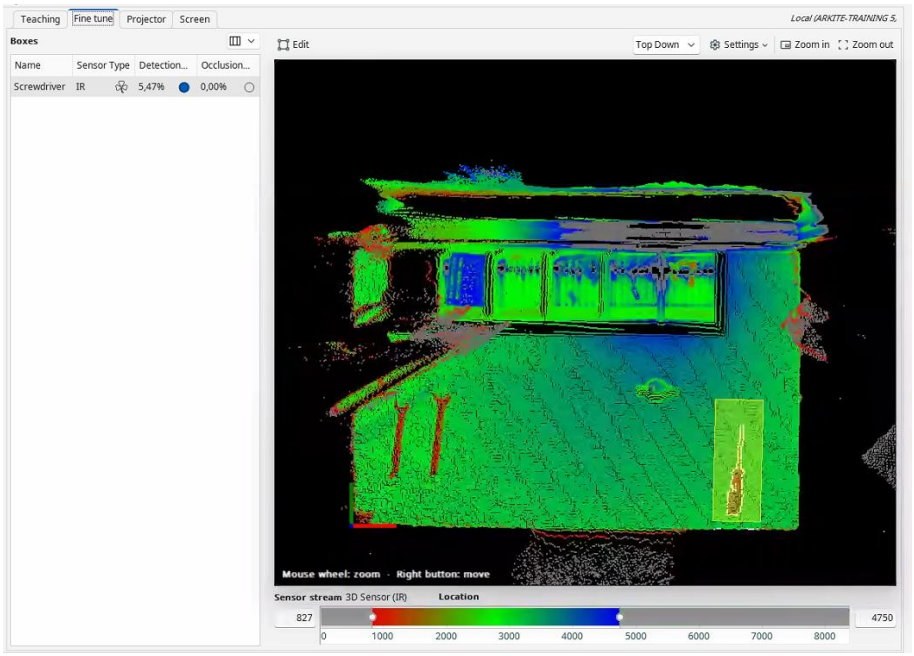
   Note
   *You can also remove all teaching data by right-clicking the detection and selecting **Clear Teaching.***



5.6.5 Finetune a detection

After a detection has been taught, you can tweak the boxes that have been generated in the **Finetune** subtab.

1. Under **Detections**, select the detection you want to finetune.
2. In the **Finetune** subtab, select the box you want to finetune.
3. Click the **Edit** button above the image.
4. Change the size and position of the box by dragging the angular points with your mouse. The same can be achieved by updating the coordinates in the **Properties** panel.



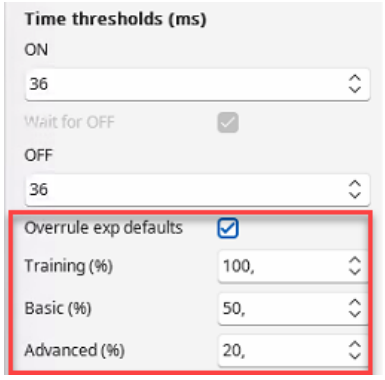
Tip

*Right-click an angular point and select **Remove Point** to delete it or **Insert Point** to add an additional angular point.*

5. Update the properties of the box in the **Properties** panel:

Property	Description
Z-Coordinates	Change the position of the box in relation to the workbench by updating the **Base** z-coordinate or the **Height** of the box.
Calculation properties for detections taught with detection mode Depth or IR	
Use IR	Check the checkbox to use the **Infrared** stream. Uncheck the checkbox to use the **Depth** stream.
Image processing	Select an image processing filter if needed: – **Background subtraction (depth stream)**: filters away all background, so only the moving foreground objects are visible. The **Reset Background** button sets the current situation to be the background. – **Edge detection (IR stream)**: the sensor will look for edges, in other words for the boundaries of an object where there is a high contrast with the adjacent pixels. It can be used to detect objects regardless their color or variations in infrared lighting conditions.
IR intensity	Define the minimum and maximum IR intensity in the box that is required to consider an object as present.
Detection rate	Define the minimum and maximum percentage of fill rate (or edge rate when Edge detection is used) of the box that is required to consider an object as present.

Property	Description
Calculation properties for detections taught with the Arkite Vision sensor or the Arkite RGB sensor	
Calculation Type	Algorithm that is used for calculating the detection: – **Shape based:** use the shape-based matching algorithm. This algorithm will detect the edges of an object, in order to recognize its shape. Color and material is mostly ignored. Use this option to create a quality check that works for the same object in different colors. – **Texture based:** use the correlation-based matching algorithm, based on a Normalized Cross-Correlation (NCC) model. This algorithm looks for patterns in grey values. As opposed to the shape based algorithm, it uses all grey values and not only clear edges. Different color and material will have an impact on the quality check. – **Histogram based:** use the histogram-intersection based matching algorithm. Use this option for checking: – Non-rigid shapes, for example a cable that can be bent in various ways or a dot of glue of which the shape and size may vary. – Objects with different shades/colors. ***Note*** *Not all colors can be distinguished – the infrared intensity value should differ sufficiently. For example, yellow and orange are typically hard to tell apart.* – **Barcode (QR/DM):** used for barcode detections – **OCR:** used for text detections
Taught code type	For barcode detections. The type of barcode that was taught. To read other code types, create a different detection and teach it with the required code.
Detection rate	Define the minimum and maximum percentage of fill rate of the box that is required to consider an object as present. For Quality checks, this is the minimum percentage of similarity that is needed to classify the detection as correct.
Position X (mm)	Define the difference in position in the 'X' direction that is acceptable. Only available when **Verify Position** has been selected when teaching the detection.
Position Y (mm)	Define the difference in position in the 'Y' direction that is acceptable. Only available when **Verify Position** has been selected when teaching the detection.
Orientation (°)	Define the difference in orientation that is acceptable. Only available when **Verify Orientation** has been selected when teaching the detection.
Symmetric Orientations	Define how many symmetric orientations should be accepted when validating the detection. The value for this property can be between 1 and 360, with default value 1. ***Example*** *If you are validating an object that is symmetrical along the X axis, the software will sometimes perceive its orientation as 0° and sometimes as 180° since both orientations look identical. In this case, set 2 as a value to accept both orientations.* *In case you are checking a 36-teeth gear, set 36 as a value to accept all possible orientation values. Be aware you should also decrease the default value of 5° for the **Orientation Tolerance** property in this case, otherwise you will end up accepting any orientation.* Only available when **Verify Orientation** has been selected when teaching the detection.

Property	Description
Occlusion	Check the **Occlusion logic** checkbox if you want the system to verify if the object is hidden. The **Min Rate** dropdown list contains the minimal percentage of occlusion that is required to consider the object as hidden.
Time thresholds (ms)	Define for how long an object needs to be within (ON) or outside (OFF) the range defined in the fill rate properties to change the detection state. For activities, you can disable the OFF setting. To do so, uncheck the **Wait for OFF** checkbox. In this case, an activity does not need to be deactivated to proceed to the next step when used in a process. In case the **Experience Level** functionality is enabled, a percentage of these time thresholds is used for every specific experience level. ***Note*** *For more information on the **Experience Level** functionality, refer to chapter **8 Experience levels** on page 233.* To overrule the default percentages for this specific detection, check the **Overrule exp defaults** checkbox. After checking this checkbox, the system will fall back to 100% for level 1, 50% for level 2 and 20% for level 3. You can update these percentages as required.  To change the default percentages for all detections, refer to section **8.3 Experience level settings** on page 234.
Coordinates	Change the position of the box by updating the X and Y coordinates of the angular points. ***Note*** *You can also change the size and position of the box by dragging the angular points with your mouse.*

5.6.6 Managing visualizations for the detection

What?

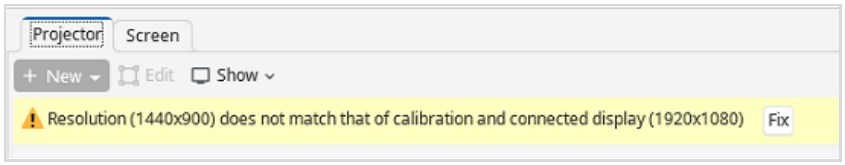
After a detection has been added, you can add visualizations to it on any of the available display subtabs. You will see a subtab for every display that is available:

- By default one **Projector** and one **Screen** subtab will be available.

  Note

  - *To add more displays, refer to section **7.2.3 Adding a display** on page 195.*

- *When you have created the project before calibrating, a Projector display is added with the screen resolution of the device you are working on (e.g. your laptop). This may not match with the actual projector resolution, used during calibration. This will cause projections to be faulty. A message on the Projector subtab will warn you for this. Click Fix and update the Resolution property to the correct value in the projector's Properties pane.*



The system will automatically generate a draw shape visualization on the **Projector** subtab:

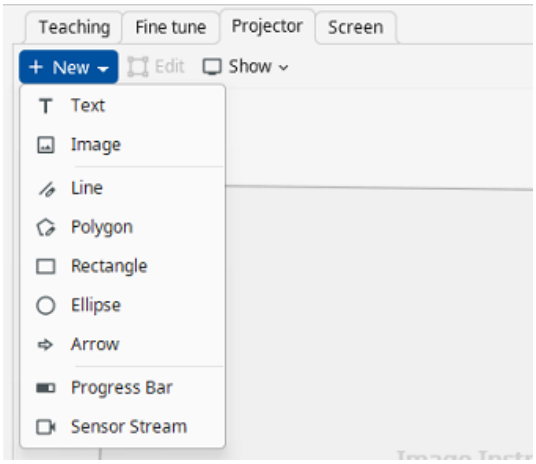
- When using a **Validate** or **Vision** license (= with Arkite 3D sensor), this is done when teaching the detection.
- When using a **Guide** license (= without sensor), this is done when adding the detection.

You can edit and update that visualization.

Adding visualizations

To add a new visualization, execute the steps below:

1. Under **Detections**, select the detection for which you want to add a visualization.
2. Click the display subtab on which you want to add a visualization, **Projector** or **Screen**.
3. Click the **New** button and select the visualization of your choice.



Tip

*You can also select an existing visualization and click **Duplicate**.*

4. Define the size and position of the visualization using your mouse.

Tip

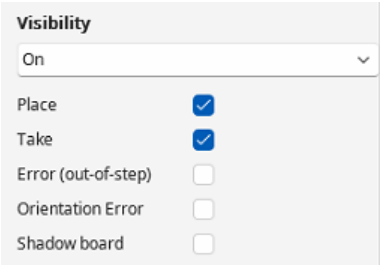
While dragging the visualization with your mouse, visual guidelines will appear that will allow you to easily align the visualization. The guidelines below will appear:

- *Guidelines towards the **middle of the screen** will appear when you reach the middle of the screen with an edge or with the middle of the visualization, both horizontally and vertically.*

- ***Object to object alignment lines*** *appear to help you to align two visualizations with each other, both horizontally and vertically. This way, you can align the top, the bottom or the middle of the visualizations with each other.*
- *Arrows appear to help you to position two visualizations at the* ***same distance from the edge or the center*** *of the screen.*
- *When you have more than two visualizations, arrows appear when the distance between visualizations 1 and 2 is the same as between visualizations 2 and 3. This helps you to ensure there is the same distance between visualizations.*
- *When you are changing the size of an object, lines appear inside the objects when they have the same length or width. This helps you to make the objects have the* ***same size****.*

5. In the **Properties** pane under **Visibility**, indicate when the visualization should appear:

Property	Description
Visibility	Select one of the options below from the **Visibility** dropdown list: – **On**: the visualization will always be visible – **Blink**: the visualization will blink. Use the **Blink Interval** property to define how fast the visualization will blink. – **Based on Variable**: the visibility of the visualization will be based on the current state of the variable provided in the **Variable for visibility** property.
Variable for visibility	Only available if the **Based on Variable** option is selected in the **Visibility** dropdown list. The state of this variable will determine whether the visualization will be visible: – If the state of the variable is ON, TRUE or 1, the visualization will be visible. – If the state of the variable is OFF, FALSE or 0, the visualization will be hidden. – If the state of the variable is BLINK, the visualization will blink. Use the **Blink Interval** property to define how fast the visualization will blink. – If the variable has another state than the ones mentioned above, the visualization will be visible.
Blink Interval (ms)	Only available if the **Blink** option is selected in the **Visibility** dropdown list. Interval in ms with which the visualization will blink.

Property	Description
Detection state checkboxes	Depending on the detection type, different checkboxes are available for the different states the checkbox can have: – Place (Objects, Tools, Picking bins, Objects with Quality Check in case of a **Vision** license) – Take (Objects, Tools, Picking bins, Objects with Quality Check in case of a **Vision** license) – Pick (Picking bins) – Press (Virtual Buttons) – Perform (Activities, Quality Checks in case of a **Vision** license) – Shadowboard (Tools, Objects, Objects with Quality Check in case of a **Vision** license) A shadowboard visualization is projected at the location of the tool or object while it is being used. This way, the operator always knows where to put the tool back after use. When checking this option, the checkboxes for all other options are cleared automatically. – Error (out-of-step) (Objects, Tools, Picking bins, Activities) The visualization will be shown when the detection is an error state, regardless of the process step. – Error (in-step) (Quality Checks, Objects with Quality Check, **Vision** license only) The visualization will be shown when the detection is an error state and in a process step in which the detection is active. – Picking Error (Picking bins) – Orientation Error (Tools and Objects with Vision license only, only available when **Verify Orientation** has been selected when teaching the detection) – Position Error (Tools and Objects with Vision license only, only available when **Verify Position** has been selected when teaching the detection)  Check the checkbox(es) for the states in which you want the visualization to appear Depending on the detection type, different checkboxes are checked by default. For example: – For an object, the **Place** and **Take** checkboxes are checked by default. However, particularly for larger objects, it can be valuable to make different **Place** and **Take** visualizations that match the empty space on the workstation and the top surface of the object respectively. – For a picking bin, the **Place**, **Take, Pick, Detection Error** and **Picking Error** checkboxes are checked by default. However, it can be valuable to make a visualization for **Place**, **Take** and **Detection Error**, matching the entire picking bin size, and a different smaller visualization for **Pick** and **Picking error**, matching the picking bin picking entrance.
Rotate for verify position (Vision license only)	Check this option to rotate the visualization depending on the actual position of the linked detection compared to the expected position. You can use this for example to display an arrow that indicates in which direction the detection should be moved in order to be on the expected position. Only available when **Verify Position** has been selected when teaching the detection.

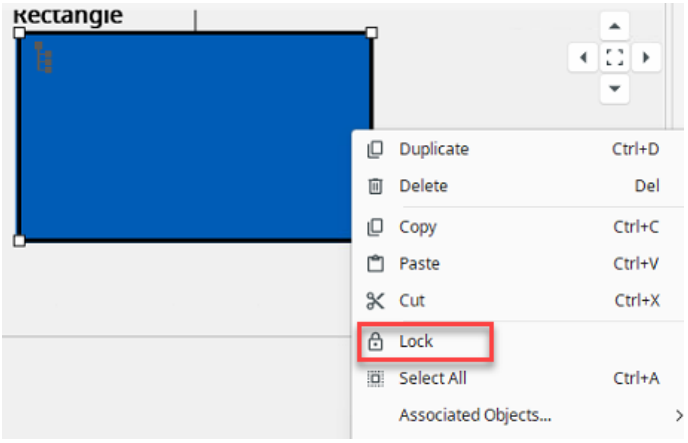
Note

*By default, the brush color of a visualization that is linked to a detection and that is shown on a **Projector** display changes depending on the state of the detection, based on the **Detection colors** template. To manage the behavior and the colors that are used, refer to **15.2 The Detection Colors template** on page 276.*

6. In the **Properties** pane, update the other properties of the visualization if needed.
   The properties that are available depend on the type of visualization you have added. For an overview of visualization types and their properties, refer to sections **4.4.1 Visualization types** on page 83 and **4.4.2 Visualization properties** on page 84.

   Note

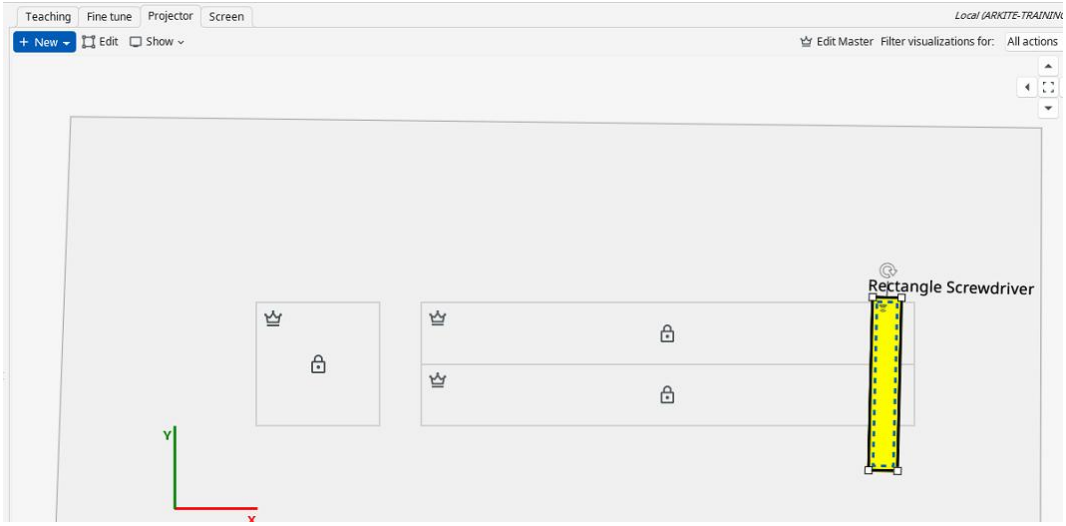
   *To protect a visualization from being moved or edited by accident, right-click it and select **Lock**.*



Editing visualizations

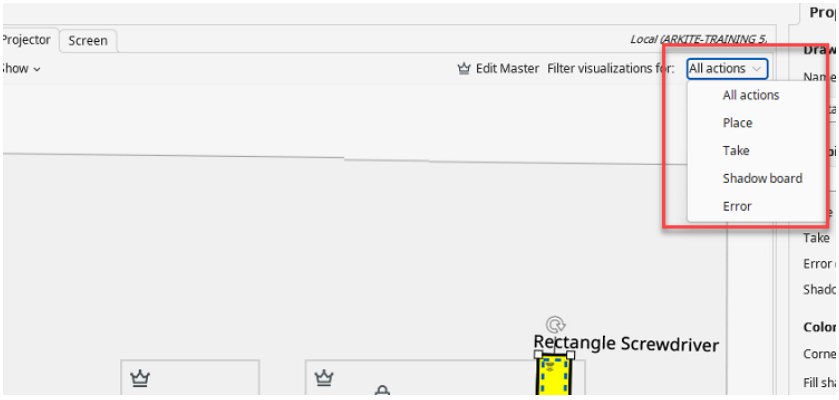
To edit an existing visualization, execute the steps below:

1. Under **Detections**, select the detection for which you want to edit a visualization.
2. Click the display subtab on which you want to edit a visualization, **Projector** or **Screen**.
3. Select the visualization you want to edit.

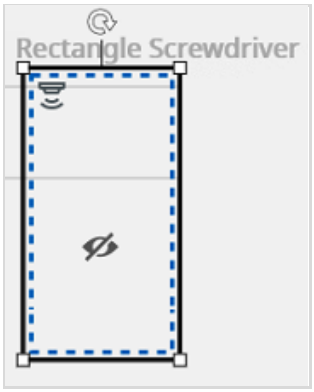


Note

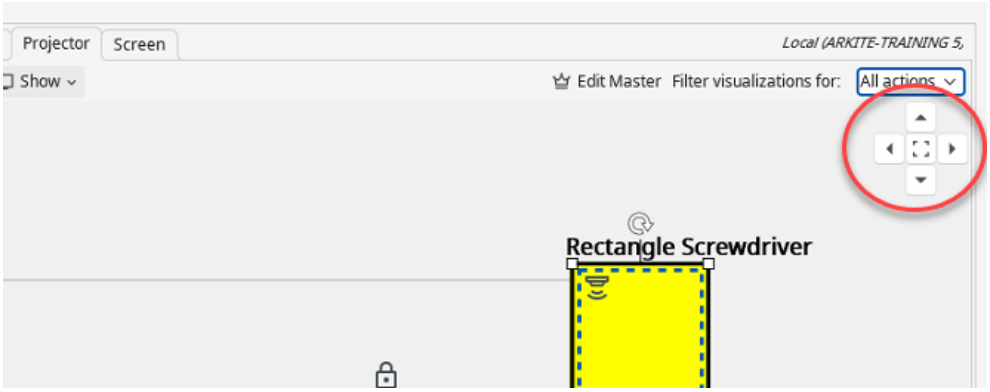
- *You can filter the visualizations for the different state a detection can have:*



A visualization that is selected, but not visible because of the selected filter will be rendered like this:



- *You can consult multiple visualizations by holding the **Ctrl** button and clicking the visualizations of your choice with your mouse, or by holding the left mouse button and dragging a rectangle over the visualizations of your choice. Press **Ctrl-A** to select all visualizations.*
- *You can zoom in and out using your mouse wheel. When zoomed in, arrows appear to navigate around in your display. Click the button in the center of the arrows to reset the zoom to 100%.*



4. For draw shapes:
   - Change the size and position of the visualization by dragging the shape or the angular points with your mouse.

- To change the shape of the visualization, click the **Edit** button above the image and move the angular points.

5. For all shapes, update the properties of the visualization in the **Properties** pane if needed.

   Note

   *The properties that are available depend on the type of visualization you have added. For an overview of visualization types and their properties, refer to sections **4.4.1 Visualization types** on page 83 and **4.4.2 Visualization properties** on page 84.*

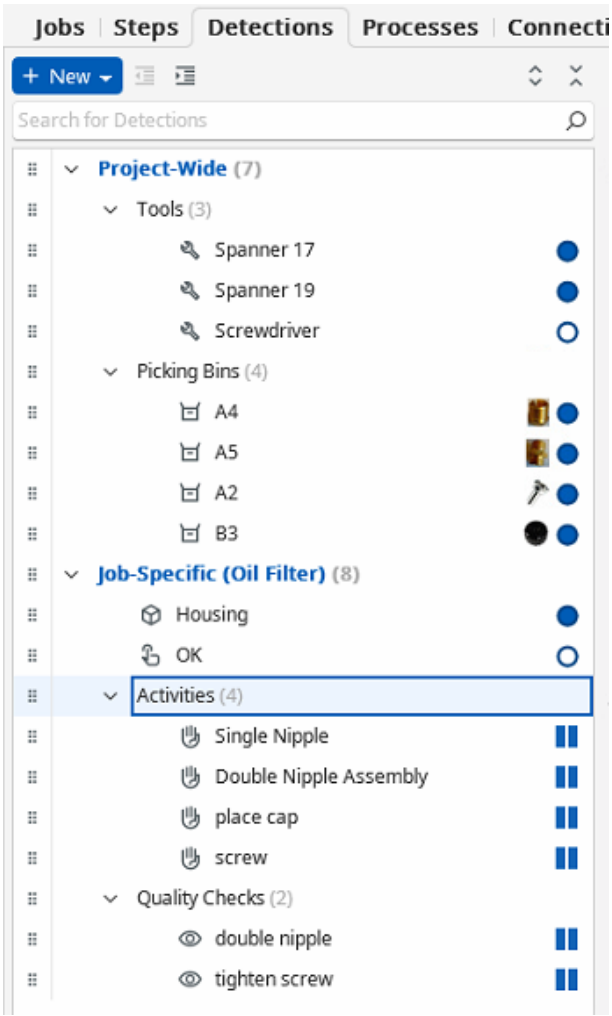
To delete an existing visualization, select it and click **Delete**.

By default, the visibility and the color of a visualization that is linked to a detection and that is shown on a **Projector** display changes depending on the status of the detection, based on the **Detection colors** template. To manage the behavior and the colors that are used, refer to section **15.2 The Detection Colors template** on page 276.

5.7 Managing detections

What?

After having added the required detections, you can manage and organize them to have a better overview.



How to manage detections

On the **Detections** tab, you can:

- Right-click a detection and select **Duplicate** to duplicate it.
- Right-click a detection and select **Delete** to delete it.
- Select a detection and drag it with your mouse to change its order.
- Organize your detections:
  - Click **New** and select **Group**.
    A group is created. Any new detection you add while this group is selected will be added under the group. You can also drag existing detections into the group.
  - Select one or more detections and click the button. This way, a group can also be created.

  Tip

  Grouping detections allows you to:

  - *Structure your list of detections, providing you with a better overview*
  - *Aligning visualizations for the detections in the group: if you select the group, you will see the visualizations for all detection in that group on the Projector or Screen tab and you will be able to align them easily.*
  - *Enabling and disabling all detections within the group using one variable.*
- Click the button to expand all groups of detections.
- Click the button to collapse all groups of detections.

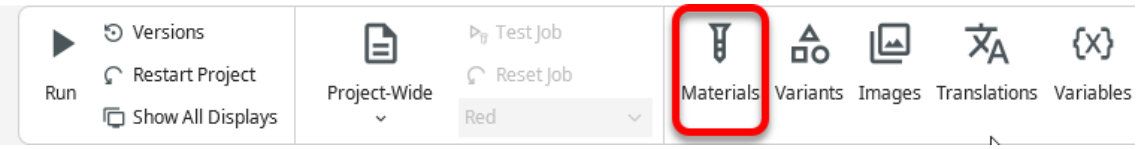
5.8 Managing materials

What?

When creating a picking bin, you can either leave it empty, associate it with a new material or select an existing material. These materials can still be managed afterwards.

How to manage materials

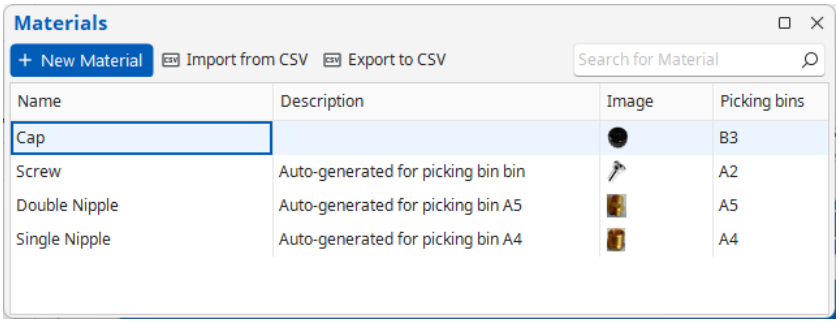
In the ribbon, click the **Materials** button.



The **Materials** window appears. For every existing material, you can update the existing information by clicking in the column of your choice:

- **Name:** change the material's name
- **Description:** change the material's description
- **Image**: add an image of the material. This image will be used on the **Detections** tab and in the **Image Instructions** for the job.

- **Picking bins:** associate the material to one or more picking bins. You can only select picking bins that haven't been associated with a material yet.



Click the **New Material** button to add a new material to the list.

To delete materials from the list, select the row(s) for the materials you want to delete, right-click your selection and select **Delete**. Use Shift-Click or Ctrl-Click to select multiple rows.

Importing and exporting materials

To export your list of materials to a CSV file, click the **Export to CSV** button. The **Name** and **Description** column will be exported. The generated CSV file uses a **comma (,)** as the separator.

You can also add materials to your list by importing a CSV file. To do so:

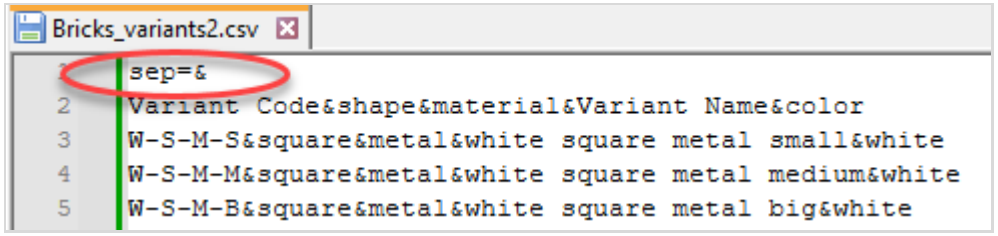
1. Click the **Import from CSV** button.
2. Select the CSV file of your choice.

   Note

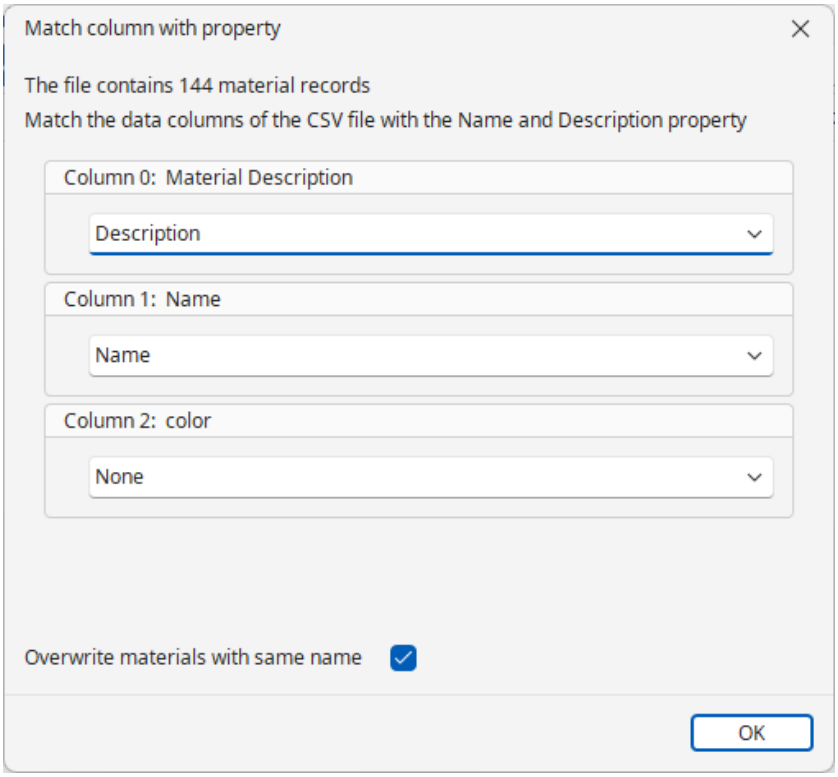
   The CSV import automatically detects the following separators:

   - *Comma (,)*
   - *Semicolon (;)*
   - *Colon (:)*
   - *Pipe (|)*
   - *Tab*

   If your CSV file uses a different separator, open the file in a text editor first and add ***sep=[separator]*** *as the first line of the file. Arkite will then import the file correctly. In the example below, & is defined a separator.*



3. In the **Match column with property** window, you can indicate which column from the CSV file you want to use for the material **Name** and **Description**.



Note

- *If the top row of the CSV file contains a value **Name** and/or **Description**, the Arkite software will suggest to match these columns with the corresponding property.*
- *If you check the **Overwrite materials with same name** checkbox, existing materials will not be duplicated. The information in the **Description** column will be overwritten with the information from the CSV file.*
  *If you don't check the **Overwrite materials with same name** checkbox, materials with a name that already exists will be added. A **(2)** is added behind the name of the newly added material.*

4. Click **OK**.

The materials from the CSV file will be imported into your project.

5.9 Working with smart tools

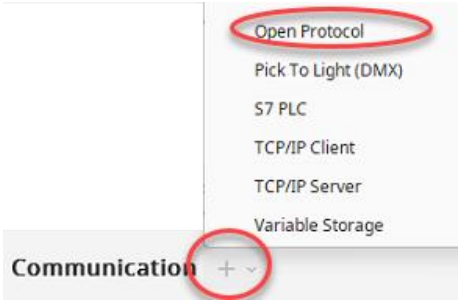
What

Smart tools are tools that can communicate with external systems, such as the Arkite software. Through the Arkite software, they can be configured in such a way that they can only be used during the correct step in the assembly process. When using a **Validate** or **Vision** license (= with Arkite 3D sensor), the system will also ensure the smart tool can only be used at the correct location.

How to configure smart tools using Open Protocol communication

To configure a smart tool using Open Protocol communication, execute the following steps:

1. Make sure your smart tool is on the same network as your workstation.
2. In the **Connectivity** tab under **Communication**, click the **Add** button and select **Open Protocol**.



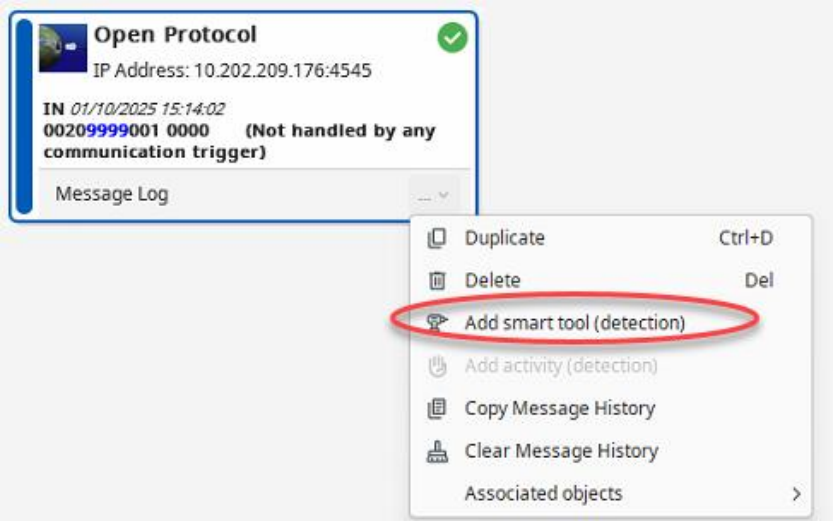
A new card for the **Open Protocol** communication module is added to the list of communication modules.

3. In the **Properties** pane, configure the properties for the communication module.

Property	Description
Name	Name of the communication module ***Note*** *Next to the **Name** label, the object ID for the communication module is shown. You can copy it to use it when working with the Arkite API.*
Log Table	The table name in the logging database in which all incoming and outgoing messages and exceptions will be logged.
Automatic Smart Tool Settings	If this checkbox is enabled, the required variables for communication with a smart tool will be generated and configured.
IP Address	The IP address of the remove device, for example: 10.0.11.12.
Device Type	– When using Atlas Copco's MicroTorque Controller MTF6000, select **MTFocus 6000**. – For all other devices, select **Generic**.
Port	The TCP/IP port to use. Default for open protocol communication is 4545.
Start communication revision	When an open protocol communication actuator is first sparked, a 'start communication' message will be sent to the controller. This message contains a 3 digit revision. For most controllers, this revision is '001'. Check the manual of your controller to check for divergent revisions.
End communication revision	When the module is paused, an 'end communication' message will be sent to the controller. This message contains a 3 digit revision. For most controllers, this revision is '001'. Check the manual of your controller to check for divergent revisions.
Stop String Type	Every message will be ended by a special 'ending character'. The default value is null (null character). Other types include a CRLF (carriage return line feed) and #.
Log 'Keep Alive' messages	Every incoming, outgoing and exception message will be logged. However, to keep the connection with the controller open, 'keep alive' messages will be sent (and received) every ten seconds. In most situations, you don't want to see these messages cluttering your message stream. However, if you would like to see these messages, please check this checkbox.

Property	Description
Open Protocol Device Test	Before an open protocol device can be used, a parameter set must always be selected. Provide a parameter set in the text input field (typically consisting of three digits, e.g. 000, 001, 002....) and click **Set Parameter**. Once you have done that: – Click the **Enable Tool** button to enable the tool. – Click the **Disable Tool** button to disable the tool. By clicking these buttons, the communication towards the tool can be tested.

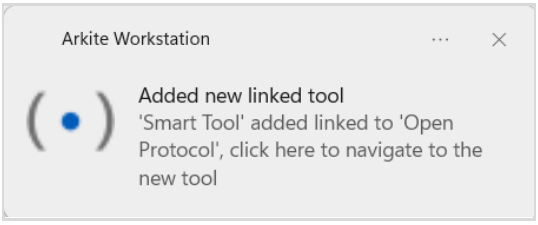
4. Select the card for the **Open Protocol** module, click the **...** button in the lower-right corner of the card and select **Add Smart Tool (detection)**.



Note

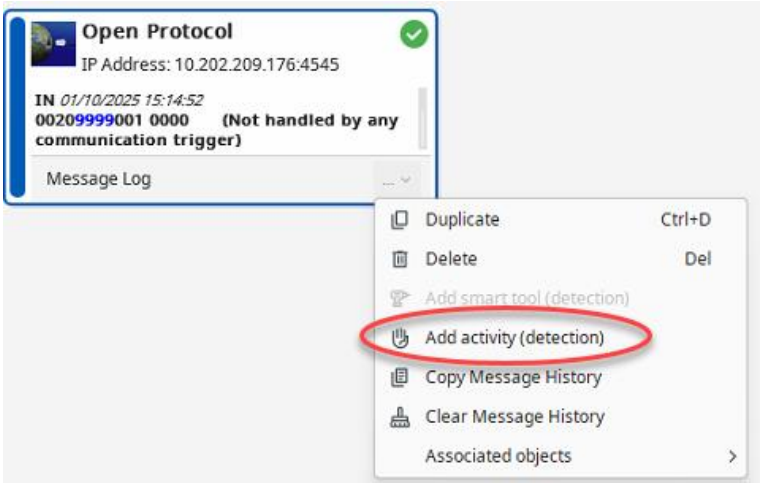
*Only **one** tool can be linked to a single **Open Protocol** module. To use multiple tools, create multiple **Open Protocol** modules.*

5. A smart tool linked to the **Open Protocol** module is created on the **Detections** tab. Click the toast message in the lower-right corner of the screen to navigate to it.

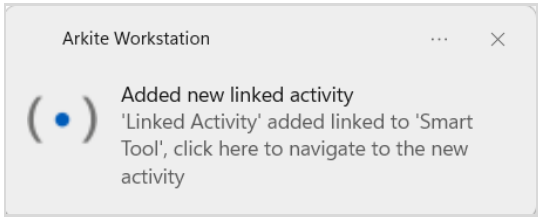


6. (optional) When using a **Validate** or **Vision** license, teach the tool as described in section **Teach an object, picking bin, tool or virtual button** on page 135. This way the system can detect when the tool is taken away or placed back.

7. On the **Connectivity** tab, select the card for the **Open Protocol** module, click the **...** button in the lower-right corner of the card and select **Add Activity (detection)**.



8. An activity linked to the smart tool you set up in step 5 is created on the **Detections** tab. Click the toast message in the lower-right corner of the screen to navigate to it.

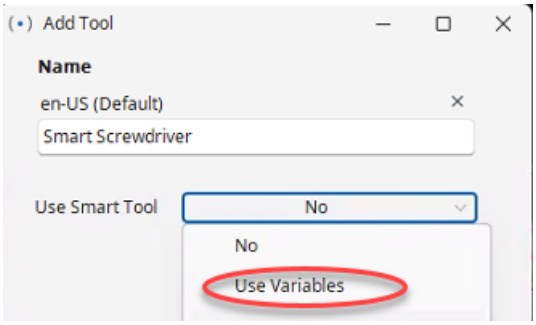


9. (optional) When using a **Validate** or **Vision** license, teach the activity as described in section **Teach an activity** on page 142.
   This way the system can detect when the tool is being used at the desired location.
10. Add more activities as needed.

How to configure smart tools using variables

To configure a smart tool using variables, execute the following steps:

1. Navigate to the **Detections** tab.
2. Click the **New** button and select **Tool.**
3. Select **Use Variables**.



4. (optional) When using a **Validate** or **Vision** license, teach the tool as described in section **Teach an object, picking bin, tool or virtual button** on page 135.

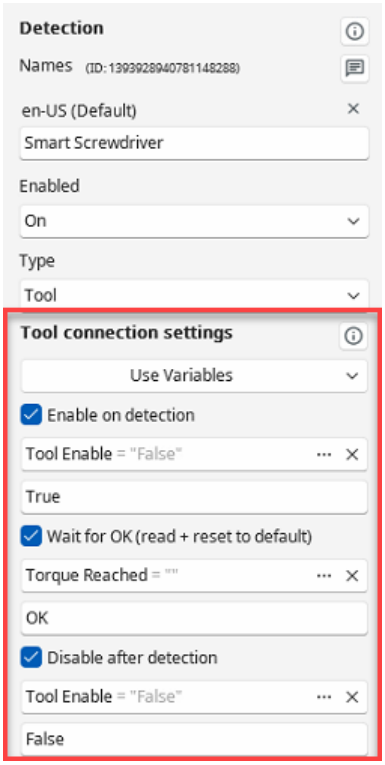
This way the system can detect when the tool is taken away or placed back.

5. Select the tool in the **Detections** tab and configure the tool connection settings in the **Properties** pane:

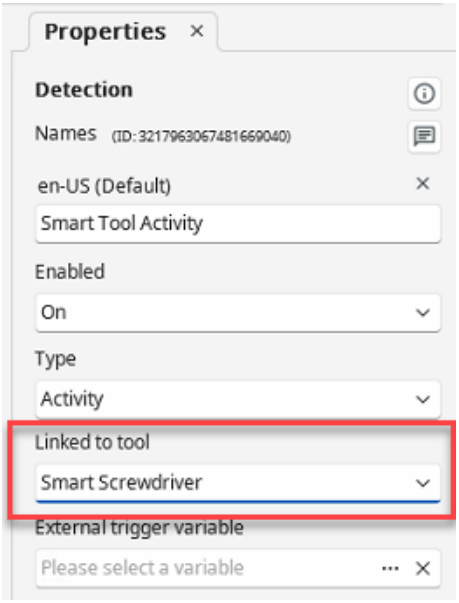
Note

*If you are using an **Open Protocol** communication module, the required variables and settings will be preconfigured by the Open Protocol module.*

Property	Explanation
Enable on detection/step	The selected variable will be set to the specified value when: – the associated activity detection becomes active (**Validate** or **Vision** license) – the associated process step is reached (**Guide** license) ***Example*** *Set the variable **tool enable** to **true** when the activity on time or process step is reached.*
Wait for OK (Validate license only)	When this setting is configured, the activity will only be regarded as completed when the selected variable reaches the selected state. Afterwards, the variable will be set to its default value, making sure that one positive trigger of the variable can only confirm one activity. ***Example*** *Wait until the variable **torque reached** is **true** to complete the activity.*
Disable after detection/step	The selected variable will be set to the specified value when: – the associated activity detection becomes inactive (**Validate** or **Vision** license) – the associated process step is completed (**Guide** license) ***Example*** *Set the variable **tool enable** to **false** when the activity off time is reached or when the activity process step is completed.*



6. Link the tool to an activity:
   - Add the activity as described in section **5.5 Adding detections** on page 129.
   - (optional) When using a **Validate** or **Vision** license, teach the activity as described in section **Teach an activity** on page 142.
     This way the system can detect when the tool is being used at the desired location.
   - Select the activity in the **Detections** tab.
   - In the **Properties** pane, link the activity to the tool:



5.10 Working with reflective tape (Validate and Vision license only)

What

In some cases, objects are difficult to detect due to a low contrast or changing light conditions in the working area. This can be solved by replacing the object with a piece of reflective tape with a high and constant IR value.

How to use reflective tape – option 1

1. Replace the object on your working bench with a piece of reflective tape of the same size.
2. Teach the system the location of this reflective tape, using the **Sensor Mode** Reflective tape.
   For more information, refer to section **5.6.2 Teaching detections** on page 135.
3. On the **Finetune** tab, change the **Fill Rate** fields:
   When working with reflective tape, the object is present when the reflective tape cannot be detected. For example, if a 70-100% range is used to detect if the reflective tape is present, you should change the range to 0-70% in order to let the system detect correctly the reflective tape is covered by the present object.
   For more information, refer to section **5.6.5 Finetune a detection** on page 149.

How to use reflective tape – option 2

1. Replace the object on your working bench with a piece of reflective tape of the same size.
2. Teach the system the location of this reflective tape, using the **Sensor Mode** Reflective tape.
   Use the snapshot with the tape in the **Without** image.
   Use the snapshot without the tape in the **With** image.
   For more information, refer to section **5.6.2 Teaching detections** on page 135.

5.11 Working with stacked picking bins (Validate and Vision license only)

What?

In many cases, several picking bins will be stacked upon each other on the workbench. Except for the top row, the picking bins will therefore be hidden from the sensor's view and the system won't be able to see them. In this chapter, a workaround for this issue is explained.

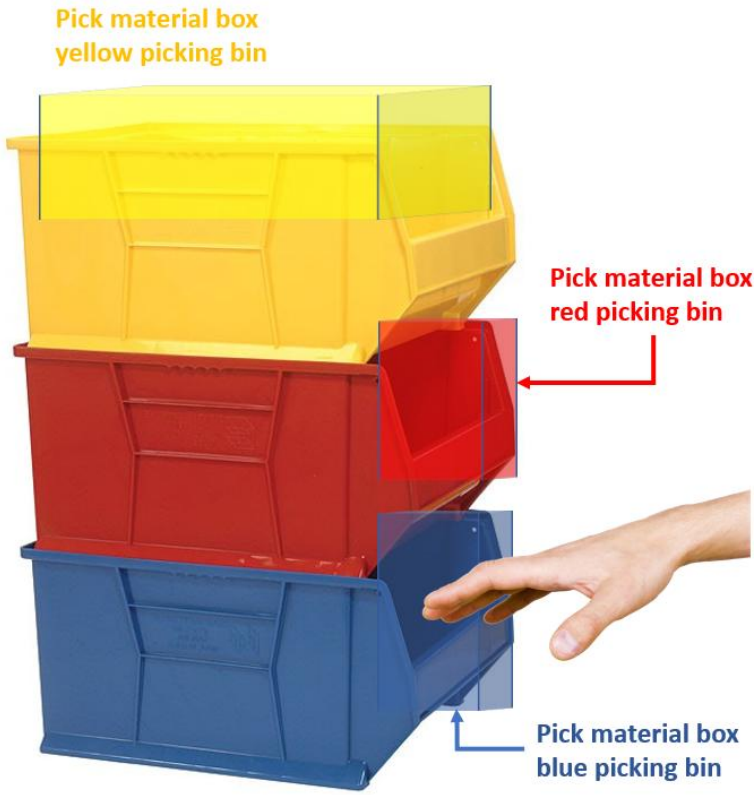
How to work with stacked picking bins

When a picking bin is taught, the system automatically creates two boxes:

- A box to detect if the picking bin is present
- A box to detect right above this box to detect the operator's hand when picking material from the picking bin.

When working with stacked picking bins, the boxes for all picking bins will by default be all on top of each other, and the system will only be able to detect the upper row of picking bins. This is not really a problem for detecting picking bin presence, since they will only be removed very rarely.

However, to successfully detect picking materials, you should move the corresponding boxes **in front of** the actual picking bins and make them a lot **smaller**. You should set them in such a way that they form a **small cuboid gate** at the entrance of the picking bin. As a result, the boxes won't be occluded to the sensor and the system can successfully detect the operator's hand going into the correct picking bin and picking a material. For more information on how to edit the pick material boxes, refer to section **5.6.5 Finetune a detection** on page 149.



5.12 Working with mobile picking racks (Validate and Vision license only)

What?

In complex assembly processes, many materials are required to complete an assembly. In such a case, picking bins are often organized in different mobile picking racks that are exchanged during the assembly so that the relevant materials are available at all times.

It is important the Arkite Operator Guidance system only calculates the presence of the picking bins in the active picking rack, to avoid error visualizations for picking bins that are not there.

How to work with mobile picking racks

1. Put a mobile picking rack on its in-use location and teach all the picking bins in the picking rack.
2. Repeat this for all mobile picking racks.
3. Add a detection group for each picking rack, and move all picking bins into the applicable group.
4. Create a variable for each picking rack.
5. Set the **Enabled** property for the detection group to **Based on variable**.
6. In the **Based on variable** property, select the variable you created for the picking rack the picking bin belongs to.
7. In your assembly process, each time you exchange picking racks, set the value of the variable for the active picking rack to ON and the value of the variable for all other picking racks to OFF.

5.13 CAD/CAM detections (Validate and Vision license only)

5.13.1 About CAD/CAM detections

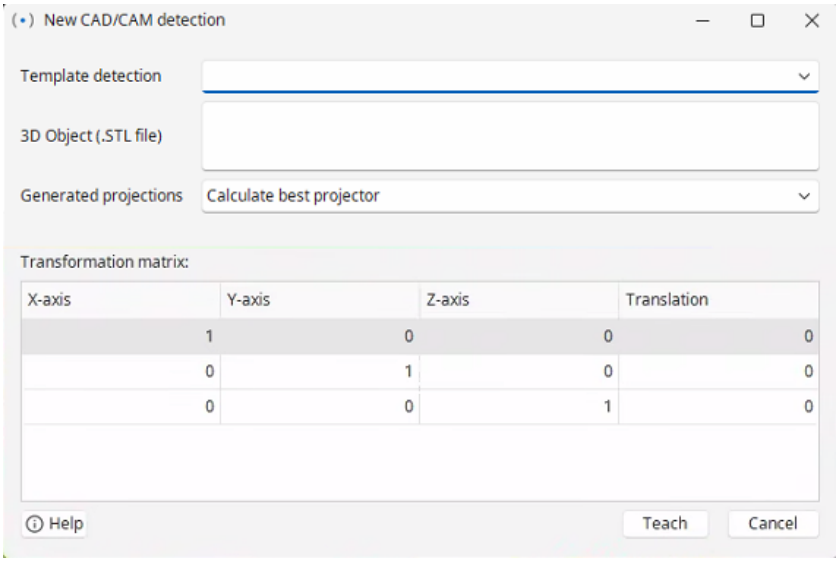
CAD/CAM detections allow to create a detection where the size and location in 3D space is based on a CAD model.

CAD/CAM detections are just like other detections (objects, tools …), but their location in 3D space is provided by both a 3D model (.STL file) and a transformation matrix that positions the CAD model in space. Such a transformation is typically provided by the CAD/CAM system. All other properties (such as which sensor to use) are taken from the template.

5.13.2 Adding CAD/CAM detections

To add a CAD/CAM detection, execute the steps below:

1. Navigate to the **Detections** tab.
2. Click the **New** button and select **Object.**
3. Teach the object as described in section **Teach an object, picking bin, tool or virtual button** on page 135.
4. Click the **New** button and select **CAD/CAM Detection**.
   The **New CAD/CAM Detection** window appears.



5. Provide the input below:

Field	Description
Template detection	Select the object you have taught in step 3. All properties of the CAD/CAM detection will be identical to the one selected as template, except its location in 3D space. Such properties include which sensor to use, fill rate thresholds, whether it is depth or IR etc.
3D object	The 3D object, which is provided as an STL file (Standard Triangle Language). STL is a simple and commonly used file format, to which any CAD program can export. To select an STL file, click the 3 dots. In the **Please select an image** window, select **From File** in the **New** menu.

Field	Description
Transformation matrix	Specifies how the 3D object is positioned in 3D space, relative to the coordinate system that was defined during calibration. Such a matrix is usually provided by the CAD/CAM system, and allows to translate, rotate and scale the object as needed. The first three columns specify the XYZ-coordinates of where the X-axis, Y-axis and Z-axis of the 3D object should end up respectively. The last column specifies the translation, i.e. where the origin of the 3D object is moved to. The default values specify an identity matrix, which means the 3D object is used as-is, without transforming it.
Generate projections	The locations of the projections are calculated based on the provided 3D object and the accompanying transformation matrix. If multiple projectors are used, the most suitable projector is by default chosen for the projections (**Calculate best projector** option). Say you have two projectors that cover the left and right areas. A detection on the left area will use the left projector, and a detection on the right will automatically pick the right projector. If the **Preserve projector(s) from template** option is selected, the locations of the projections are still calculated, but they will use the same projector as the template. This option is more suitable if you want fine-grained control on which projector will be used, or if you want to use all projectors in which case the template should also define projections for all projectors.

6. Click **Teach**.

Result

Your detection is updated with the information from the STL file.

5.13.3 Example 1: no transformation

Template detection

Create a detection of type **Object**, and use all default settings.

Teach the object as described in section **Teach an object, picking bin, tool or virtual button** on page 135.

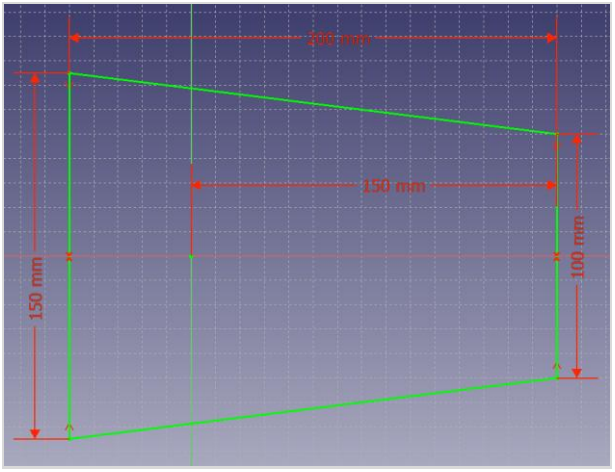
3D object

Use the below CAD model as input it can be found here:

C:\Program Files\Arkite\Arkite Workstation\Help\CAD.

Select the **block.stl** file in the **3D object** field.

The origin is in the middle of the shape, the red X-axis points right and the green Y-axis points up.



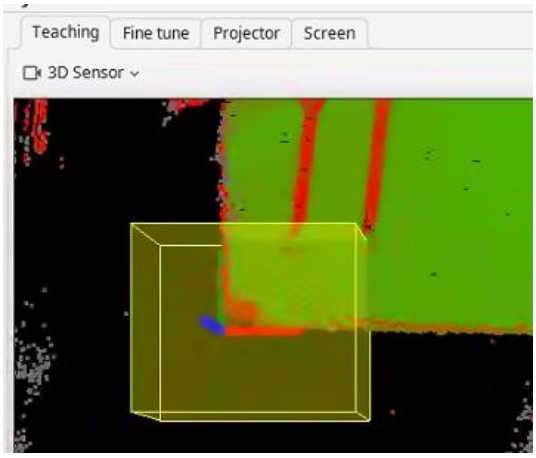
Transformation

Just use the default identify transformation.

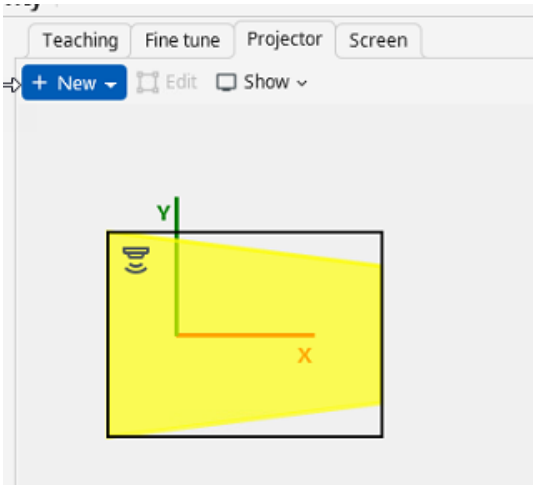
Result

This is what you see in the **Detections** tab. Notice how the axis origin specified during calibration (XYZ = RGB) aligns with the origin of the 3D object.

This is what you see in **Teaching** subtab.



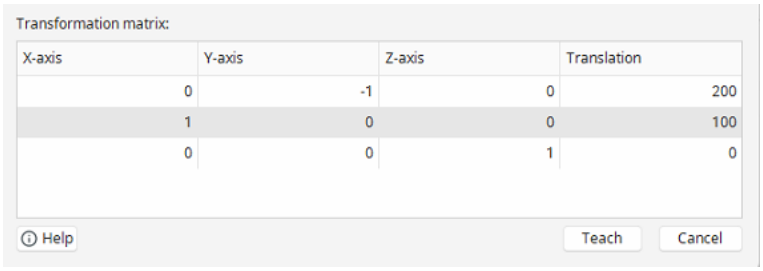
This is what you see in **Projector** subtab.



5.13.4 Example 2: rotate and translate

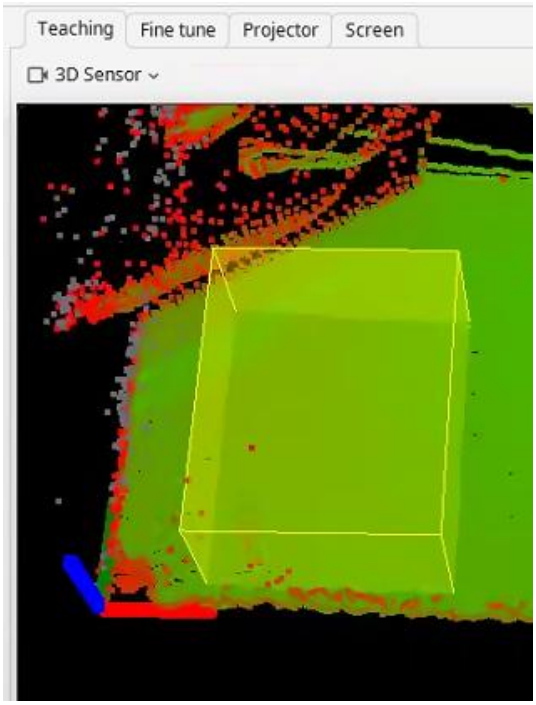
Inputs

The same inputs are used as for the previous example, but this time the 3D object is rotated 90 degrees counterclockwise in the XY plane, and translated with (200, 100, 0) for XYZ. So the transformation matrix becomes:

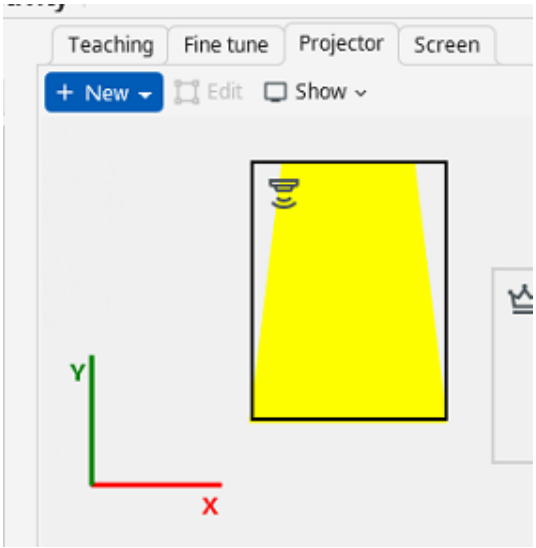


Result

This is what you see in **Teaching** subtab. Notice how the object is rotated, and it is translated 200mm to the right and 100mm upwards.



This is what you see in the **Projector** subtab:



6. Processes

6.1 About processes

What?

Next to your job assembly, you can configure different processes in the Arkite software. Processes run in parallel with job execution.

When you create a project, by default one process called **Operator Job Selection** will be available that is started when the project is loaded. This process takes care of job selection and execution.

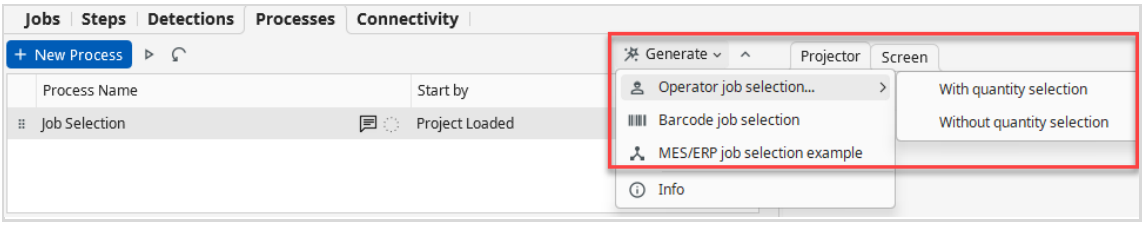
Additionally, you may need more processes to, for example::

- Log data
- Send data to a communication module
- Initialize parameters when a project is started
- Configure the behavior of a touch screen
- ...

6.2 Generating Job Selection processes

What?

With the **Generate** menu, you can add a job selection process yourself.

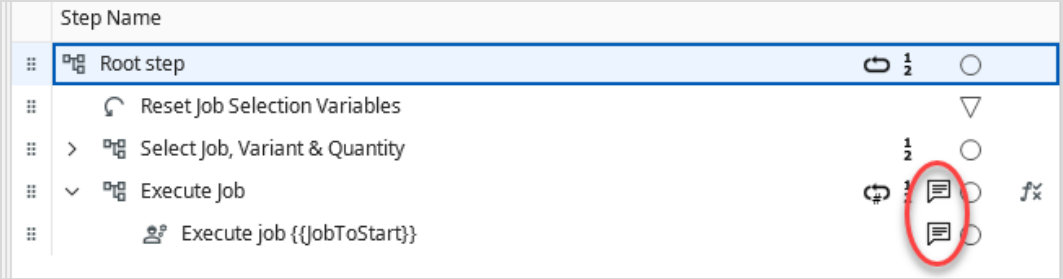


You can use the **Generate** menu to:

- Add the **Operator Job Selection** process to an existing project, made in a software version before 2025.1
- Bring back the **Operator Job Selection** process if you have accidentally removed it
- Replace the default **Operator Job Selection** process with another job selection option:
  - Operator job selection without quantity
  - Barcode job selection
  - MES/ERP job selection example

  Note

  - *Only one of these processes should be active at a time. Therefore, if you add another option, the **Project Loader** trigger of the existing process will be removed.*
  - *For a better understanding of the different steps in the job selection processes, read the comments that are added in the software.*



Operator job selection with quantity selection

The **Operator job selection with quantity selection** process consists of three major steps in which an operator:

- Selects a job from a gallery on the touch screen
- Selects a variant from a gallery on the touch screen
- Enters a quantity on the touch screen

After the selection, the selected job variant is executed the specified number of times. After the last iteration, the job selection is automatically started again.

When you add new jobs or new job variants, these are automatically added to the galleries on the selection screens.

Warning

If you haven't configured any job variants, the job will be executed skipping all optional steps (if there are any).

Operator job selection without quantity selection

The **Operator job selection without quantity selection** process consists of two major steps in which an operator:

- Selects a job from a gallery on the touch screen
- Selects a variant from a gallery on the touch screen

After the selection, the selected job variant is executed in a loop until the operator stops the execution using the **Close** button on the **Job** screen. After that, the job selection is automatically started again.

When you add new jobs or new job variants, these are automatically added to the galleries on the selection screens.

Warning

If you haven't configured any job variants, the job will be executed skipping all optional steps (if there are any).

Barcode job selection

Note

- *For the barcode job selection process to work, you should first connect a barcode scanner to the workstation through USB.*
- *The Arkite software expects each barcode to end with a carriage return/line feed suffix. Make sure to correctly configure this in your barcode scanner's settings.*

The **barcode job selection** process consists of a step in which an operator scans a barcode. The scanned barcode should correspond with the **Variant Code** of a job variant.

After scanning the barcode, the corresponding job variant is launched. When the operator has finished the assembly, the system will automatically ask to scan a new barcode.

When adding this process, an **Barcode/QR Code Scanner** communication module and a variable for storing the barcode are added at the same time.

MES/ERP job selection example

The **MES/ERP job selection example** process consists of a steps in which a job order containing job, variant and quantity is retrieved from a MES/ERP system.

When adding this process, an HTTP communication module and an example job is added at the same time.

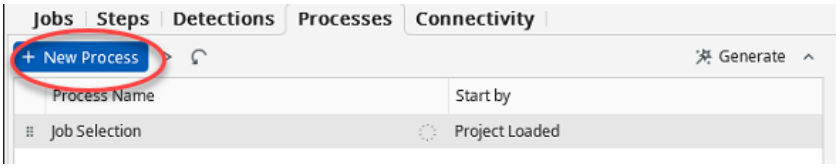
In the first step of the process, Arkite connects to the Arkite MES and retrieves a JSON file. You will need to update the URL in the **Properties** of that step to have it connect to your own MES.

In the subsequent steps, the job, variant and quantity is retrieved from the work order and written into variables in Arkite. Depending on the input from your MES, you may have to update these steps as well.

6.3 Adding a new process

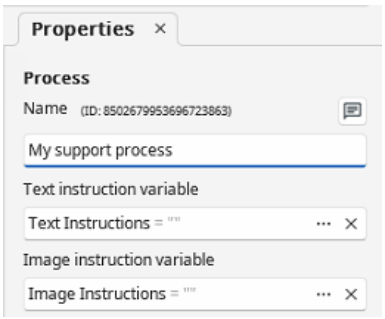
To add a new process, execute the steps below:

1. On the **Processes** tab, click **New Process**.



   The process is added to the list of processes.

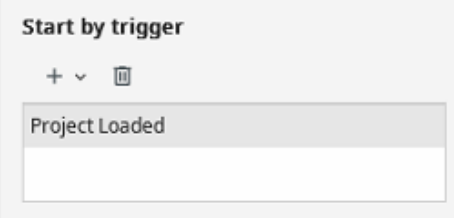
2. In the **Properties** pane, update the name of your process to something meaningful.
3. By default, one text instructions and image instructions variable is generated for all instructions for all processes. If you want to define a specific variable for the process you add, you can change the variable here.



Note

When you define specific variables, you will also have to add a specific text and image visualizations to your process that displays the state of those variables.

4. By default, a process is started when the project is loaded. Optionally, update the trigger(s) by which the process is started:
   - Under **Start by trigger**, click the + button to add a trigger.



   - Select the appropriate trigger.
   - Configure the trigger properties.

   Note
   In the next section, you can find an overview of all process triggers and their properties.

Result

Your process is set up now. You can start configuring process steps to build the contents of your process.

6.4 Overview of process start triggers

6.4.1 Alarm Clock

About the Alarm Clock trigger

The **Alarm Clock** trigger allows you to start a process every day at a specified point in time. This point in time can be defined as a static value or as a variable.

Note
If there is no variable selected or if the format is incorrect, the time will be 00:00:00 by default.

Properties

Property	Description
Name	The name for the trigger. ***Note*** *Next to the **Name** label, the object ID for the trigger is shown. You can copy it to use it when working with the Arkite API. For more information about the Arkite API, refer to chapter **18 The Arkite API** on page 292.*
Use variables for time	Check this checkbox to use the value of a variable rather than a fixed time.
Time	The time at which the process should start.
Variable for time	The variable in the time at which the process should start is stored.

6.4.2 Project Loaded

About the Project Loaded trigger

The **Project Loaded** trigger allows you to start a process as soon as your project is loaded. This is the default trigger for a process.

Properties

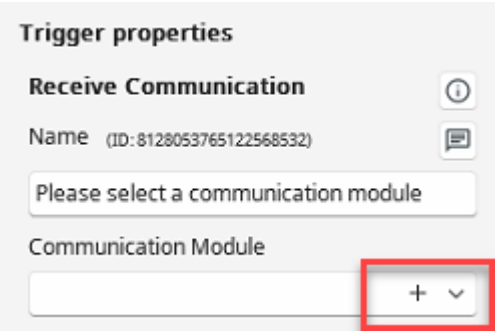
Property	Description
Name	The name for the trigger. ***Note*** *Next to the **Name** label, the object ID for the trigger is shown. You can copy it to use it when working with the Arkite API. For more information about the Arkite API, refer to chapter **18 The Arkite API** on page 292.*
Disable	Check this checkbox to temporarily disable the trigger, e.g. while you are still configuring your project.

6.4.3 Receive Communication

About the Receive Communication trigger

The **Receive Communication** trigger allows you to start a process within your project when data is received from an external device, such as a PLC, a smart tool, a database, a barcode scanner or another workstation.

The communication is done through a communication module. Before you can use a receive communication trigger, you first need to add and configure the appropriate communication module. Click the **+** icon in the **Communication** drop-down list.



Note

*For more information regarding the next steps, refer to the section **7.3.3 Adding a communication module** on page 214.*

Common properties

For every **Receive Communication** trigger, the properties below need to be configured:

Property	Description
Name	Name for your **Receive Communication** trigger. By default, the name of the communication module you have selected is used, but you can change it. ***Note*** *Next to the **Name** label, the object ID for the trigger is shown. You can copy it to use it when working with the Arkite API. For more information about the Arkite API, refer to chapter **18 The Arkite API** on page 292.*
Communication Module	Communication module that is used to receive the communication.

The other properties depend on the communication module you have selected for communication. They are listed below for every available communication module.

Inter-Workstation

Property	Description
Variables	When the variable(s) you indicate here are received from the other workstation, the process will be started. Check the **All Variables** checkbox to start the process when receiving any variable.

Modbus TCP

Property	Description
Variable	The variable that will be used to store the received messages.
Address Type	The object type of the read action. It is possible to read: – **Discrete inputs:** read-only entity with a size of 1 bit (ON/OFF) – **Coils**: read-write entity with a size of 1 bit (ON/OFF) – **Input registers:** read-only entity with a size of 16 bits (0-65 535) – **Holding registers:** read-write entity with a size of 16 bits (0-65 535)
Address	The address of the object from which will be read. The process will be started when a message is received from the address specified here, with the correct type.

Open Protocol

Property	Description
MID to filter on	The MID (which is character 5 to 8 of the message header) to filter the messages on. If this field is filled in, only messages with a matching MID will be captured and the process will only be started when the matching MID is received. If this field is left empty, all messages will be captured. The process will started whenever a message is received.
Variable for message	The variable that will be used to store the received messages.

6.4.4 Timer

About the timer trigger

The **Timer** trigger allows you to start a process with a certain interval.

Example

With a combination of a Timer trigger and a Counter actuator you can create a clock.

Properties

Property	Description
Name	The name for the trigger. ***Note*** *Next to the **Name** label, the object ID for the trigger is shown. You can copy it to use it when working with the Arkite API. For more information about the Arkite API, refer to chapter **18 The Arkite API** on page 292.*
Milliseconds in between sparks	The interval with which the process should start.

6.4.5 Watchdog

About the Watchdog trigger

The **Watchdog** trigger allows you to let a process run as long as no errors in hardware of software are detected. The watchdog will check the system with a certain interval and flag possible malfunctions.

Note

Combined with an ethernet communication module and external hardware like a light or buzzer, it can be used to visualize the state of your system.

Properties

Property	Description
Name	The name for the trigger. ***Note*** *Next to the **Name** label, the object ID for the trigger is shown. You can copy it to use it when working with the Arkite API. For more information about the Arkite API, refer to chapter **18 The Arkite API** on page 292.*
Milliseconds between ticks	The interval with which the watchdog checks the system's hardware and software features.
Minimum FPS	The minimum frames per second the sensor should have.
Variable for message	The variable that will store the state of the watchdog (OK or FAILURE). The process will only start when the state of this variable is OK.
Variable for failure	The variable that will store the watchdog's error message when its state is FAILURE

6.4.6 Variable Changed

About the Variable Changed trigger

The **Variable Changed** trigger allows you to start a process as soon as one or more variables reach a certain state or are changed.

Properties

Property	Description
Name	The name for the trigger. ***Note*** *Next to the **Name** label, the object ID for the trigger is shown. You can copy it to use it when working with the Arkite API. For more information about the Arkite API, refer to chapter **18 The Arkite API** on page 292.*
Any change	Check this checkbox to start the process at any change in state. In this case, there is no desired state.
Compare with	Check this checkbox to use the state of another variable as the desired state.
Variables	Define the variable(s) you want to monitor for state change.
Comparison	Select a compare operator
Other variable	Define the variable you want to compare the variable you monitor with.
Desired state	Define which state the variable(s) should reach to start the process
Type	– Select AND if all variables need to reach the desired state. – Select OR if only one of the variables need to reach the desired state. – Select Probing to use the probing algorithm. More information about can be found below this table.
State to set (probing)	The probing algorithm will select 1 variable. The state of this variable will be set to the value in this property for 1 ms. After that, it returns to its previous state. You can use this change to make a change to another variable.
Premature Probing Spark	If not checked, the probing algorithm will wait until all variables have returned to their default values. Then it will select the variable that has been in the desired state for the longest time. If checked, the probing algorithm will select a variable when it has reached desired state for a significantly longer time than any other variable.
Premature Spark Time	Set the minimum time in milliseconds before the probing algorithm selects a variable.
Premature Spark Fraction	Define how much longer a variable needs to be in the desired state than any other variable for the probing algorithm to select it.

Probing

In case you select probing, there are two possibilities for the start of your process:

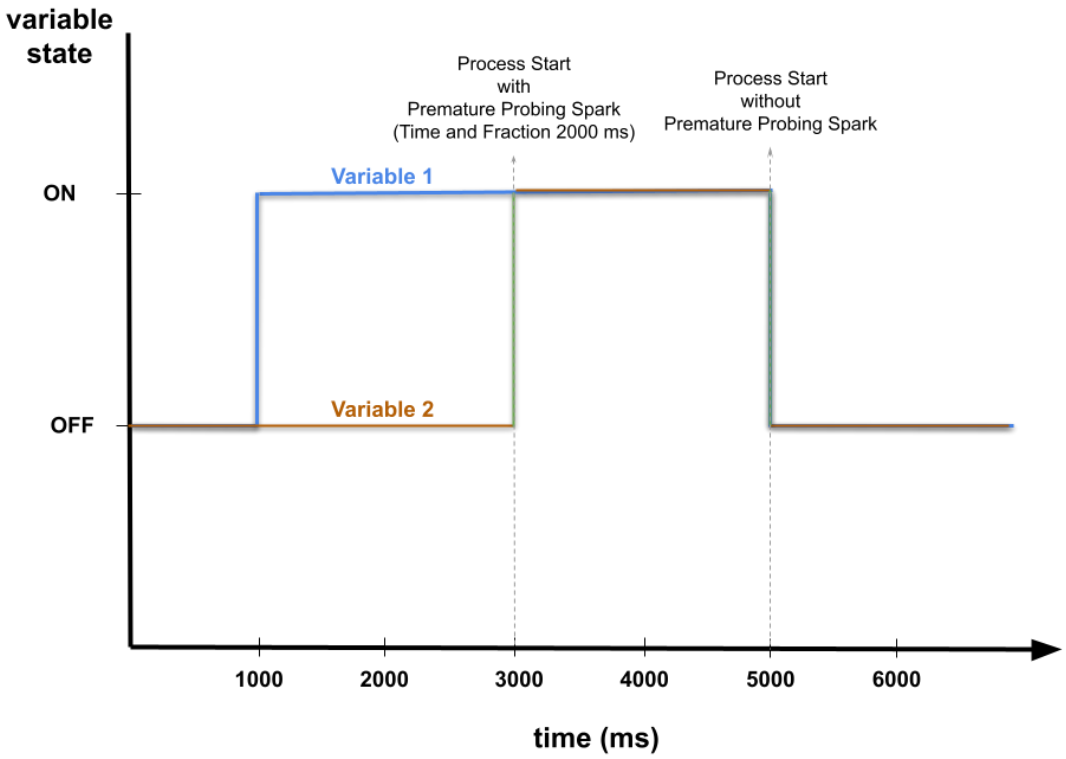
- When **Premature Probing Spark** is not checked, your process will be started when all variables have reached the desired state at one point ***and*** returned to their default state afterwards.
- When **Premature Probing Spark** is checked, your process will be started:
  - When the time set in the **Premature Spark Time** property has passed, AND
  - When one variable was in the desired state significantly longer than any of the other variables. To define what is significantly longer, the time set in the **Premature Spark Fraction** is taken into account.

In both cases, the variable that was in the desired state for the longest time is changed to the value defined in the property **State to Set (probing)** for 1 ms. After that, it returns to its previous state.

Example

Suppose we have:

- 2 variables with default state OFF.
- Probing is selected, with desired state ON and state to set PROBING
- Variable 1 changes to state ON after 1 second
- Variable 2 changes to state ON after 4 seconds
- Both variables go back to their default state OFF after 5 seconds.



In case **Premature Probing Spark** is not checked, the behaviour will be as in the table below.

The process is started after 5 seconds, when both variables are back in their default state.

The probing algorithm selects variable 1 because it has been in state ON for the longest time and changes its state to PROBING for 1 ms.

Time (ms)	Variable 1	Variable 2	Process status
0	OFF	OFF	Not started
1000	ON	OFF	Not started
2000	ON	OFF	Not started
3000	ON	OFF	Not started
4000	ON	ON	Not started
5000	OFF	OFF	Started

Time (ms)	Variable 1	Variable 2	Process status
5001	PROBING	OFF	Running
5002	OFF	OFF	Running
6000	OFF	OFF	Running

In case **Premature Probing Spark** is checked with both **Premature Spark Time** and **Premature Spark Fraction** set at 2000 ms, the behaviour will be as in the table below.

The process is started after 3 seconds, when variable 1 has been in state ON for 2 seconds longer than variable 2.

The probing algorithm selects variable 1 and changes its state to PROBING for 1 ms.

Time (ms)	Variable 1	Variable 2	Process status
0	OFF	OFF	Not started
1000	ON	OFF	Not started
2000	ON	OFF	Not started
3000	ON	OFF	Started
3001	PROBING	OFF	Running
3002	ON	OFF	Running
4000	ON	ON	Running
5000	OFF	OFF	Running
6000	OFF	OFF	Running

6.5 Configuring process steps

6.5.1 Process step types

A vast amount of process steps types are available to you to build your process. They are divided in the categories below:

- **General steps**: some basic, often used process steps.
- **Variables**: process steps that use the values of a variable as an input or an output.
- **Detections**: process steps related to the detections you have added
- **Materials:** process steps related to the materials you have configured.
- **Screen buttons:** process steps related to the buttons you have added to the touch screen.
- **Communication:** process steps related to the communication modules you have configured.
- **Extra**: a number of advanced, less often used process steps.

The available step types are mostly the same as the ones that are available for your job steps on the **Steps** tab.

For more information, refer to section **4.3.3 Step types** on page 70.

The sections below contain an overview of process types that are available additionally on the **Processes** tab.

General steps

Process step type	Description
Execute Job	Launches a job, specifying a job variant or values for the different job options. The step will remain active until the job is completed, and is used to calculate the job's cycle time. When the process containing the **Execute Job** step is paused, the Cycle Time will also be paused. The cycle time is stored in the variables **Cycle Time** (format HH:mm:ss) and **Cycle Time Sec** (format ss). This allow you to easily display the Cycle Time on a **Projector** or a **Screen** using a **Text** visualization. The cycle time is also used in the Cycle Time dashboard. For more information about dashboarding, refer to chapter **12 Dashboarding** on page 257.

Extra

Process step type	Description
Open/Close Display	Open or close a particular display.
Project Loader	Restart the current project or load another project.
Switch Language	Sets the language in which instructions are displayed. ***Note*** *By default, instructions are displayed in the user language of the user that is logged in. If no instructions are available in that language, the system will look for a similar language. For example, if the user language is en-GB, en-US will be used. If no similar language is available, the first instruction language in the project will be used.*
Switch User	Logs out the current user. You can indicate a new user to sign in with, or show a login screen.
System Command	Issues one of the system commands below: – **Shut down:** shut down your workstation – **Reboot:** reboot your workstation – **Restart Arkite:** close and restart the Arkite software – **Project:** open the **Projects** overview

6.5.2 Adding process steps

Steps

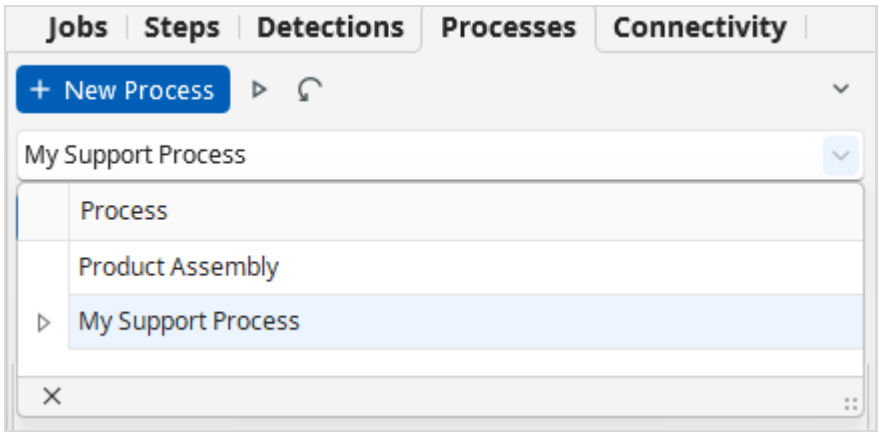
To add a new process step, execute the steps below:

1. Navigate to the **Processes** tab.
2. Under **Processes**, select the process in which you want to add a new process step.

   Note

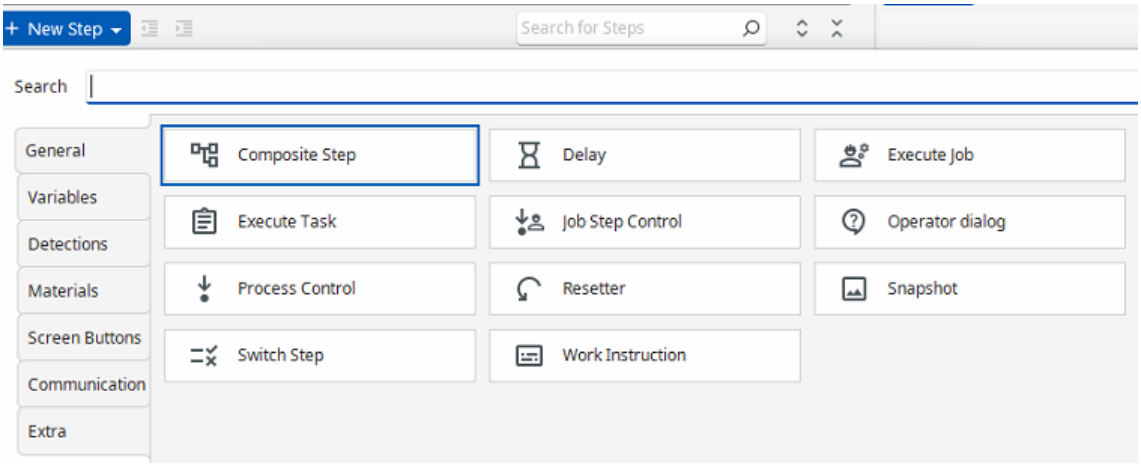
   *For a better overview and more space when adding process steps, you can collapse the **Processes** section by clicking the ⌃ button.*

   You can still select the active process in the dropdown list.

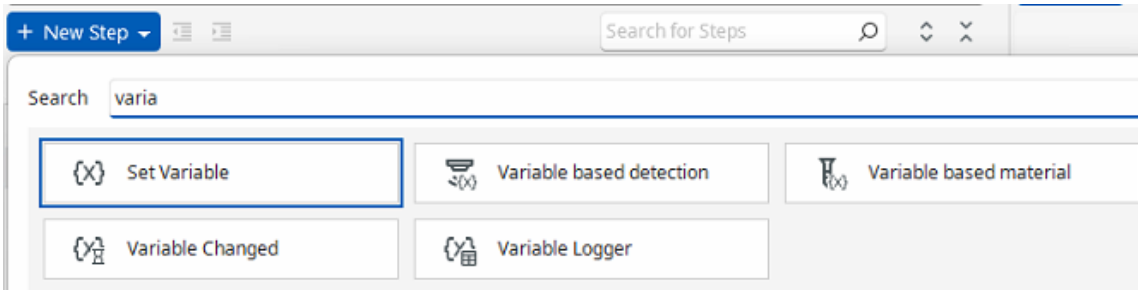


*To expand the **Processes** section again, click the ⌄ button.*

3. Under **Steps**, click **New Step** and select the process type you want to add.



You can also enter the process type in the **Search** field to find it.



Result

The process step is added to the list of process steps.

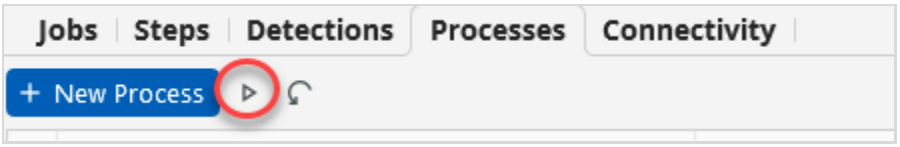
6.5.3 Organizing your process steps

You can organize your process steps in exactly the same way as your job steps on the **Steps** tab.

For more information, refer to section **How to organize your steps** on page 69.

6.5.4 Testing your process and process steps

You can test your process by selecting the process and clicking the **Play** button.



To reset the process and start again from the beginning, click the button.

Under steps, you can:

- click the **Next** button to force the process to continue.
- click the **Previous** button to return to the previous step.
  Going to a **Previous** step is only possible if that previous is step is a detection step or work instruction.
- select a step and click the **Go To** button to go to a specific step.

The icons for every step indicate the progress of your process.

For more information, refer to section **4.3.4 Testing your job** on page 80.

6.6 Managing visualizations for your process steps

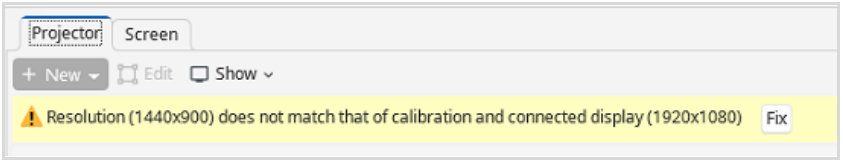
What?

After creating your process steps, you can manage the visualizations that will be displayed when those steps are active, on a connected projector or screen.

You will see a subtab for every display that is available. By default, a **Projector** subtab will always be available.

Note

- *When you have created the project before calibrating, a **Projector** display is added with the screen resolution of the device you are working on (e.g. your laptop). This may not match with the actual projector resolution, used during calibration. This will cause projections to be faulty. A message on the **Projector** subtab will warn you for this. Click **Fix** and update the **Resolution** property to the correct value in the projector's **Properties** pane.*



- *To add more displays, refer to section **7.2.3 Adding a display** on page 195.*

When creating a process, the Arkite software will automatically generate visualizations to display text and image instructions.

Additionally, you may also create more visualizations to display.

When the **Experience Level** functionality is enabled, you can add experience level buttons to a **Screen** display to allow an operator to select his experience level.

How to manage visualizations

Visualizations for your process steps can be managed in exactly the same way as visualizations for your job steps, as configured on the **Steps** tab.

For more information, refer to section **4.4 Managing visualizations for your job steps** on page 82.

6.7 Working with conditions

What?

Conditions can be used to specify whether a process step should be executed or not. Any individual or composite process step can be linked to one or more conditions. Only if at least one of the conditions holds, the step will be executed.

A condition can consist of one or more subconditions.

You can use conditions in your processes in the same way you use them in your jobs on the **Steps** tab. For more information, refer to section **4.5 Working with conditions** on page 118.

6.8 Working with tasks

What?

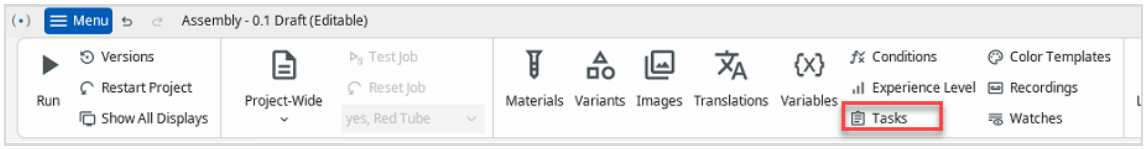
Tasks allow you to easily re-use a step or a group of steps across multiple jobs.

Whenever you make changes to a task, those changes will automatically be applied at every location where you have used the task.

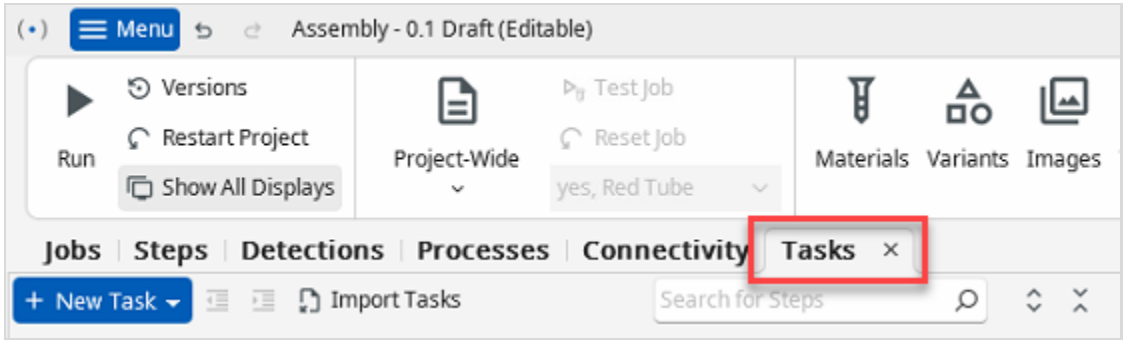
Adding tasks

To add a task, execute the steps below:

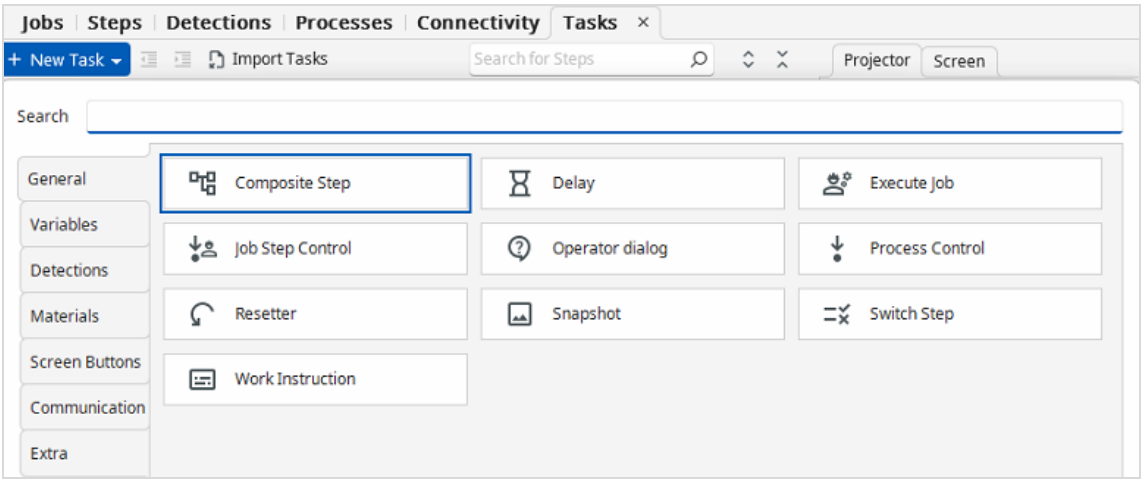
1. In the ribbon, click the **Tasks** button.



The **Tasks** tab is added to the list of tabs.



2. To add a task, click **New Task** and select the step type you want to add.



3. Organize your process steps in the same way you would in a normal process.
   For more information, refer to section **6.5.3 Organizing your process steps** on page 184.

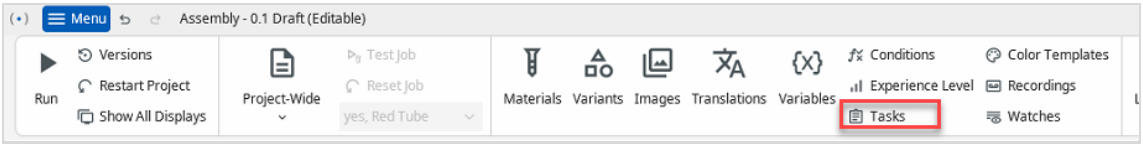
   Note

   *Alternatively, you can also create a task based upon a composite step in an existing process. To do so, right-click the step and select **Convert to Task**.*

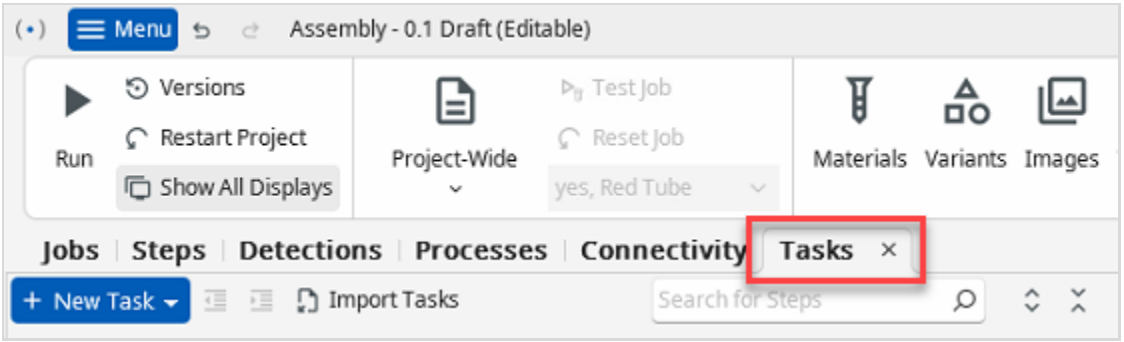
Importing tasks from another project

You can also add tasks to your project by importing them from another project. To do so, execute the steps below:

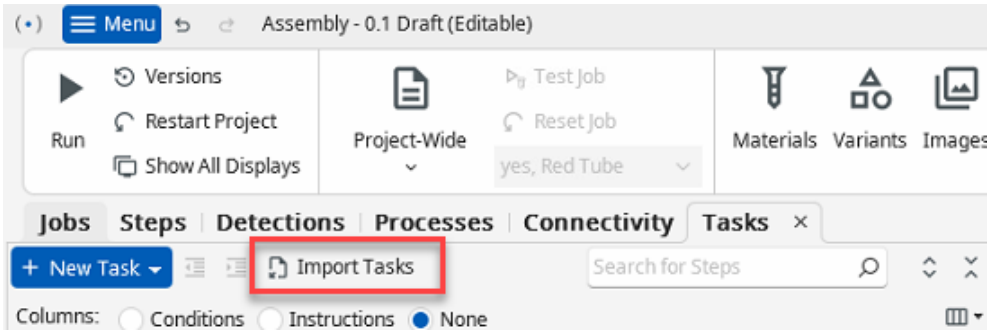
1. In the ribbon, click the **Tasks** button.



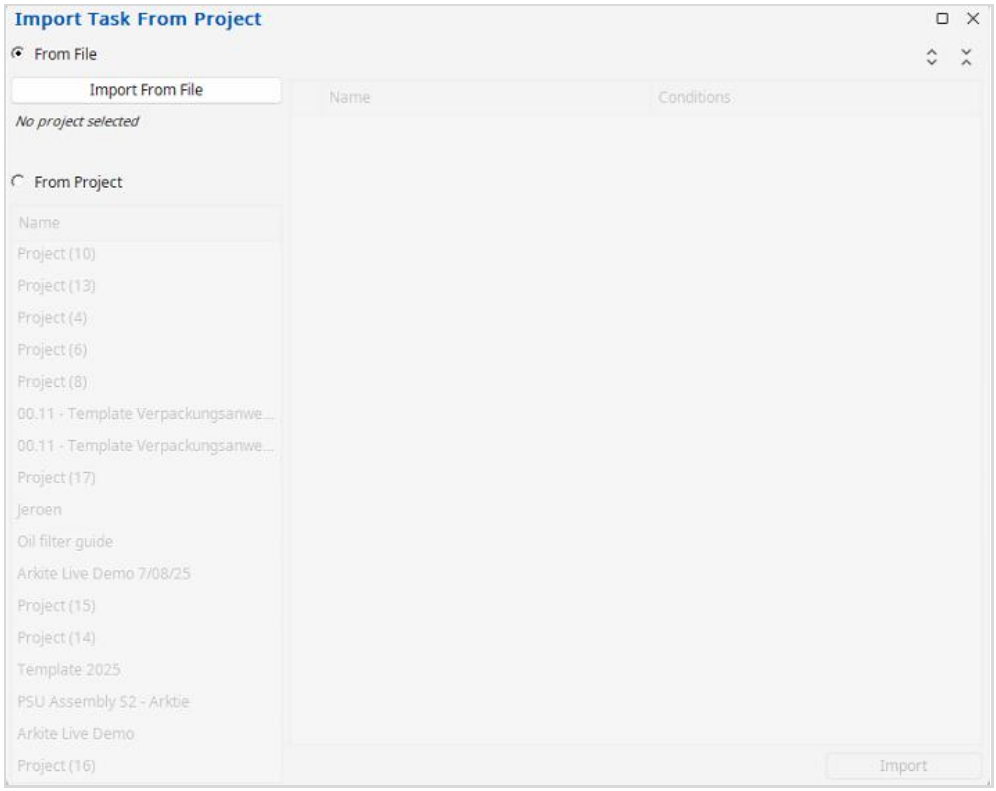
   The **Tasks** tab is added to the list of tabs.



2. Click the **Import Tasks** button.



The **Import Task From Project** window opens.



3. To import tasks from a project on your workstation, select the **From Project** radio button and select the project of your choice.

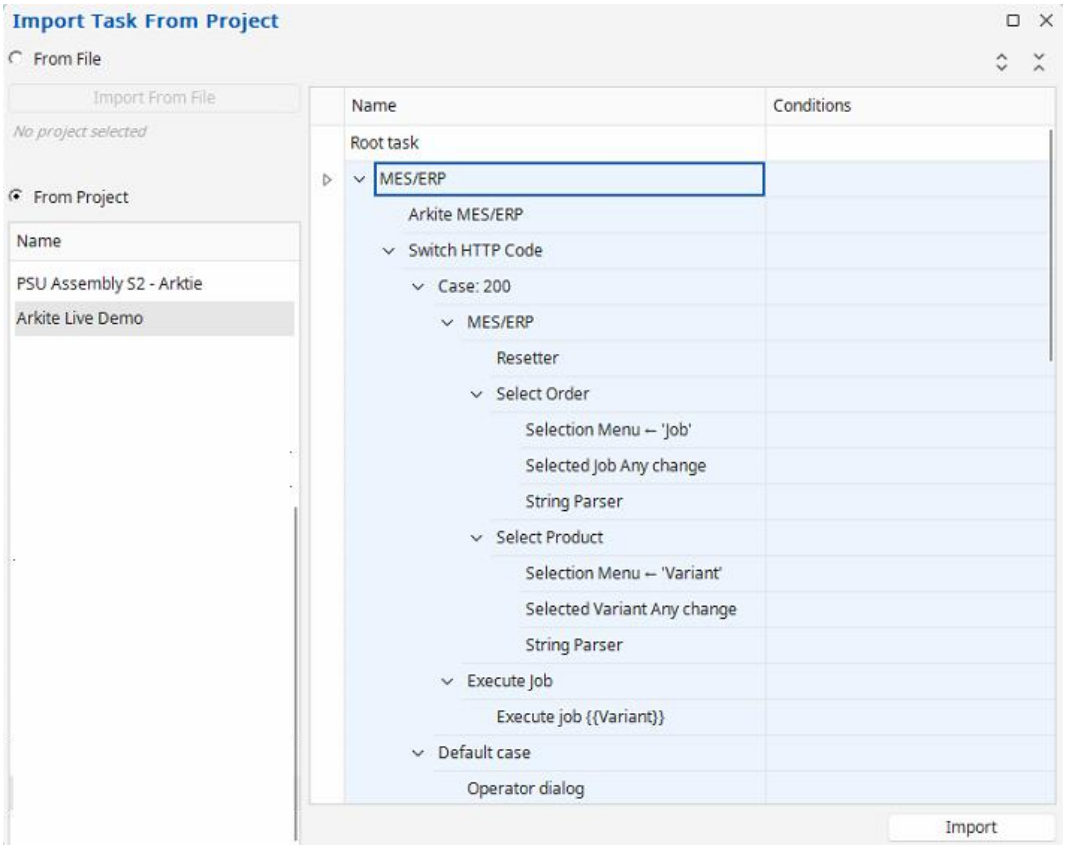
   To import tasks from a project file:

   - Select the **From File** radio button
   - Click the **Import From File** button
   - Select the .hef file of your choice and click **Open**.

   Note

   *To be able to import tasks from a project file, you should export the project first. For more information, refer to section **3.3 Managing projects** on page 34.*

4. Select the task(s) you want to import into your project.

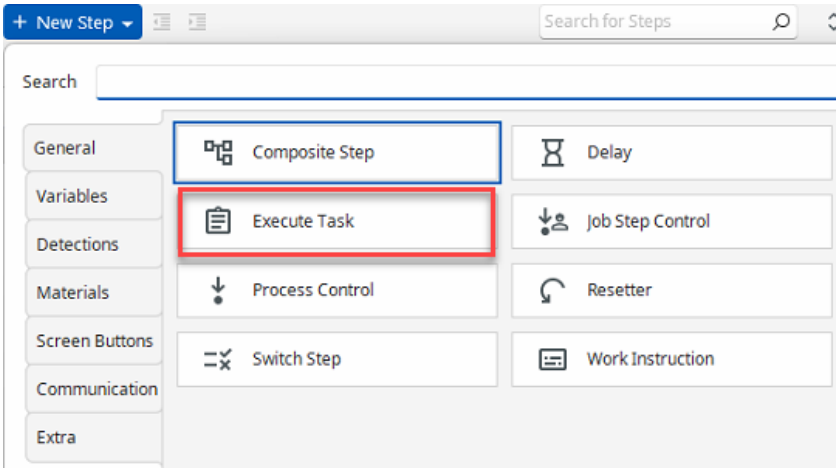


5. Click **Import**.
   The task(s) you selected are imported into your project.

Using tasks in your job or process

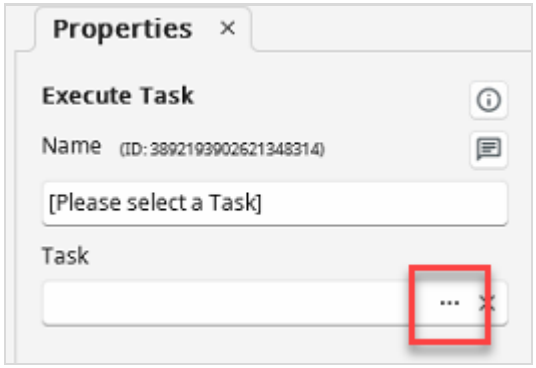
To use a task in your job or process, execute the steps below:

1. Navigate to the **Steps** tab to use a task in a job, or to the **Processes** tab to use a task in a process.
2. In the **Steps** section, click the **New Step** button and on the **General** tab, select **Execute Task**.

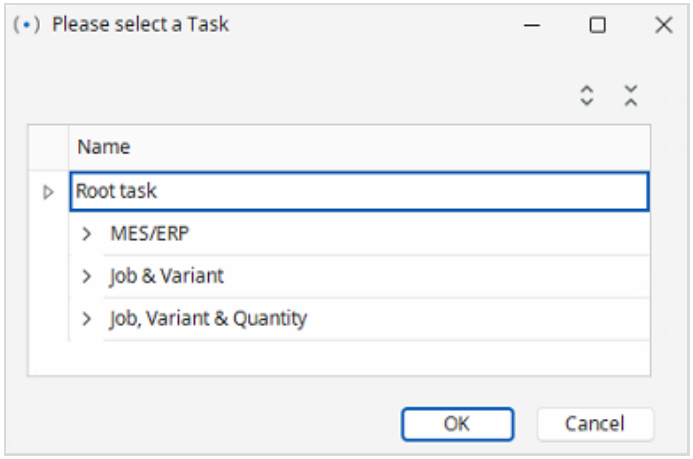


An **Execute Task** step is added to your list of steps.

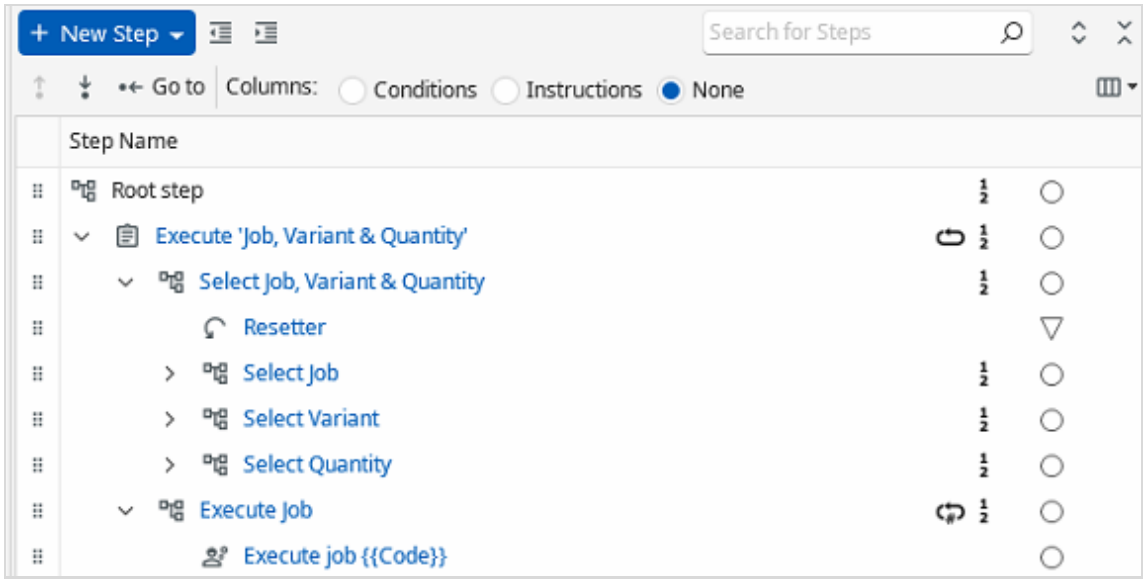
3. Select the task step in your list of process steps, and in the **Properties** pane, click the **...** button behind the **Task** field.



4. Select a task and click **OK**.

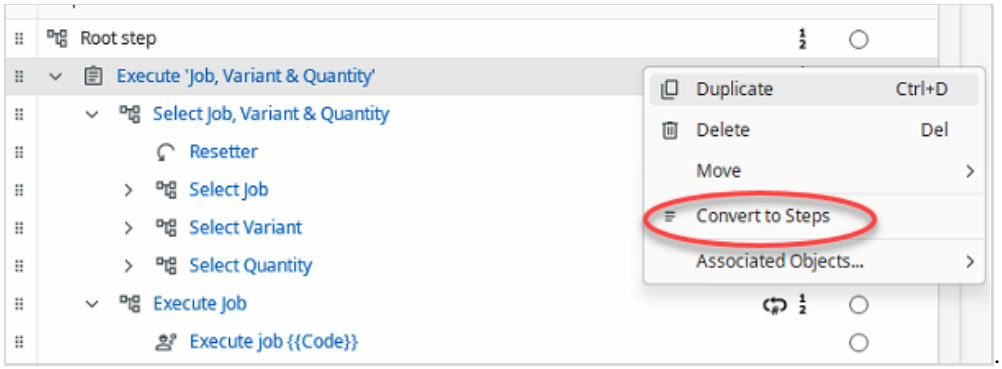


The task step is added to the list of steps. It is rendered in blue and is read-only, which means you cannot make any changes to the steps. If you want to make any changes, you need to update the task itself. However, be aware that your changes will be applied at every location where you have used the task step.



Note

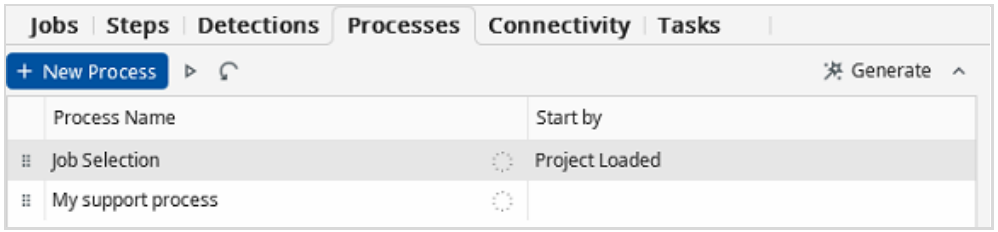
If you want to make any changes that only apply for that particular process, you can unlink the task step from its task. To do so, right-click the task step and select ***Convert to Steps****. Any future changes to the task will no longer be applied in this process.*



6.9 Managing processes

What?

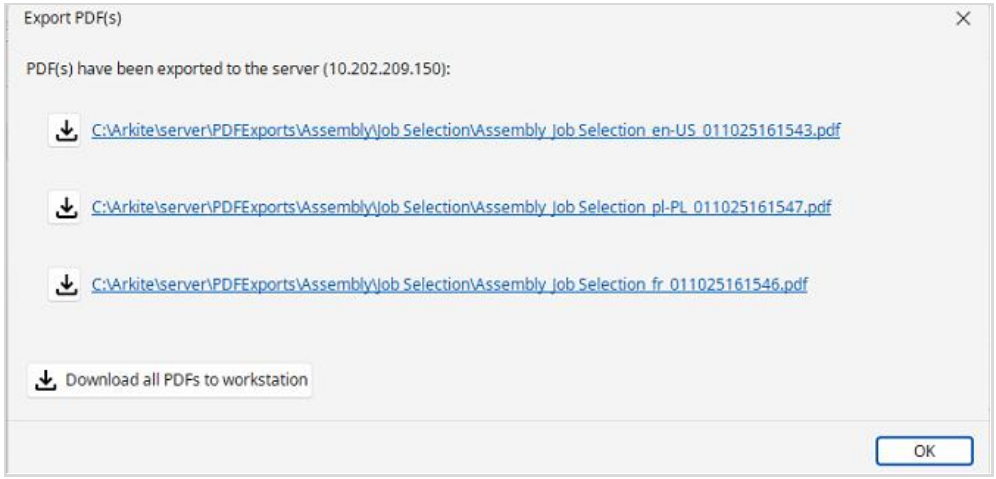
After having set up all of your processes, you can manage them on the **Processes** tab.



How to manage processes

On the **Processes** tab, you can:

- Select a process and click to pause it or to start/continue it
- Select a process and click to reset it.
  The process will start from the first step again.
- Right-click a process and select **Duplicate** to duplicate it.
- Right-click a process and select **Delete** to delete it.
- Right-click a process and select **PDF export** to create a pdf document containing all process steps with text and image instructions. If your project contains variants, a separate pdf document will be generated for every variant, as well as a pdf document that includes all steps of all variants.
  For every text and image visualization in your project, you can indicate if you want to show it in the pdf export or not.
  After the generation of the PDF exports, a dialog will appear with all the documents that have been created.

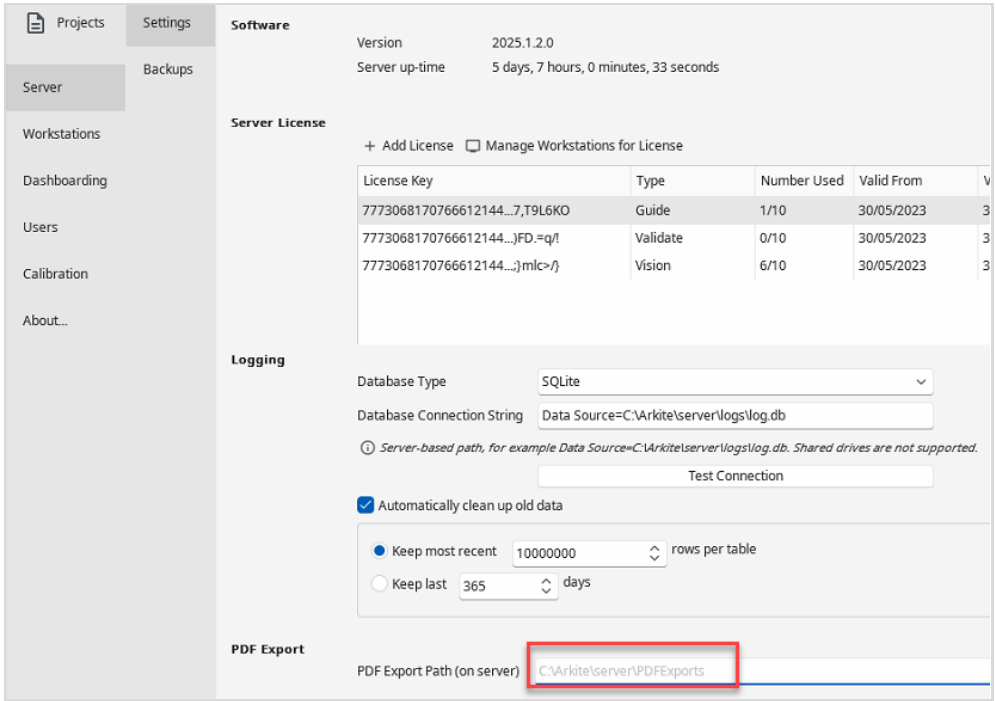


- To view a document, click the hyperlink for that document
- To download a document to your workstation, click the button in front of the document.
- To download all documents at once, click the **Download all PDFs to workstation** button. A ZIP file containing all PDFs will be downloaded.

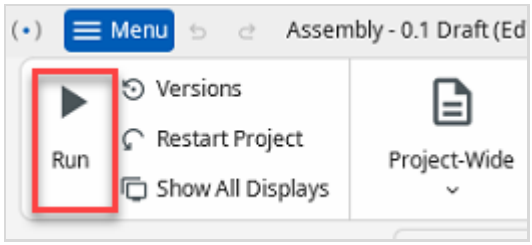
Note - PDF Export Settings

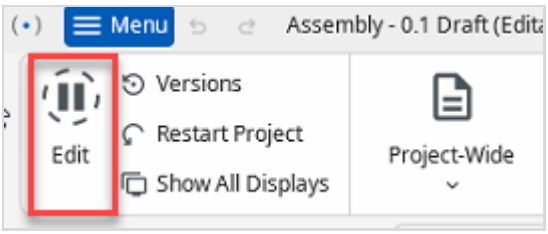
The pdf document is created in a folder on the server. By default, this is C:\Arkite\server\PDFExports. To change this folder:

- *Click **Menu**.*
- *Click **Server** and then **Settings**.*
- *Under **PDF Export**, change the **PDF Export Path (on server)***



On the ribbon you can switch between **Run** and **Edit** mode.





When you make any changes to your project, it automatically switches to **Edit** mode.

When a project is (re)loaded or a process is started, it automatically switches to **Run** mode.

In **EDIT** mode:

- Processes are not running
- Projection of currently selected component or step is visible
  This makes it possible to finetune projections on the spot.

In **RUN** mode:

- Processes are running
- Projection of currently active step is visible

Note

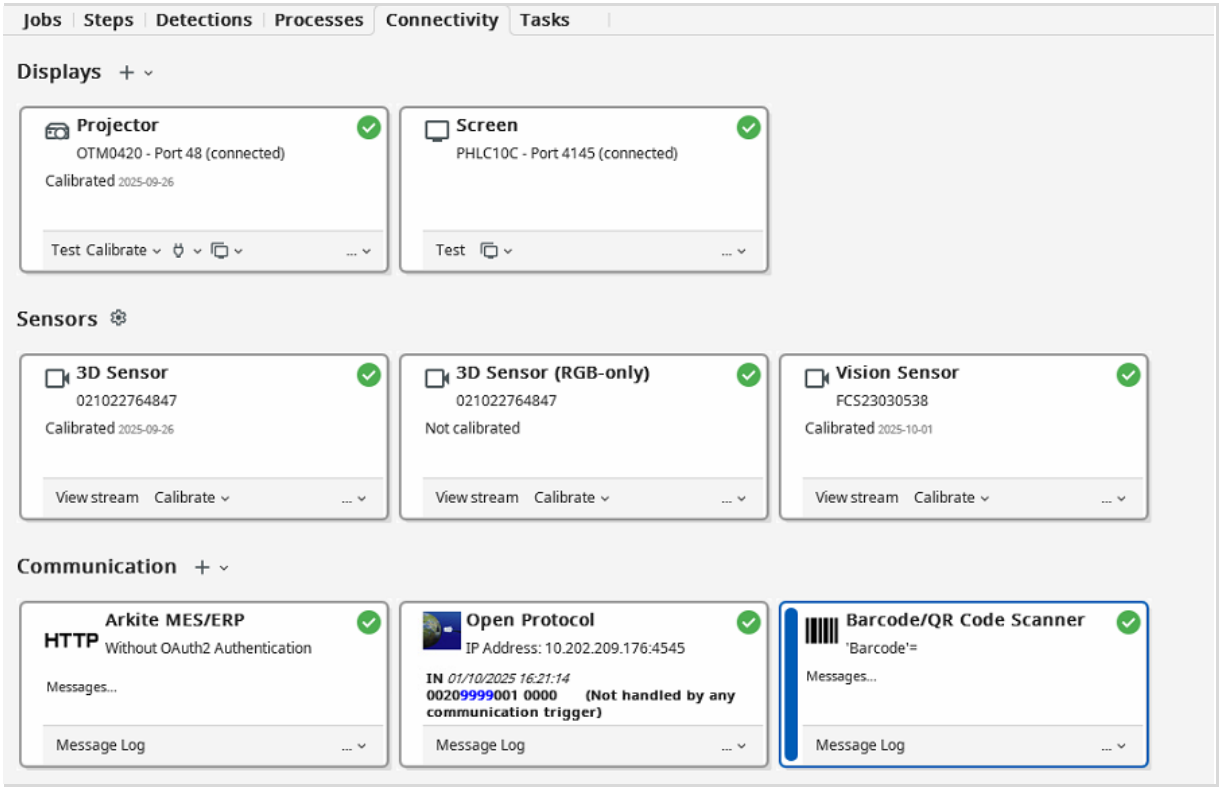
*On an editor, i.e. a workstation without a valid **Guide**, **Validate** or **Vision** license, you will be able to run a project for a maximum of 15 minutes. After that, it will automatically revert to **Edit** mode.*

7. Connectivity

7.1 About connectivity

The Arkite software allows to connect a variety of peripheral devices:

- Displays, i.e. projectors and (touch)screens that are connected.
- Sensors, i.e. the Arkite 3D sensor, the Arkite Vision sensor (when using a **Vision** license) and, if applicable, an RGB camera or webcam that is connected
- Communication modules, enabling the system to communicate to external devices.



7.2 Displays

7.2.1 What?

Once your project has been configured, the Arkite software will be able to provide your operators with real-time picking and assembly instructions through Augmented Reality. These instructions are provided on one or more displays.

7.2.2 Display types

Two types of displays can be distinguished:

- **Projector**:
  One or more projectors are mounted above the workbench. These projectors enable you to show instructions on the workbench at the time the operator needs them.

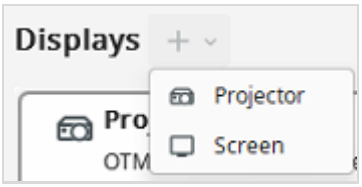
- **Screen**:
  A (touch)screen can be connected to the workstation. On the screen, information and instructions can be displayed, as well as buttons to interact with the operator.

7.2.3 Adding a display

Steps

To add a display, execute the steps below:

1. Navigate to the **Connectivity** tab.
2. Under **Displays**, click the **Add** button and select the display type you want to add.



   A new display card is added.
3. In the **Properties** pane, configure the properties for the display.

Projector Properties

Note

*It is also possible to communicate with a PJLink supported projector by adding a **Projector** process step to your process. The PJLink protocol uses an ethernet connection. It can for example turn the projector on or off, or switch the input source.*

*For instructions on installing and configuring an Optoma ZH420 projector (our standard projector), please consult our **Arkite hardware installation guide** on our customer portal.*

*For instructions on installing and configuring an Optoma ZW400 projector, you can download the procedure **How to configure the Optoma ZW400 projector for use with Arkite** from our customer portal.*

*For instructions on configuring a Casio projector, you can download the procedure **How to configure Casio XJ-F210WN For PJ link Control** from our customer portal.*

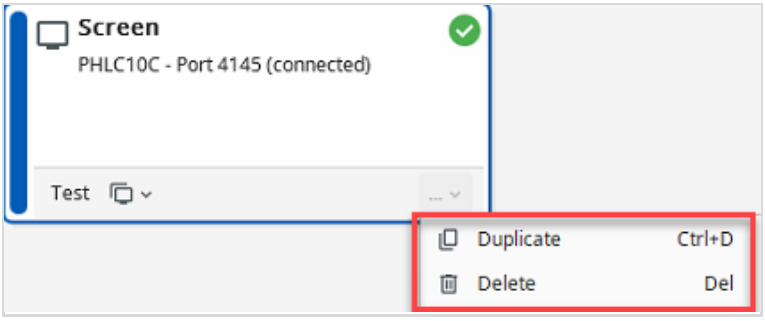
Property	Description
Name	Name of the projector. ***Note*** *Next to the **Name** label, the object ID for the projector is shown. You can copy it to use it when working with the Arkite API. For more information about the Arkite API, refer to chapter **18 The Arkite API** on page 292.*
Auto Show	When checked, this display will be shown when the project is opened.
Background Color	The display's background color.
Resolution	Select the resolution for the display. ***Note*** *When you have created the project before calibrating, a **Projector** display is added with the screen resolution of the device you are working on (e.g. your laptop). This may not match with the actual projector resolution, used during calibration. This will cause projections to be faulty. Update the **Resolution** property to the correct value to fix this.*
Display Device	Select the display device on which the projector will be shown.
Log Table	The table name in the logging database in which all incoming and outgoing messages and exceptions will be logged. ***Note*** *The logging database is configured in the server settings. For more information, refer to section **13.3 Changing the logging database, the logging database and the cleanup of old data** on page 264.*
Projector IP address	The IP Address of the connected PJLink supported projector.
Port	The Port of the connected PJLink supported projector. PJLink has a default port of 4352.

Screen properties

Property	Description
Name	Name of the screen. ***Note*** *Next to the **Name** label, the object ID for the screen is shown. You can copy it to use it when working with the Arkite API. For more information about the Arkite API, refer to chapter **18 The Arkite API** on page 292.*
Auto Show	When checked, this display will be shown when the project is opened.
Background Color	The display's background color.
Resolution	Select the resolution for the display.
Display Device	Select the display device on which the screen will be shown.

7.2.4 Managing displays

- To duplicate a display, select it, click the **...** button in the lower-right corner of the card and select **Duplicate**.
- To delete a display, select it, click the **...** button in the lower-right corner of the card and select **Delete**.



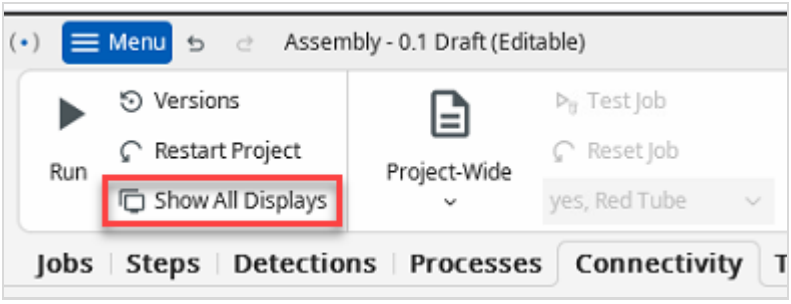
- Select a display and click the **Show** icon to immediately display it.

Tip

If you're not working at the workbench location, you can choose to show the display on the screen of your pc.

Note

*You can show all displays at any time by clicking the **Show All Displays** button in the ribbon.*



- Select a display and click **Test** to verify which display is connected. The display will be bright yellow for 5 seconds to help you differentiate it from other displays.
- To turn a **Projector** display on or off, select it, click the power icon and select **Turn on** or **Turn off**.

7.3 Communication modules

7.3.1 What?

The Arkite software allows to configure a variety of communication modules, enabling the system to communicate to external devices.

Note

A lot of communication modules use the workstation's ethernet ports. Always make sure the workstation's IP address and the IP address of the device you're trying to communicate with, are within the same range. Remember that the workstation has 2 separate network cards, each with its own settings.

7.3.2 Communication module types

Barcode / QR Code Scanner

The Barcode / QR Code Scanner communication module is used to connect barcode and QR code scanners to the workstation through USB. Such scanners are widely used for product recognition. They convert a barcode or QR code to a specified combination of letters and numbers, or in other characters. This communication module can also be used as a keyboard logger or to connect an NFC reader, a foot pedal or hardware button. In the table below, you can find the properties you can configure for this communication module type.

Note

- *A barcode scanner technically works in the same way as a keyboard. If the scanned barcode appears incorrectly, please verify whether the Windows keyboard layout matches with the scanner.*
- *The Arkite software expects each barcode to end with a carriage return/line feed suffix. Make sure to correctly configure this in your barcode scanner's settings.*

Property	Description
Name	Name of the communication module ***Note*** *Next to the **Name** label, the object ID for the communication module is shown. You can copy it to use it when working with the Arkite API. For more information about the Arkite API, refer to chapter **18 The Arkite API** on page 292.*
Log Table	The table name in the logging database in which all incoming and outgoing messages and exceptions will be logged. ***Note*** *The logging database is configured in the server settings. For more information, refer to section **13.3 Changing the logging database, the logging database and the cleanup of old data** on page 264.*
Variable for barcode / QR code	A variable that contains the last incoming text message from the scanner.

COM (serial port)

The COM communication module allows to establish serial communication with an external device. In the table below, you can find the properties you can configure for this communication module type.

Property	Description
Name	Name of the communication module ***Note*** *Next to the **Name** label, the object ID for the communication module is shown. You can copy it to use it when working with the Arkite API. For more information about the Arkite API, refer to chapter **18 The Arkite API** on page 292.*
Log Table	The table name in the logging database in which all incoming and outgoing messages and exceptions will be logged. ***Note*** *The logging database is configured in the server settings. For more information, refer to section **13.3 Changing the logging database, the logging database and the cleanup of old data** on page 264.*
Receive Variable	The variable that will be used to store the received messages.

Property	Description
COM port	The COM port on which the external device is connected.
Baud Rate	The baud rate of the serial connection. Check the manual of the connected device to know the Baud Rate.
Data bits	The information to transmit, usually one of the values below: – 5 for baudot code – 7 for true ASCII – 8 for most kinds of data since this is the size of a byte
Stop bits	The stop bits allow the receiving signal hardware to detect the end of a character and to resynchronize with the character stream.
Parity	The party is a method of detecting errors in transmission: – **None**: no parity bit is sent at all. – **Odd**: Counts the given set of bits with value 1, sets the parity bit so the count is odd. – **Even**: Counts the given set of bits with value 1, sets the parity bit so the count is even. – **Mark**: Set the parity bit to 1. – **Space**: Set the parity bit to 0.
Handshake	Handshaking is an automated process for negotiation between two entities before communication is possible. – **None:** No control is used for the handshake – **XOnXOff:** the XON/XOFF software control protocol is used. – **RTS:** Request to send hardware flow control is used. – **RTS/XOnXOff:** Both RTS and XOnXOff software controls are used.
Receive Append Until Symbol	If unchecked, every incoming message is used to update the selected variable and spark subsequent components. If checked, incoming data is concatenated until a newline character appears, even if this data is spread across multiple messages.
Receive New Line Symbol	Select the new line symbol to use for receiving: – CR LF (\r\n): carriage return + line feed – CR (\r): carriage return – LF (\n): line feed – Custom: specify a new line symbol of your choice
Send Append Symbol	If checked, the 'Send New Line Symbol' is appended at the end of a message when sending.
Send New Line Symbol	Select the new line symbol to use for sending. – CR LF (\r\n): carriage return + line feed – CR (\r): carriage return – LF (\n): line feed – Custom: specify a new line symbol of your choice

Database

The Database communication module is used to communicate with a MySQL, MS SQL Server, Oracle, MariaDB or SQLite database. A variable is used to represent a query which will be executed

against the database. If the query is a select statement, you can either choose to store the first row of the result in a number of variables or to store all rows as a JSON array.

Property	Description
Name	Name of the communication module ***Note*** *Next to the **Name** label, the object ID for the communication module is shown. You can copy it to use it when working with the Arkite API. For more information about the Arkite API, refer to chapter **18 The Arkite API** on page 292.*
Log Table	The table name in the logging database in which all incoming and outgoing messages and exceptions will be logged. ***Note*** *The logging database is configured in the server settings. For more information, refer to section **13.3 Changing the logging database, the logging database and the cleanup of old data** on page 264.*
Variable for last message type	A variable storing the message type of the last message: – 'INFO' – 'WARNING' – 'ERROR'
Variable for last message	A variable storing the last message for this communication module
Database Type	Database type. Currently supported types are: – MS SQL Server – MySQL – Oracle Database – SQLite – Maria DB
Server	The name or URL of the server running the database.
User	The user who is used to query the database. This user should have the proper rights to access the table defined below.
Password	The password used to authenticate the user with. ***Note*** *You can click the  button to make the password you've entered visible and verify if you haven't made any typing mistakes.*
Database	The name of the database on the server. ***Note*** *In case of SQLite, this should be the path to the database file on the server's file system.* *For example: **C:\Arkite\server\mydatabase.db*** *Shared drives are not supported.*

Pick To Light (DMX)

The Pick To Light (DMX) communication module is used to connect DMX lights, either over the Art-Net ethernet protocol or through an Enttec Open DMX USB interface (SKU 70303). You can then link your detections to a DMX light, and use the DMX light for operator guidance instead of a projection. Alternatively, you can also control a DMX light through a **Send Communication** step.In the table below, you can find the properties you can configure for this communication module type.

Property	Description
Name	Name of the communication module ***Note*** *Next to the **Name** label, the object ID for the communication module is shown. You can copy it to use it when working with the Arkite API. For more information about the Arkite API, refer to chapter **18 The Arkite API** on page 292.*
Log Table	The table name in the logging database in which all incoming and outgoing messages and exceptions will be logged. ***Note*** *The logging database is configured in the server settings. For more information, refer to section **13.3 Changing the logging database, the logging database and the cleanup of old data** on page 264.*
Type	Select how you want to connect DMX lights: – **Art-Net:** the Art-Net ethernet protocol allows you to control multiple DMX universes from a single device. – **DMX USB:** connect your DMX lights through an Enttec Open DMX USB interface (SKU 70303). You will need a separate cable for each DMX device.
IP Address	Only available if **Art-Net** is selected in the **Type** property. The IP address of the ethernet controller you use to control your DMX lights.
DMX device	Only available if **DMX USB** is selected in the **Type** property. The serial number of the Enttec Open DMX USB interface that is linked to the communication module. By default, the first unlinked controller that is found will be linked automatically.
Configure Lights	Click this button to configure different lights on the connected DMX device. For more information, refer to section **7.3.4 Configuring lights for a Pick To Light (DMX) communication module** on page 216.

Epson Label Printer

The Epson Label Printer communication module is used to communicate with Epson label printers through the EPOS protocol over ethernet. It is possible to print logo's, variables, barcodes, QR codes. The module has been tested with the Epson TM-T88VI. In the table below, you can find the properties you can configure for this communication module type.

Property	Description
Name	Name of the communication module ***Note*** *Next to the **Name** label, the object ID for the communication module is shown. You can copy it to use it when working with the Arkite API. For more information about the Arkite API, refer to chapter **18 The Arkite API** on page 292.*
Log Table	The table name in the logging database in which all incoming and outgoing messages and exceptions will be logged. ***Note*** *The logging database is configured in the server settings. For more information, refer to section **13.3 Changing the logging database, the logging database and the cleanup of old data** on page 264.*
IP address	The IP Address of the connected Epson label printer. ***Note*** *Epson label printers have "192.168.192.168" as the default IP address.*
Device ID	The id of the target printer to be used. The default value is **local_printer**.

Property	Description
Time Out (ms)	The time out specified in the printer. The default value is **60000**.

Inter-Workstation

The Inter-Workstation communication module is used to send events between two workstations. The module needs to be configured on both workstations to be able to communicate properly. In the table below, you can find the properties you can configure for this communication module type.

Property	Description
Name	Name of the communication module ***Note*** *Next to the **Name** label, the object ID for the communication module is shown. You can copy it to use it when working with the Arkite API. For more information about the Arkite API, refer to chapter **18 The Arkite API** on page 292.*
Log Table	The table name in the logging database in which all incoming and outgoing messages and exceptions will be logged. ***Note*** *The logging database is configured in the server settings. For more information, refer to section **13.3 Changing the logging database, the logging database and the cleanup of old data** on page 264.*
Remote Workstation IP Address	The IP address of the 'other' workstation. You can receive messages and send messages to this workstation, using the same communication module. Click the **Find workstations** button to find workstations on the same network. ***Tip*** *If you want to know the IP address(es) of a workstation, press CTRL-ALT-A on that workstation.*
Local Port	The TCP/IP port to use on your workstation.
Remote Port	The TCP/IP port to use on the remote workstation.

Important

*For the Inter-Workstation communication to work, the communication module needs to be configured on both workstations. The **Local Port** on the first workstation needs to be the same one as the **Remote Port** on the second workstation, and vice versa.*

HTTP

The HTTP communication module is used to communicate with external web services through the get, post, put, patch and delete request methods. You can send data from your workstation to the web service, and process JSON responses. In the table below, you can find the properties you can configure for this communication module type.

Property	Description
Name	Name of the communication module ***Note*** *Next to the **Name** label, the object ID for the communication module is shown. You can copy it to use it when working with the Arkite API. For more information about the Arkite API, refer to chapter **18 The Arkite API** on page 292.*

Property	Description
Log Table	The table name in the logging database in which the information below will be logged for each HTTP request: – the URL – the Response Code – the first 1000 characters of the message ***Note*** *The logging database is configured in the server settings. For more information, refer to section **13.3 Changing the logging database, the logging database and the cleanup of old data** on page 264.*
Use OAuth2 Authentication	If checked, OAuth2 Authentication will be used. Additional properties will appear (see below). Based on these properties, an OAuth2 Access Token will be generated, which will be sent in the headers with each request.
OAuth2 Client ID	The OAuth2 Client ID
OAuth2 Client Secret	The OAuth2 Client Secret
OAuth2 Token URL	The URL which will be used to generate the Access Token using the client id and secret

Modbus TCP

The Modbus TCP communication module is used to communicate (read/write) with an external device through the Modbus TCP/IP protocol. In the table below, you can find the properties you can configure for this communication module type.

Property	Description
Name	Name of the communication module ***Note*** *Next to the **Name** label, the object ID for the communication module is shown. You can copy it to use it when working with the Arkite API. For more information about the Arkite API, refer to chapter **18 The Arkite API** on page 292.*
Log Table	The table name in the logging database in which all incoming and outgoing messages and exceptions will be logged. ***Note*** *The logging database is configured in the server settings. For more information, refer to section **13.3 Changing the logging database, the logging database and the cleanup of old data** on page 264.*
Modbus Device IP address	The IP address of the Modbus device.
Port	The port of the Modbus device. The default port for Modbus protocol is 502.
Polling rate	The interval in milliseconds between readings. The default polling rate is 250ms.
Slave Address	The address of the specific slave device. Typical value is 1.

MQTT

The MQTT communication module is used to communicate through MQTT (Message Queue Telemetry Transport), a lightweight and flexible network protocol that supports asynchronous communication between parties. The MQTT protocol defines two types of network entities: a message broker and a number of clients. An MQTT broker is a server that receives all messages from the clients and then routes the messages to the appropriate destination clients. Your workstation(s) will act as MQTT client(s). In the table below, you can find the properties you can configure for this communication module type.

Property	Description	
Name	Name of the communication module ***Note*** *Next to the **Name** label, the object ID for the communication module is shown. You can copy it to use it when working with the Arkite API. For more information about the Arkite API, refer to chapter **18 The Arkite API** on page 292.*	
Log Table	The table name in the logging database in which all incoming and outgoing messages and exceptions will be logged. ***Note*** *The logging database is configured in the server settings. For more information, refer to section **13.3 Changing the logging database, the logging database and the cleanup of old data** on page 264.*	
Variables	Variables that will be synced with the MQTT broker. For each variable, the information below can be configured:	
	Variable	Name of the variable
	Topic	Information on the broker and the clients are organized in a hierarchy of topics. Topic on the MQTT broker where the value will be read from or written to.
	Publish/Subscribe	– **Publish:** any change to the state of the variable will be sent to the MQTT broker and stored in the relevant topic. – **Subscribe:** any change to the topic value on the MQTT broker will be stored in the variable's state.
	State	The current value of the variable.
Broker Address	The IP address of the MQTT broker.	
Port	The port on which to connect to the MQTT broker. The default port for the MQTT protocol is 1883, or 8883 when TLS is used.	
Use credentials	If checked, a username and password are used to authenticate when connecting to the MQTT broker.	
Username	The user to connect to the MQTT broker.	
Password	The password used to authenticate the user with. ***Note*** *You can click the  button to make the password you've entered visible and verify if you haven't made any typing mistakes.*	
Use TLS	If checked, the communication is encrypted using the TLS (Transport Layer Security) protocol.	

Property	Description
Client ID	ID that is used to identify the workstation when connecting to the MQTT broker. It is advised to choose something meaningful to you, e.g. the workstation name (assembly station, packing table...) In case you don't fill in a client ID, Arkite will randomly generate one. ***Note*** *In some MQTT brokers, you need to configure the client IDs that can connect. Those MQTT brokers won't accept randomly generated client IDs. Please consult your MQTT broker's documentation for more information.*

OPC UA

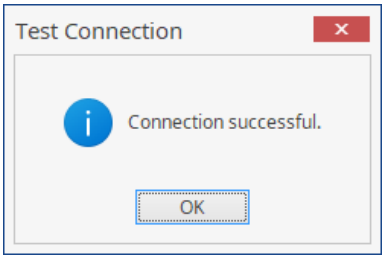
The OPC UA communication module is used to communicate using a Open Platform Communications (OPC) standard. More specifically, this module can synchronize variables on a workstation with a table on an OPC UA server containing a column for the variable name and one for the variable state.

In the table below, you can find the properties you can configure for this communication module type.

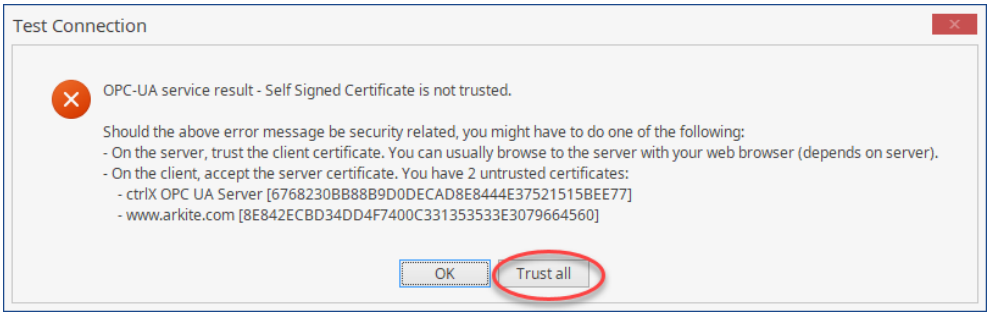
Property	Description
Name	Name of the communication module ***Note*** *Next to the **Name** label, the object ID for the communication module is shown. You can copy it to use it when working with the Arkite API. For more information about the Arkite API, refer to chapter **18 The Arkite API** on page 292.*
Log Table	The table name in the logging database in which all incoming and outgoing messages and exceptions will be logged. ***Note*** *The logging database is configured in the server settings. For more information, refer to section **13.3 Changing the logging database, the logging database and the cleanup of old data** on page 264.*
Server	The name of the OPC UA Server to communicate with. Should be in the format **opc.tcp://[ip address]:[port number].**
Security Settings	– **Auto (default):** Arkite will automatically select the **Security Mode** and **Policy** with the highest protection that is supported for communicating with the OPC UA server – **Manual:** Allows you to select the **Security Mode** and **Policy** yourself.
Security Mode	Only available when the **Manual** option has been selected under **Security Settings**. Specify whether the exchanged data should be signed and/or encrypted.
Policy	Only available when the **Manual** option has been selected under **Security Settings**. Specify the security policy to be used for signing and encrypting data.
Authentication	Indicate how to authenticate when connecting to the OPC UA server: – **Anonymous:** no authentication required – **Username:** authentication with user name and password
User Name	Only available when the **Username** option has been selected under **Authentication**. The user who is used to access the table on the OPC UA server.

Property	Description
Password	Only available when the **Username** option has been selected under **Authentication**. The password used to authenticate the user with. ***Note*** *You can click the button to make the password you've entered visible and verify if you haven't made any typing mistakes.*

When connecting to an OPC UA server, you can test your connection by clicking the **Test Connection** button. If communication is successful, you will get the popup below:



In case there are any untrusted server certificates on the client, there will be a **Trust All** button. Click the button to trust all untrusted server certificates.



Alternatively, you can also cut the files from **C:\Arkite\client\OPC\CertificateStores\RejectedCertificates** and paste them to **C:\Arkite\client\OPC\CertificateStores\UA Applications\certs**

You might also have trust the client certificate on the server. You can usually browse to the server with your web browser.

Open Protocol

The Open Protocol communication module uses the TCP/IP protocol to control an external Open Protocol compliant device (called a controller), for example a smart tool.

Note

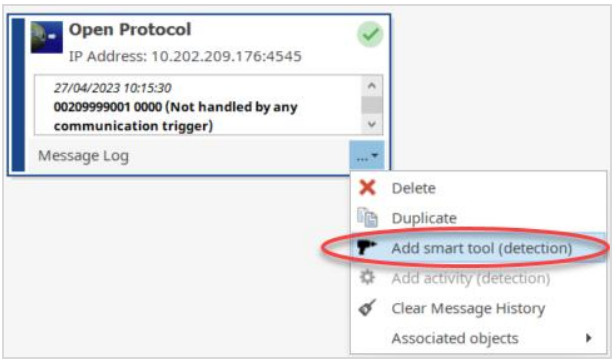
This module can only be used to communicate with devices on the same network.

In the table below, you can find the properties you can configure for this communication module type.

Property	Description
Name	Name of the communication module ***Note*** *Next to the **Name** label, the object ID for the communication module is shown. You can copy it to use it when working with the Arkite API. For more information about the Arkite API, refer to chapter **18 The Arkite API** on page 292.*
Log Table	The table name in the logging database in which all incoming and outgoing messages and exceptions will be logged. ***Note*** *The logging database is configured in the server settings. For more information, refer to section **13.3 Changing the logging database, the logging database and the cleanup of old data** on page 264.*
Automatic Smart Tool Settings	If this checkbox is enabled, the required variables for communication with a smart tool will be generated and configured.
IP Address	The IP address of the remove device, for example: 10.0.11.12.
Device Type	– When using Atlas Copco's MicroTorque Controller MTF6000, select **MTFocus 6000**. – For all other devices, select **Generic**.
Port	The TCP/IP port to use. Default for open protocol communication is 4545.
Start communication revision	When an open protocol communication actuator is first sparked, a 'start communication' message will be sent to the controller. This message contains a 3 digit revision. For most controllers, this revision is '001'. Check the manual of your controller to check for divergent revisions.
End communication revision	When the module is paused, an 'end communication' message will be sent to the controller. This message contains a 3 digit revision. For most controllers, this revision is '001'. Check the manual of your controller to check for divergent revisions.
Stop String Type	Every message will be ended by a special 'ending character'. The default value is null (null character). Other types include a CRLF (carriage return line feed) and #.
Log 'Keep Alive' messages	Every incoming, outgoing and exception message will be logged. However, to keep the connection with the controller open, 'keep alive' messages will be sent (and received) every ten seconds. In most situations, you don't want to see these messages cluttering your message stream. However, if you would like to see these messages, please check this checkbox.
Open Protocol Device Test	Before an open protocol device can be used, a parameter set must always be selected. Provide a parameter set in the text input field (typically consisting of three digits, e.g. 000, 001, 002....) and click **Set Parameter**. Once you have done that: – Click the **Enable Tool** button to enable the tool. – Click the **Disable Tool** button to disable the tool. By clicking these buttons, the communication towards the tool can be tested.

Once you've configured the open protocol module, you can set up a smart tool that uses the module:

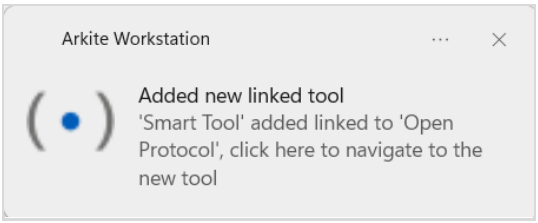
1. Select the card for the **Open Protocol** module, click the **...** button in the lower-right corner of the card and select **Add Smart Tool (detection)**.



Note

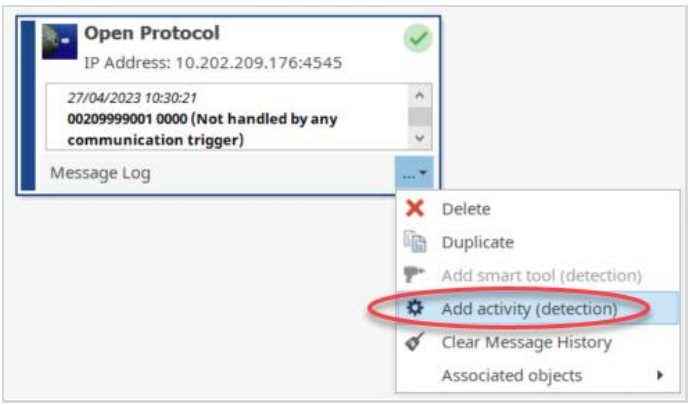
*Only **one** tool can be linked to a single **Open Protocol** module. To use multiple tools, create multiple **Open Protocol** modules.*

2. A smart tool linked to the **Open Protocol** module is created on the **Detections** tab. Click the toast message in the lower-right corner of the screen to navigate to it.

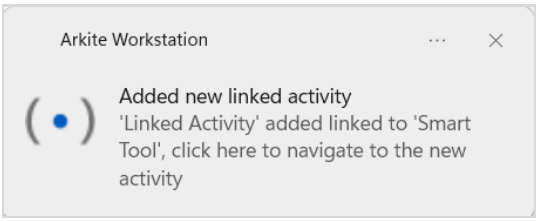


3. (optional) When using a **Validate** or **Vision** license, teach the tool as described in section **Teach an object, picking bin, tool or virtual button** on page 135. This way the system can detect when the tool is taken away or placed back.

4. On the **Connectivity** tab, select the card for the **Open Protocol** module, click the **...** button in the lower-right corner of the card and select **Add Activity (detection)**.



5. An activity linked to the smart tool you set up in step 5 is created on the **Detections** tab. Click the toast message in the lower-right corner of the screen to navigate to it.



6. (optional) When using a **Validate** or **Vision** license, teach the activity as described in section **Teach an activity** on page 142.
   This way the system can detect when the tool is being used at the desired location.
7. Add more activities as needed.

S7 PLC

The S7 PLC communication module allows to communicate with an external S7 compliant PLC device. It makes use of the TCP/IP network protocol.

Supported S7 devices are: Siemens Logo 0BA8, S7-200, S7 200 Smart, S7-300, S7-400, S7-1200, S7-1500.

Note

This module can only be used to communicate with devices on the same network.

In the table below, you can find the properties you can configure for this communication module type.

Property	Description	
Name	Name of the communication module ***Note*** *Next to the **Name** label, the object ID for the communication module is shown. You can copy it to use it when working with the Arkite API. For more information about the Arkite API, refer to chapter **18 The Arkite API** on page 292.*	
Log Table	The table name in the logging database in which all incoming and outgoing messages and exceptions will be logged. ***Note*** *The logging database is configured in the server settings. For more information, refer to section **13.3 Changing the logging database, the logging database and the cleanup of old data** on page 264.*	
Variables	Variables that will be synced with the PLC. For each variable, the information below can be configured:	
	Variable	Name of the variable
	DB	Data Block in the PLC where the value will be read from or written to.

Property	Description	
	Data Type	Can be any of the following: – **Bit:** Boolean value. Can be TRUE or FALSE, 0 or 1 – **Byte**: bit string of 8 bits. – **Word:** bit string of 16 bits. – **DWord:** bit string of 32 bits. – **Int:** signed integer with a length of 16 bits. Values can be -32 768 to +32 767. – **DInt:** signed integer with a length of 32 bits. Values can be -2 147 483 648 to +2 147 483 647. – **Real:** signed floating number with a length of 32 bits. The precision defined by the IEEE754 standard is stored. Additionally specified decimals are rounded off. – **LReal:** signed floating number with a length of 64 bits. The precision defined by the IEEE754 standard is stored. Additionally specified decimals are rounded off. – **String:** ASCII character. – **S7String:** ASCII character string incl. special characters. The length of the string can be defined, and can contain up to 254 characters. – **S7WString:** Unicode character string. The default length is 254 characters, but you can define a length up to 16382 characters. – **Timer:** contains a timer operation – **Counter:** contains a counter operation – **DateTime:** stores date and time information in BCD format (year-month-day-hour:minute:second:millisecond) – **DateTimeLong:** stores date and time information in a predefined structure (Year-Month-Day-Hour:Minute:Second.Nanoseconds)
	Offset	Relative address in the datablock where the value will be read from or written to. Addresses in a datablock are defined by a byte and a bit. – For the data type **Bit**, use the format **byte.bit.** For example, "10.4" will read from or write to bit 4 of byte 10. – For the data types **String** or **WString**, use the pattern **byte.bit[length].** For example, "20.0[100]" will read or write 100 bytes starting from byte 20. If you don't provide a length, the default length of the data type will be used. – For all other data types, use the format **byte.** For example: "54" will start reading from or writing to byte 54. The number of bytes that will be written or read depends on the length of the data type. In case the data type is dint, the length is 4 bits. Bytes 54 to 57 will be read or written.
	I/O	Indicates if the variable will be written to or read from the PLC.
	State	The current value of the variable. Accepted variables depend on the selected **Data Type**.
	Log Message	If checked, the sent or received data will be logged in the Message History.
	Note *For more detailed information, consult the documentation of your PLC programming application.*	

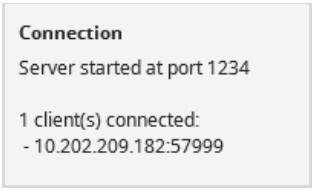
Property	Description
S7 Type	The type of the connected S7 device.
IP Address	The IP address of the connected S7 device.
Rack number	The rack number of the connected S7 device. Default value is 0.
Slot number	The slot number of the connected S7 device. Default value for S7-200, S7-300 and S7-400 is 2, S7-1200 and S7-1500 is 1.
Polling rate	The interval in ms with which the variables will be synced with the PLC.

TCP/IP Server

The TCP/IP Server module is used to send messages to and receive messages from TCP/IP clients over the TCP/IP protocol.

In the table below, you can find the properties you can configure for this communication module type.

Property	Description
Name	Name of the communication module ***Note*** *Next to the **Name** label, the object ID for the communication module is shown. You can copy it to use it when working with the Arkite API. For more information about the Arkite API, refer to chapter **18 The Arkite API** on page 292.*
Log table	The table name in the logging database in which all incoming and outgoing messages and exceptions will be logged. ***Note*** *The logging database is configured in the server settings. For more information, refer to section **13.3 Changing the logging database, the logging database and the cleanup of old data** on page 264.*
Receive variable	The variable that will be used to store the received messages.
Encoding	Defines how the data being sent or received is represented: – **String:** data is interpreted as a text string, using UTF encoding. – **Hex:** data is represented in hexadecimal format, where each byte is shown as a two-digit hex number (0-9, a-f)
Message boundary	Defines how the end of a message is determined: – **TCP/IP packet**: each TCP packet is interpreted as a different message – **Delimiter**: a message ends with a specific character/sequence. For **String** encoding, the delimiter is defined in the **Delimiter** property. For **Hex** encoding, the delimiter is 0. – **Fixed Length:** a message consists of a fixed number of bytes.
Delimiter	The delimiter that defines the end of a message: – Space – Tab – New line – Carriage return – Carriage return + new line Only available if **String** is selected in the **Encoding** property and **Delimiter** is selected in the **Message Boundary** property. For **Hex** encoding, the delimiter is 0.
Fixed length (bytes)	The message length in bytes. Only available if **Fixed Length** is selected in the **Message Boundary** property.

Property	Description
Padding character (left)	Character that is added on the left of a sent message that is shorter than the defined fixed length. Only available if **String** is selected in the **Encoding** property and **Fixed Length** is selected in the **Message Boundary** property. For **Hex** encoding, the padding character is 0.
Port	The TCP port to listen on. The client device should send its messages to this port.
Connection	The status of the TCP/IP server and the clients that are connected to it will be listed here: 

TCP/IP Client

The TCP/IP client module is used to send messages to and receive messages from a TCP/IP server over the TCP/IP protocol.

In the table below, you can find the properties you can configure for this communication module type.

Property	Description
Name	Name of the communication module ***Note*** *Next to the **Name** label, the object ID for the communication module is shown. You can copy it to use it when working with the Arkite API. For more information about the Arkite API, refer to chapter **18 The Arkite API** on page 292.*
Log table	The table name in the logging database in which all incoming and outgoing messages and exceptions will be logged. ***Note*** *The logging database is configured in the server settings. For more information, refer to section **13.3 Changing the logging database, the logging database and the cleanup of old data** on page 264.*
Receive variable	The variable that will be used to store the received messages.
Encoding	Defines how the data being sent or received is represented: – **String:** data is interpreted as a text string, using UTF encoding. – **Hex:** data is represented in hexadecimal format, where each byte is shown as a two-digit hex number (0-9, a-f)
Message boundary	Defines how the end of a message is determined: – **TCP/IP packet**: each TCP packet is interpreted as a different message – **Delimiter**: a message ends with a specific character/sequence. For **String** encoding, the delimiter is defined in the **Delimiter** property. For **Hex** encoding, the delimiter is 0. – **Fixed Length:** a message consists of a fixed number of bytes.

Property	Description
Delimiter	The delimiter that defines the end of a message: – Space – Tab – New line – Carriage return – Carriage return + new line Only available if **String** is selected in the **Encoding** property and **Delimiter** is selected in the **Message Boundary** property. For **Hex** encoding, the delimiter is 0.
Fixed length (bytes)	The message length in bytes. Only available if **Fixed Length** is selected in the **Message Boundary** property.
Padding character (left)	Character that is added on the left of a sent message that is shorter than the defined fixed length. Only available if **String** is selected in the **Encoding** property and **Fixed Length** is selected in the **Message Boundary** property. For **Hex** encoding, the padding character is 0.
IP address	The IP address of the TCP/IP server you want to connect to.
Port	The port the TCP/IP Server listens on.
Connection	The status of the connection to the TCP/IP server is shown here. 

Variable Storage

The Variable Storage communication module is a communication module used to communicate with a database. More specifically, this module can synchronize variables with a table containing a column for the variable name and one for the variable state. In the table below, you can find the properties you can configure for this communication module type.

Property	Description
Name	Name of the communication module ***Note*** *Next to the **Name** label, the object ID for the communication module is shown. You can copy it to use it when working with the Arkite API. For more information about the Arkite API, refer to chapter **18 The Arkite API** on page 292.*
Log Table	The table name in the logging database in which all incoming and outgoing messages and exceptions will be logged. ***Note*** *The logging database is configured in the server settings. For more information, refer to section **13.3 Changing the logging database, the logging database and the cleanup of old data** on page 264.*

Property	Description
Database Type	Database type. Currently supported types are: – MS SQL Server – MySQL – Oracle Database – SQLite – Maria DB
Server	The name or URL of the server running the database.
User	The user who is used to query the database. This user should have the proper rights to access the table defined below.
Password	The password used to authenticate the user with. ***Note*** *You can click the*  *button to make the password you've entered visible and verify if you haven't made any typing mistakes.*
Database	The name of the database on the server. ***Note*** *In case of SQLite, this should be the path to the database file on the server's file system.* *For example:* ***C:\Arkite\server\mydatabase.db*** *Shared drives are not supported.*
Table	The table on the database, which will contain the event values.
Variable Name Column	The column of the table containing the name of the variable(s)
Variable State Column	The column of the table containing the state of the variable(s)

7.3.3 Adding a communication module

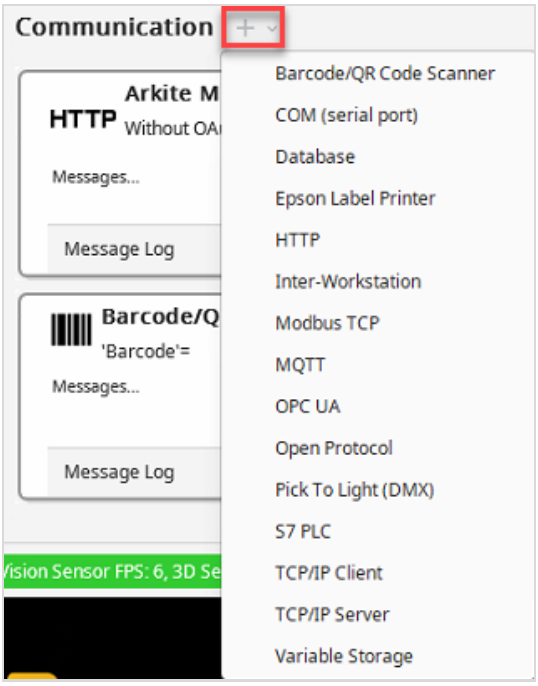
What?

Before you can start using a communication module, you need to add and configure it in the Arkite software.

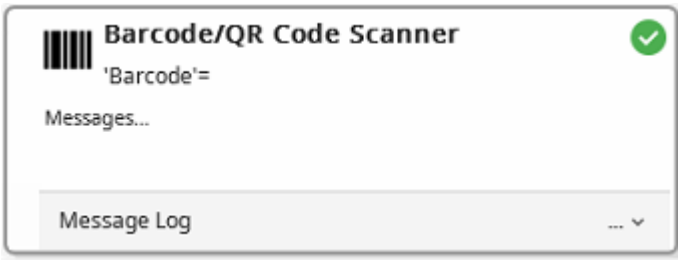
Steps

To add a communication module, execute the following steps:

1. In the **Connectivity** tab under **Communication**, click the **+** button and select the communication module you want to add.



   A new card for the communication module is added to the list of communication modules.



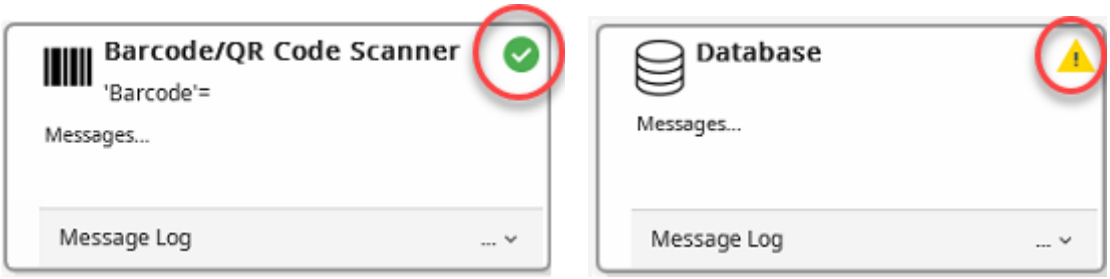
2. In the **Properties** pane, configure the properties for the communication module.

   Note

   The properties that need to be configured depend on the communication module type you have added. For an overview of each property and its meaning, refer to section ***7.3.2 Communication module types*** *on page 198.*

Result

The communication module has been configured and will immediately start running. Data will be received through the communication module. The status icon in the upper right corner of the card indicates if the module is correctly configured and running.



For the communication module to send data, you need to add a **process step**, defining which data to send. You can also configure a **trigger** for the communication module to start a process when receiving data.

7.3.4 Configuring lights for a Pick To Light (DMX) communication module

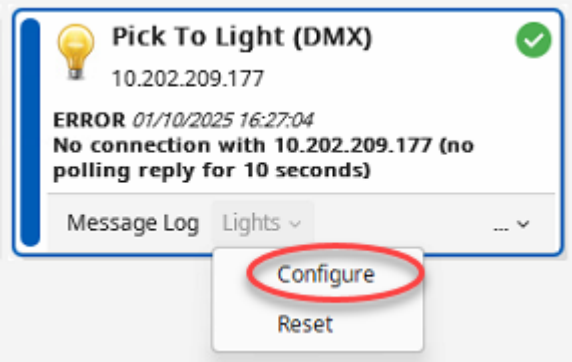
What

After you have connected a DMX device through a Pick To Light (DMX) communication module, you can configure different lights on that device by defining which channels to use.

After configuring a light, you can control it by linking it to a detection. This way, you can use the DMX light for operator guidance instead of a projection. Alternatively, you can also control the light through a **Send Communication** process step.

Steps

1. On the **Connectivity** tab, click the **Configure Lights** button in the **Properties** tab or click **Ligths** and then **Configure** on the communication module card.



2. In the **Configure Lights** window, click the ➕ button.
   A light is added with default name **L1**.
3. Edit the properties for the DMX Light:

Property	Description
Name	Name of the DMX light ***Note*** *Next to the **Name** label, the object ID for the communication module is shown. You can copy it to use it when working with the Arkite API. For more information about the Arkite API, refer to chapter **18 The Arkite API** on page 292.*

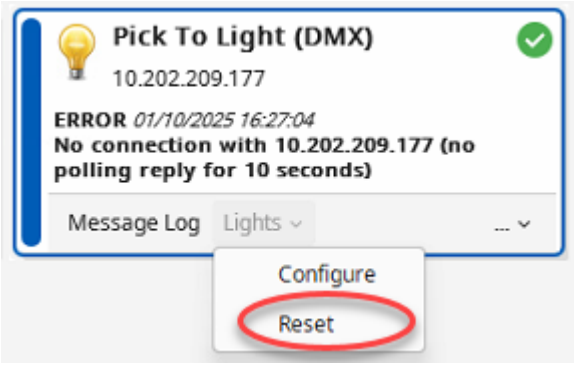
Property	Description
Universe	Only available if you are connecting to the DMX lights over the Art-Net ethernet protocol, allowing you to control multiple DMX universes, each consisting of 512 channels.
Start Address	The first channel on the DMX device that will be used to control the light. By default, the first unused channel will be selected. ***Note*** *When using Art-Net, channels are within the range 0-511.* *When using DMX USB, channels are within the range 1-512.*
Number of channels	The number of channels on the DMX device that will be used to control the light. By default, 4 channels will be used.
Channel Mapping	Depending on the **Start Address** and the **Number of Channels**, a table is shown with a row for each channel. Select what each channel will be used for in the configuration of an RGB color value and light intensity (Dimmer). For channels that should not be used, select **None**.
Max Dimmer Range	Configure the value that corresponds to the light's maximum light intensity. By default 255.

Test your configuration

After configuring your light, you can use each channel using the **R, G** and **B** toggle buttons. The light will emit the corresponding color until the button is pressed again, the window is closed or another light is selected To test the **Dimmer** channel, use the slider next to the toggle buttons. Use the **Off** button to completely turn off the light manually.

Reset Lights

To reset all lights linked to the Pick To Light (DMX) module to their default value, typically off, click **Lights** and then **Reset** on the communication module.



7.3.5 Configuring a step for the communication module to send data

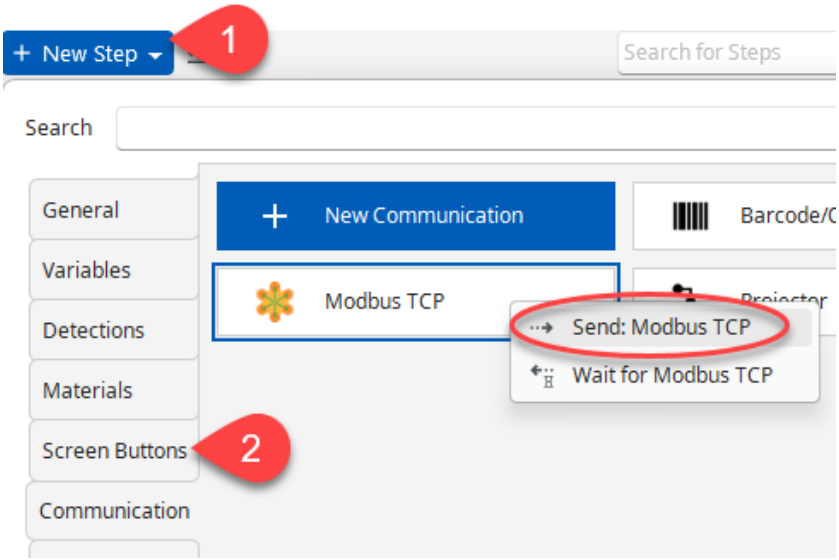
What?

In order to enable the communication module to send data, you need to configure a process step, defining when which data will be sent.

Steps

1. Navigate to the **Steps** or **Processes** tab.

2. Click the **New Step** button.
3. Select the **Communication** subtab, click the communication module you want to wait for and select the **Send:** action from the context menu.



Note

When using Open Protocol communication, you can select the action you want to execute from the context menu. The correct message will then be preconfigured. The available options are:

- *Enable Tool*
- *Disable Tool*
- *Select Parameter*
- *Select Job*
- *Other...: this will allow you to specify your own message*

*Before an open protocol device can be used, a parameter set must always be selected. If the tool does not react after sending an **Enable Tool** message, make sure a parameter set is selected. Send a **Select Parameter** message first.*

Note

*When using Pick To Light (DMX) communication, you can select the light you want to control from the context menu. In most use cases, you will control a DMX light by linking it to a detection. For more information, refer to section **5.5 Adding detections** on page 129.*

Note

*Apart from the communication modules you have added in the **Connectivity** tab, you can also select a **Projector** display that is available in the **Displays** section of the **Connectivity** tab here.*

4. Configure the properties for the step.
   The properties that need to be configured depend on the communication module type. You can find an overview of properties per communication module type below. By clicking the **Comment**

icon next to the **Name** property, you can add a few words of explanation regarding the use of the step.

- **COM (serial port)**

Property	Description
Message to send	The message you would like to send. To insert dynamic content, click the {x} icon at the location where you want to include the content and select the variable containing the dynamic content. Alternatively, you can also type the variable name between double braces, i.e. *{{Variable name}}*. ***Tip*** *For long texts, click the*  *icon to enlarge the text box.*

- **Database**

Property	Description
Query	Enter the query you want to run on the database. To use the value of a variable as query, click the {x} icon and select the variable containing (part of) the query. Alternatively, you can also type the variable name between double braces, i.e. *{{Variable name}}*. You can also use a variable within a variable. For example, if the value of a variable **Index** is 01, then *{{Variable{{Index}}}}* will resolve into the value of the variable **Variable01**. When communicating with an SQL server database, you can also call stored procedures in that database. ***Tip*** *For long texts, click the*  *icon to enlarge the text box.*
Result	– **All Rows:** all rows that are returned are stored in a JSON array, with a maximum of 10 000 rows. – **First Row Only:** only the first row in the result is taken into account. For every column in the result, a variable with the same name is created or updated. In the **Prefix for column variables** property, you can optionally specify a prefix for those variable names. ***Note*** *For queries where no result is expected, for example an UPDATE statement, use the **All Rows** option. If you select the **First Row Only** option, the query will also execute successfully, but a warning message will be generated because no result was returned.*
Variable for result	Only available when the **All Rows** option is selected. The variable in which the result of the query will be stored as a JSON array.
Prefix for column variables	Only available when the **First Row Only** option is selected. Names of newly created variables (as a result of a 'select' query) will be prefixed with this value

- **Pick To Light (DMX)**

Property	Description
DMX Light	The DMX light you want to control.
Configure Lights	Click this button to configure different lights on the connected DMX device. For more information, refer to section **7.3.4 Configuring lights for a Pick To Light (DMX) communication module** on page 216.

Property	Description
Color	The color you want the DMX light to show. This color will be shown until the color of the light is changed or until the light is put out by another **Send Communication** step or when a detection linked to the light is activated. ***Note*** *You can either set a fixed color, select two alternating colors, define the color based on the value of a variable or define the color based on a template. For more information, refer to section **Managing visualization or DMX light colors** on page 112.*

Note

*In most use cases, you will control a DMX light by linking it to a detection. For more information, refer to section **5.5 Adding detections** on page 129. If you are adding a **Send Communication** step to control a light that is linked to a detection, a warning message will be shown.*

- **Epson Label Printer**

Property	Description
Variables	The variables that are going to be printed. Check the **All Variables** checkbox to print all variables.
Print Variable Name	When checked, the variable names are printed together with the current state.
Divider Text	The text between the variable name and its current state.
Print Logo at Top	When checked, the logo is printed, specified as key1=32 key2=32 defined in the printer.
Print Barcode	When checked, a barcode is printed.
Barcode	The text that will be encoded and printed as a barcode. To insert dynamic content, click the {x} icon at the location where you want to include the content and select the variable containing the dynamic content. Alternatively, you can also type the variable name between double braces, i.e. *{{Variable name}}*. ***Tip*** *For long texts, click the icon to enlarge the text box.* ***Note*** *The barcode is restricted up to 15 numeric characters.*
Print QR	When checked, a QR code is printed.
QR Code	The text that will be encoded and printed as a QR code (QR Code Model 2). Valid characters include alphanumeric, _, -, brackets... To insert dynamic content, click the {x} icon at the location where you want to include the content and select the variable containing the dynamic content. Alternatively, you can also type the variable name between double braces, i.e. *{{Variable name}}*. ***Tip*** *For long texts, click the icon to enlarge the text box.* ***Note*** *Some special characters are not supported.*
Cut at Bottom	When checked, the printer will cut the label at the bottom.

- **Inter-Workstation**

Property	Description
Variables	When the variable(s) you indicate here are received from the other workstation, the process will be started. Check the **All Variables** checkbox to start the process when receiving any variable.

- **HTTP**

Property	Description
URL	The URL of the web service to communicate with, for example: http://myservice.com/logger To use the value of a variable in the URL, click the {x} icon and select the variable containing (part of) the URL. Alternatively, you can also type the variable name between double braces, i.e. *{{Variable name}}*. ***Tip*** *For long texts, click the*  *icon to enlarge the text box.*
Variables for Query	The variables you specify here are encoded in the query part of the URL. If the **Variables for Query** list contains one element **variableName** the request could look like this: http://myservice/logger?variableName=variableValue
Resulting URL	The requests that results from the contents of the **URL** and **Variable for Query** properties.
Request Type	The HTTP request type. You can select one of the values below: – **Get:** this method is used to retrieve the representation of a resource – **Post:** this method is used to create new resources – **Put:** this method is typically used to update capabilities of a resource. The request body contains the newly-updated representation of the original resource. – **Patch:** this method is typically used to modify capabilities of a resource. The request body contains the changes to the resource, not the complete resource. – **Delete**: this method is used to delete a resource identified by a URI
Body Content Type	Only visible when the **Request Type** is **Post, Put** or **Patch**. Determines the content type of the body. There are currently two options: – **Raw** (=default option): You can specify the body of the request in the **Raw Body Content** property. – **FormData:** The variable names and their current states you specify in the **Variables for Form Data** property will be sent in FormData format.
Variables for Form Data	When the **Body Content Type** is **FormData**, the variables you specify here are encoded in the body of the request in FormData format.
Content Type	When **Body Content Type** is **Raw**, the content type of the raw data should be specified. By default, this is 'application/json'

Property	Description
Raw Body Content	When **Body Content Type** is **Raw,** you can compose the body of the request here. To enter the value of a variable in the body content, click the {x} icon and select the variable you want to insert. Alternatively, you can also type the variable name between double braces, i.e. *{{Variable name}}*. ***Tip*** *For long texts, click the*  *icon to enlarge the text box.*
Extra Headers	Pass additional information by including extra headers in your request. An HTTP header consists of its case-insensitive name followed by a colon (:), then by its value. In the **Extra Headers** property you can add one or more names with their corresponding values. You don't need to add the colons. To use the value of a variable in the **Value** column, click the {x} icon and select the variable you want to insert. Alternatively, you can also type the variable name between double braces, i.e. *{{Variable name}}*.
Response Variable	The variable in which the response data will be stored.
Variable for Response Code	The variable in which the response code (HTTP statusCode) will be stored, allowing you to verify the successful execution of the request.

- **Modbus TCP**

Property	Description
Address Type	The object type of the write action. It is possible to write: – **Coils**: read-write entity with a size of 1 bit (ON/OFF) – **Holding registers:** read-write entity with a size of 16 bits
Address	The address of the object which will be changed to the specified state.
State to set	Only available when the **Address Type** is **coil**. The state to which the object will be changed. The state can be either on or off.
Value	Only available when the **Address Type** is **holding register**. The value to which the object will be changed. The value can be a number between 0 and 65 535.

- **Open Protocol**

Property	Description
Header	Depending on the option you have selected in the context menu when adding the process step, the header will be prefilled with the Open Protocol message that is required to complete the action of your choice: – Enable Tool: 00200043 (Generic) / 0023022500100000000 008 (MTFocus 6000) – Disable Tool: 00200042 (Generic) / 0023022400100000000 008 (MTFocus 6000) – Select Parameter: 00230018001 (Generic and MTFocus 6000) – Select Job: 00220038001 (Generic and MTFocus 6000) – Other...: the header remains empty, you can specify a message yourself. ***Note*** *Before an open protocol device can be used, a parameter set must always be selected. If the tool does not react after sending an **Enable Tool** message, make sure a parameter set is selected. Send a **Select Parameter** message first.* To have a full overview of all message types that are supported by Open Protocol, you can consult the **Open Protocol specification documents** embedded below. **Error! Objects cannot be created from editing field codes.** MTFocus 6000 Open Protocol messages consist of 20 digits (for example. 00200001000000000000 for communication start). – Digit 1 – 4: Length – Digit 5 – 8: MID – Digit 9 – 11: Revision – Digit 12: No ack flag – Digit 13 – 14: Station ID – Digit 15 – 16: Spindle ID – Digit 17 – 20: Spare ***Note*** *If the header is less than 20 digits, it will be padded with spaces until it contains 20 digits before being sent.* *If the header is more than 20 digits, the excess will be considered data.*
Data	Insert the data you want to be appended to the header. Together with the 'ending character', this will be the full message to be sent to the controller. To insert dynamic content, click the {x} icon at the location where you want to include the content and select the variable containing the dynamic content. Alternatively, you can also type the variable name between double braces, i.e. *{{Variable name}}*. ***Tip*** *For long texts, click the icon to enlarge the text box.* ***Note*** *The ending character is defined by the property **Stop String Type** configured in the communication module. Refer to section **Open Protocol** on 206 for more information.*

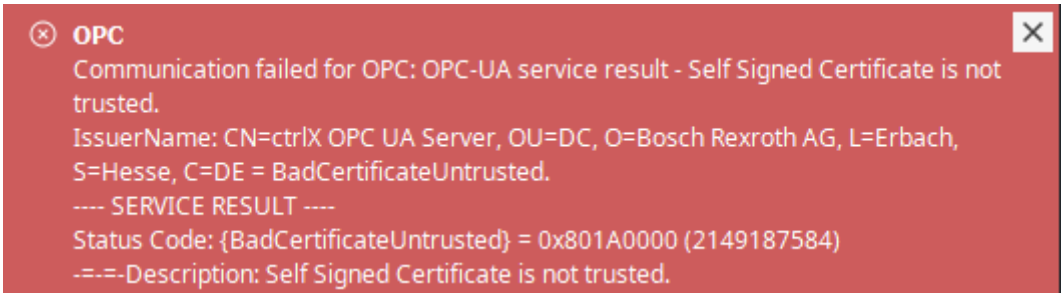
- **OPC UA**

Property	Description
Type	The action you want to do on the OPC UA Server: – Read – Write
Variable for result	When **reading** a node/item, the value of this variable will be updated with the value from node/item on the OPC UA server.
Data to write	When **writing** to a node/item, enter the data to write to the node/item on the OPC UA server. To insert dynamic content, click the {x} icon at the location where you want to include the content and select the variable containing the dynamic content. Alternatively, you can also type the variable name between double braces, i.e. *{{Variable name}}*. ***Tip*** *For long texts, click the icon to enlarge the text box.*
Node id	The data item inside the OPC UA server. Click the **Find Node** button to select a node.

Note

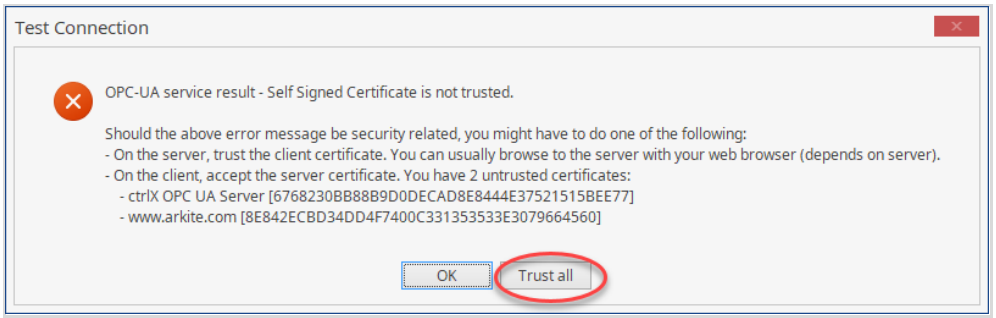
- ***Trusting Server Certificates on the client***

  If you are getting a popup or a notification about missing or non-trusted certificates, you are missing OPC Server Certificates.



  *On the **Connectivity** tab, select the **OPC UA** communication module and click the **Test Connection** button in the **Properties** pane.*

  *Click the **Trust All** button to trust all server certificates on the client.*



  *Alternatively, you can also cut the files from **C:\Arkite\client\OPC\CertificateStores\RejectedCertificates** and paste them to **C:\Arkite\client\OPC\CertificateStores\UA Applications\certs***

- ***Trusting Client certificates on server***

*If you get an exception **The server may have rejected the connection because it does not trust the client**, you should trust the client certificate on the server. You can usually browse to the server with your web browser.*

- **Projector**

Property	Description
Projector Command	The commands below can be sent to the projector: – **Power On:** turns on the projector – **Power Off:** turns off the projector – **Switch To VGA:** switches the active input of the projector to VGA – **Switch To HDMI1:** switches the active input of the projector to HDMI1 – **Switch To HDMI2:** switches the active input of the projector to HDMI2 – **Mute On:** mutes the video and audio of the projector – **Mute Off:** stops muting the video and audio of the projector – **Get Current State:** returns the (power) state of the projector – **Get Current Input:** returns the active input of the projector – **Get Error Code:** returns the error code of the projector (returns "000000" when there is no error) – **Get Lamp Hours:** returns the number of hours the projector has been active
Variable for answer	The variable in which the responses of the projector will be stored.

- **TCP/IP Server**

Property	Description
Data to send	The message you would like to send to the connected TCP/IP clients. To insert dynamic content, click the {x} icon at the location where you want to include the content and select the variable containing the dynamic content. Alternatively, you can also type the variable name between double braces, i.e. *{{Variable name}}*. ***Tip*** *For long texts, click the*  *icon to enlarge the text box.*

- **TCP/IP Client**

Property	Description
Data to send	The message you would like to send to the TCP/IP server. To insert dynamic content, click the {x} icon at the location where you want to include the content and select the variable containing the dynamic content. Alternatively, you can also type the variable name between double braces, i.e. *{{Variable name}}*. ***Tip*** *For long texts, click the*  *icon to enlarge the text box.*

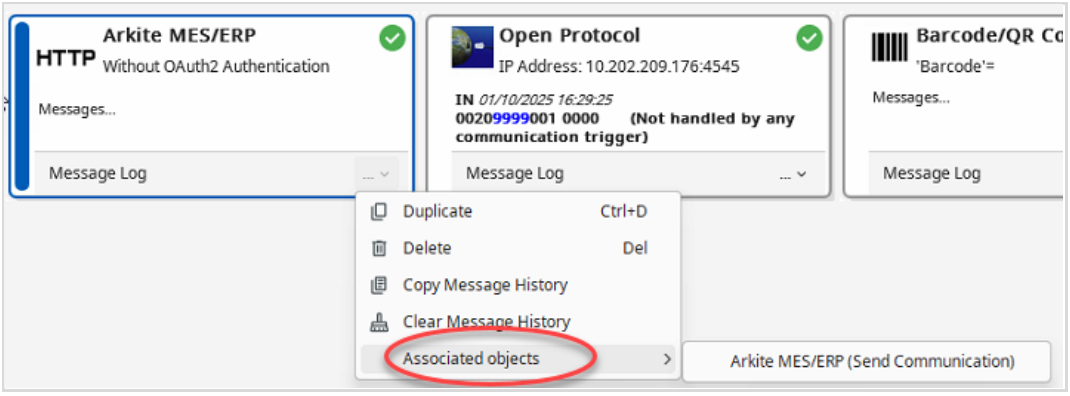
- **Variable Storage**

Property	Description
Variables	The variables that you want to fetch, send or remove to/from the database. Check the **All Variables** checkbox to insert/update/delete all variables.
Type	The action that will be performed on the database: – **Select:** Read variables from the database and set the state of the project variables to the values present in the database – **InsertOrUpdate:** Insert the project variables into the database, or, if they are already present in the database, update their value – **Remove:** Remove the specified variables from the database

Result

The communication module has been configured to send data.

On the **Connectivity** tab, the outgoing communication will be listed in the **Associated objects** for the communication module. Click the **...** button in the lower-right corner of the card and select **Associated objects.** The screenshot below shows an example for the **HTTP** communication module.



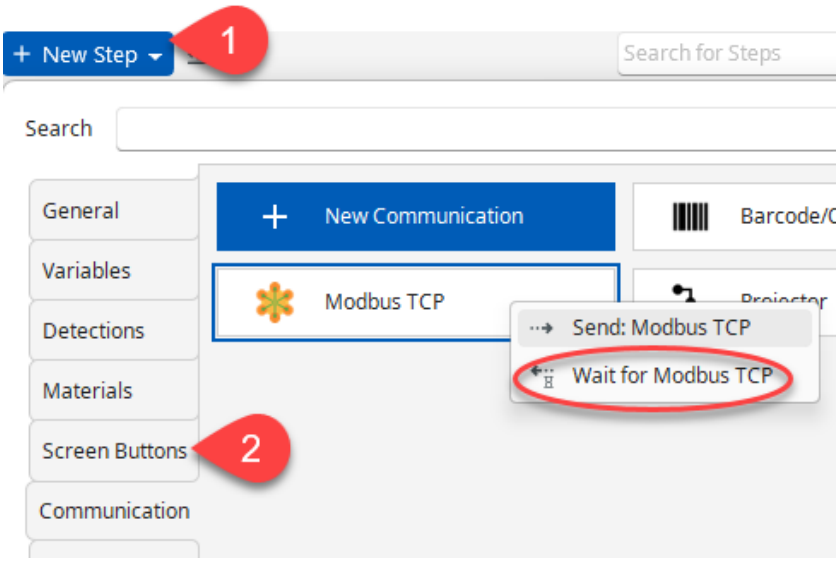
7.3.6 Configuring a step to wait for the receival of data from the communication module

What?

You can add a step in your job or process that pauses your job or process until data is received from a communication module. For example, you can configure the process to wait for a barcode being scanned.

Steps

1. Navigate to the **Steps** or **Processes** tab.
2. Click the **New Step** button.
3. Select the **Communication** subtab, click the communication module you want to wait for and select the **Wait for…** action from the context menu.



4. Configure the properties for the step.

   The properties that need to be configured depend on the communication module type. You can find an overview of properties per communication module type below. By clicking the **Comment** icon next to the **Name** property, you can add a few words of explanation regarding the use of the step.

   - **Open Protocol**

Property	Description
MID to filter on	The MID (which is character 5 to 8 of the message header) to filter the messages on. If this field is filled in, only messages with a matching MID will be captured and the job or process will only continue when the matching MID is received. If this field is left empty, all messages will be captured. The process will continue whenever a message is received.
Variable for message	The variable that will be used to store the received messages.

   - **Inter-Workstation**

Property	Description
Variables	When the variable(s) you indicate here are received from the other workstation, the job or process will continue. Check the **All Variables** checkbox to continue the job or process when receiving any variable.

   For the modules below, no additional properties need to be configured. The job or process will continue as soon as any message is received on the corresponding communication module:

   - Barcode Scanner
   - COM (serial port)
   - Modbus TCP
   - TCP/IP Server
   - TCP/IP Client
   - MQTT
   - S7 PLC

Result

The process or job has been configured to wait for the receival of data from the communication module.

On the **Connectivity** tab, the incoming communication will be listed in the **Associated objects** for the communication module. Click the **...** button in the lower-right corner of the card and select **Associated objects.**

7.3.7 Configuring a trigger for the communication module to start a process when receiving data

What?

You can configure a process to start when data is received through the communication module you have added. In order to do so, you need to configure a communication trigger.

Steps

1. On the **Processes** tab, select the process you want to trigger by the receival of communication.
2. In the **Properties** pane, under **Start by trigger**, click the + button.
3. Select **Receive Communication**.
4. Under **Trigger Properties**, select the module you want to configure as a trigger from the **Communication Module** dropdown list.
5. Configure the properties for the trigger.

   The properties that need to be configured depend on the communication module type. You can find an overview of properties per communication module type below.

   - **Inter-Workstation**

Property	Description
Variables	The variables that are going to be received. Check the **All Variables** checkbox to send all variables.

   - **Modbus TCP**

Property	Description
Variable	The variable that will be used to store the received messages.
Address Type	The object type of the read action. It is possible to read: – **Discrete inputs:** read-only entity with a size of 1 bit (ON/OFF) – **Coils**: read-write entity with a size of 1 bit (ON/OFF) – **Input registers:** read-only entity with a size of 16 bits (0-65 535) – **Holding registers:** read-write entity with a size of 16 bits (0-65 535)
Address	The address of the object from which will be read. The process will be started when a message is received from the address specified here, with the correct type.

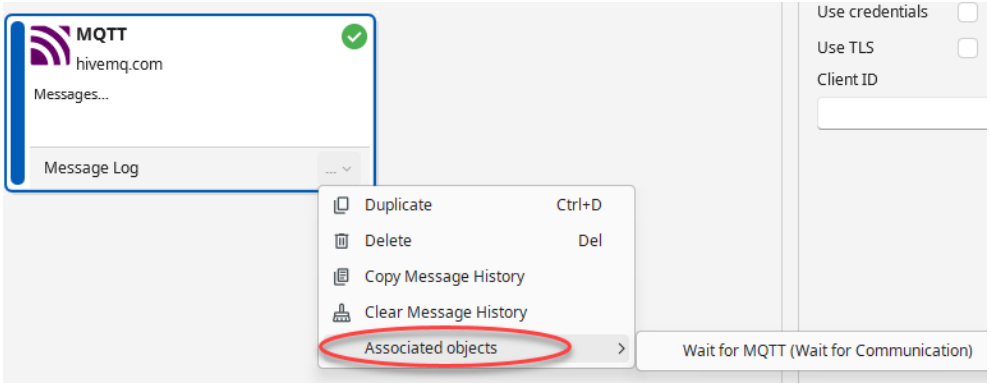
   - **Open Protocol**

Property	Description
MID to filter on	The MID (which is character 5 to 8 of the message header) to filter the messages on. If this field is filled in, only messages with a matching MID will be captured and the process will only be started when the matching MID is received. If this field is left empty, all messages will be captured. The process will started whenever a message is received.
Variable	The variable that will be used to store the received messages.

Result

The communication module has been configured to receive data.

On the **Connectivity** tab, the incoming communication will be listed in the **Associated objects** for the communication module. Click the **...** button in the lower-right corner of the card and select **Associated objects.** The screenshot below shows an example for the **MQTT** communication module.



7.3.8 Following up on the communication that has been exchanged

What?

Once you have correctly set up the communication module and the associated triggers or actuators, you can follow up on the exchanged information.

How to follow up on exchanged information

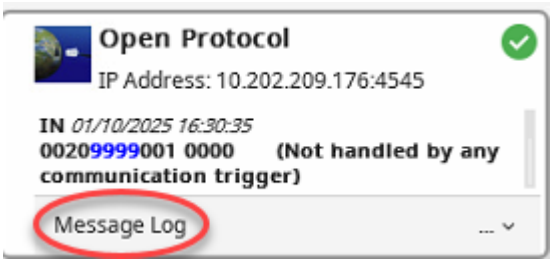
On the **Connectivity** tab, the latest outgoing and incoming communication will be listed in the **Messages...** field on the card for the communication module.



Three types of messages can appear:

- **IN:** Incoming messages
- **OUT:** Outgoing messages
- **ERROR:** Error messages, specifying what is wrong

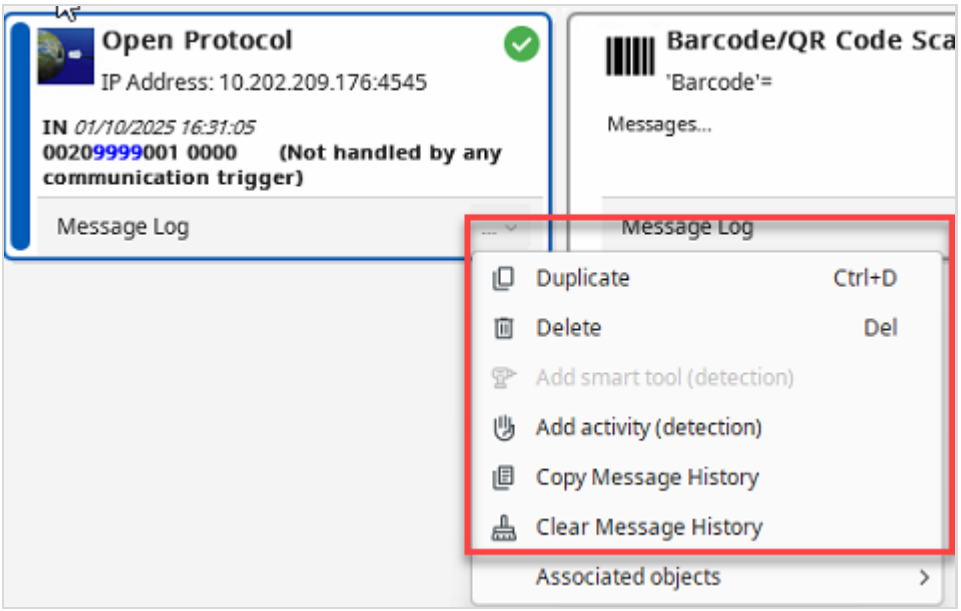
To view more messages, click the **Message Log** button.



7.3.9 Managing communication modules

- To duplicate a communication module, select it, click the **...** button in the lower-right corner of the card and select **Duplicate**.
- To delete a communication module, select it, click the **...** button in the lower-right corner of the card and select **Delete**.
- To copy a communication module's message history, select it, click the **...** button in the lower-right corner of the card and select **Copy Message History**.

  The message history is copied to your clipboard.
- To clear a communication module's message history, select it, click the **...** button in the lower-right corner of the card and select **Clear Message History**.



7.4 Sensors

What?

The **Sensors** section in the **Connectivity** tab manages the sensors that are used in the project. Whenever a sensor is connected, it is automatically added to the list.

The items in the list show the sensors that are used in the project.

On a typical workstation with a **Validate** license, two sensors appear: the Arkite 3D sensor, a time-of-flight sensor that has the ability to measure both distance and infrared, and an RGB camera for regular color images.

If you have a **Validate** or **Guide** license and you want to be able to detect barcodes or texts through OCR recognition, you need an Arkite RGB sensor. We recommend using this sensor for making RGB product snapshots as well.

Webcams are also supported: simply plug it into an available USB port, and it appears in the list.

On a typical workstation with a **Vision** license, another sensor appears: the Arkite Vision sensor.

If sensors are at some point not available, they remain to be part of the project. For example, if a project is edited on a laptop, the sensors are not connected but remain part of the project. This allows to continue editing without having an actual sensor connected.

Note

The Arkite 3D and Arkite Vision sensors, the Arkite RGB sensor and webcams are detected by default when connected. You can enable a few other sensor types by clicking the button. However, these other types of sensor are all experimental. If you want to use one of them, please contact support@arkite.com to verify which features are supported.

Specifying Sensor Serial Number

If a project uses multiple sensors of the same type, you have to specify their serial number, so the software knows without ambiguity which connected sensor corresponds to which logical sensor of the project.

For example, if two Arkite RGB sensors are used to make product snapshots, and one is called 'Left RGB sensor' and the other is called 'Right RGB sensor', you have to specify which serial number corresponds to the 'Left RGB sensor', and do the same for the right camera. You have to specify this for every unit that uses the project, as the serial numbers of the respective sensors will be different. Whenever action is required, a warning appears in the status bar.

To specify the sensor serial number, select the sensor card. In the **Properties** pane, select the correct serial number in the **Specified Serial** dropdown list. In the **Name** field, you can specify a meaningful name for the sensor.

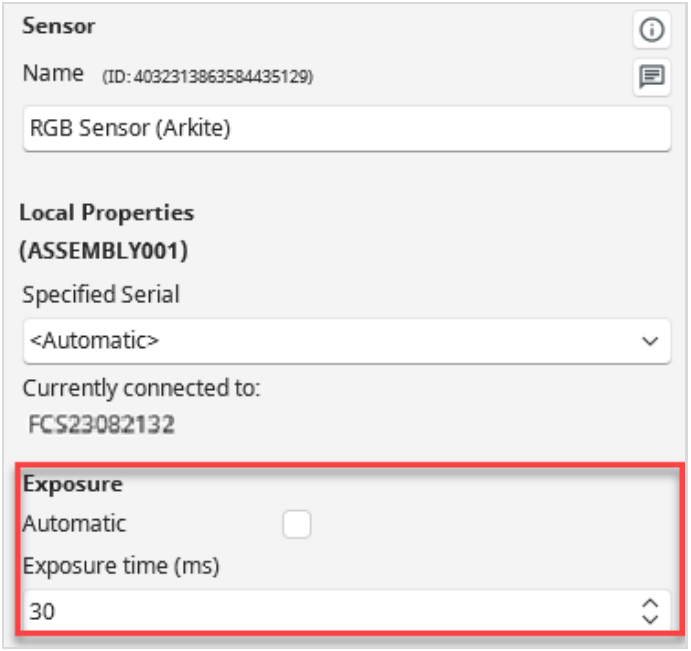
Note

*In a typical case, where only one sensor of a certain type is used, there is no need to specify the serial numbers and the default **<Automatic>** can be left untouched. The software will automatically use the attached sensor of the same type.*

Setting the exposure time for an Arkite RGB sensor

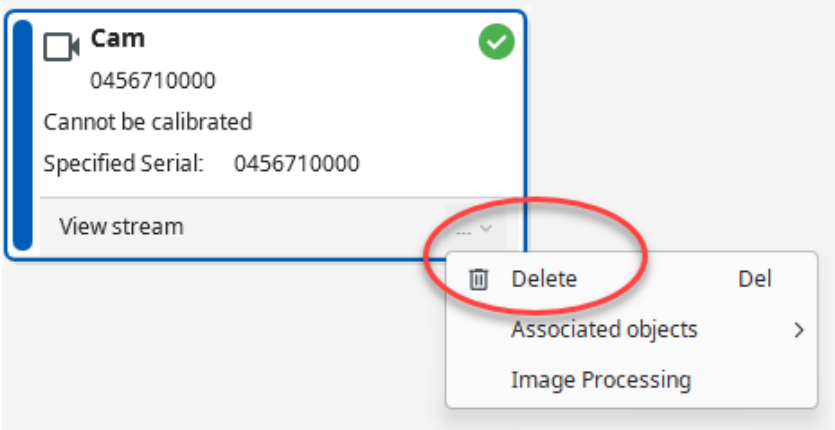
When connecting an Arkite RGB sensor, the exposure time is set by default to 30 milliseconds.

To set another exposure time, select the sensor card. In the **Properties** pane under **Exposure**, uncheck the **Automatic** checkbox and in the **Exposure time (ms)** property, set the exposure time of your choice.



Removing a sensor

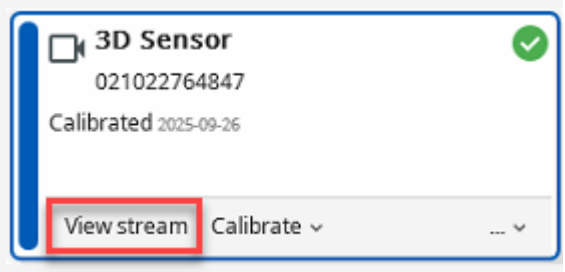
To remove a sensor, click the **...** button in the lower-right corner of the senor's card and select **Delete.**



An example use case is when an external webcam was required in a project was, which was at one point connected and hence a webcam sensor appeared in the list. If, later on, the webcam is no longer needed, it can be permanently removed.

Viewing the sensor stream

To view images from the sensor on your screen, click the **View Stream** button in the sensor card.



8. Experience levels

8.1 About experience levels

The **Experience level** functionality allows you to customize your project and change its behavior based on the operator's level of experience:

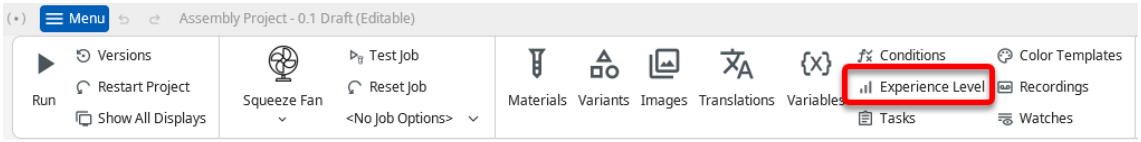
- When the **Experience Level** functionality is disabled, the project behavior will be the same for all operators.
- When the **Experience Level** functionality is enabled, by default three operator experience levels become available.
  Depending on the experience level, you can customize:
  - Detection settings (Validate and Vision license only)
  - Projection settings
  - Display of visualizations

8.2 Managing experience levels

Enabling experience levels

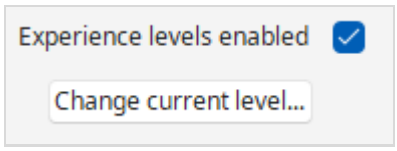
To enable the **Experience level** functionality, execute the steps below:

1. In the ribbon, click the **Experience Level** button.



   The **Experience Level** window opens.

2. Check the **Experience levels enabled** checkbox.



   The **Experience level** functionality is enabled. The functionality in the **Experience Level** window becomes available.

Available experience levels

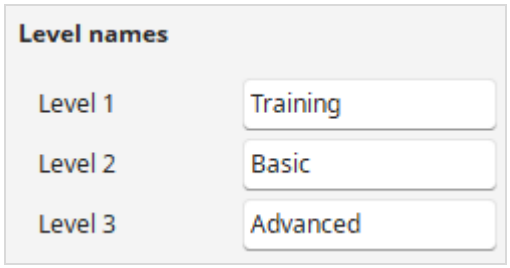
By default, three experience levels are available:

- Level 1: Training
- Level 2: Basic
- Level 3: Advanced

After enabling the **Experience level** functionality, by default level 1 will be used.

Modifying experience level names

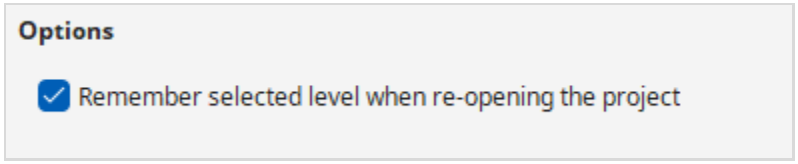
To change the name for an experience level, update its label in the **Experience Level** window under **Level names.** The experience level name will be updated throughout the application.



Experience level options

In the **Experience Level** window under **Options**, you can specify whether:

- the selected experience level should be retained after restarting the project.
- the experience level should be reset to the default Level 1 every time the project is restarted.



8.3 Experience level settings

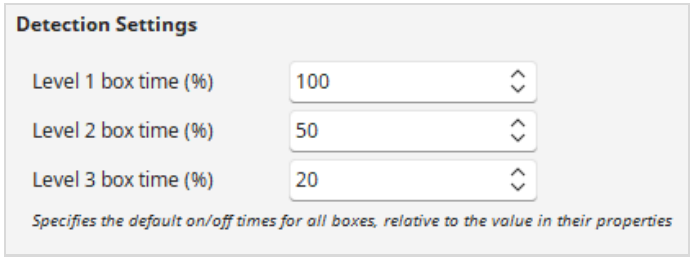
Detection settings (Validate and Vision license only)

For every detection in your project, there are time threshold properties that define for how long an object needs to be within (ON) or outside (OFF) the range defined in the fill rate properties to change detection state.

In the **Experience Level** window under **Detection Settings**, you can set a percentage of these time thresholds that will be used for every specific experience level.

By default, these percentages are set to:

- 100% for level 1
- 50% for level 2
- 20% for level 3



This means that for a detection with a time threshold of 40 ms, the following thresholds will be used for each specific level:

- 40 ms for level 1

- 20 ms for level 2
- 8 ms for level 3

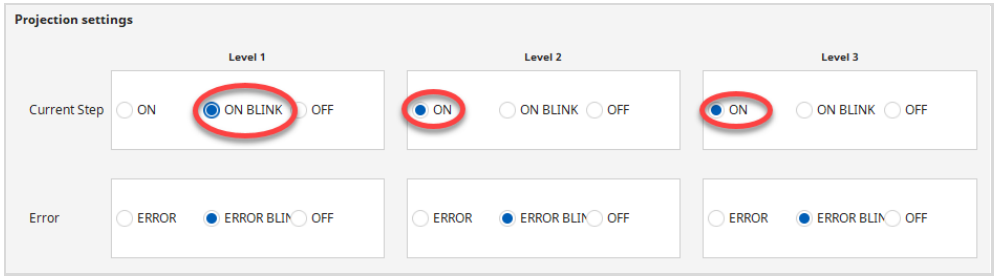
You can change the percentages for every level as required. It is also possible to overrule these default percentages for a specific detection. For more information, refer to section **5.6.5 Finetune a detection** on page 149.

Projection settings

In the **Experience Level** window under **Projection Settings**, you can customize the projection settings independently for each experience level.

The options are:

- For the current step in your process:
  - **ON:** a projection will be shown
  - **ON BLINK:** a projection will be shown and it will blink
  - **OFF:** no projection will be shown
- For errors during your process (Validate and Vision license only):
  - **ERROR:** a projection will be shown
  - **ERROR BLINK:** a projection will be shown and it will blink
  - **OFF:** no projection will be shown



Example

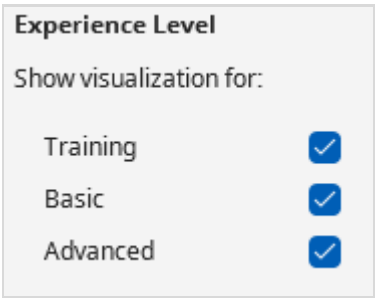
In a training situations for novice users, you may choose to use blinking projections for the Current Step (ON BLINK) for Level 1. Blinking projections attract attention more quickly but can also quickly become distracting or irritating for an operator. Therefore, we recommend using them only in training scenarios.

Note

*To change the colors of these projections, click the **Color Templates** button in the ribbon and select the **Detection colors** template. There you can update the colors for the projection settings mentioned above. For more information, refer to section **15.2 The Detection Colors template** on page 276.*

Display of visualizations

In the **Properties** of a visualization, there is a new **Experience Level** property.



This way, you can define whether the visualization will be displayed for a specific experience level or not. By default, every visualization is shown for all experience levels.

Example

*You can choose to show text instructions to users with **Level 1 (Training)** or **Level 2 (Basic)** but hide them for users with **Level 3 (Advanced)**, since there is no real added value for those users.*

Note

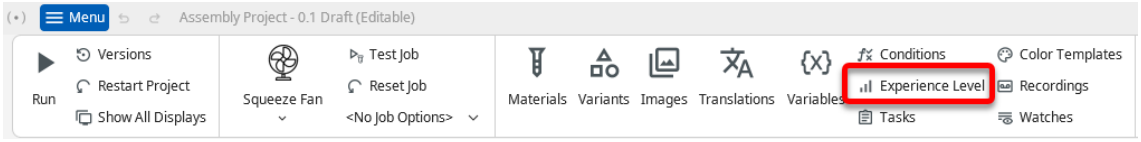
*For more information on the properties of a visualization, refer to sections **4.4.1 Visualization types** on page 83 and **4.4.2 Visualization properties** on page 84.*

8.4 Selecting the experience level

Selection by the project administrator

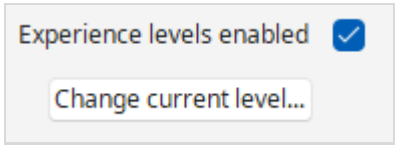
To change the experience level in the Arkite software, execute the steps below:

1. In the ribbon, click the **Experience Level** button.



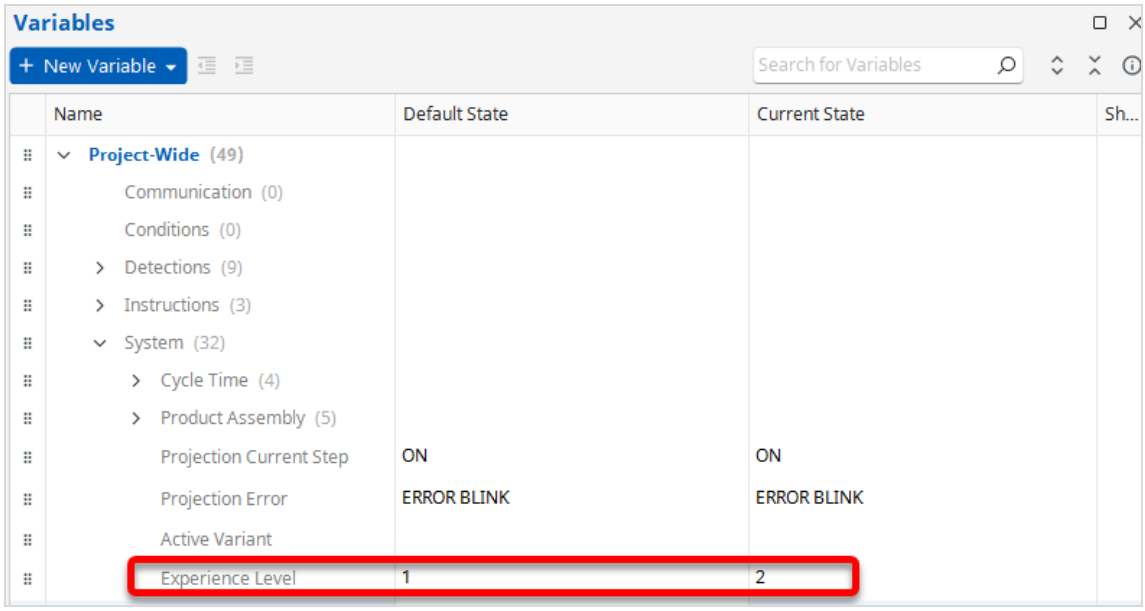
   The **Experience Level** window opens.

2. Make sure the **Experience levels enabled** checkbox is checked.



3. Click the **Change current level...** button.
   The **Variables** window opens. The variable **Experience Level** is selected.

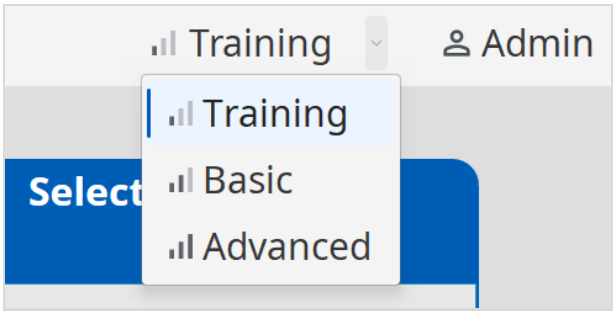
4. Change the value in the column **Current State** to the experience level of your choice (1, 2 or 3).



The experience level has been changed.

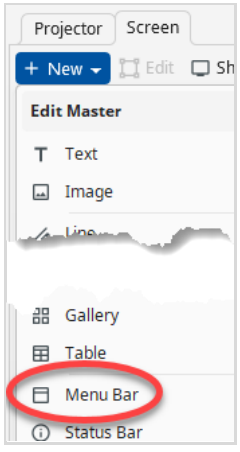
Selection by the operator

By default, an Arkite project contains a **Menu Bar** on the **Screen** display, in which the operator can select the experience level from a dropdown list if the feature is enabled.



Note

In projects made in a software version before 2025.1, you will still need to add the menu bar (typically on the master).



9. Variables

9.1 About variables

Variables are data items that can have different values or states during your assembly process.

Some variables are automatically generated by the Arkite software, for example detection and system variables. Other variables are created by the user.

The state of a variable can be used to log information or to trigger something in the process.

In the **Variables** overview, you can see:

- All project-wide variables, i.e. variables that are available across jobs
- All job-specific variables for the currently selected job

Within each section, variables are grouped per variable type.

9.2 Variable types

In the table below, you can find a list of available variable types:

Detection type	Description
Detection variables (Validate and Vision license only)	Variables generated by the Arkite software for every detection that is added. Detection variables can be **Project-Wide** variables or **Job-Specific**, depending on the detection type. – For every detection, there is a variable that indicates if the detection is present (ON) or not (OFF). – For every detection, there is an **occluder_** variable that indicates if the detection is occluded (ON) or not (OFF). – For every detection, there is a **Vis_** variable that indicates whether visualizations for the detection are projected or not – For every detection, there is a **Sb_** variable that indicates whether shadowboards for the detection are projected or not – For every detection, there are **PositionX_** and **PositionY_** variables that indicate the position of the detection – For every detection, there is an **OrientationAngle_** variable that indicates the orientation of the detection – For every detection, there are **VisPositionError_** and **VisOrientationError _** variables that indicate whether error visualizations for the position or orientation of the detection are projected or not – For every picking bin, there is a **pick_** variable that indicates whether material is being picked from the picking bin (ON) or not (OFF). – For every picking bin, there is a **_material** variable that indicates the material that is linked to the picking bin
Instruction variables	Variables generated by the Arkite software for displaying instructions. Instruction variables are **Project-Wide** variables: – Text instructions variable, containing the currently displayed text instruction – Image instructions variable, containing the currently displayed image instruction – Error instructions variable, containing the currently displayed error message (in case of a Quality Check or Object with Quality Check)

Detection type	Description
Jobs variables	Variables generated by the Arkite software containing information about jobs. Jobs variables are **Project-Wide** variables: – Cycle Time – Cycle Time Sec – Expected Cycle Time – Expected Cycle Time Sec – Jobs: contains a JSON array with all available jobs with their variants and options – Current Job – Current Job Status – Current Job Name – Current Job Options – Current Job Variant Name – Current Job Variant Code – Current Job Active Numbered Steps: if more Numbered Steps are active at the same time, they are all stored in the variable, separated by a semicolon.
System variables	Variables generated by the Arkite software. All system variables are **Project-Wide** variables: – For each process in the project: Current step [process], Status [process], [Process] Step Start Time, [Process] Step StopTime, [Process] Step Duration – Projection Current Step – Projection Error – Experience Level – Step Info – IP addresses – MAC addresses – System Serial Number – Sensor Serial Number – Software Version – Current Workstation – Current Project Name – Current Project Description – Current Project Status – Current User – Current User Language – Current User Authentication Method – Instruction Language – Current User Roles – Date and Time – Server Status – Project Status
Communication variables	Variables that are generated by the Arkite software for communication with external devices through a communication module. All communication variables are **Project-Wide** variables**.**
Condition variables	For each condition that is added by the user, a variable is generated. The value of this variable reflects the current status of the condition (**True** of **False**). Condition variables can be **Project-Wide** variables or **Job-Specific**, depending on the condition type.
Manual variables	Variables that are added by the user. Manual variables can be **Project-Wide** variables or **Job-Specific**, depending on the user's choice. By default, an Arkite project contains some manual variables to make the job selection process work.

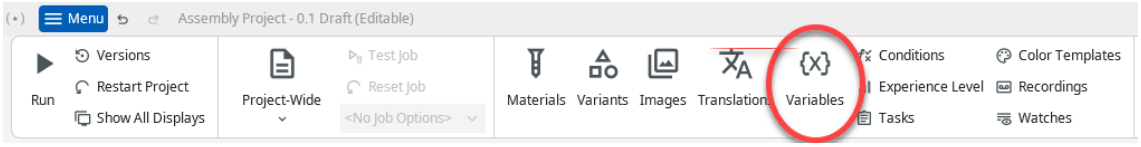
Note

Only manual variables can be updated by the user. All other variables can only be updated by the Arkite software.

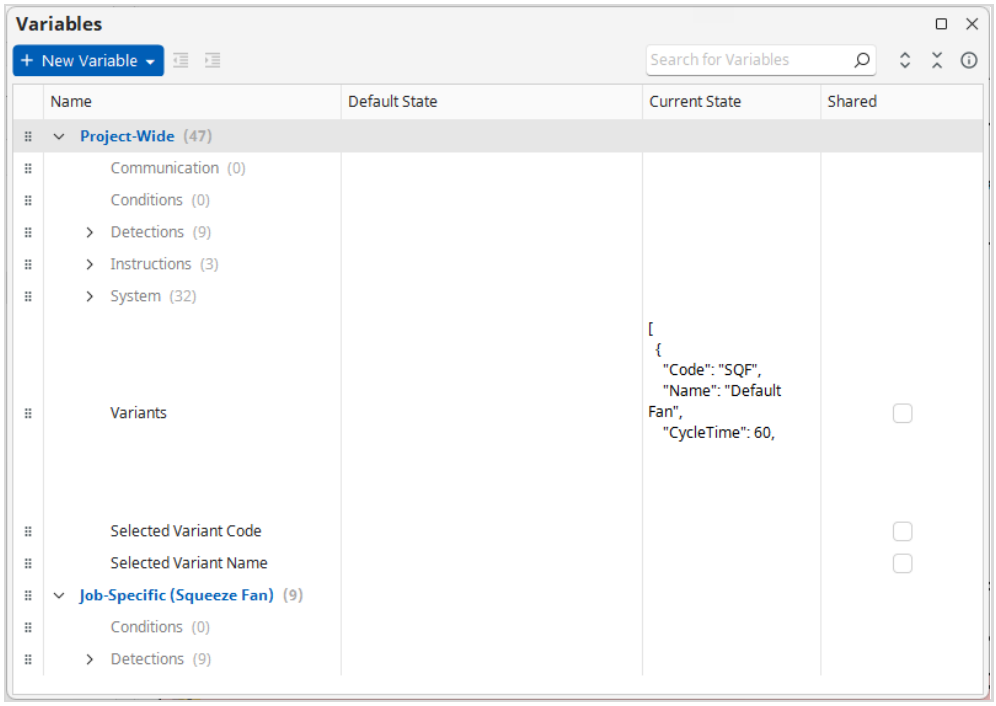
9.3 Managing variables

Consulting Variables

In the ribbon, click the **Variables** button.



The **Variables** window appears.



All **Project-Wide** variables and all variables for the job that is currently active are displayed. Within each group, variables are organized per variable type.

For every variable, the information below is available

- **Name**
  For manual variables, the name can be changed.
- **Default state of the variable:**
  Initial state of the variable, when the project is started. For manual variables, the default state can be edited.
- **Current state of the variable:**
  When the project is restarted, the current state is always reset to the default state.

- **Shared:**
  If checked, the value of the manual variable is kept in sync across all workstations connected to the same server.
  If unchecked, the value of the manual variable can be changed independently for every workstation.

  Note
  - *You can share variables in different projects too. To make things work, make sure to create the variable in both of the projects and mark them as shared accordingly.*
  - *For more information, refer to section **Sharing variables** on page 51.*

Adding and deleting manual variables

You can add manual variables yourself:

1. To add a project-wide variable, select the **Project-Wide** group.
   To add a job-specific variable, first select the relevant job in the ribbon and then select the **Job-Specific** group.
2. Click **New** and select **Variable**.

You can organize manual variables into variable groups. To create a new group, click **New** and select **Group**. Drag and drop your variables in the group of your choice.

To delete a manual variable, right-click the variable and select **Delete**.

9.4 Watching variables

About Watches

In the **Watches** panel, you can add a number of variables you want to keep track of. It shows the variable name along with its current state. This way, you can easily make a selection of variables to monitor, and put them together in a comprehensive overview.

Opening and closing the Watches panel

To open or close the **Watches** panel, click **Watches** in the ribbon.

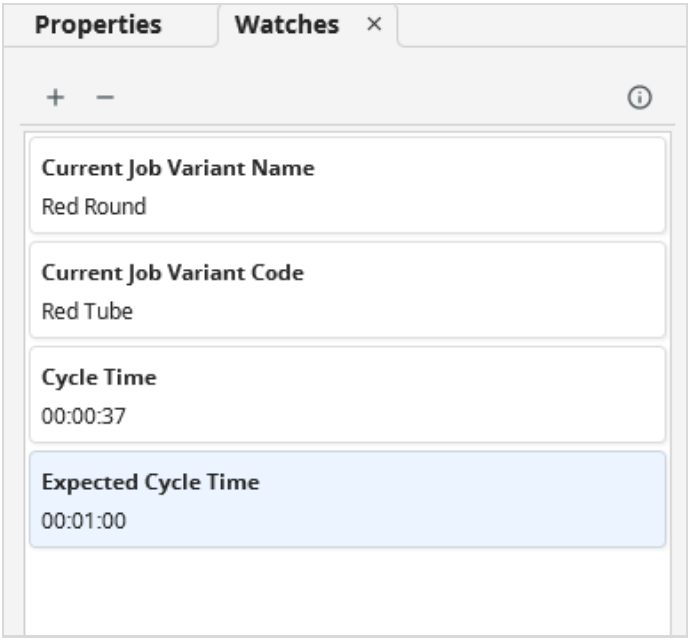


The **Watches** panel will open in the right side of the screen, as a second tab next to the **Properties** panel.

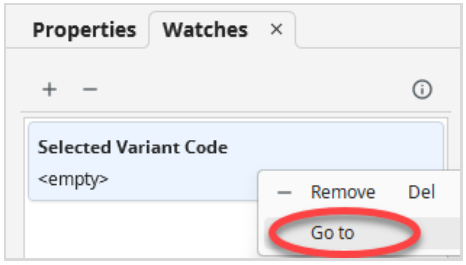
Adding and removing variables to the Watches panel

Click the **+** icon to add a variable and its state to the **Watches** panel.

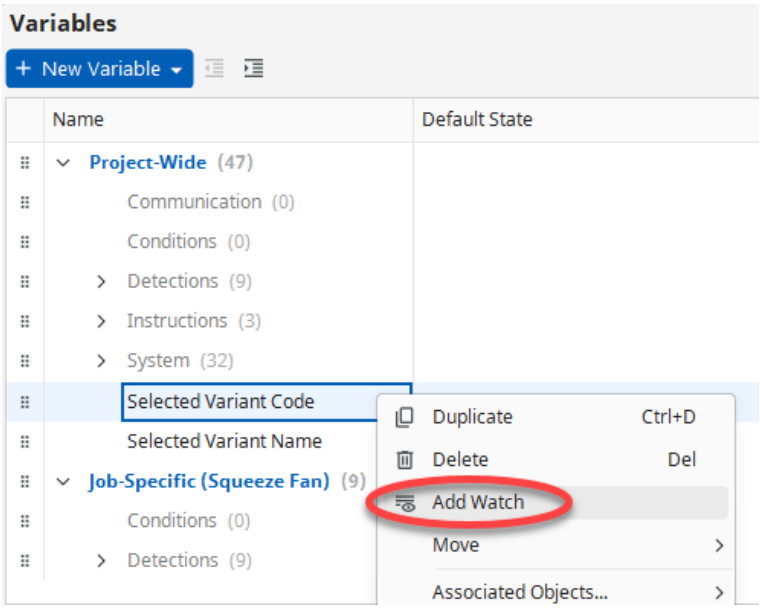
Select a variable and click the **–** icon to remove it from the **Watches** panel, or right-click the variable and select **Remove**.



To go to the variable in the **Variables** window, right-click the variable and select **Go To**. For manual variables, i.e. variables you created yourself, you can edit the variable's state or default state here.



You can also add a variable to your **Watches** panel from the **Variables** window: right-click the variable and select **Add Watch**.



10. Recordings (Validate and Vision license only)

10.1 About recording

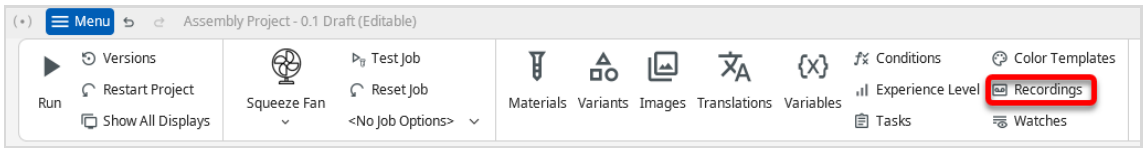
The Arkite software allows you to record the sensor stream. These recordings can be sent to Arkite for support, or you can use them to add detections from the comfort of your desk.

10.2 Creating recordings

You can create recordings in several ways:

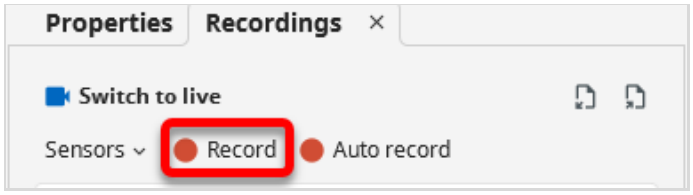
Record manually

1. In the ribbon, click **Recordings**.



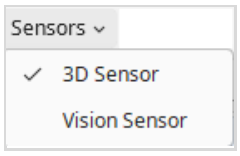
   The **Recordings** panel will open in the right side of the screen, as a second tab next to the **Properties** panel.

2. Click **Record** to start the recording.



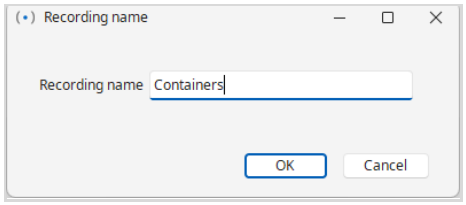
   Note

   *To change the sensor(s) that is/are recorded, click the **Sensors** menu and select the sensor(s) of your choice.*



3. Perform the activities you want to record.
4. Click the **Record** button again to stop the recording.

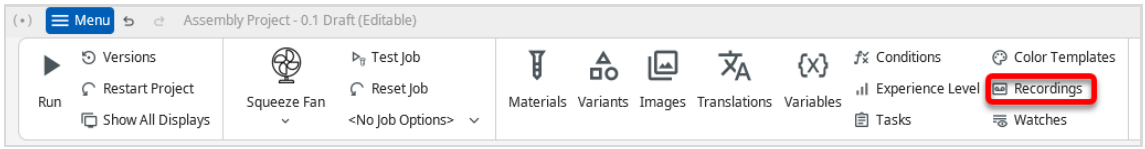
5. Provide a name for your recording and click **OK**.



Your recording is saved.

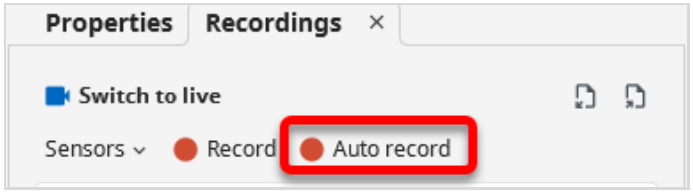
Auto Record

1. In the ribbon, click **Recordings**.



The **Recordings** panel will open in the right side of the screen, as a second tab next to the **Properties** panel.

2. Click the **Auto Record** button.

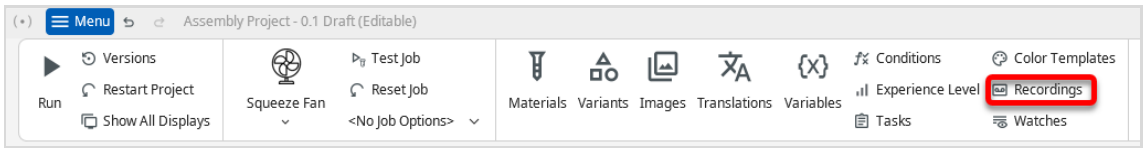


A recording is made for every completed step involving a detection. Nothing is recorded during idle time.

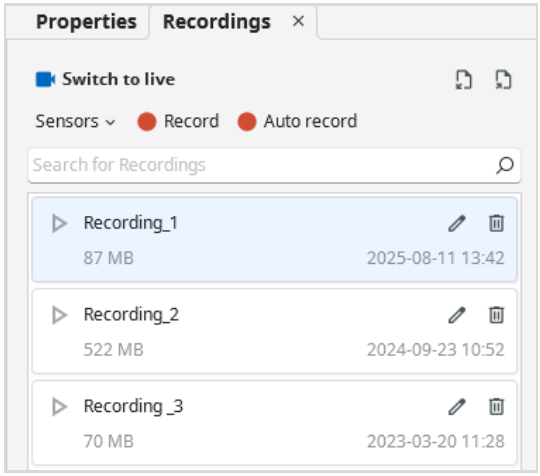
3. Click the **Auto Record** button again to deactivate auto recording.

10.3 Managing recordings

In the ribbon, click **Recordings**.



The **Recordings** panel will open in the right side of the screen, as a second tab next to the **Properties** panel.



Right-click a recording and select **Delete** or click the button to delete it.

To export the recordings to a .hrec file, click the button.

To import a recording from another project, click the button and select the .hrec file of your choice.

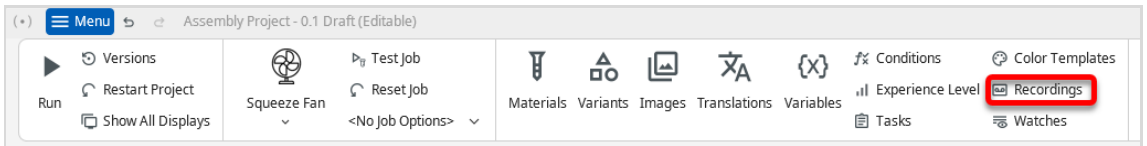
10.4 Using recordings

Recordings can be used by Arkite for troubleshooting and support.

You can also use recordings yourself, for example for adding detections while you are not at the workbench.

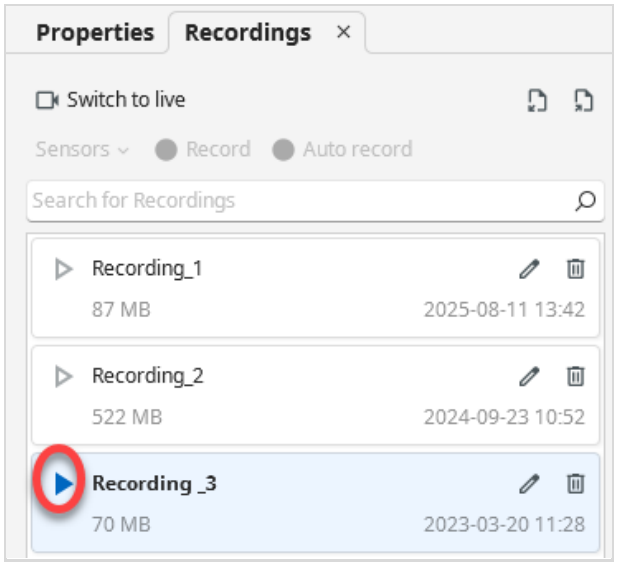
To do this, execute the following steps:

1. At the workbench, make a recording of the sensor stream and make sure to remove all objects, tools and picking bins and put them back at the location of your choice one by one.
   Perform all activities for which you want to create a detection.
2. Open the project on your pc.
   Make sure you also import the recording.
3. In the ribbon, click **Recordings**.

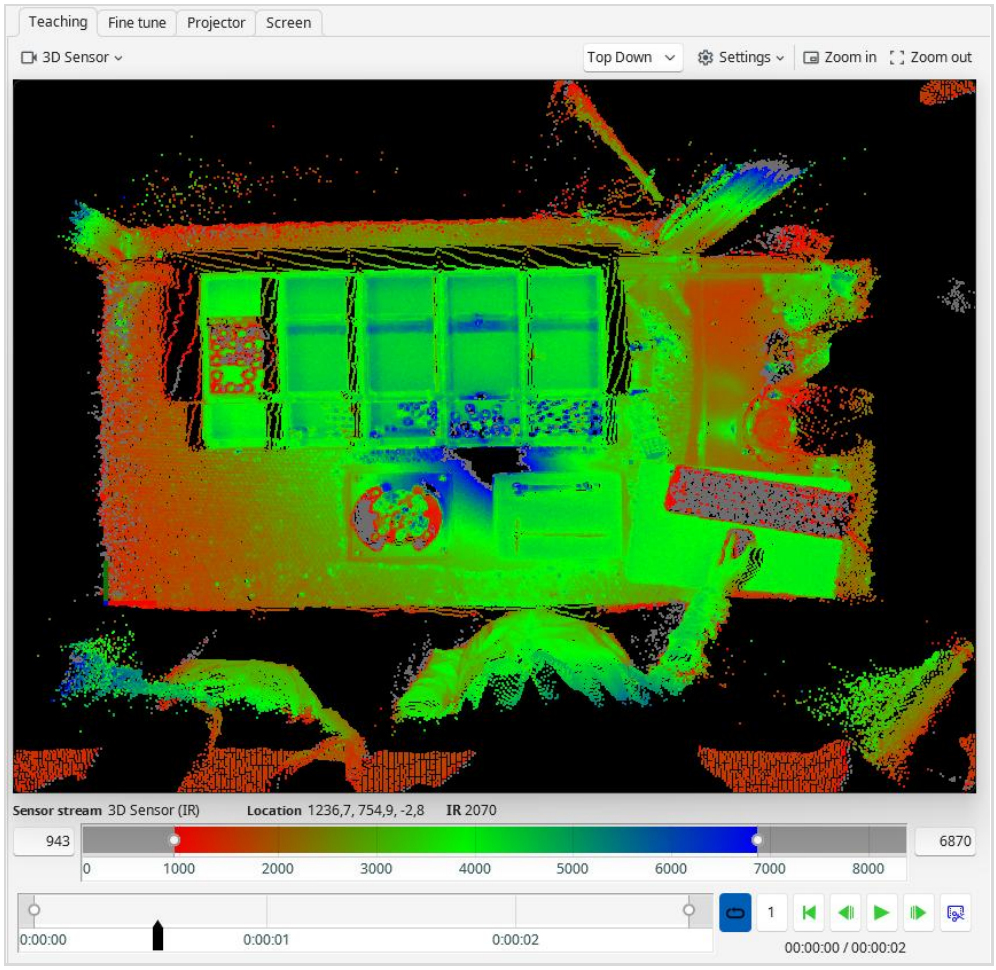


   The **Recordings** panel will open in the right side of the screen, as a second tab next to the **Properties** panel.

4. Select the recording of your choice and click the **Play** button.



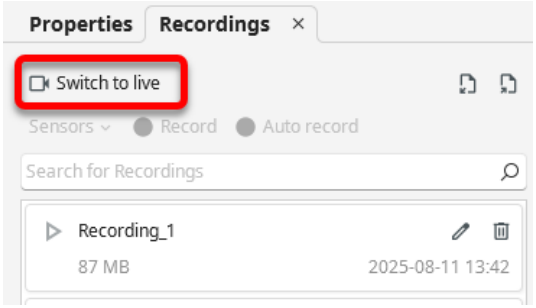
The recording starts to play in the **Teaching** subtab of the **Detections** tab



You can drag the round handles to set the start and end time of the recording and use the arrow marker to select the current time.

Use the buttons to play and pause the recording, to browse through the recording frame by frame or to trim the recording.

5. To add detections, teach them by:
   - pausing the recording at a frame where the object is not at its place and take a **Without object** snapshot
   - pausing the recording at a frame where the object is at its place and take a **With object** snapshot
6. To return to the live feed of the sensor, click the **Switch to live** button in the **Recordings** pane.



11. Server management

11.1 About server management

The Arkite software is split in a workstation and a server component. The server, containing all data and resources for multiple clients and projects, should preferably run on a separate system to obtain higher security, maintainability and data integrity.

From within the software on any workstation connected to the server, you can consult an overview of all workstations on that server, manage licenses and manage backups.

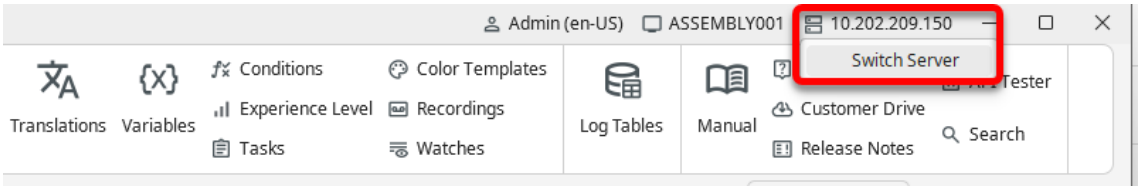
11.2 Connecting to a server

What?

When opening the Arkite software on a workstation the first time, you need to specify to which server you want to connect. Afterwards, the system automatically connects to that server again when starting the software. At all times however, you can connect to a different server.

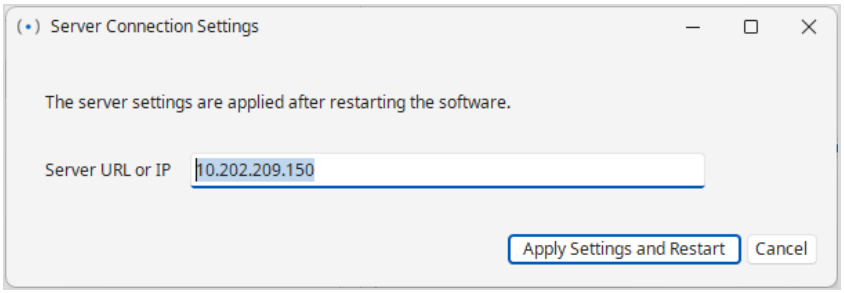
How to connect to a different server

1. In the top bar, click the **[server]** button and then **Switch Server.**.



   The **Server Connection Settings** popup appears.

2. In the **Server URL or IP** field, fill in the URL or IP address of the server.



3. Click the **Apply Settings and Restart** button.

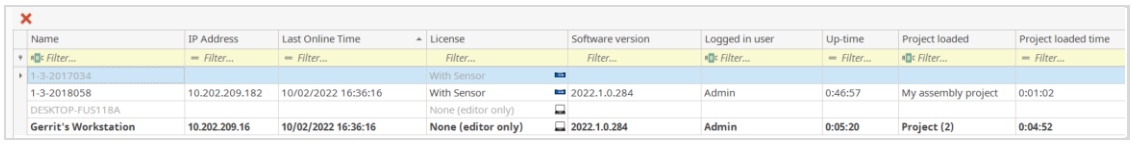
11.3 Workstation overview

What?

From any workstation connected to the server, you can consult information about all workstations that are connected to the same server, or have been connected to it in the past.

Overview of workstations connected to the server

To consult the overview of workstations, click **Menu** and then **Workstations**.



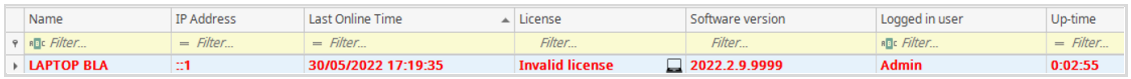
Name	IP Address	Last Online Time	License	Software version	Logged in user	Up-time	Project loaded	Project loaded time
1-3-2017034			With Sensor					
1-3-2018058	10.202.209.182	10/02/2022 16:36:16	With Sensor	2022.1.0.284	Admin	0:46:57	My assembly project	0:01:02
DESKTOP-FUS118A			None (editor only)					
Gerrit's Workstation	**10.202.209.16**	**10/02/2022 16:36:16**	**None (editor only)**	**2022.1.0.284**	**Admin**	**0:05:20**	**Project (2)**	**0:04:52**

In the overview table, you can see for every connected workstation:

- the workstation's **Name**.
  The name can be changed by clicking in the cell and entering a new name.
- the workstation's **IP address**
- the server **License** type that is used for the workstation:
  - Vision (= with Arkite 3D sensor and with Arkite Vision Sensor)
  - Validate (= with Arkite 3D sensor)
  - Guide (= without sensor)
  - none:
    in this case, the workstation is an editor. It can be used to prepare and configure projects, but not to run them. For testing purposes, you will be able to run your project for a maximum of 15 minutes.

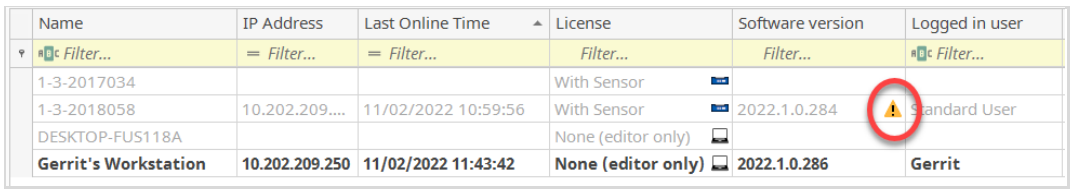
  Note

  Workstations with an invalid license, for example a license that is only valid for a previous software version, are marked in red.



Name	IP Address	Last Online Time	License	Software version	Logged in user	Up-time
LAPTOP BLA	**::1**	**30/05/2022 17:19:35**	**Invalid license**	**2022.2.9.9999**	**Admin**	**0:02:55**

- the **Software version** that is running on the workstation.
  A warning sign is displayed when the software version is not the same as the server software version.



Name	IP Address	Last Online Time	License	Software version	Logged in user
1-3-2017034			With Sensor		
1-3-2018058	10.202.209....	11/02/2022 10:59:56	With Sensor	2022.1.0.284	Standard User
DESKTOP-FUS118A			None (editor only)		
Gerrit's Workstation	**10.202.209.250**	**11/02/2022 11:43:42**	**None (editor only)**	**2022.1.0.286**	**Gerrit**

- The **user** hat is logged in on the workstation
- The workstation's **up-time**, i.e. the time it has been connected to the server
- The **name of the project** that is currently loaded on the workstation

- The **time** this project was loaded on the workstation
- **Vision Ready**: indicates if the workstation is ready to be used with an Arkite Vision license.
- **Host ID's**: ID that is required to make the workstation Vision Ready.

  Note

  *If you have a Vision license, but your workstations are not **Vision Ready** yet, follow the steps described in the procedure **How to make your workstation Vision Ready** (available on the customer portal). The number of workstations you can make **Vision Ready** is limited to the number of workstations for which you have bought a Vision license. If you already have reached this number, your request will be denied.*

- The **workstation id**

  You can copy this id to use it when working with the Arkite API. For more information about the Arkite API, refer to chapter **18 The Arkite API** on page 292.

The workstation from which the overview is consulted is displayed in **bold**.

Workstations that are currently not connected to the server are displayed in **gray**. These workstations can be turned off or no longer connected to the server. When hovering over a gray line, a tooltip is shown mentioning how long ago the workstation disconnected from the server. Workstations that are displayed in gray can be removed from the overview by selecting them and clicking **Delete Workstation**.

11.4 Managing licenses

What?

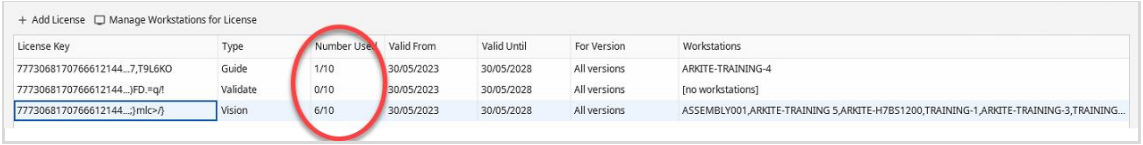
When using the workstation for the first time, you need to provide a server **License Key**. A license key is valid for a particular number of workstations. From the application, you can manage which workstations are using the license.

How to consult licenses

To consult licenses, click **Menu**, then **Server** and then **Settings**.

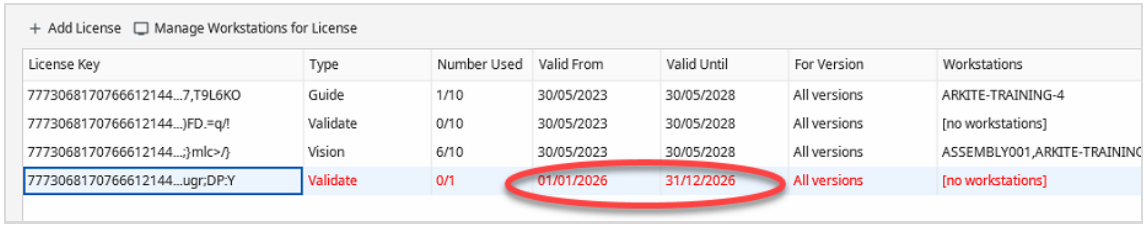
Under **Server License**, you can see an overview of the license key(s) used for the server, their validity period, the number of workstations that can use the license and the workstations that are currently using the license.

In the example below, the **Guide**, **Validate** and **Vision** license keys can all be used for a maximum of ten workstations is available on the server. For the **Vision** license, 6 out of 10 slots are used, while for the **Guide** license, one slot is used by ARKITE-TRAINING-4.



Note

A license that is no longer valid or not valid yet is marked in red. In the example below, the license is only valid from 1/1/2026 (screenshot taken in October 2025).

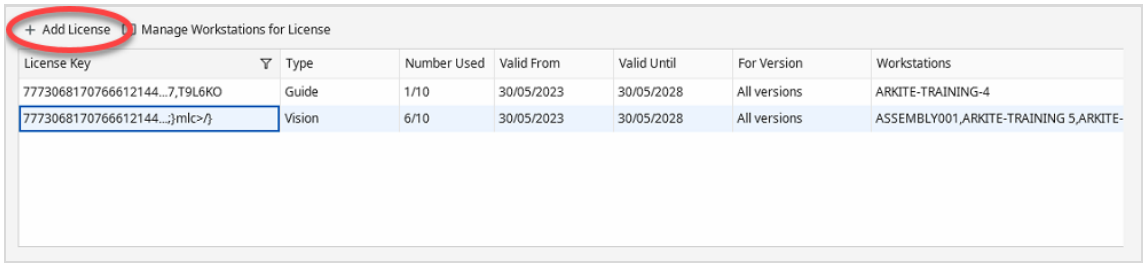


How to add a license key

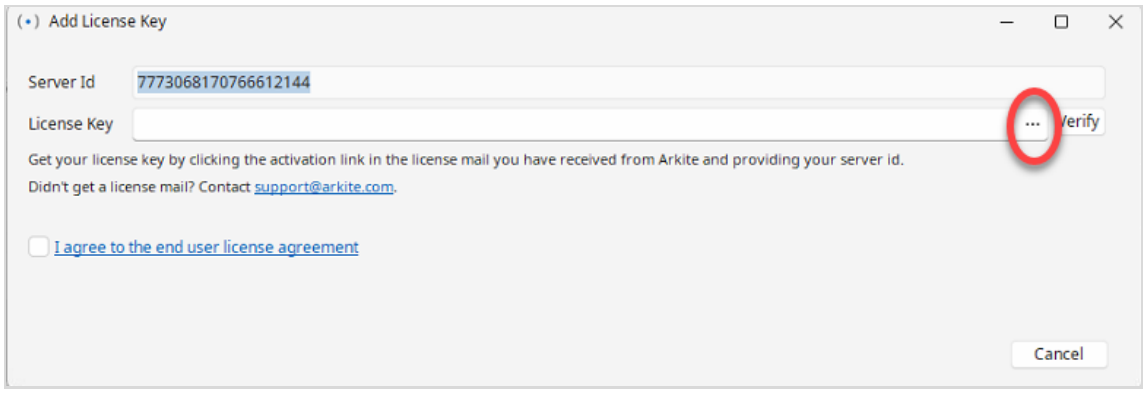
You can use several license keys on the same server.

To add a new **Server License Key**:

1. Click the **Add License** button.



2. In the **Add License Key** window, in the **License Key** field fill in your license key or click the ... button to select the file containing your license key.

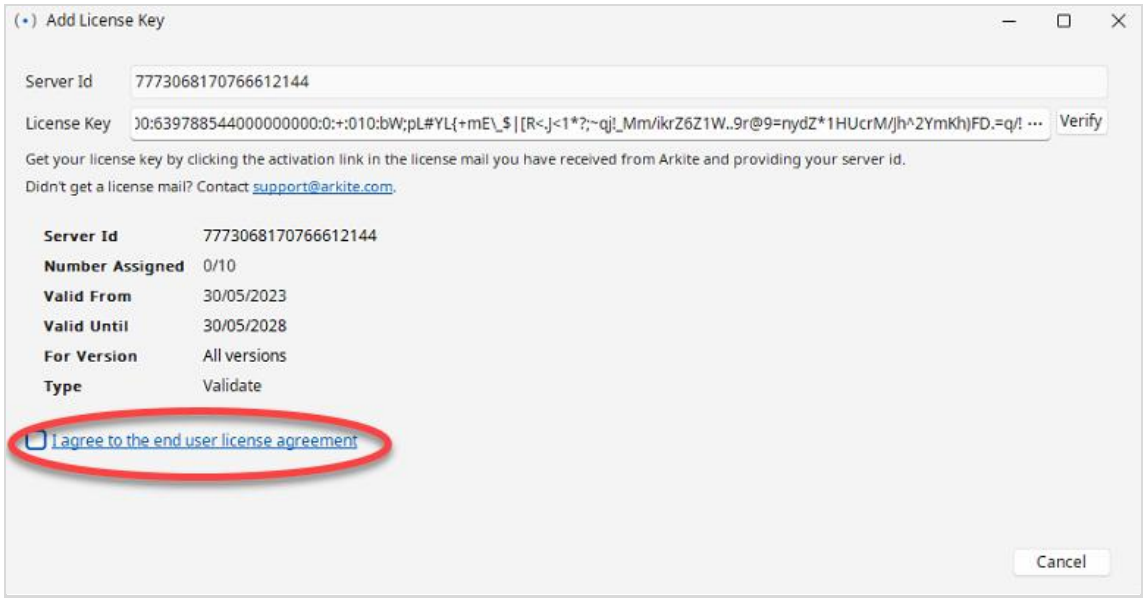


3. Click **Verify**.



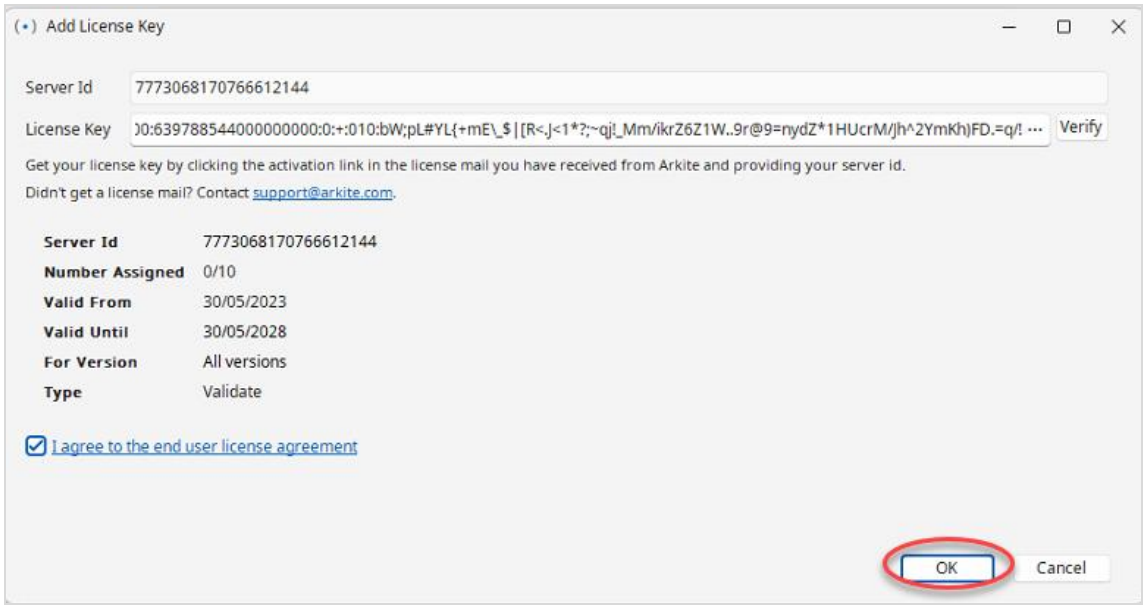
The details and validity date of your license is displayed.

4. Check the **I agree to the end user license agreement** checkbox.

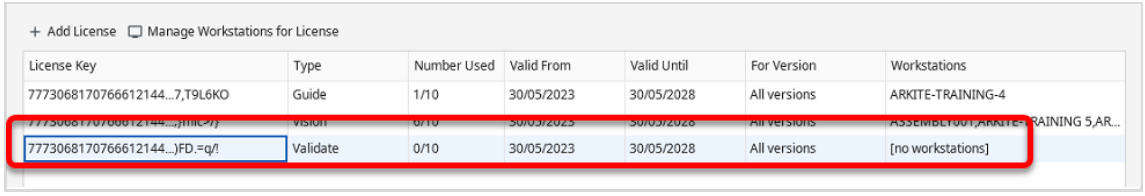


An **OK** button will appear.

5. Click **OK** to confirm.



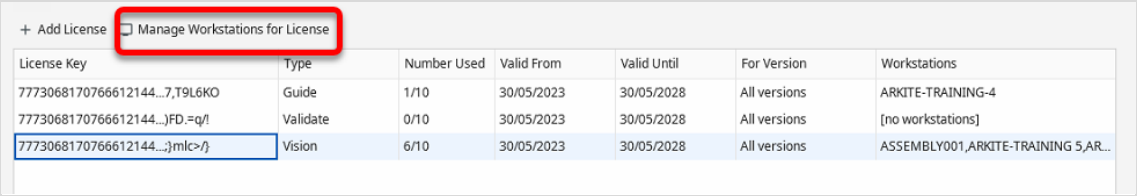
The license key is added to the overview.



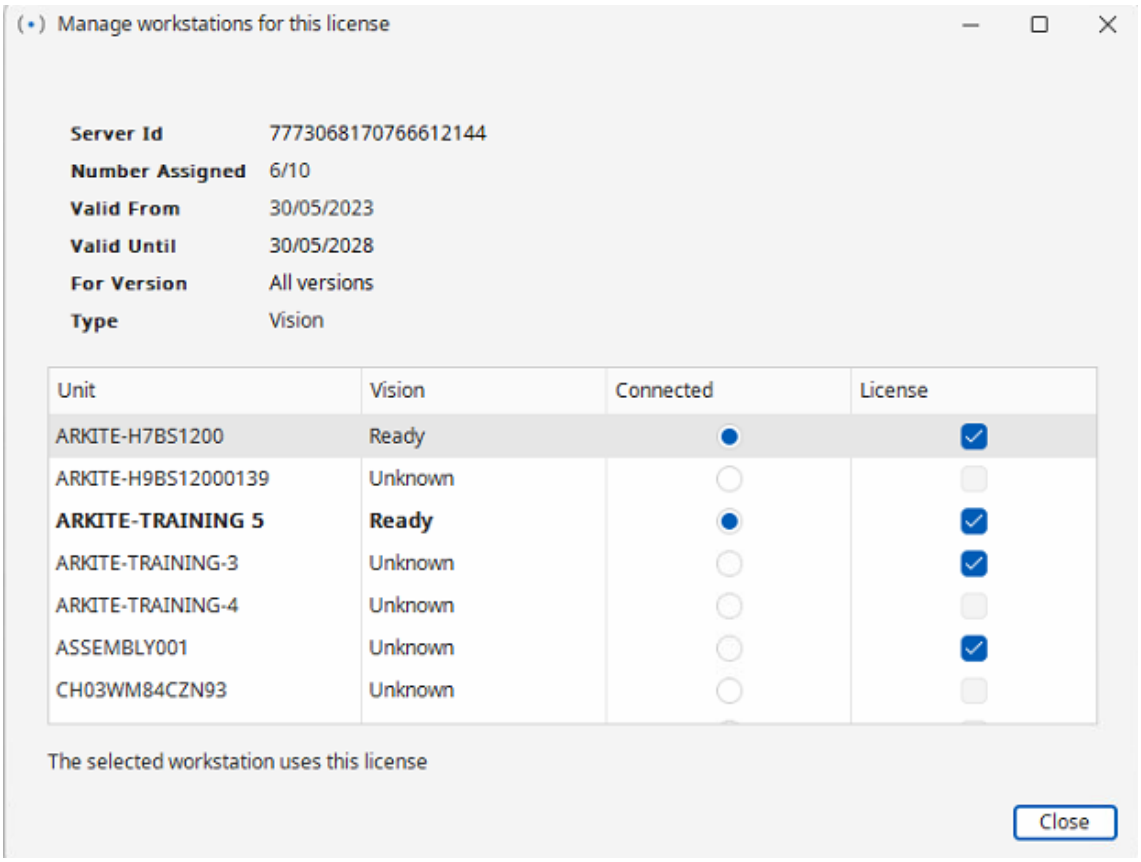
How to manage workstations

To manage the workstations on which the license key is used:

1. Select the license key and click the **Manage Workstations for license** button.



2. In the **Manage workstations for this license** window:
   - Check the checkbox(es) in the **License** column for the workstation(s) on which you want to use the license
   - Uncheck the checkbox(es) in the **License** column for the workstation(s) on which you no longer want to use the license



3. Click **Close** to apply your changes.

Note

- *If you have a Vision license, you will only be able to add workstations to it that are **Vision Ready**. If your workstations are not **Vision Ready** yet, follow the steps described in the procedure **How to make your workstation Vision Ready** (available on the customer portal). The number of workstations you can make **Vision Ready** is limited to the number of workstations for which you have bought a Vision license. If you already have reached this number, your request will be denied.*
- *Workstations for which you have removed the license will lose the license after they have rebooted.*

- *Workstations that are connected to the server but are not assigned to a license will function as an editor. You will be able to prepare and configure projects on them, but you cannot use them to run a project. For testing purposes, you will be able to run your project for a maximum of 15 minutes.*

How to renew a license key

Between three and two months before your license expires, you will receive a quotation from Arkite for the renewal of your contract. Once you have placed the purchase order, you will receive a new license key. You can then add this new license key as described above, without assigning any workstations to it yet.

Once your license key expires, the system will automatically check if another valid, comparable license key with open slots is available. If there is, it will automatically transfer the workstations to the new license key.

How to remove a license key

To remove a **Server License Key,** right-click the license and select **Remove**.

11.5 Managing server backups

What?

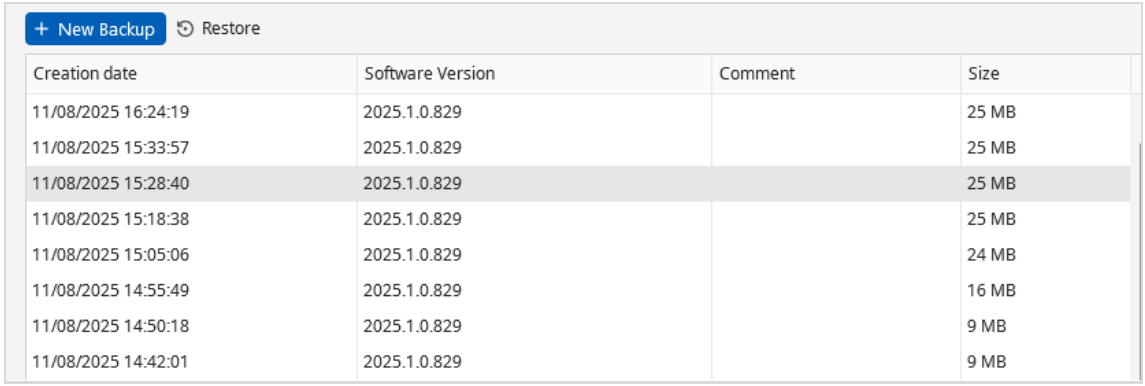
By default, a backup of the server databases is taken every hour (provided there have been any changes during this timeframe). You can select another frequency, or create a backup yourself at any time.

The 10 most recent backups are always available. Additionally, all backups that were created right before a server version upgrade also remain available, as well as a selection of backups from the past. Other backups are removed automatically. You can consult all available backups and restore a previous version if required.

How to consult server backups

To consult server backups, click **Menu**, then **Server** and then **Backups**.

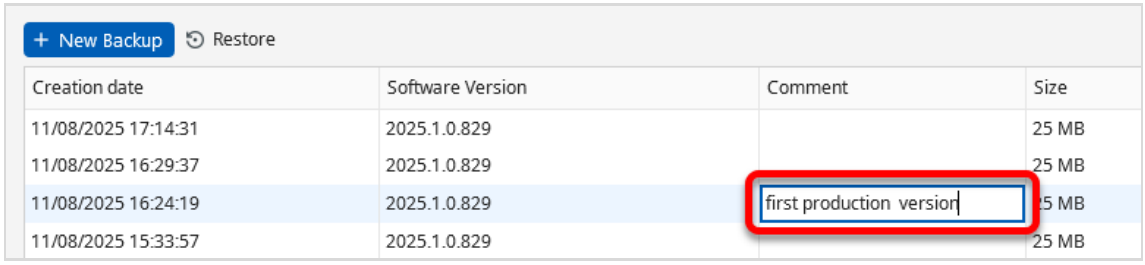
An overview of all available backups is shown.



For every backup, the information below is displayed:

- The backup's **creation date**
- The **software version** with which the backup was created
- **Comments** regarding the backup.

You can add a comment for a backup by clicking in the corresponding cell in the **Comments** column and start typing.

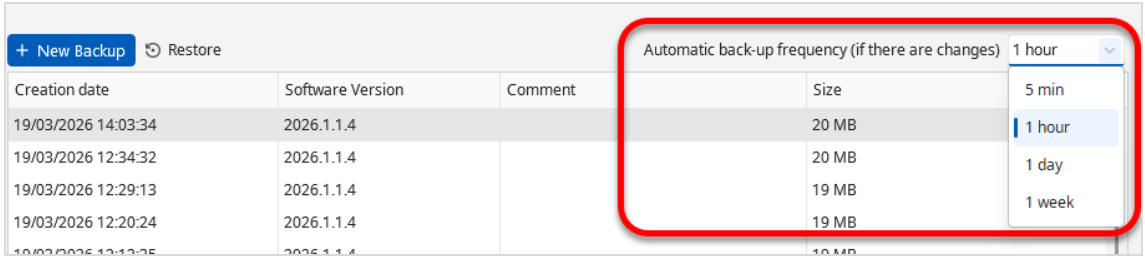


- The backup's **size**

How to change the automatic backup frequency

While you are working in Arkite, by default the system automatically takes a backup every hour.

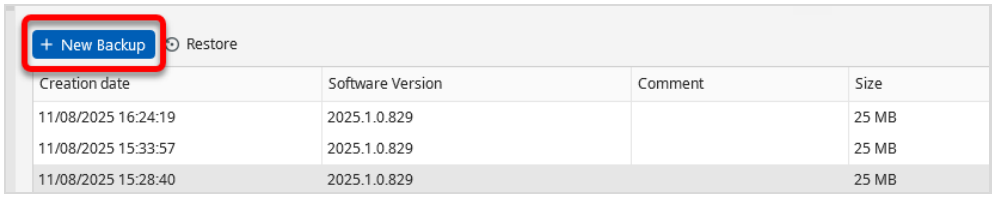
You can select a different frequency in the **Automatic back-up frequency** dropdown list.



How to create a server backup

On top of the automatic backups, you can also create a backup yourself at any time.

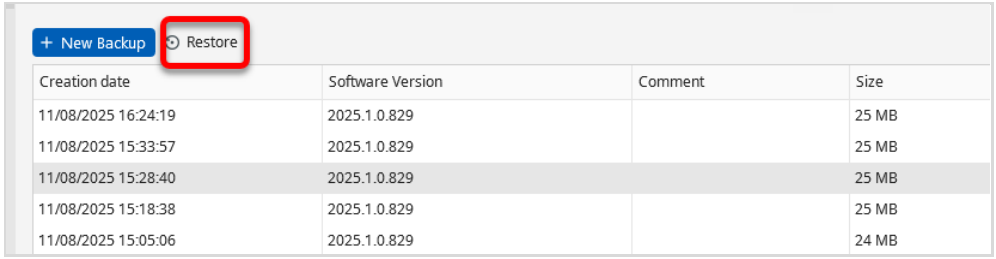
To do so, click the **New backup** button on top of the backup overview.



How to restore a server backup

In case of issues, for example a corrupted database, you can put a previous version of your server database back in place by restoring a backup.

To do so**,** select the backup and click the **Restore** button.



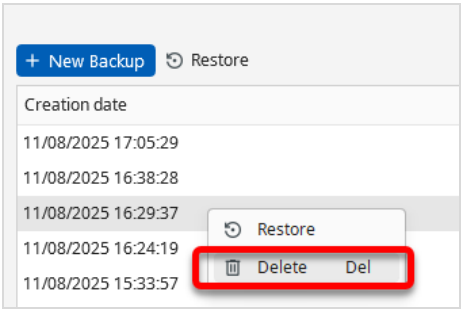
Warning

- *Server backups are backups of the entire server database. Restoring a server backup affects all versions of all projects on all workstations connected to the server. All workstations will need to restart their running project after the restore.*
- *Resources, such as image files, are not part of the server backups. This means that when you are restoring a server backup in which a resource file is used that has been removed from the project since, this resource file will no longer be available. An error message will be shown to inform you about the missing file(s).*

How to delete a server backup

Older backups are automatically deleted by the system. Additionally, you can also delete server backups yourself.

To do so, right-click the backup and select **Delete**.



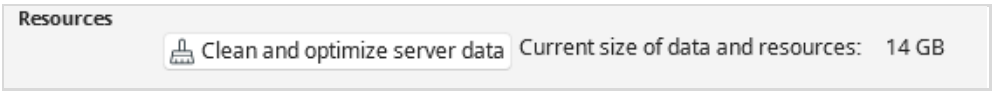
11.6 Cleaning up server resources

What?

When your disk is (almost) full, a clean up mechanism will remove all unused resources (e.g. image files). You an also trigger this mechanism yourself and free up disk space in this way.

Steps

1. Click **Menu** and then **Server**.
2. Under **Resources**, click the **Clean and optimize server data** button.



Result

- Unused resources such as image files will be removed.
- The server database will be compacted and defragmented.

12. Dashboarding

12.1 About dashboarding

Dashboarding provides you with insights on various aspects of your assembly process and the status of your system through a graphical representation of data.

> ***Note***
> *Dashboarding is only available for users with the **View Dashboard Tab** role.*

12.2 Consulting dashboards

To consult dashboards, click **Menu** and then **Dashboarding**.

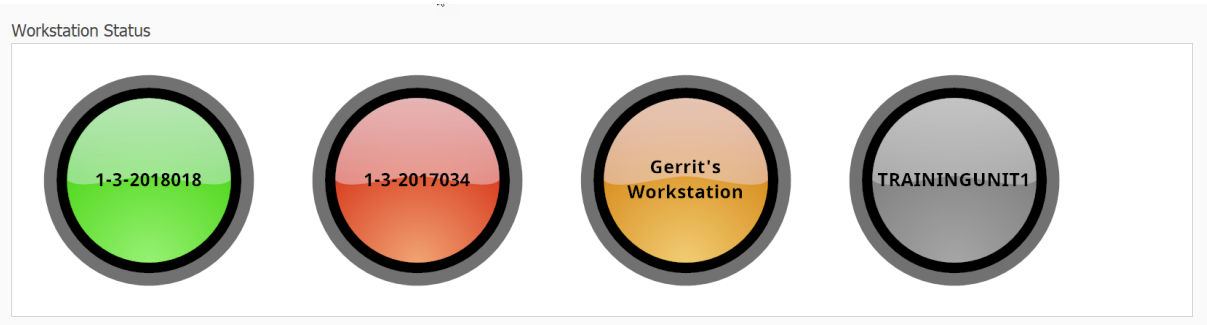
The dashboards below are currently available in the Arkite software:

- **Workstation status:**
  provides you with information on the current status of each workstation that is connected to your server. Only workstations with a license are shown.
- **Job output:**
  provides you with information on the number of jobs that have been executed during a specific period.
- **Cycle Times:**
  provides you with information on the time the execution of a job has taken.
- **Current Step Duration:**
  provides you with real-time information about the current assembly on the workstations connected to your server. Only workstations with a license are shown.

  > ***Note***
  > *A lot more information concerning assembly is logged. You can use those to easily create your own dashboards in a dedicated tool.*

12.3 Available dashboards

Workstation status

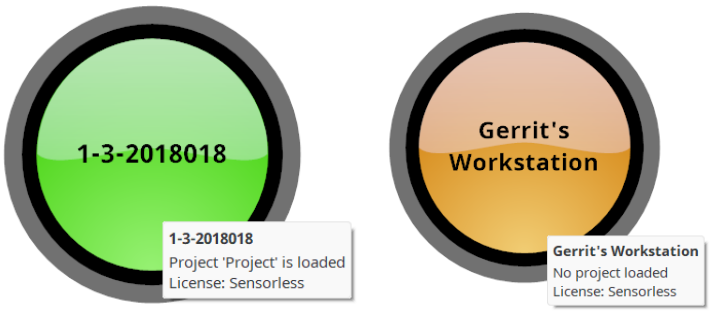


The workstation status dashboard provides you with information on the current status of each workstation connected to your server. Only workstations with a valid license are shown. In other words, editors are not displayed in the overview.

Each workstation is represented by a circle and can have four different colors:

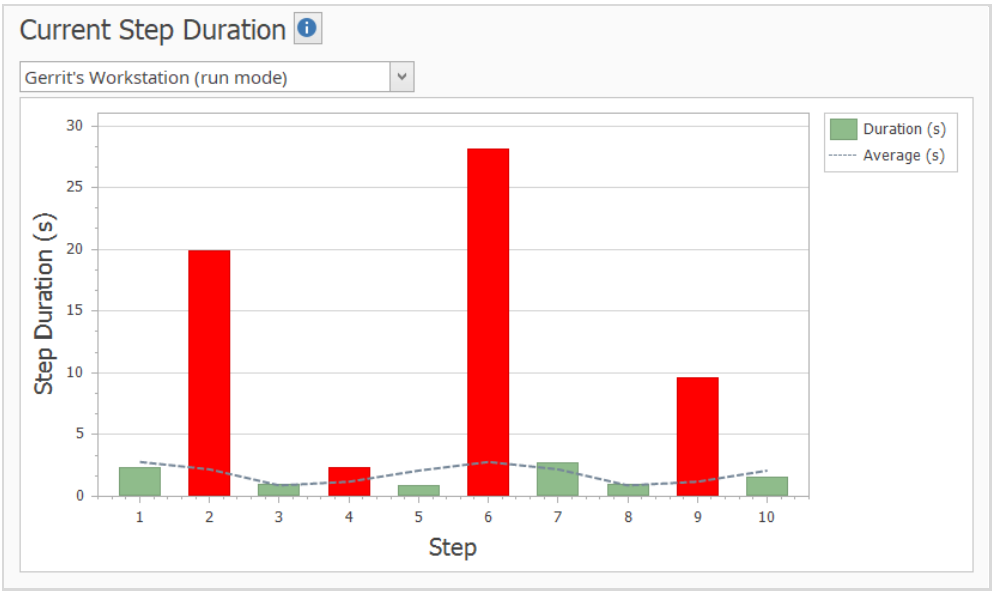
- **Green:**
  the workstation is online, and a project is loaded.
- **Orange:**
  the workstation is online, but currently no project is loaded.
- **Red:**
  the workstation is online, but its software version does not match the software version on the server. It is not possible to load a project
- **Gray:**
  the workstation is currently not connected to the server.

  Tip
  When you hover over a circle with your mouse, more information on the status and the workstation's license type is displayed in a tool tip.



Current step duration

The **Current Step Duration** dashboard provides you with real-time information about the current assembly on the workstations connected to your server. For each workstation with a license, the dashboard shows the duration of the step that is currently executed and the nine previous steps. Both job steps and process steps are taken into account. Steps without operator interaction are ignored.



In the dropdown list above the dashboard, you can select the workstation for which you want to see the step duration.

For every step, you can see:

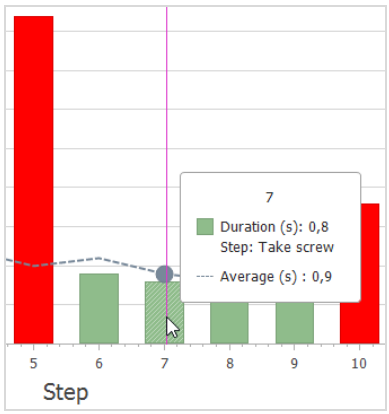
- The **average duration** of that step. The average is calculated based on the last 30 executions of that step. Outliers are not taken into account.
- The **duration in seconds** of the last time the step has been executed (or for the current step: the ongoing duration).

  In case the duration deviates significantly from the average, the bar for that step is shown in red.

  If the deviation from the averages is within range, the bar for that step is shown in green.

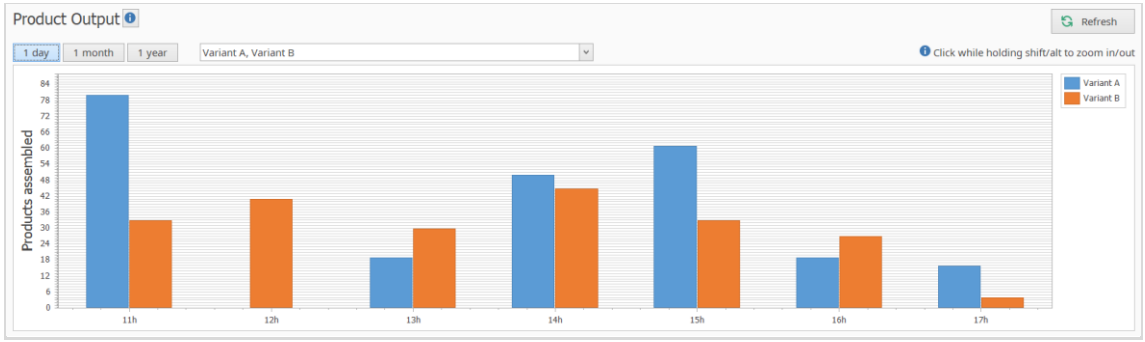
  Tip

  When you hover over a circle with your mouse, the step name and the exact duration and average duration are displayed in a tool tip.



Job Output

The **Job Output** dashboard provides you with information on the number of jobs that have been executed during a specific period.



Click the **1 day** button to display the data for the last 24 hours. Data are split per hour.

Click the **1 month** button to display the data for the last month. Data are split per day.

Click the **1 year** button to display the data for the last 12 months. Data are split per week.

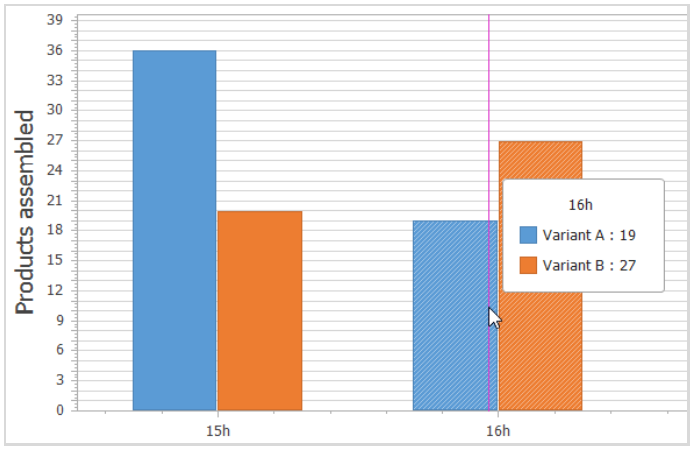
In the **Jobs** dropdown list, you can select which jobs and job variants you want to see. By default, all jobs that have been executed on a workstation connected to your server are shown.

- When the **Group variants/options** checkbox is checked, the data for all job variants and options is aggregated and one bar per job will be shown.

- When the **Group variants/options** checkbox is not checked, data will be rendered separately for each job variant or combination of job options.

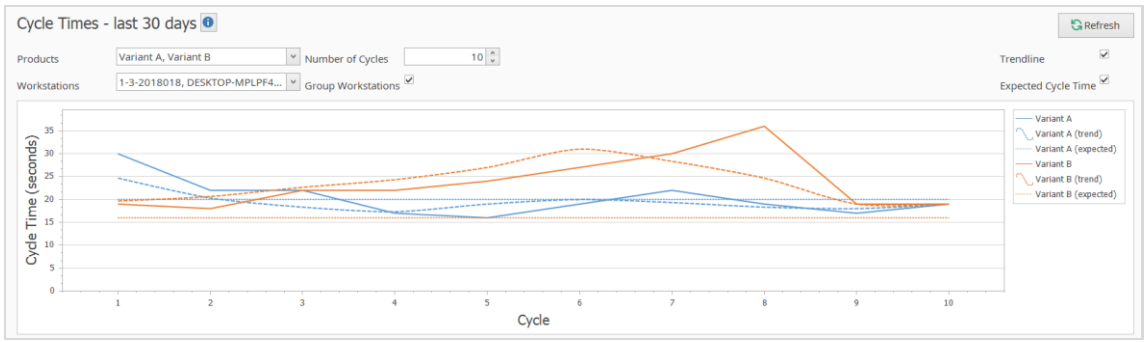
Note

- *Click the **Refresh** button to update the dashboard with the latest data.*
- *If you still have data from before the introduction of jobs in the Arkite software, you can show these data by selecting PRODUCT_LOG in the **Based on** dropdown list.*
- *When you hover over the graph with your mouse, the exact numbers for each bar in the graph are displayed in a tool tip.*



Cycle Times

The **Cycle Times** dashboard provides you with information on the time the execution of a job has taken. By default, the last 10 executions of a job are shown. Only the last 30 days are taken into account.



In the **Jobs** dropdown list, you can select which jobs and job variants you want to see. By default, all jobs that have been executed on a workstation connected to your server are shown.

- When the **Group variants/options** checkbox is checked, the data for all job variants and options is aggregated and one line per job will be shown.
- When the **Group variants/options** checkbox is not checked, data will be rendered separately for each job variant or combination of job options.

In the **Workstations** dropdown list, you can select for which workstations you want to see the cycle times. By default, data for all workstations connected to your server are shown.

- When the **Group Workstations** checkbox is checked, one line for each job (variant) will be shown. You will see the last cycle times for the job (variant), but you won't see on which workstation these assemblies have taken place.
- When the **Group Workstations** checkbox is not checked, a separate line for every workstation will be shown for each job (variant). This allows you to compare cycle times between workstations for the same job (variant).

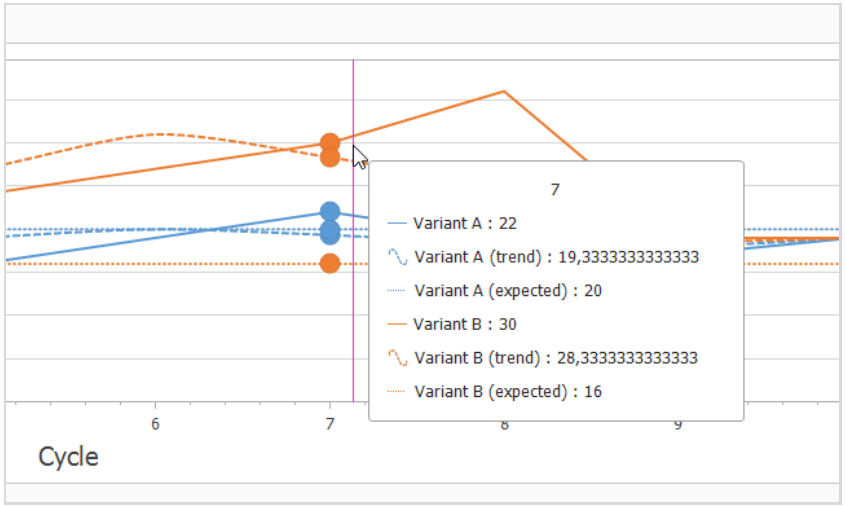
In the **Number of Cycles** dropdown list, you can select how many cycles are shown in the graph.

When the **Trendline** checkbox is checked, for each job (variant) or job (variant)/workstation combination a moving average is shown.

When the **Expected Cycle Time** checkbox is checked, for each job (variant) the expected cycle time is shown. You can set the expected cycle time in the properties of a job or job variant.

Note

- *Click the **Refresh** button to update the dashboard with the latest data.*
- *If you still have data from before the introduction of jobs in the Arkite software, you can show these data by selecting PRODUCT_LOG in the **Based on** dropdown list.*
- *When you hover over the graph with your mouse, the exact numbers for each line in the graph are displayed in a tool tip.*



13. Logging

13.1 About logging

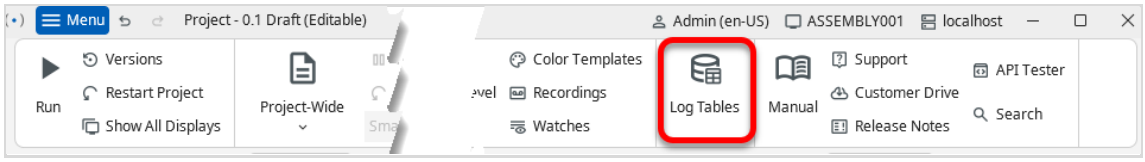
The server contains a database in which information about the platform and its projects are logged. Logging can be used to troubleshoot or reproduce issues, or to provide you with insights on various aspects of your assembly process and the status of your system through a graphical representation of data in dashboards.

A lot of information is logged by default. Additionally, you can also add logging yourself.

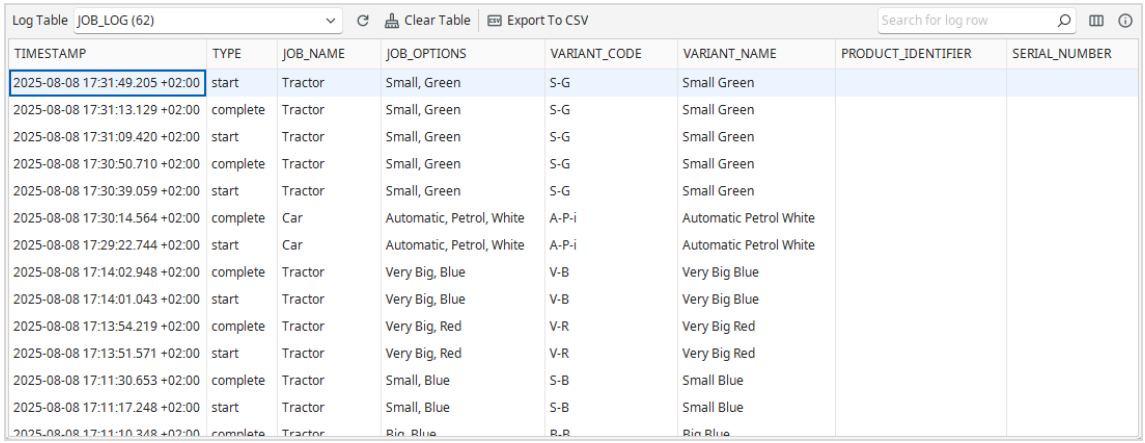
> ***Note***
> *Logging is only visible in the software for users with the **View Log Tables** role.*

13.2 Consulting log tables in the Arkite software

To consult the log tables, click the **Log Tables** button in the ribbon.



The **Log Tables** window appears.



TIMESTAMP	TYPE	JOB_NAME	JOB_OPTIONS	VARIANT_CODE	VARIANT_NAME	PRODUCT_IDENTIFIER	SERIAL_NUMBER
2025-08-08 17:31:49.205 +02:00	start	Tractor	Small, Green	S-G	Small Green		
2025-08-08 17:31:13.129 +02:00	complete	Tractor	Small, Green	S-G	Small Green		
2025-08-08 17:31:09.420 +02:00	start	Tractor	Small, Green	S-G	Small Green		
2025-08-08 17:30:50.710 +02:00	complete	Tractor	Small, Green	S-G	Small Green		
2025-08-08 17:30:39.059 +02:00	start	Tractor	Small, Green	S-G	Small Green		
2025-08-08 17:30:14.564 +02:00	complete	Car	Automatic, Petrol, White	A-P-i	Automatic Petrol White		
2025-08-08 17:29:22.744 +02:00	start	Car	Automatic, Petrol, White	A-P-i	Automatic Petrol White		
2025-08-08 17:14:02.948 +02:00	complete	Tractor	Very Big, Blue	V-B	Very Big Blue		
2025-08-08 17:14:01.043 +02:00	start	Tractor	Very Big, Blue	V-B	Very Big Blue		
2025-08-08 17:13:54.219 +02:00	complete	Tractor	Very Big, Red	V-R	Very Big Red		
2025-08-08 17:13:51.571 +02:00	start	Tractor	Very Big, Red	V-R	Very Big Red		
2025-08-08 17:11:30.653 +02:00	complete	Tractor	Small, Blue	S-B	Small Blue		
2025-08-08 17:11:17.248 +02:00	start	Tractor	Small, Blue	S-B	Small Blue		
2025-08-08 17:11:10.348 +02:00	complete	Tractor	Big, Blue	B-B	Big Blue		

In the **Log Table** dropdown list, select the table you want to consult. By default, the log tables below are available:

Log table	Description
GLOBAL_LOG	Records are logged for error messages and default actions. ***Note*** *To change the default actions that are logged, refer to section **13.5 Changing logging settings** on page 266.*
HEARTBEAT	Every 10 seconds, a record is logged when the workstation is online.

Log table	Description
STEP_LOG	A record is logged for every job step that is executed. You can also add logging for process steps here. To do so, navigate to the **Processes t**ab and select the process of your choice. In the **Properties** pane, check the **Log Steps** checkbox.
AUDIT_LOG	When audit logging is enabled, a record is logged for every change in the project or the settings. ***Note*** *For more information on enabling audit logging, refer to section **13.4 Enabling audit logging** on page 265.* In the **Changes** column of the table, you'll find a detailed description of the change in JSON format. Audit records are append-only and protected by a hash chain. Each record has a record hash, and in the changes JSON, the record hash of the previous record is mentioned. ***Example of a JSON change record*** *{"added":{},"removed":{},"changed":{"Model.Entities.Project, Model, Version=2026.1.0.332, Culture=neutral, PublicKeyToken=null":[{"PreviousChangeId":6633368674942531451,"ChangeId":8518132837971970129,"SoftwareRevisionNumber":"26376","SoftwareVersionDate":"05/02/2026","SoftwareVersion":"2026.1.0.332","Description":null,"Details":null,"CreationDate":"2026-02-05T16:00:31.5485285+01:00","VersionDate":"2026-02-05T16:01:17.6852222+01:00","Status":0,"IsTemplate":false,"ParentProjectId":0,"ProjectBaseId":4707583786906154066,"VersionInfo":null,"ChangeNotes":null,"ChangedById":0,"ChangedByUserName":"Admin","ProjectVersion":"0.1","IsBackup":false,"IsHistoricalVersion":false,"GoldenVersion":false,"AutoGenerated":false,"Id":8306425898705230721,"RemovedId":0,"Imported":false,"UnitId":0,"JobId":0,"ProjectId":0,"Name":"new project","Comment":null}],"Model.Entities.Jobs.Job, Model, Version=2026.1.0.332, Culture=neutral, PublicKeyToken=null":[{"ImageResourceId":0,"DefaultExpectedCycleTime":"00:01:00","ProductIdentifier":null,"Index":100,"Parent":{"IncludeInCrossReference":false,"Index":0,"Parent":null,"ParentJobId":0,"Id":679485491207232092,"RemovedId":0,"Imported":false,"UnitId":0,"JobId":0,"ProjectId":8306425898705230721,"Name":"Root Job","Comment":null},"ParentJobId":679485491207232092,"Id":2079730693800180542,"RemovedId":0,"Imported":false,"UnitId":0,"JobId":0,"ProjectId":8306425898705230721,"Name":"my first job","Comment":null}]}}*
API_LOG	A record is logged for every communication with the Arkite API.
JOB_LOG	A record is logged every time a job is started, finished or cancelled before it's finished. The records in this table are used for generating the **Job Output** and **Cycle Time** dashboards. ***Note*** *For more information about dashboarding, refer to chapter **12 Dashboarding** on page 257.*
COMMUNICATION	Default log table for all communication modules in the project. A record is logged for every incoming and outgoing message, as well as error messages related to the communication module. ***Note*** *For more information about communication modules, refer to section **7.3 Communication modules** on page 197.*

Click **Refresh** to update the information in the log table.

Click the icon to define which columns from the log table you want to display.

Click **Clear Table** to remove the records from a log table.

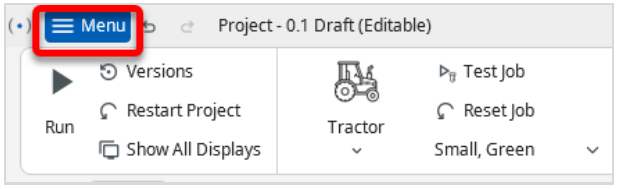
Warning

This action cannot be undone.

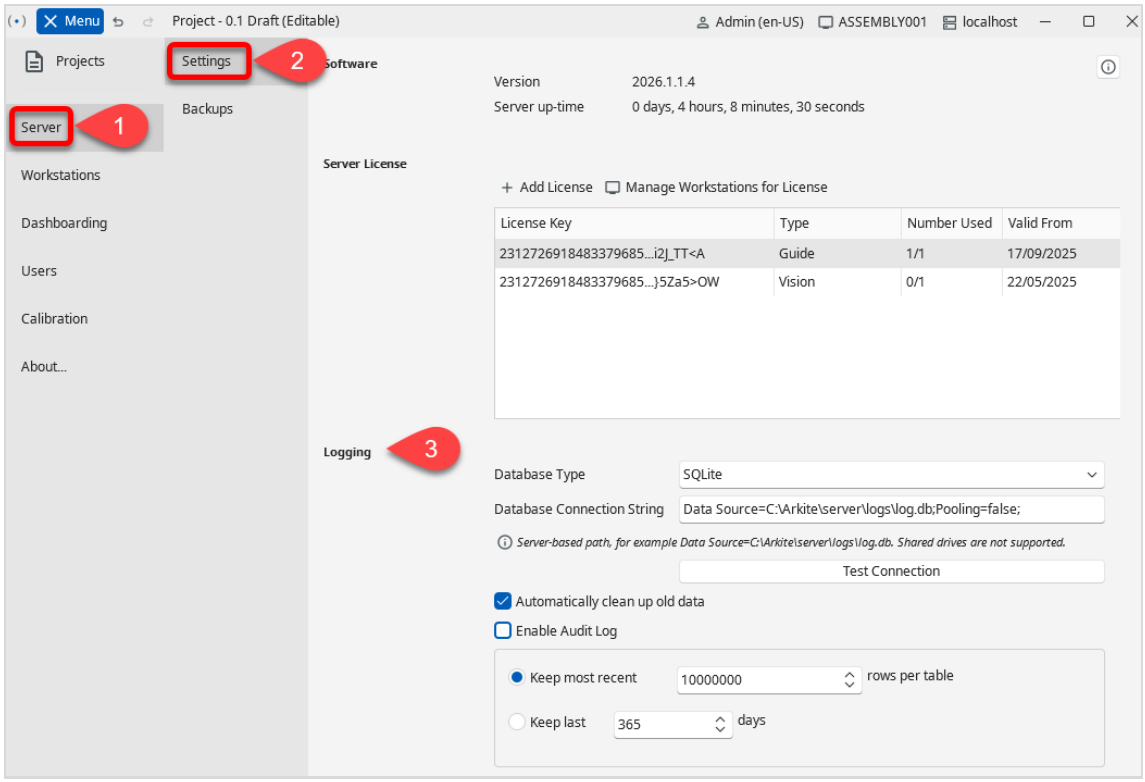
13.3 Changing the logging database, the logging database and the cleanup of old data

To change the logging database and/or the logging database type, execute the steps below:

1. Click **Menu**.



2. Click **Server** and then **Settings**.
3. Under **Logging**, select the **Database Type** and enter the **Database Connection String**.

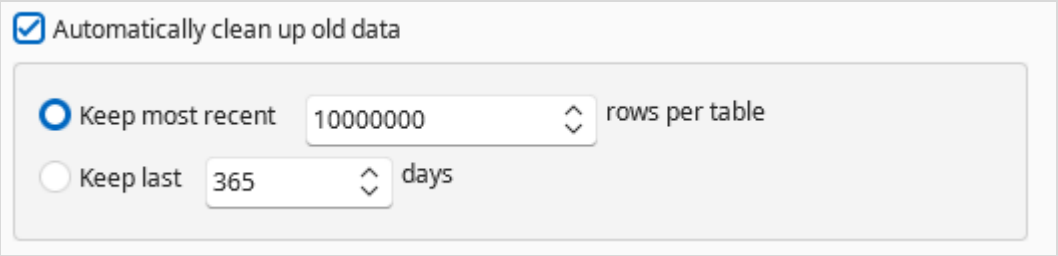


Note

- *Using MySQL may result in performance issues due to the workload involved. We recommend to use MariaDB instead.*
- *When using SQLite, specify a path on the server, for example **C:\Arkite\server\someDb.db**. Shared drives are not supported.*

4. Click **Test Connection** to verify your configuration.

By default, 10 million rows per table are stored. If the number of rows exceeds this threshold, the oldest rows are cleaned up automatically. You can set a different number of rows between 10 000 and 10 million, or a number of days to keep old data.



Note

You can also disable the automatic clean up mechanism, but be aware that in that case logs keep on growing and you will run out of disk space eventually.

13.4 Enabling audit logging

What?

Arkite supports full traceability by tracking all actions and changes performed within the software, including the user responsible, the timestamp of each change, and the affected data with its previous and updated values. Audit logging is disabled by default.

Note

Only admin users can enable audit logging.

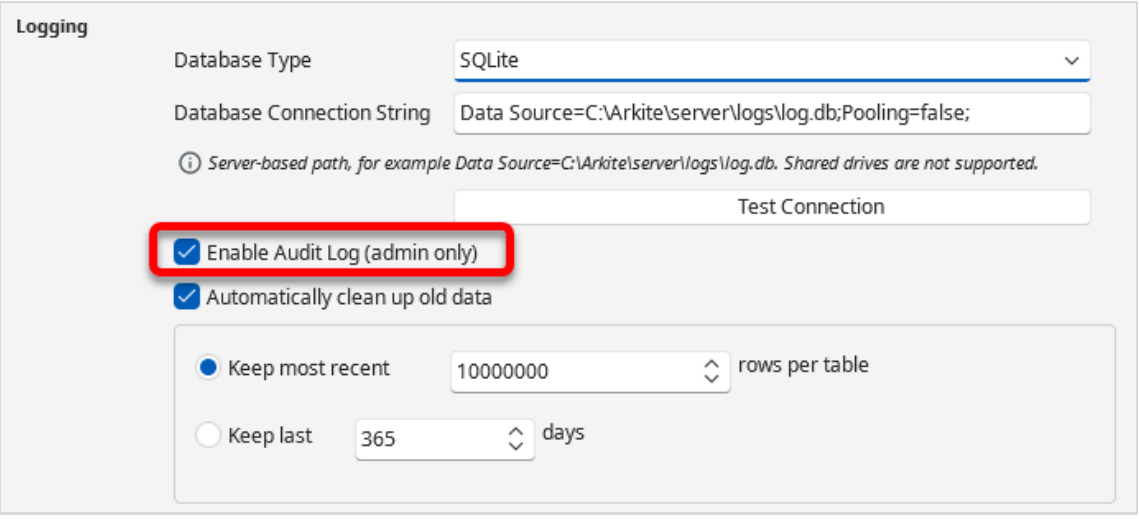
Steps

To enable audit logging, execute the steps below:

1. Click **Menu**.



2. Click **Server** and then **Settings**.
3. Under **Logging**, check the **Enable Audit Log** checkbox.



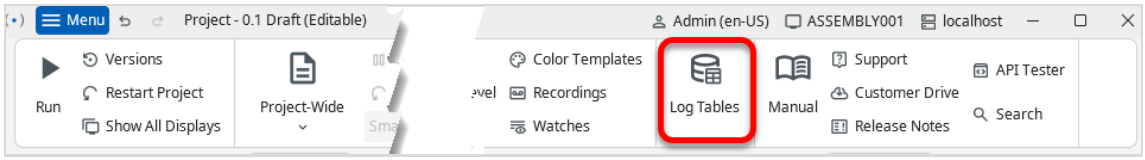
13.5 Changing logging settings

What?

You can change the default actions that are logged in the **Global_LOG** table and the steps that are logged in the **STEP_LOG** table.

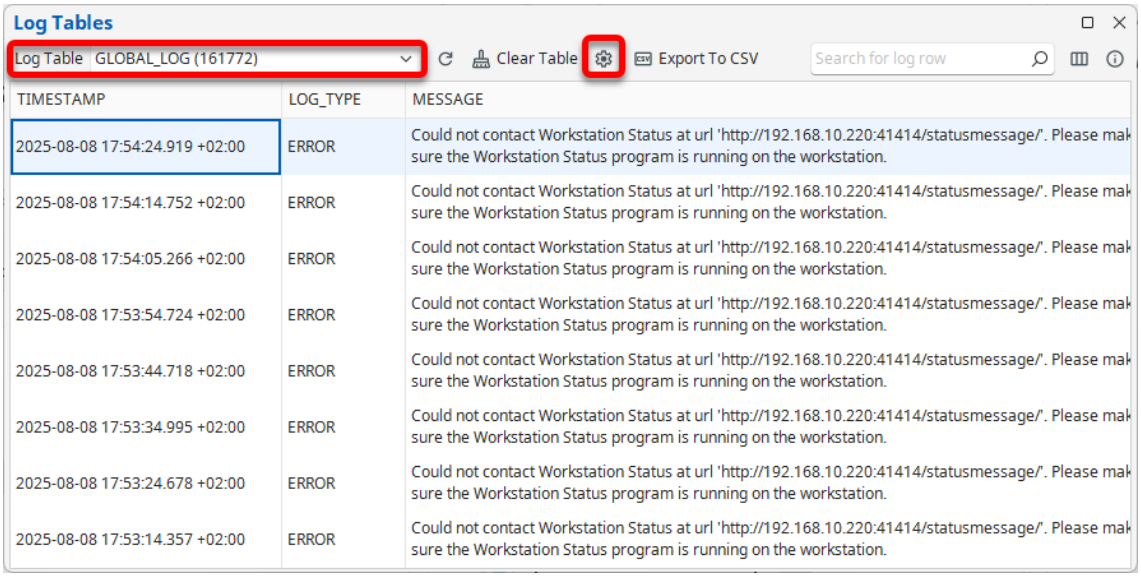
For the GLOBAL_LOG table

1. Click the **Log Tables** button in the ribbon.

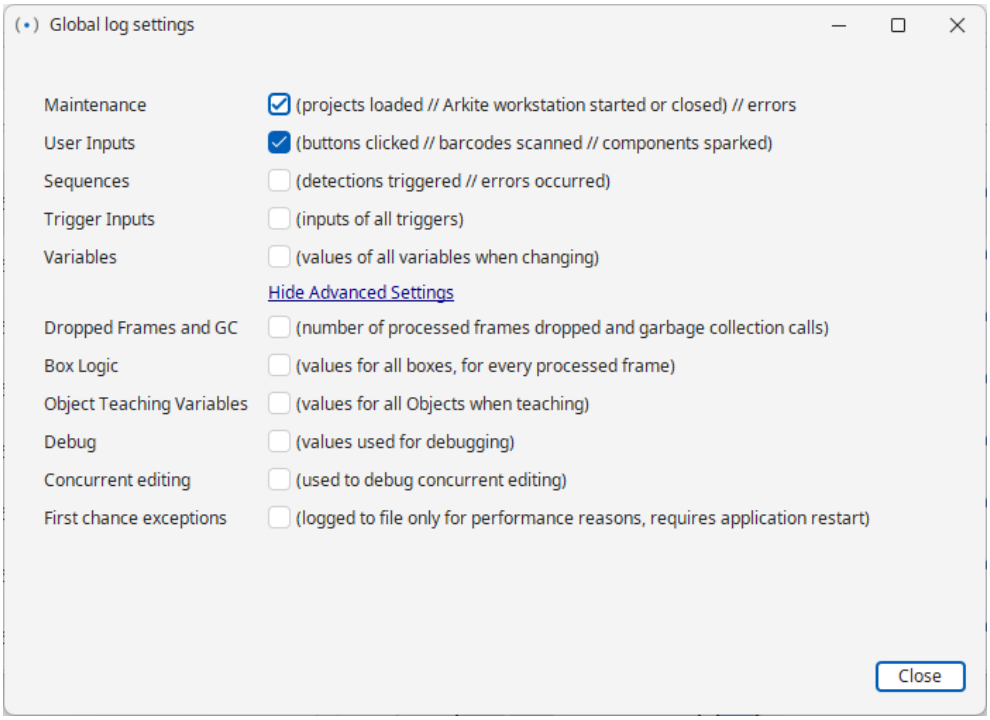


   The **Log Tables** window appears.

2. In the **Log Table** dropdown list, select the **Global_LOG** table.
3. Click the ⚙ button.



4. Check the checkboxes for the actions you want to log in the **Global_LOG** database.



5. Click **Close**.

For the STEP_LOG table

To add logging for the steps of particular process:

1. Navigate to the **Processes** tab.
2. Select a process.
3. In the **Properties** pane, check the **Log Steps** checkbox.

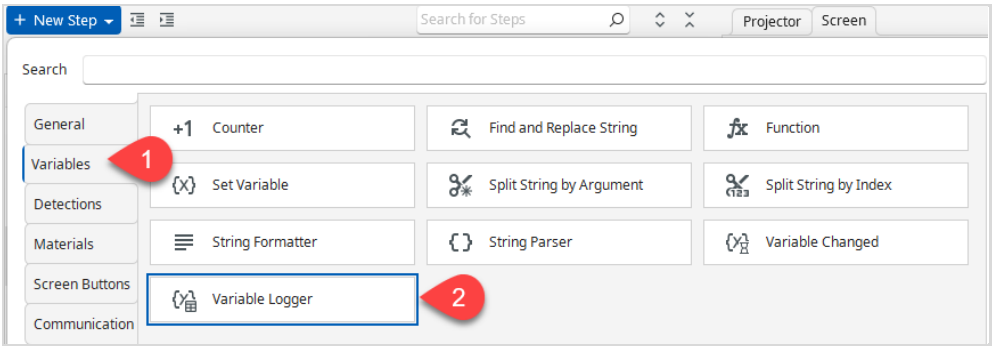
13.6 Adding logging yourself

Add a Variable Logger step

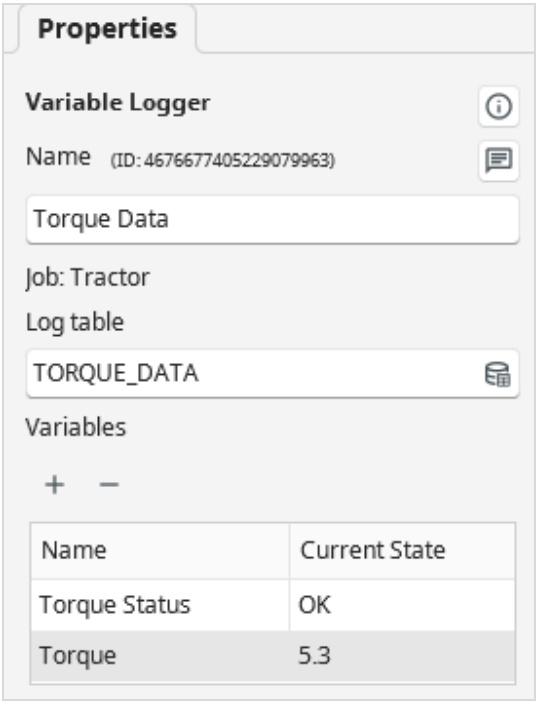
At any point in your process or job, you can add a step that logs the state of one or more variables:

1. Navigate to the **Steps** or **Processes** tab.
2. Click the **New Step** button.

3. Select the **Variables** subtab and click **Variable Logger**.



4. In the **Properties** pane, add the variables for which you want to log the state and specify the **Log Table.**



Configure logging for a communication module

When configuring a communication module, you can specify a log table in which all incoming and outgoing messages and exceptions will be logged. Specify the log table of your choice in the **Log Table** property of the communication module.

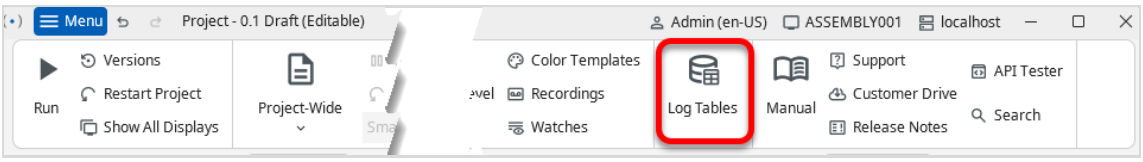
13.7 Exporting log data

What?

Log data can be exported to a csv file for further analysis in Microsoft Excel for example.

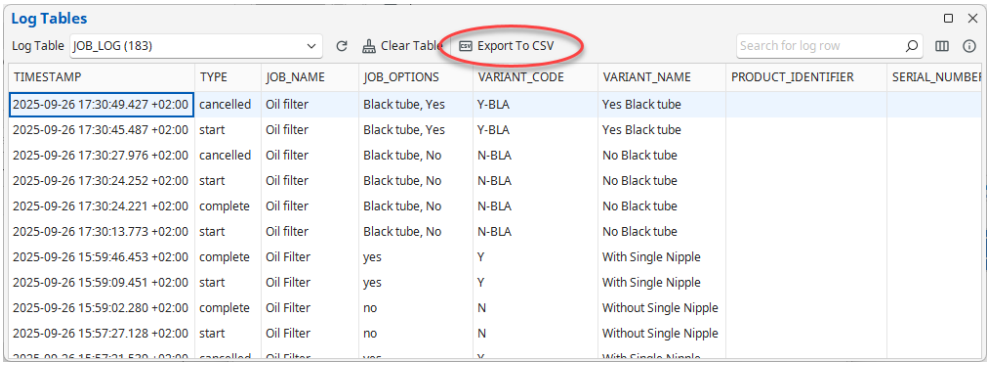
Steps

1. Click the **Log Tables** button in the ribbon.

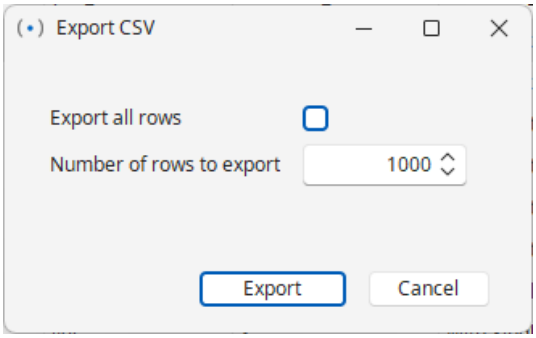


The **Log Tables** window appears.

2. In the **Log Table** dropdown list, select the table you want to export.
3. Click the **Export to CSV** button.



4. In the **Export CSV** window, indicate the number of rows you want to export or check the **Export all rows** checkbox to export everything.



Note

If you select a number of rows, the most recent records in the database will always be exported.

5. Click **Export.**
6. In the **Export CSV** window, navigate to the folder of your choice and click **Save.**
   The selected table is exported to the directory of your choice.

14. Images

14.1 About images

On several locations in the Arkite software, images and videos can be used, for example:

- Shadowboards for an object or tool
- Images linked to a detection
  These are projected on the workbench or shown on the screen when the detection is active.
- Image instructions linked to a process step
  These are projected on the workbench or shown on the screen when the process step is active.
- Images linked to a material
  These are displayed as an icon for picking bins containing this material
- ...

Project-Wide versus Job-Specific images

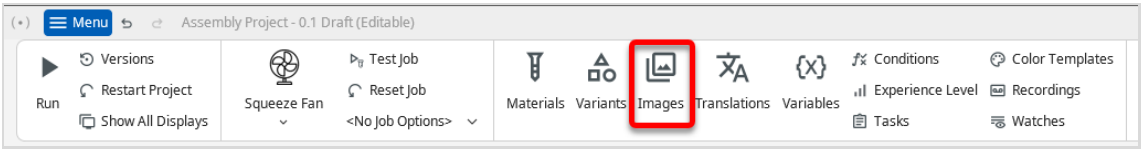
Images can be project-wide or job-specific.

Project-wide images can be used in every job in your project, while job-specific images can be used in one job only, and will only be visible if that job is selected.

14.2 Managing images

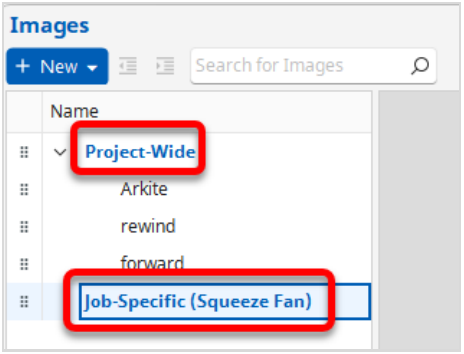
Adding images or videos

1. In the ribbon, click the **Images** button.

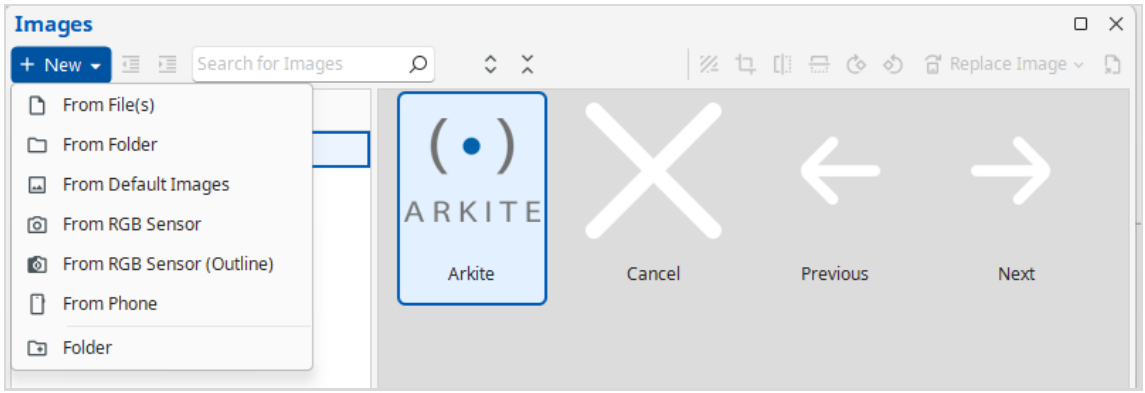


   The **Images** window appears.

2. To add a project-wide image, select the **Project-Wide** group.
   To add a job-specific image, first select the relevant job in the ribbon and then select the **Job-Specific** group.

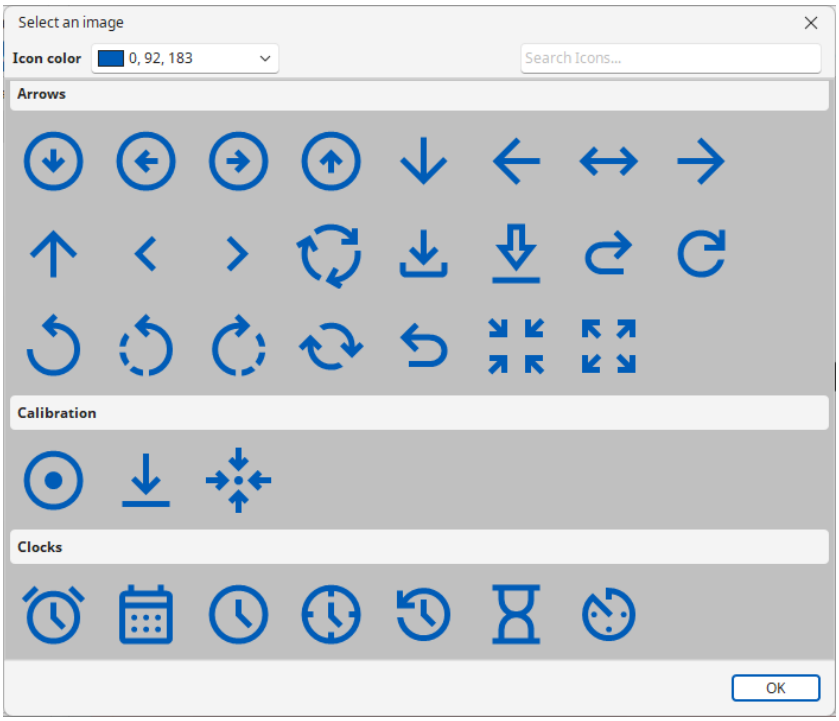


3. Click the **New** button.



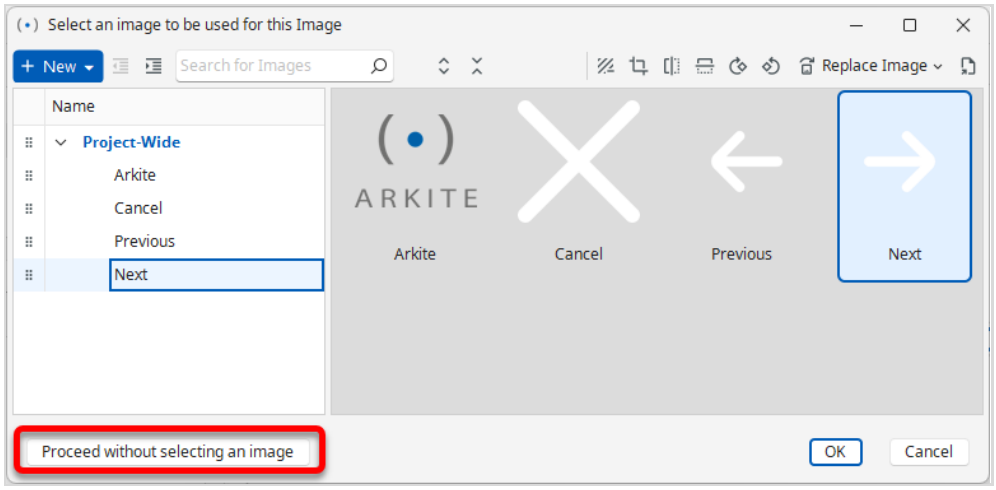
Several options are available:

- **From File(s):**
  Allows you to add image files from your pc or a server for use in the Arkite software.
  Supported file types are: JPEG (.jpg, .jpeg), GIF (.gif), Bitmap (.bmp), PNG (.png), TIFF (.tiff, .tif), SVG (.svg), STA (.sta) and STL (.stl) – to be used for a CAD/CAM detection
  For video files, .mp4 and .mov are supported
- **From Folder:**
  Allows you to add a folder from your pc. All image and video files from that folder will be added for use in the Arkite software.
- **From default images:**
  Allows you to add images from a library available to the Arkite software by default. It contains images that are commonly used, e.g. tools, warnings, etc...
  The images are in SVG format and can be used in different colors.



- **From RGB sensor:**
  Allows you to take a picture with any connected RGB camera, for example the one from the Arkite 3D sensor or a webcam. You have the option to crop the image before you save it.
- **From RGB sensor (outline):**
  Allows you to take a before and an after snapshot with any connected RGB camera, for example the one from the Arkite 3D sensor or a webcam. An outline image will be created containing only the detected changes.
- **From Phone:**
  Allows you to connect an Android phone to your workstation and directly send images or videos from your phone to the Arkite software using the Arkite Companion app. When selecting this option, a pop up with a QR code is displayed. With this QR code, you can download the Arkite Companion app from the Google Play store. Once the Arkite Companion app is installed on your phone, you can use the same QR code to connect your phone to the workstation. As long as your phone is connected, you can keep sending images or videos to Arkite.

  Note
  *You can also use a variable to define which image or video to show in the **Image** visualization. To do so, click **Proceed without selecting an image**. In the properties of the **Image** visualization, you can define the variable to use. For more information, refer to section **4.4.2 Visualization properties** on page 84.*



Organizing your images and videos

A number of buttons are available to help you organize your images and videos:

- Right-click an image or video and select **Delete** to delete it.
- Select an image or video and drag it with your mouse to change its order.
- Select an image or video and click the button to create a subfolder and move the image or video into it.
  Once the subfolder exists, you can rename it and drag other images or videos into it.
- Select an image or video and click the button to move it up one folder.
- Click the button to expand all folders and subfolders.
- Click the button to collapse all folders and subfolders.

Editing images and videos

A number of buttons are available to edit images and videos you have added:

- Select an image and click to mirror it horizontally or click to mirror it vertically.
- Select an image and click to rotate it clockwise or click to rotate it counterclockwise.
- Select an image and click **Replace Image** to replace it.
- Select an image or video and click to adjust the cropping.
  The **Adjust cropping** window appears. Set the zone you want to be visible and click **OK**.

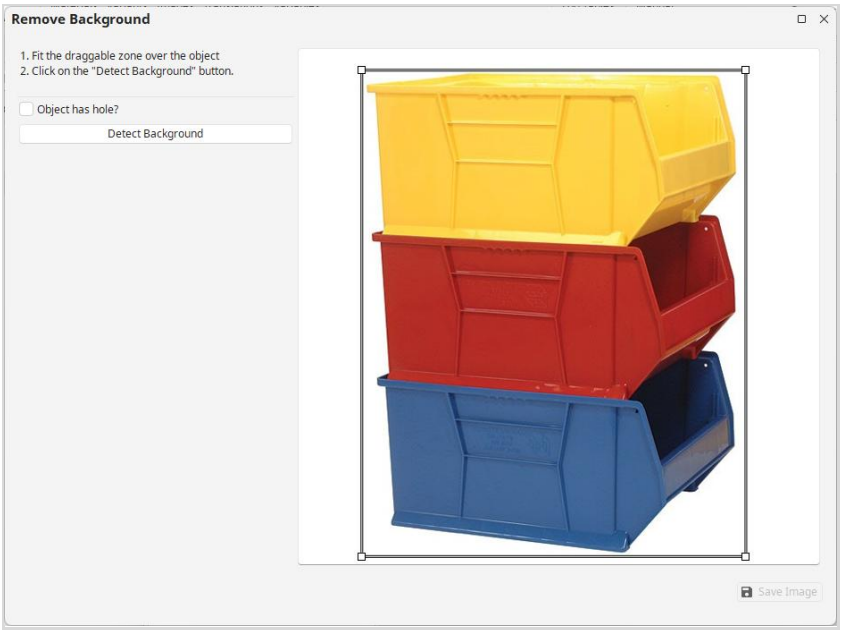


When you add the image, only the selected zone will be visible.

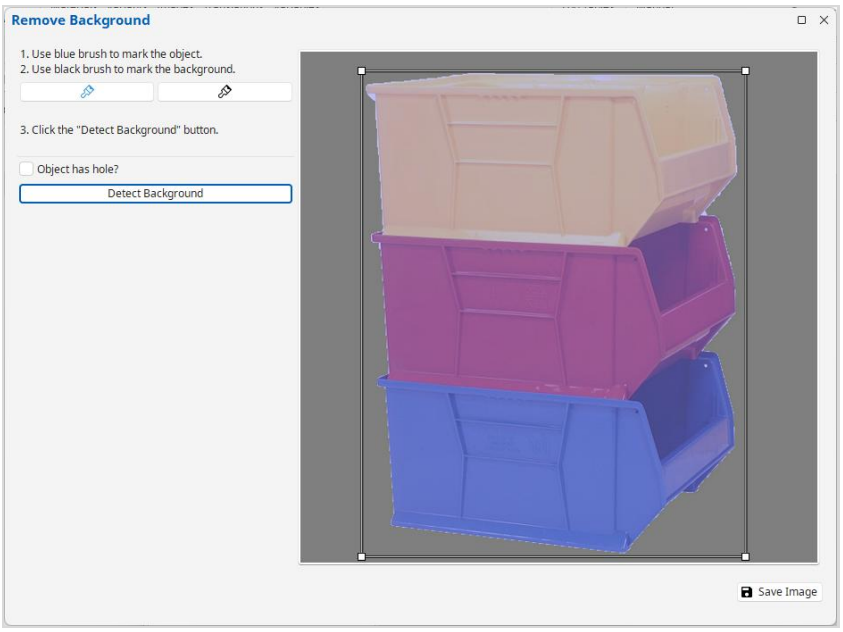
Later, you can still change or remove the cropping. That change will be applied on all locations where the image is used.

- Select an image and click to remove its background.

  The **Remove background** window appears. Set the zone from which you want to remove the background and click **Detect background**.

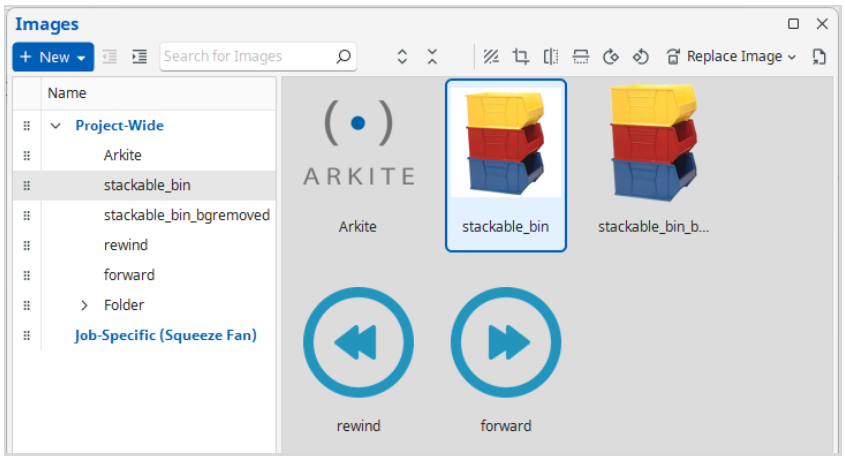


  Afterwards you can finetune the background detection. Click the **black brush** button and indicate an area that is also part of the background but is not detected yet. Click the **blue brush** button and indicate an area that is part of the object but was mistakenly detected as background.



  When you are satisfied with the result, click **Save Image**.

  A copy of the image without background is saved with the extension **_bgremoved** added to the filename.



Exporting images and videos

Select an image or video and click to export it.

Your image or video will be exported to the folder of your choice.

15. Color templates

15.1 About color templates

Color templates can be used to define a visualization's color based on the value of a particular variable. When changing or updating the template, all visualizations linked to the template will be updated automatically.

Note

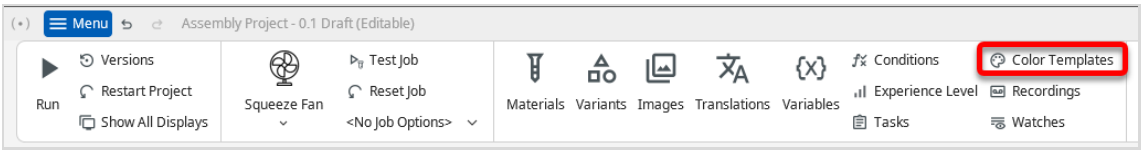
You can also define a visualization's color based on a variable without using templates. In that case, changes need to be applied for every visualization separately. For more information, refer to section ***Defining the color based on the value of a variable*** *on page 114.*

15.2 The Detection Colors template

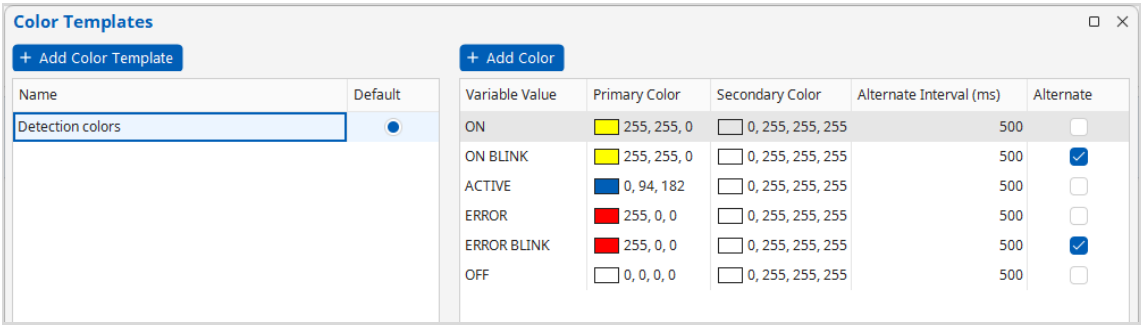
By default, one color template called **Detection colors** is available in the Arkite software. This template is used to define the colors of visualizations on a **Projector** display linked to a detection during the execution of your assembly process.

To consult the Detection Colors template:

1. In the ribbon, click the **Color Templates** button.



2. In the **Color Templates** window, in the left pane, select **Detection Colors**.
3. In the right pane, you can see the colors as defined for each status of the detection.



Note

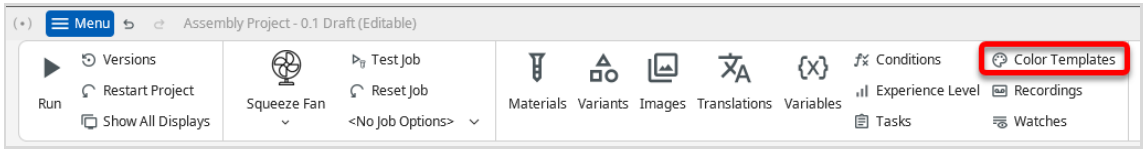
- *In case the* ***Alternate*** *checkbox is checked, the color of the projection will alternate between the* ***Primary Color*** *and the* ***Secondary Color****. By default, the* ***Secondary Color*** *is transparent.*
- *In case the* ***Alternate*** *checkbox is not checked, the* ***Primary Color*** *will always be used.*

The detection's status is defined by the active step in the assembly..

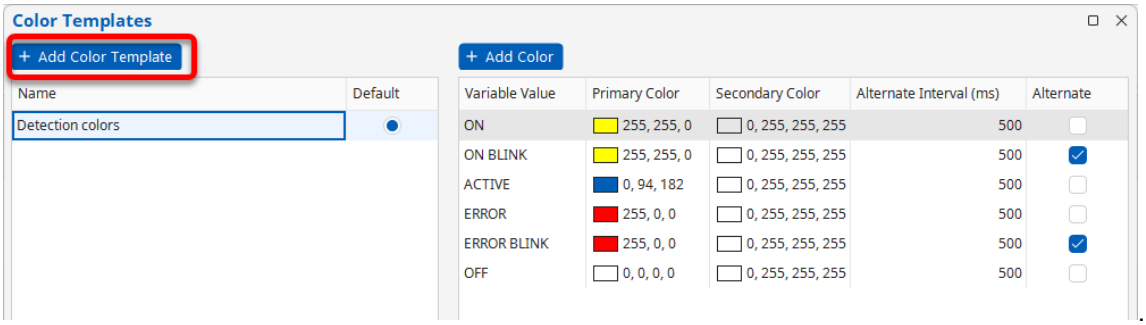
15.3 Managing color templates

Adding color templates

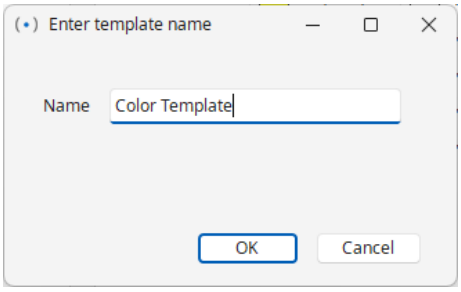
1. In the ribbon, click the **Color Templates** button.



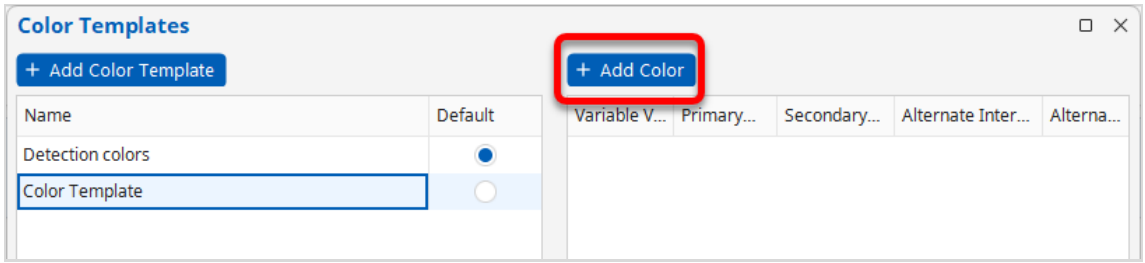
2. In the **Color Templates** window, click **Add Color Template**.



3. In the **Enter template name** window, enter a name for your template and click OK.



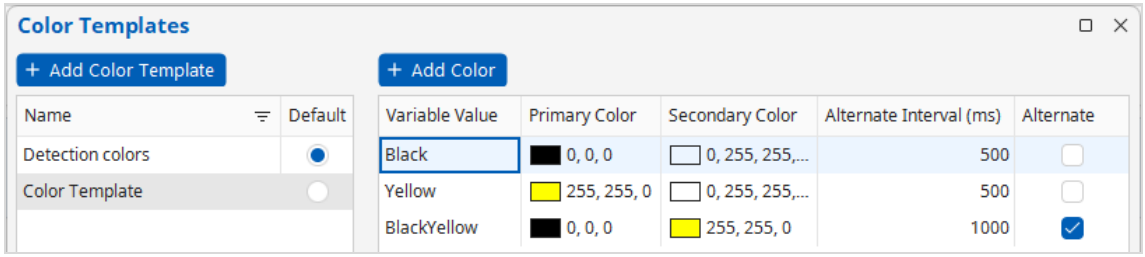
4. Click **Add Color** in the right pane.



5. Enter the information below:
   - In the **Variable Value** column, enter a value for which you want to define a color.
   - In the **Primary Color** column, specify the color you want to use.
   - If you want to alternate between two colors, check the **Alternate** checkbox, specify another color in the **Secondary Color** column and select an **Alternate Interval** (250, 500 or 1000 milliseconds). If you don't check the **Alternate** checkbox, the values in the **Secondary Color** and **Alternate Interval** columns are ignored.

6. Repeat step 5 for all required values.

   In the example below, the visualization's color will be yellow when the value of the **Color** variable is *Yellow* and black when the value of the **Color** variable is *Black*. When the value of the **Color** variable is *BlackYellow*, the color will alternate between black and yellow every second.

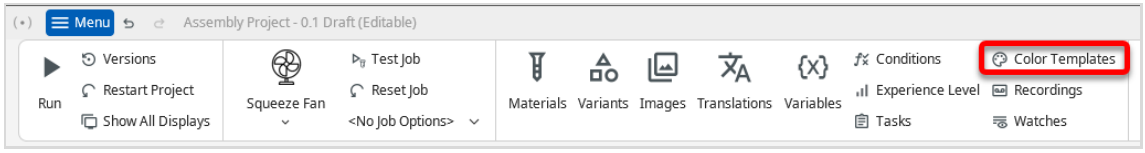


Note

For each color property where you want to use this template, you will have to specify a variable that contains the values defined in the template. For more information, refer to section ***Defining the color based on a template*** *on page 115.*

Editing color templates

1. In the ribbon, click the **Color Templates** button.



2. In the **Color Templates** window, in the left pane, select the template you want to edit.
3. In the right pane, add, remove or modify the variable values of your choice.

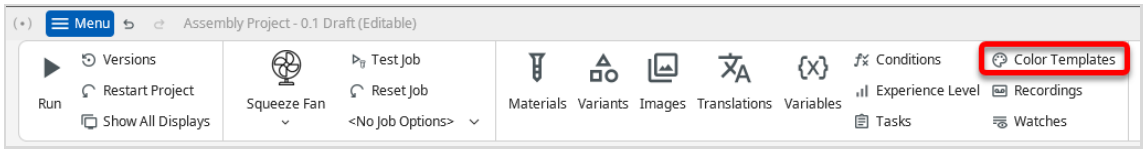
Setting a color template as default

By default, the **Detection colors** template is used to define the colors of visualizations on a **Projector** display linked to a detection during the execution of your assembly process.

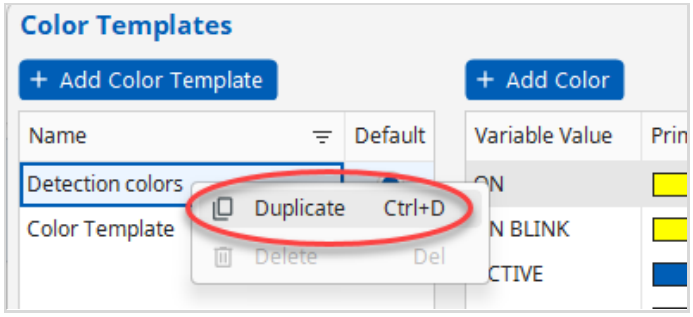
You can configure another template to use for this purpose. It is important that all values used by the projection settings are available in this template. Therefore, it is advised to duplicate the **Detection Colors** template and make the required changes in there.

Execute the steps below:

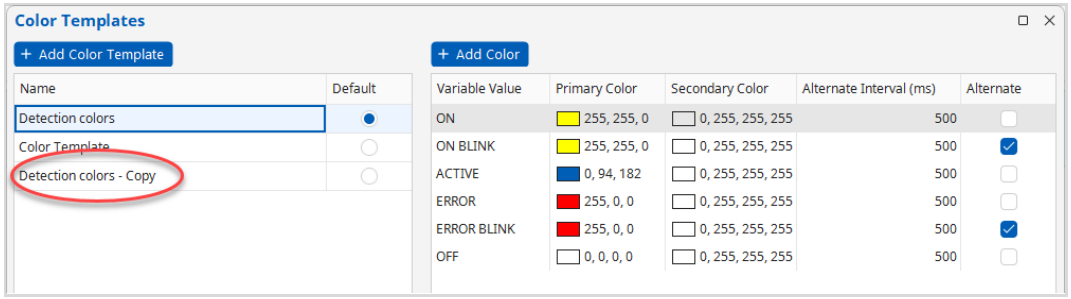
1. In the ribbon, click the **Color Templates** button.



2. In the **Color Templates** window, in the left pane, right-click **Detection Colors** and select **Duplicate**.



3. In the left pane, rename the **Detection Colors – Copy** template if required.

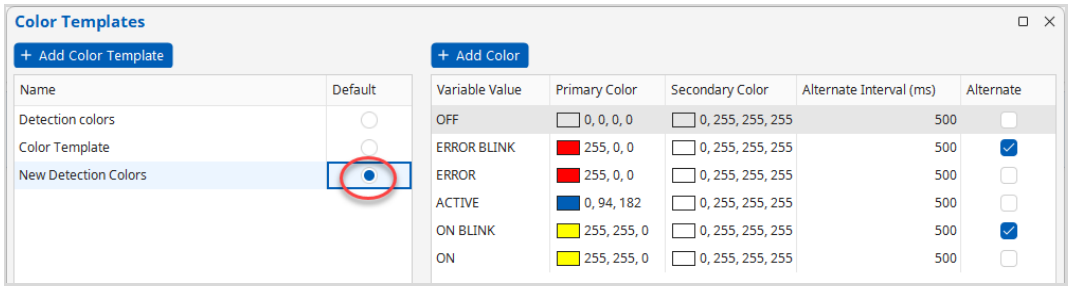


4. In the right pane, modify the colors for the required variable values.

   Important

   For the detection to work properly, do not remove any of the variable values from this template.

5. In the left pane, select the radio button in the **Default** column for the new template.



The new template will now be used for visualizations on a **Projector** display linked to new detections. Visualizations linked to existing detections will still use the original **Detection colors** template. If you want to also use the new template for these visualizations, update their **Brush Color** property, as explained in section **Defining the color based on a template** on page 115.

16. Languages and Translations

16.1 About languages and translations

The Arkite software allows you to define instructions, button labels and all other text elements in different languages. This way, every operator can be supported in his or her own language while executing a process.

By default, the Arkite software allows you to add instructions in English (en-us), but you can add as many languages as you want.

> ***Note***
>
> *By default, instructions will be displayed in the language of the user that is running the project. If no instructions are available in that language, the system will look for a similar language. For example, if the user language is en-GB, en-US will be used. If no similar language is available, the default language in the project will be used. For more information on defining the user language, refer to section **17.4 Managing local users** on page 288.*

You can translate all instructions and text elements in your project using OpenAI. Translations are always done from the default language. To translate from another language, first set that language as your default language.

> ***Warning***
>
> *The translation feature works through an integration with OpenAI. This integration should be enabled. To do so, navigate to **Menu – Server – Settings – AI**.*
>
> *Enter the required configuration data:*
>
> - *OpenAI compatible server URL*
> - *AI model*
> - *API key – you can use your own or contact support@arkite.com to obtain one.*
> - *Temperature: the default value 0 gives the best results for consistent translation of your instructions. However, it is not supported by all models.*

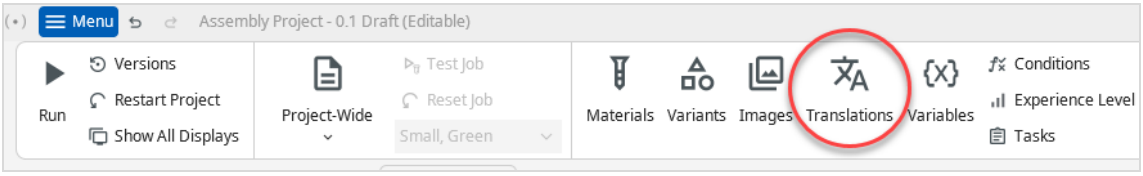


16.2 Managing languages

Adding a language

To add a language:

1. Click the **Translations** button in the ribbon.



2. Click the **New Language** button.
3. Select a language from the list.

   Tip

   *To quickly find a language, use the upper row to filter the list based on the language's **Name**, **Native Name** or **Language Code**.*

4. Select how the texts for the new language should be generated:
   - **Default:** for detection steps, default text instructions will be generated. For other steps and elements, the text fields will be empty.

     Note

     Default instructions are available for English, French, German, Dutch, Spanish and Italian.
   - **Copy:** text elements will be copied from the language you select in the dropdown list.
   - **Translate:** text elements will be translated from the language you select in the dropdown list.
5. Click **OK**.

Deleting a language

To delete a language, right-click the column header for that language and select **Delete Language**.

Warning

- *All texts for that language will be deleted. If you add the language again later, you will have to re-implement any changes you made to default instructions or automatic translations.*
- *You cannot delete the default language..*

Setting a default language

To set the default language, select the language code from the dropdown list in the upper right corner.

The default language is used for showing instructions to users that have a user language that is not available in the project and to generate automatic translations to other languages.

16.3 Managing Translations

Consulting and editing texts in your project

In the ribbon, click the **Translations** button.

The **Translations** window appears. It contains all text elements within the project in all configured languages: text instructions, button labels, text visualizations,…

You can directly edit them in this overview.

Translating texts

You can translate all instructions and text elements in your project using OpenAI. Translations are always done from the default language. To translate from another language, first set that language as your default language.

Several options are possible:

- To translate a specific field, right-click the field in the table and select **Translate from default**.
- To translate only the empty fields, click the **Translate** button, select **Empty fields** and then the language you want to translate**.**
- To translate everything, click the **Translate** button, select **All fields** and then the language you want to translate**.**

Warning

Existing translations can be overwritten. If the field for the default language is empty, the fields for the other languages will also be erased.

Fields for which the translation has been changed will be marked in orange until the **Translations** window is closed.

Tip

*You can also translate a text in the **Properties** of the corresponding control by clicking the*  *button.*

Importing and exporting texts

You can export texts to let them translate outside of the Arkite software, for example by an external party.

To export a language:

1. Click the **Export** button.
2. Select a **Folder** to export to.
3. Select a **Language** to export, or select all languages.
4. Click **OK**.

For each exported language, a file with extension .hdb is created. This file can be opened and edited in Notepad++ or another text editor. After translation, you can import the file into the Arkite software again.

1. Click the **Import** button.
2. Select a **File** to import.
3. Select the correct **Language**.
4. Click **OK**.

The file is imported, and the instructions for the corresponding language in the project are updated.

17. User management

17.1 About user management

Access to the Arkite software is granted through a username and password. That can either be a local user created in the Arkite software, or an Active Directory user if Active Directory authentication has been configured.

The actions that can be performed in the Arkite software depend on the role that is assigned to the user or the Active Directory User Group. A role is a set of permissions.

> ***Important***
> *Users and roles are managed on server level. This means that a user that has been configured can log in to the Arkite software on each workstation that is connected to the server.*

17.2 Managing Authentication Settings

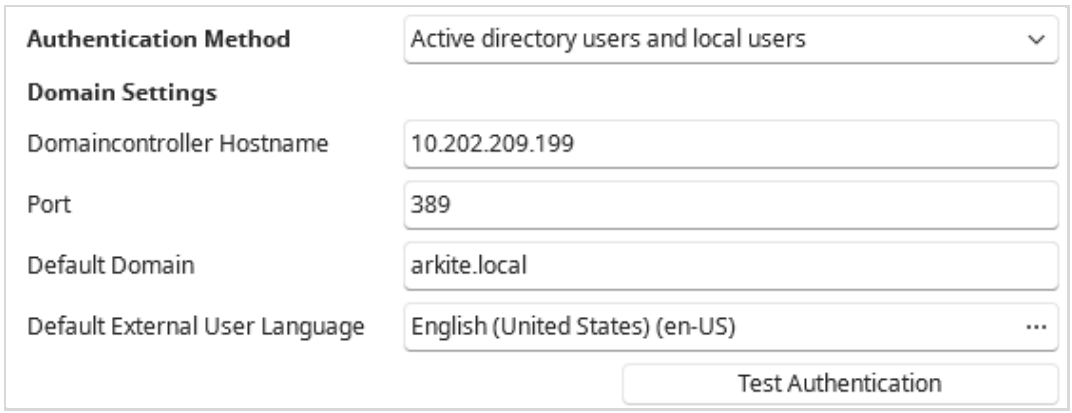
What?

By default, users are created and managed locally in the Arkite software. However, you can also allow authentication with an Active Directory user.

Configuring Active Directory Authentication

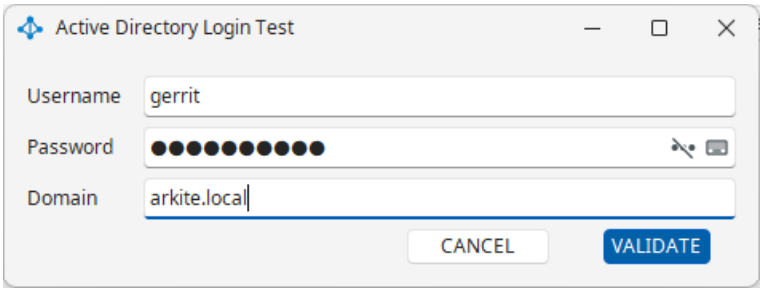
To configure Active Directory, execute the steps below:

1. Click the **Menu** button.
2. Click **Users** and then under **Authentication** click **Settings**.
3. From the **Authentication Method** dropdown list, select **Active directory users and local users**.
4. Specify the domaincontroller hostname, port, default domain and default external user language for your Active Directory setup.

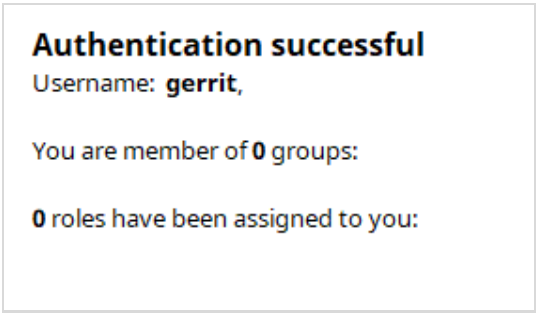


> ***Note***
> *If a language is assigned to the user in Active Directory, this language will also be used in the Arkite software. If no language is assigned in Active Directory, the language specified in the* ***Default External User Language*** *field will be used.*

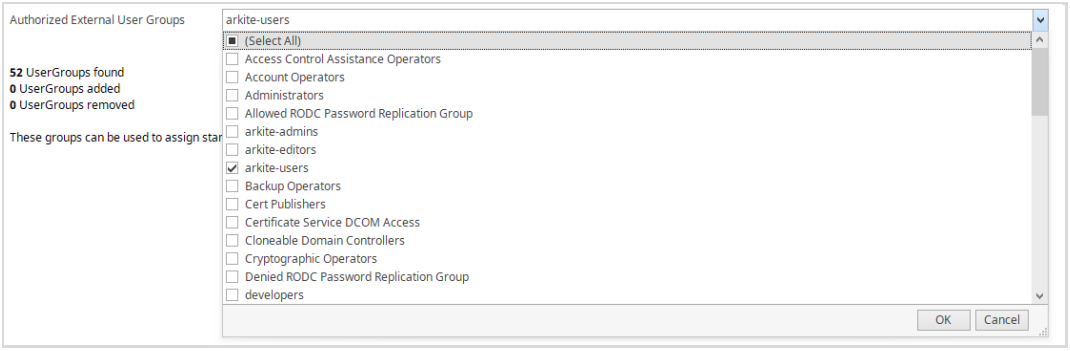
5. To test the connection, click the **Test Authentication** button.
6. Provide an Active Directory user name and password and click **Validate**.



If the connection was successful, an authentication successful message will be shown.



7. To fetch the user groups from your Active Directory, click **Update External User Groups**.
8. Provide an Active Directory user name and password and click **Validate**.
   A message is shown with the number of found user groups.
9. In the **Authorized External User Groups** list, indicate the user groups you want to grant access to the Arkite software and click **OK**.



Result

All Active Directory users that are part of an authorized user group will be able to login to the Arkite software using their Active Directory login and password.

By default, an authorized Active Directory user will have read-only access to projects in status **Production**, similar to the Operator user**.** To grant more permissions to the Active Directory user, assign the external user group to a role.

17.3 Managing roles

Default roles

By default, four roles are configured in the Arkite software:

- **System Admin:**
  users with this role can edit settings and manage projects and users.
- **User Manager:**
  users with this role can configure users and roles.
- **Editor**:
  users with this role can edit projects and move them from **Draft** to **Review** and from **Review** to **Production** (if no approval is required).
- **Approver**:
  users with this role can do the same as editors, but can also approve or reject a project.
- **View Sensor Streams:**
  users with this role can view, record and take snapshots of the streams of the sensors (3D sensor, Vision sensor, RGB camera…) in the Arkite software.
- **View Dashboarding Tab:**
  users with this role can view the **Dashboarding** tab in the Arkite software.
- **View Log Tables:**
  users with this role can view the log tables in the Arkite software.

The table below gives an overview of the permissions of these default roles:

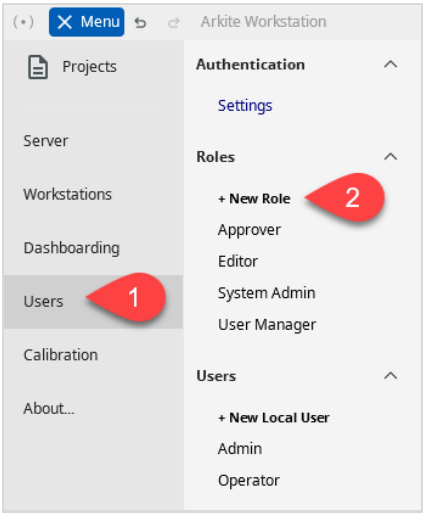
Permission	Description	System Admin	User Manager	Editor	Approver	View Sensor Streams	View Dashboarding Tab	View Log Tables
Manage Users	Add, modify and delete users and roles. Assign roles to users and permissions to roles.	✓	✓	-	-	-	-	-
Edit Settings	Edit settings of the server.	✓	✓	-	-	-	-	-
Manage Projects	Add, modify and delete projects	✓	-	✓	✓	-	-	-
Approve Projects	Approve or reject a project	-	-	-	✓	-	-	-
View Sensor Streams	View and record the sensor streams inside the software	-	-	-	-	✓	-	-

Permission	Description	System Admin	User Manager	Editor	Approver	View Sensor Streams	View Dashboarding Tab	View Log Tables
View Dashboard Tab	View the Dashboarding tab in the software	-	-	-	-	-	✓	-
View Log Tables	View the log tables in the software	-	-	-	-	-	-	✓

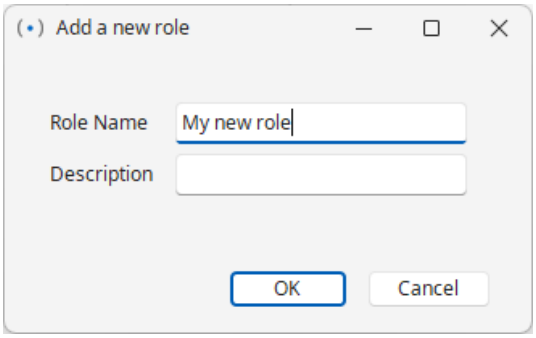
Adding roles

To add a role, execute the steps below:

1. Click **Menu**.
2. Click **Users** and then click **New Role**.



3. Provide a name and (optionally) a description for the new role and click **OK**.

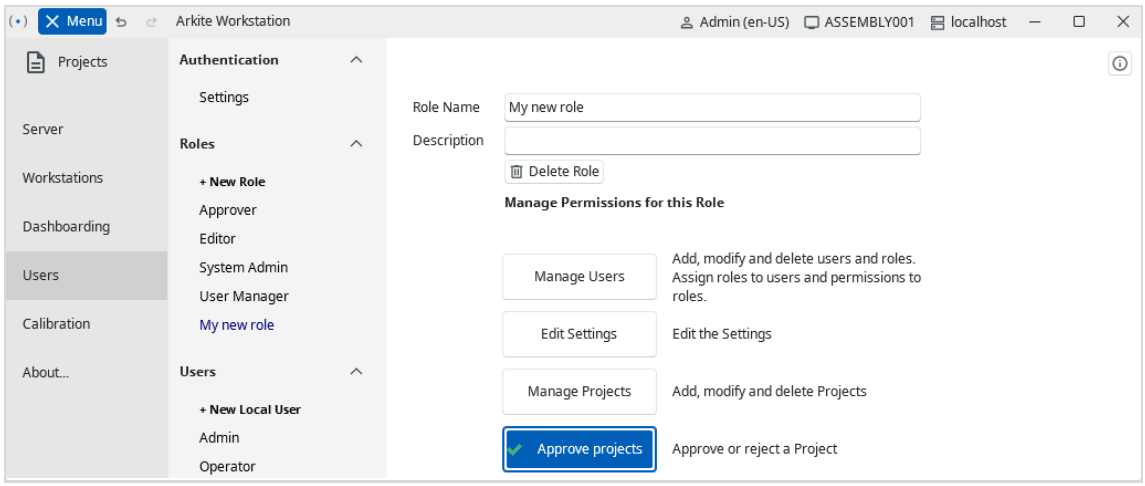


4. (optional) If Active Directory configuration has been configured, you can assign the role to one or more Active Directory user groups in the **User Groups** dropdown list.

   Note

   *An Active Directory user will only have access to the software if the selected user group or another user group he is part of is marked as an **Authorized External User Group** in the authentication **Settings**.*

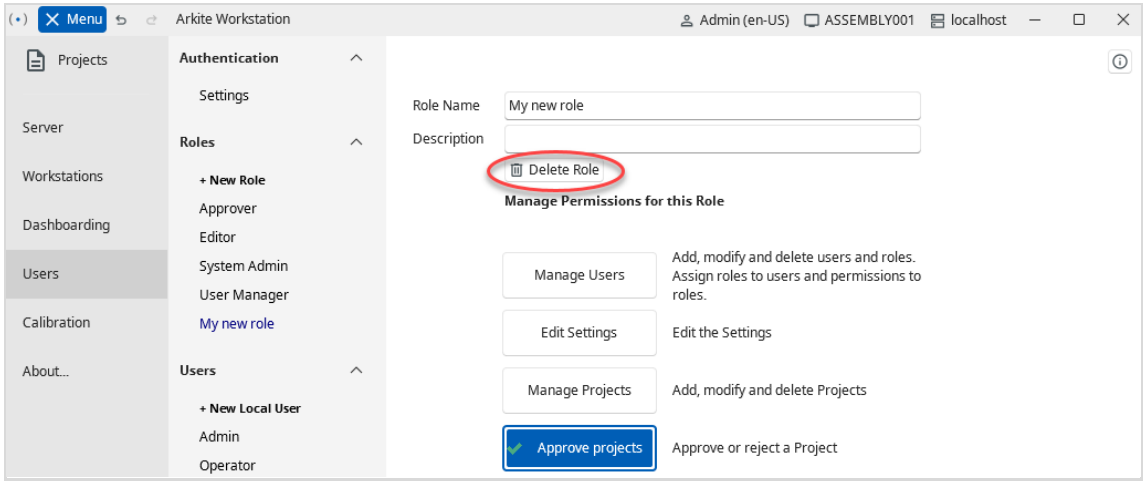
5. Click the permissions you want to assign to the role.
   In the example below, only the permission **Approve Projects** has been assigned to the role.



Editing roles

To edit an existing role, execute the steps below:

1. Click **Menu**.
2. Click **Users.**
3. Click the role you want to edit.
4. Update the name, description and permissions of the role as required.
   If you want to delete the role, click the **Delete Role** button.



Note

The default roles cannot be deleted nor edited. This way, you cannot accidentally lock yourself out of the application.

17.4 Managing local users

Default local users

By default, two local users are configured in the Arkite software:

- **Admin:**
  Can do everything, except approving or rejecting projects.
- **Operator:**
  Has read-only access to projects in status **Production**. No password nor roles have been assigned to this user.

  Note

  *If you have configured Active Directory authorization, be aware that by default it is still possible to log in with this user and get read-only access to the software. If you do not want this, you should set a password for the **Operator** user.*

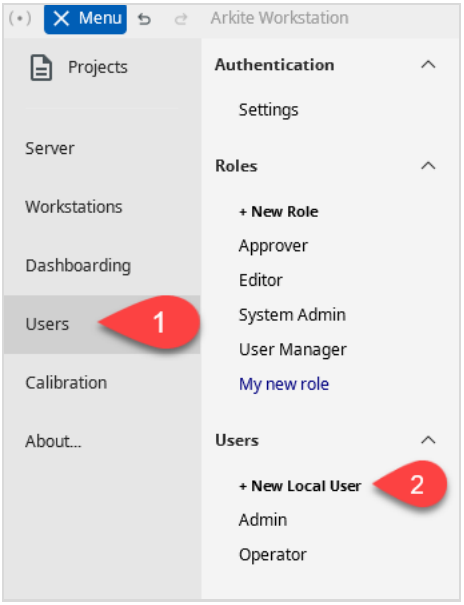
The table below gives an overview of the roles of these default local users:

Role	Admin	Operator
System Admin	✓	-
User Manager	✓	-
Editor	-	-
Approver	-	-
View Sensor Streams	✓	-
View Log Tables	✓	-
View Dashboard Tab	✓	-

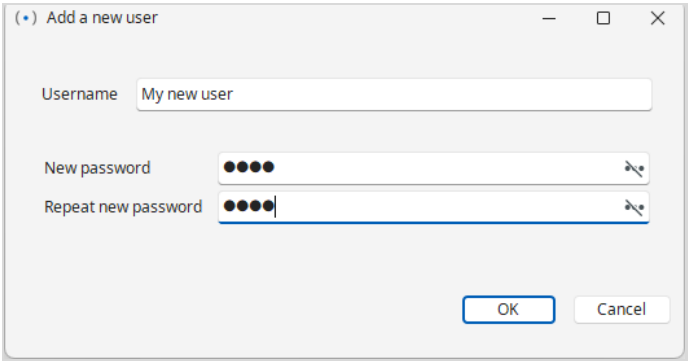
Creating local users

To add a user, execute the steps below:

1. Click **Menu**.
2. Click **Users** and then click **New Local User**.



3. Provide a username and (optionally) a password for the new user and click **OK**.



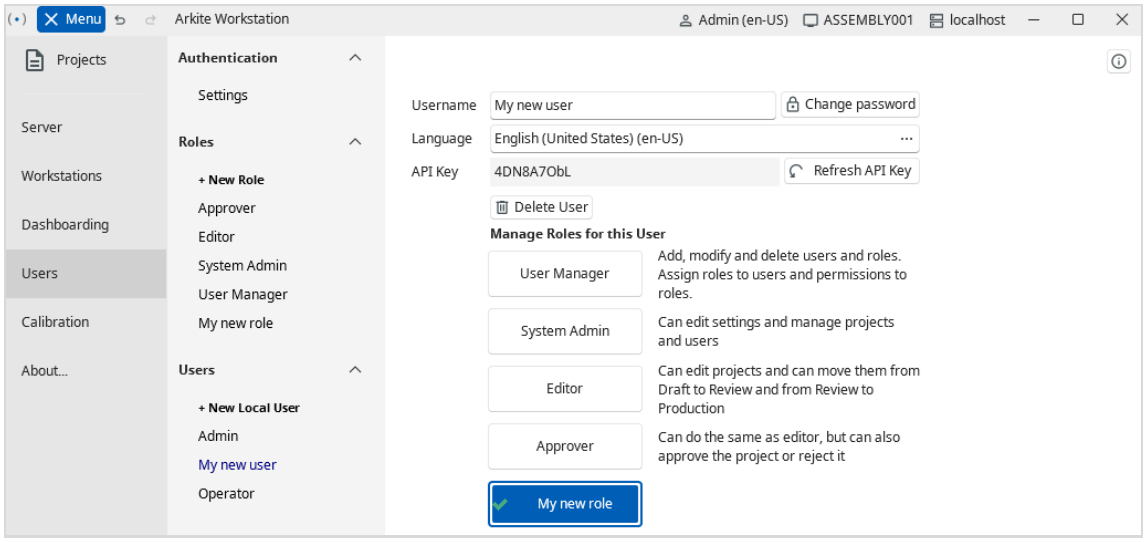
4. Select a language for the user.
   Text instructions from all projects will be shown in this language when this user is logged in.

   Note

   *If no instructions are available in the user's language, the system will first check if there are instructions available in a language with the **same primary language subtag**. If this is not the case, the system will fall back to **en-US** as a default language.*

   *For example, if user's language is **fr-BE** and no instructions are available in that language, instructions will be shown in **fr-FR** if available. Otherwise, **en-US** will be used.*

5. Click the roles you want to assign to the user.
   In the example below, only the new role we created in the section above has been assigned to the user.



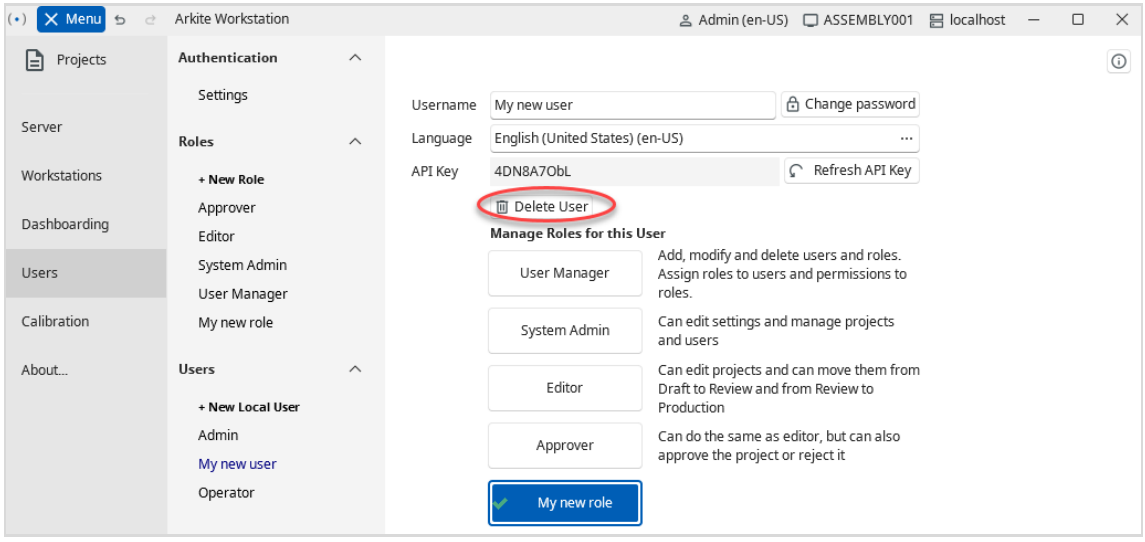
Note

For every user that is added, the system automatically generates an API key. This key is used to authenticate when accessing the Arkite API. You can let the system generate a new key by clicking the ***Refresh API Key*** *button. When you do so, existing functionality will not work anymore though.*

Editing users

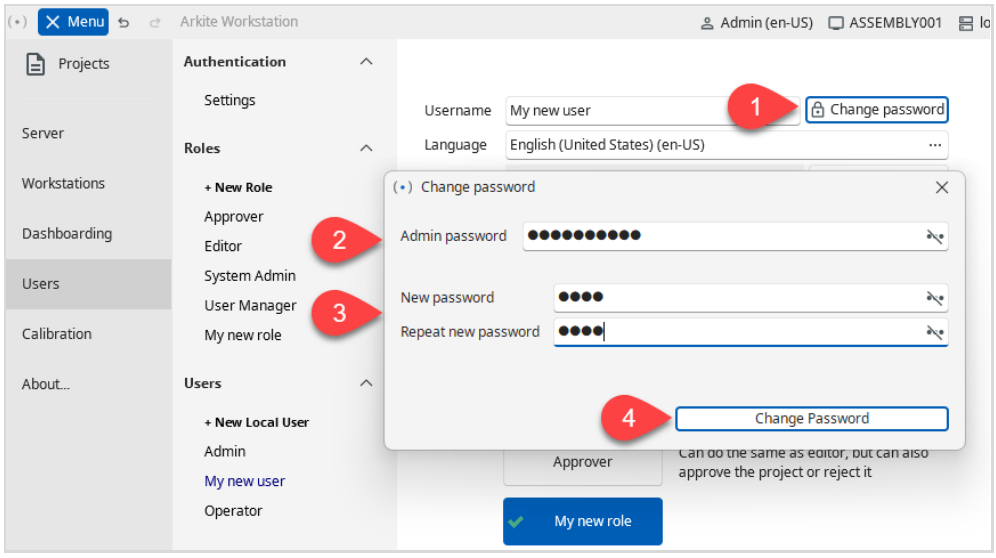
To edit an existing user, execute the steps below:

1. Click **Menu**.
2. Click **Users**.
3. Click the user you want to edit.
4. Update the name, language and roles of the user as required.
   If you want to delete the user, click the **Delete User** button.



To change a user's password:

- click the **Change Password** button.
- Provide your own password, the new password for the user, and click **Change Password**.



Note

- *You can click the button to make the password you've entered visible and verify if you haven't made any typing mistakes.*
- *You can't remove any of the default users.*
- *It is also not possible to edit the roles of the **Admin** user. This way, you cannot accidentally lock yourself out of the application.*

18. The Arkite API

18.1 About the Arkite API

The Arkite software has a flexible Application Programming Interface (API) that allows swift communication with other software applications in the software ecosystem of the manufacturer like MES and ERP systems.

This API can be used to push data from MES into the Arkite software in order to enable the on-demand flexible production. This integration will avoid stock pile-up and can benefit from the fact that the Arkite software is flexible to be programmed to deal with product variants on the fly.

It contains rest calls for retrieving and modifying project data, as well as a number of calls to run operations on a currently running workstation.

18.2 API examples

The Arkite API can be used for example to:

- Schedule tasks on a workstation when an order is received by your ERP system
- Load a project on a specific workstation when a new workday starts
- Update the picking bin content from your WMS system
- Get the status of a detection
- ...

Note

*For most entities in the Arkite software, an object ID is displayed next to the label of the **Name** property on the **Properties** pane. In the **Workstations** overview, the workstation's object ID is shown in the **Workstation ID** column. You can copy it to use it when working with the Arkite API.*

18.3 More information

Extended documentation of the Arkite API can be found on SwaggerHub:

https://app.swaggerhub.com/apis-docs/Arkite/ArkiteAPI/2026.1

For testing, you can use the Open API Tester. To access this page, click the **API Tester** button in the ribbon of the software.

